Introduction to Mathematical Logic

FOURTH EDITION

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To Arlene

Contents

Preface

This is a compact introduction to some of the principal topics of mathematical logic. In the belief that beginners should be exposed to the easiest and most natural proofs, I have used free-swinging set-theoretic methods. The significance of a demand for constructive proofs can be evaluated only after a certain amount of experience with mathematical logic has been obtained. If we are to be expelled from 'Cantor's paradise' (as non-constructive set theory was called by Hilbert), at least we should know what we are missing.

The major changes in this new edition are the following.

1. In Chapter 2, a section has been added on logic with empty domains, that is, on what happens when we allow interpretations with an empty domain.
2. In Chapter 4, Section 4.6 has been extended to include an outline of an axiomatic set theory with urelements.
3. The subjects of register machines and random access machines have been dropped from Section 5.5 Chapter 5.
4. An appendix on second-order logic will give the reader an idea of the advantages and limitations of the systems of first-order logic used in Chapters 2–4, and will provide an introduction to an area of much current interest.
5. The exposition has been further streamlined, more exercises have been added, and the bibliography has been revised and brought up to date.

The material of the book can be covered in two semesters, but, for a one-semester course, Chapters 1–3 are quite adequate (omitting, if hurried, Sections 1.5, 1.6 and 2.10–2.16). I have adopted the convention of prefixing a D to any section or exercise that will probably be difficult for a beginner, and an A to any section or exercise that presupposes familiarity with a topic that has not been carefully explained in the text. Bibliographic references are given to the best source of information, which is not always the earliest paper; hence these references give no indication as to priority.

I believe that the essential parts of the book can be read with ease by anyone with some experience in abstract mathematical thinking. There is, however, no specific prerequisite.

This book owes an obvious debt to the standard works of Hilbert and

Bernays (1934; 1939), Kleene (1952), Rosser (1953) and Church (1956). I am grateful to many people for their help and would especially like to thank the following people for their valuable suggestions and criticism: Richard Butrick, James Buxton, Frank Cannonito, John Corcoran, Newton C.A. da Costa, Robert Cowen, Anil Gupta, Eric Hammer, Bill Hart, Stephen Hechler, Arnold Koslow, Byeong-deok Lee, Alex Orenstein, Dev K. Roy, Atsumi Shimojima and Frank Vlach.

Elliott Mendelson
August 1996

Introduction

One of the popular definitions of logic is that it is the analysis of methods of reasoning. In studying these methods, logic is interested in the form rather than the content of the argument. For example, consider the two arguments:

1. All men are mortal. Socrates is a man. Hence, Socrates is mortal.
2. All cats like fish. Silvy is a cat. Hence, Silvy likes fish.

Both have the same form: All A are B. S is an A. Hence, S is a B. The truth or falsity of the particular premisses and conclusions is of no concern to logicians. They want to know only whether the premisses imply the conclusion. The systematic formalization and cataloguing of valid methods of reasoning are a main task of logicians. If the work uses mathematical techniques or if it is primarily devoted to the study of mathematical reasoning, then it may be called *mathematical logic*. We can narrow the domain of mathematical logic if we define its principal aim to be a precise and adequate understanding of the notion of *mathematical proof*.

Impeccable definitions have little value at the beginning of the study of a subject. The best way to find out what mathematical logic is about is to start doing it, and students are advised to begin reading the book even though (or especially if) they have qualms about the meaning and purpose of the subject.

Although logic is basic to all other studies, its fundamental and apparently self-evident character discouraged any deep logical investigations until the late 19th century. Then, under the impetus of the discovery of non-Euclidean geometry and the desire to provide a rigorous foundation for calculus and higher analysis, interest in logic revived. This new interest, however, was still rather unenthusiastic until, around the turn of the century, the mathematical world was shocked by the discovery of the paradoxes – that is, arguments that lead to contradictions. The most important paradoxes are described here.

1. *Russell's paradox* (1902). By a set, we mean any collection of objects – for example, the set of all even integers or the set of all saxophone players in Brooklyn. The objects that make up a set are called its members or

elements. Sets may themselves be members of sets; for example, the set of all sets of integers has sets as its members. Most sets are not members of themselves; the set of cats, for example, is not a member of itself because the set of cats is not a cat. However, there may be sets that do belong to themselves – for example, the set of all sets. Now, consider the set A of all those sets X such that X is not a member of X. Clearly, by definition, A is a member of A if and only if A is not a member of A. So, if A is a member of A, then A is also not a member of A; and if A is not a member of A, then A is a member of A. In any case, A is a member of A and A is not a member of A.

2. *Cantor's paradox* (1899). This paradox involves the theory of cardinal numbers and may be skipped by those readers having no previous acquaintance with that theory. The cardinal number $\overline{\overline{Y}}$ of a set Y is a measure of the size of the set; $\overline{\overline{Y}} = \overline{\overline{Z}}$ if and only if Y is equinumerous with Z (that is, there is a one–one correspondence between Y and Z). We define $\overline{\overline{Y}} \leqslant \overline{\overline{Z}}$ to mean that Y is equinumerous with a subset of Z; by $\overline{\overline{Y}} < \overline{\overline{Z}}$ we mean $\overline{\overline{Y}} \leqslant \overline{\overline{Z}}$ and $\overline{\overline{Y}} \neq \overline{\overline{Z}}$. Cantor proved that, if $\mathscr{P}(Y)$ is the set of all subsets of Y, then $\overline{\overline{Y}} < \overline{\overline{\mathscr{P}(Y)}}$. Let V be the universal set – that is, the set of all sets. Now, $\mathscr{P}(V)$ is a subset of V; so it follows easily that $\overline{\overline{\mathscr{P}(V)}} \leqslant \overline{\overline{V}}$. On the other hand, by Cantor's theorem, $\overline{\overline{V}} < \overline{\overline{\mathscr{P}(V)}}$. Bernstein's theorem asserts that, if $\overline{\overline{Y}} \leqslant \overline{\overline{Z}}$ and $\overline{\overline{Z}} \leqslant \overline{\overline{Y}}$, then $\overline{\overline{Y}} = \overline{\overline{Z}}$. Hence, $\overline{\overline{V}} = \overline{\overline{\mathscr{P}(V)}}$, contradicting $\overline{\overline{V}} < \overline{\overline{\mathscr{P}(V)}}$.
3. *Burali-Forti's paradox* (1897). This paradox is the analogue in the theory of ordinal numbers of Cantor's paradox and requires familiarity with ordinal number theory. Given any ordinal number, there is a still larger ordinal number. But the ordinal number determined by the set of all ordinal numbers is the largest ordinal number.
4. *The liar paradox.* A man says, 'I am lying', If he is lying, then what he says is true and so he is not lying. If he is not lying, then what he says is true, and so he is lying. In any case, he is lying and he is not lying.†
5. *Richard's paradox* (1905). Some phrases of the English language denote real numbers; for example, 'the ratio between the circumference and diameter of a circle' denotes the number π. All the phrases of the English language can be enumerated in a standard way: order all phrases that have k letters lexicographically (as in a dictionary) and then place all phrases with k letters before all phrases with a larger number of letters. Hence, all phrases of the English language that denote real numbers can

†The Cretan 'paradox', known in antiquity, is similar to the liar paradox. The Cretan philosopher Epimenides said, 'All Cretans are liars'. If what he said is true, then, since Epimenides is a Cretan, it must be false. Hence, what he said is false. Thus, there must be some Cretan who is not a liar. This is not logically impossible; so we do not have a genuine paradox. However, the fact that the utterance by Epimenides of that false sentence could imply the existence of some Cretan who is not a liar is rather unsettling.

be enumerated merely by omitting all other phrases in the given standard enumeration. Call the nth real number in this enumeration the nth Richard number. Consider the phrase: 'the real number whose nth decimal place is 1 if the nth decimal place of the nth Richard number is not 1, and whose nth decimal place is 2 if the nth decimal place of the nth Richard number is 1.' This phrase defines a Richard number – say, the kth Richard number; but, by its definition, it differs from the kth Richard number in the kth decimal place.

6. *Berry's paradox* (1906). There are only a finite number of symbols (letters, punctuation signs, etc.) in the English language. Hence, there are only a finite number of English expressions that contain fewer than 200 occurrences of symbols (allowing repetitions). There are, therefore, only a finite number of positive integers that are denoted by an English expression containing fewer than 200 occurrences of symbols. Let k be *the least positive integer that is not denoted by an English expression containing fewer than 200 occurrences of symbols*. The italicized English phrase contains fewer than 200 occurrences of symbols and denotes the integer k.
7. *Grelling's paradox* (1908). An adjective is called *autological* if the property denoted by the adjective holds for the adjective itself. An adjective is called *heterological* if the property denoted by the adjective does not apply to the adjective itself. For example, 'polysyllabic' and 'English' are autological, whereas 'monosyllabic' and 'French' are heterological. Consider the adjective 'heterological'. If 'heterological' is heterological, then it is not heterological. If 'heterological' is not heterological, then it is heterological. In either case, 'heterological' is both heterological and not heterological.
8. *Löb's paradox* (1955). Let A be any sentence. Let B be the sentence: 'If this sentence is true, then A'. So, B asserts: 'If B is true, then A'. Now consider the following argument: Assume B is true; then, by B, since B is true, A holds. This argument shows that, if B is true, then A. But this is exactly what B asserts. Hence, B is true. Therefore, by B, since B is true, A is true. Thus, every sentence is true.

All of these paradoxes are genuine in the sense that they contain no obvious logical flaws. The logical paradoxes (1–3) involve only notions from the theory of sets, whereas the semantic paradoxes (4–8) also make use of concepts like 'denote', 'true' and 'adjective', which need not occur within our standard mathematical language. For this reason, the logical paradoxes are a much greater threat to a mathematician's peace of mind than the semantic paradoxes.

Analysis of the paradoxes has led to various proposals for avoiding them. All of these proposals are restrictive in one way or another of the 'naive' concepts that enter into the derivation of the paradoxes. Russell noted the self-reference present in all the paradoxes and suggested that every object must have a definite non-negative integer as its 'type'. Then an expression 'x is a member of the set y' is to be considered *meaningful* if and only if the type of y is one greater than the type of x.

This approach, known as the theory of types and systematized and developed in *Principia Mathematica* Whitehead and Russell (1910–13), is successful in eliminating the known paradoxes,† but it is clumsy in practice and has certain other drawbacks as well. A different criticism of the logical paradoxes is aimed at their assumption that, for every property $P(x)$, there exists a corresponding set of all objects x that satisfy $P(x)$. If we reject this assumption, then the logical paradoxes are no longer derivable.‡ It is necessary, however, to provide new postulates that will enable us to prove the existence of those sets that are needed by the practising mathematician. The first such axiomatic set theory was invented by Zermelo (1908). In Chapter 4 we shall present an axiomatic theory of sets that is a descendant of Zermelo's system (with some new twists given to it by von Neumann, R. Robinson, Bernays, and Gödel). There are also various hybrid theories combining some aspects of type theory and axiomatic set theory – for example, Quine's system NF.

A more radical interpretation of the paradoxes has been advocated by Brouwer and his intuitionist school (see Heyting, 1956). They refuse to accept the universality of certain basic logical laws, such as the law of excluded middle: P or not-P. Such a law, they claim, is true for finite sets, but it is invalid to extend it on a wholesale basis to all sets. Likewise, they say it is invalid to conclude that 'There exists an object x such that not-$P(x)$' follows from the negation of 'For all x, $P(x)$'; we are justified in asserting the existence of an object having a certain property only if we know an effective method for constructing (or finding) such an object. The paradoxes are not derivable (or even meaningful) if we obey the intuitionist strictures, but so are many important theorems of everyday mathematics, and, for this reason, intuitionism has found few converts among mathematicians.

Whatever approach one takes to the paradoxes, it is necessary first to examine the language of logic and mathematics to see what symbols may be used, to determine the ways in which these symbols are put together to form terms, formulas, sentences and proofs, and to find out what can and cannot be proved if certain axioms and rules of inference are assumed. This is one of the tasks of mathematical logic, and, until it is done, there is no basis for

†Russells's paradox, for example, depends on the existence of the set A of all sets that are not members of themselves. Because, according to the theory of types, it is meaningless to say that a set belongs to itself, there is no such set A.

‡Russell's paradox then proves that there is no set A of all sets that do not belong to themselves. The paradoxes of Cantor and Burali-Forti show that there is no universal set and no set that contains all ordinal numbers. The semantic paradoxes cannot even be formulated, since they involve notions not expressible within the system.

comparing rival foundations of logic and mathematics. The deep and devastating results of Gödel, Tarski, Church, Rosser, Kleene, and many others have been ample reward for the labour invested and have earned for mathematical logic its status as an independent branch of mathematics.

For the absolute novice a summary will be given here of some of the basic notation, ideas, and results used in the text. The reader is urged to skip these explanations now and, if necessary, to refer to them later on.

A set is a collection of objects.† The objects in the collection are called *elements* or *members* of the set. We shall write '$x \in y$' for the statement that x is a member of y. (Synonymous expressions are 'x belongs to y' and 'y contains x'.) The negation of '$x \in y$' will be written '$x \notin y$'.

By '$x \subseteq y$' we mean that every member of x is also a member of y (synonymously, that x is a *subset* of y, or that x is *included* in y). We shall write '$t = s$' to mean that t and s denote the same object. As usual, '$t \neq s$' is the negation of '$t = s$'. For sets x and y, we assume that $x = y$ if and only if $x \subseteq y$ and $y \subseteq x$ – that is, if and only if x and y have the same members. A set x is called a *proper* subset of a set y, written '$x \subset y$' if $x \subseteq y$ but $x \neq y$. (The notation $x \subsetneq y$ is often used instead of $x \subset y$.)

The *union* $x \cup y$ of sets x and y is defined to be the set of all objects that are members of x or y or both. Hence, $x \cup x = x$, $x \cup y = y \cup x$, and $(x \cup y) \cup z = x \cup (y \cup z)$. The *intersection* $x \cap y$ is the set of objects that x and y have in common. Therefore, $x \cap x = x$, $x \cap y = y \cap x$, and $(x \cap y) \cap z = x \cap (y \cap z)$. Moreover, $x \cap (y \cup z) = (x \cap y) \cup (x \cap z)$ and $x \cup (y \cap z) = (x \cup y) \cap (x \cup z)$. The *relative complement* $x - y$ is the set of members of x that are not members of y. We also postulate the existence of the *empty set* (or *null set*) $\emptyset$ – that is, a set that has no members at all. Then $x \cap \emptyset = \emptyset$, $x \cup \emptyset = x$, $x - \emptyset = x$, $\emptyset - x = \emptyset$, and $x - x = \emptyset$. Sets x and y are called *disjoint* if $x \cap y = \emptyset$.

Given any objects $b_1, \dots, b_k$, the set that contains $b_1, \dots, b_k$ as its only members is denoted $\{b_1, \dots, b_k\}$. In particular, $\{x, y\}$ is a set having x and y as its only members and, if $x \neq y$, is called the *unordered pair* of x and y. The set $\{x, x\}$ is identical with $\{x\}$ and is called the *unit set* of x. Notice that $\{x, y\} = \{y, x\}$. By $\langle b_1, \dots, b_k \rangle$ we mean the *ordered k-tuple* of $b_1, \dots, b_k$. The basic property of ordered k-tuples is that $\langle b_1, \dots, b_k \rangle = \langle c_1, \dots, c_k \rangle$ if and only if $b_1 = c_1$, $b_2 = c_2, \dots, b_k = c_k$. Thus, $\langle b_1, b_2 \rangle = \langle b_2, b_1 \rangle$ if and only if $b_1 = b_2$. Ordered 2-tuples are called *ordered pairs*. The ordered 1-tuple $\langle b \rangle$ is taken to be b itself. If X is a set and k is a positive integer, we denote by X^k the set of all ordered k-tuples $\langle b_1, \dots, b_k \rangle$ of elements $b_1, \dots, b_k$ of X. In

†Which collections of objects form sets will not be specified here. Care will be exercised to avoid using any ideas or procedures that may lead to the paradoxes; all the results can be formalized in the axiomatic set theory of Chapter 4. The term 'class' is sometimes used as a synonym for 'set', but it will be avoided here because it has a different meaning in Chapter 4. If a property $P(x)$ does determine a set, that set is often denoted $\{x \mid P(x)\}$.

particular, X^1 is X itself. If Y and Z are sets, then by $Y \times Z$ we denote the set of all ordered pairs $\langle y,z\rangle$ such that $y \in Y$ and $z \in Z$. $Y \times Z$ is called the *Cartesian product* of Y and Z.

An *n-place relation* (or a *relation with n arguments*) on a set X is a subset of X^n – that is, a set of ordered n-tuples of elements of X. For example, the 3-place relation of betweenness for points on a line is the set of all 3-tuples $\langle x,y,z\rangle$ such that the point x lies between the points y and z. A 2-place relation is called a *binary* relation; for example, the binary relation of fatherhood on the set of human beings is the set of all ordered pairs $\langle x,y\rangle$ such that x and y are human beings and x is the father of y. A 1-place relation on X is a subset of X and is called a *property* on X.

Given a binary relation R on a set X, the *domain* of R is defined to be the set of all y such that $\langle y,z\rangle \in R$ for some z; the *range* of R is the set of all z such that $\langle y,z\rangle \in R$ for some y; and the *field* of R is the union of the domain and range of R. The *inverse* relation R^{-1} of R is the set of all ordered pairs $\langle y,z\rangle$ such that $\langle z,y\rangle \in R$. For example, the domain of the relation $<$ on the set ω of non-negative integers[†] is ω, its range is $\omega - \{0\}$, and the inverse of $<$ is $>$. Notation: Very often xRy is written instead of $\langle x,y\rangle \in R$. Thus, in the example just given, we usually write $x < y$ instead of $\langle x,y\rangle \in <$.

A binary relation R is said to be *reflexive* if xRx for all x in the field of R; R is *symmetric* if xRy implies yRx; and R is *transitive* if xRy and yRz imply xRz. Examples: The relation $\leqslant$ on the set of integers is reflexive and transitive but not symmetric. The relation 'having at least one parent in common' on the set of human beings is reflexive and symmetric, but not transitive.

A binary relation that is reflexive, symmetric and transitive is called an *equivalence relation*. Examples of equivalence relations are: (1) the *identity relation* I_X on a set X, consisting of all pairs $\langle x,x\rangle$, where $x \in X$; (2) given a fixed positive integer n, the relation $x \equiv y \pmod{n}$, which holds when x and y are integers and $x - y$ is divisible by n; (3) the congruence relation on the set of triangles in a plane; (4) the similarity relation on the set of triangles in a plane. Given an equivalence relation R whose field is X, and given any $y \in X$, define $[y]$ as the set of all z in X such that yRz. Then $[y]$ is called the *R- equivalence class* of y. Clearly, $[u] = [v]$ if and only if uRv. Moreover, if $[u] \neq [v]$, then $[u] \cap [v] = \emptyset$; that is, different R-equivalence classes have no elements in common. Hence, the set X is completely partitioned into the R-equivalence classes. In example (1) above, the equivalence classes are just the unit sets $\{x\}$, where $x \in X$. In example (2), there are n equivalence classes, the kth equivalence class ($k = 0, 1, \ldots, n-1$) being the set of all integers that leave the remainder k upon division by n.

A *function* f is a binary relation such that $\langle x,y\rangle \in f$ and $\langle x,z\rangle \in f$ imply $y = z$. Thus, for any element x of the domain of a function f, there is a unique y such that $\langle x,y\rangle \in f$; this unique y is denoted $f(x)$. If x is in the

[†] ω will also be referred to as the set of *natural numbers*.

domain of f, then $f(x)$ is said to be *defined*. A function f with domain X and range Y is said to be a function from X *onto* Y. If f is a function from X onto a subset of Z, then f is said to be a function from X *into* Z. For example, if the domain of f is the set of integers and $f(x) = 2x$ for every integer x, then f is a function from the set of integers onto the set of even integers, and f is a function from the set of integers into the set of integers. A function whose domain consists of n-tuples is said to be a *function of n arguments*. A *total function of n arguments on a set X* is a function f whose domain is X^n. It is customary to write $f(x_1,\ldots,x_n)$ instead of $f(\langle x_1,\ldots,x_n\rangle)$, and we refer to $f(x_1,\ldots,x_n)$ as the *value* of f for the *arguments* $x_1,\ldots,x_n$. A *partial* function of n arguments on a set X is a function whose domain is a subset of X^n. For example, ordinary division is a partial, but not total, function of two arguments on the set of integers, since division by 0 is not defined. If f is a function with domain X and range Y, then the *restriction* f_Z of f to a set Z is the function $f \cap (Z \times Y)$. Then $f_Z(u) = v$ if and only if $u \in Z$ and $f(u) = v$. The *image* of the set Z under the function f is the range of f_Z. The *inverse image* of a set W under the function f is the set of all u in the domain of f such that $f(u) \in W$. We say that f *maps* X onto (into) Y if X is a subset of the domain of f and the image of X under f is (a subset of) Y. By an *n-place operation* (or *operation with n arguments*) on a set X we mean a function from X^n into X. For example, ordinary addition is a binary (i.e., 2-place) operation on the set of natural numbers $\{0,1,2,\cdots\}$. But ordinary subtraction is not a binary operation on the set of natural numbers.

The *composition* $f \circ g$ (sometimes denoted fg) of functions f and g is the function such that $(f \circ g)(x) = f(g(x))$; $(f \circ g)(x)$ is defined if and only if $g(x)$ is defined and $f(g(x))$ is defined. For example, if $g(x) = x^2$ and $f(x) = x + 1$ for every integer x, then $(f \circ g)(x) = x^2 + 1$ and $(g \circ f)(x) = (x+1)^2$. Also, if $h(x) = -x$ for every real number x and $f(x) = \sqrt{x}$ for every non-negative real number x, then $(f \circ h)(x)$ is defined only for $x \leqslant 0$, and, for such x, $(f \circ h)(x) = \sqrt{-x}$. A function f such that $f(x) = f(y)$ implies $x = y$ is called a *1–1* (*one–one*) function. For example, the identity relation I_X on a set X is a 1–1 function, since $I_X(y) = y$ for every $y \in X$; the function g with domain ω, such that $g(x) = 2x$ for every $x \in \omega$, is 1–1; but the function h whose domain is the set of integers and such that $h(x) = x^2$ for every integer x is not 1–1, since $h(-1) = h(1)$. Notice that a function f is 1–1 if and only if its inverse relation f^{-1} is a function. If the domain and range of a 1–1 function f are X and Y, then f is said to be a $1-1$ (*one–one*) *correspondence between X and Y*; then f^{-1} is a 1–1 correspondence between Y and X, and $(f^{-1} \circ f) = I_X$ and $(f \circ f^{-1}) = I_Y$. If f is a 1–1 correspondence between X and Y and g is a 1–1 correspondence between Y and Z, then $g \circ f$ is a 1–1 correspondence between X and Z. Sets X and Y are said to be *equinumerous* (written $X \cong Y$) if and only if there is a 1–1 correspondence between X and Y. Clearly, $X \cong X$, $X \cong Y$ implies $Y \cong X$, and $X \cong Y$ and $Y \cong Z$ implies $X \cong Z$. It is somewhat harder to show that, if $X \cong Y_1 \subseteq Y$ and $Y \cong X_1 \subseteq X$, then $X \cong Y$ (see Bernstein's theorem in Chapter 4). If $X \cong Y$, one says that X and Y *have the same cardinal number*, and if X is equinumerous with a subset of Y but Y is not equinumerous with a subset of X, one says that the cardinal number of X *is smaller than* the cardinal number of Y.[†]

A set X is *denumerable* if it is equinumerous with the set of positive integers. A denumerable set is said to have cardinal number $\aleph_0$, and any set equinumerous with the set of all subsets of a denumerable set is said to have the cardinal number $2^{\aleph_0}$ (or to have the *power of the continuum*). A set X is *finite* if it is empty or if it is equinumerous with the set $\{1, 2, \ldots, n\}$ of all positive integers that are less than or equal to some positive integer n. A set that is not finite is said to be *infinite*. A set is *countable* if it is either finite or denumerable. Clearly, any subset of a denumerable set is countable. A *denumerable sequence* is a function s whose domain is the set of positive integers; one usually writes s_n instead of $s(n)$. A *finite sequence* is a function whose domain is the empty set or $\{1, 2, \ldots, n\}$ for some positive integer n.

Let $P(x, y_1, \ldots, y_k)$ be some relation on the set of non-negative integers. In particular, P may involve only the variable x and thus be a property. If $P(0, y_1, \ldots, y_k)$ holds, and, if, for every n, $P(n, y_1, \ldots, y_k)$ implies $P(n+1, y_1, \ldots, y_k)$, then $P(x, y_1, \ldots, y_k)$ is true for all non-negative integers x (*principle of mathematical induction*). In applying this principle, one usually proves that, for every n, $P(n, y_1, \ldots, y_K)$ implies $P(n+1, y_1, \ldots, y_k)$ by assuming $P(n, y_1, \ldots, y_k)$ and then deducing $P(n+1, y_1, \ldots, y_k)$; in the course of this deduction, $P(n, y_1, \ldots, y_k)$ is called the *inductive hypothesis*. If the relation P actually involves variables $y_1, \ldots, y_k$ other than x, then the proof is said to proceed by *induction on* x. A similar induction principle holds for the set of integers greater than some fixed integer j. An example is: to prove by mathematical induction that the sum of the first n odd integers $1 + 3 + 5 + \ldots + (2n-1)$ is n^2, first show that $1 = 1^2$ (that is, $P(1)$), and then, that if $1 + 3 + 5 + \ldots + (2n-1) = n^2$, then $1 + 3 + 5 + \ldots + (2n-1)$ $+(2n+1) = (n+1)^2$ (that is, if $P(n)$ then $P(n+1)$). From the principle of mathematical induction one can prove the *principle of complete induction*: If, for every non-negative integer x the assumption that $P(u, y_1, \ldots, y_k)$ is true for all $u < x$ implies that $P(x, y_1, \ldots, y_k)$ holds, then, for all non-negative integers x, $P(x, y_1, \ldots, y_k)$ is true, (Exercise: Show by complete induction that every integer greater than 1 is divisible by a prime number.)

A *partial order* is a binary relation R such that R is transitive and, for every x in the field of R, xRx is false. If R is a partial order, then the relation R' that is the union of R and the set of all ordered pairs $\langle x, x \rangle$, where x is in the field of R, we shall call a *reflexive partial order*; in the literature, 'partial order' is used for either partial order or reflexive partial order. Notice that

[†]One can attempt to define the cardinal number of a set X as the collection $[X]$ of all sets equinumerous with X. However, in certain axiomatic set theories, $[X]$ does not exist, whereas in others $[X]$ exists but is not a set.

(xRy and yRx) is impossible if R is a partial order, whereas (xRy and yRx) implies $x = y$ if R is a reflexive partial order. A (reflexive) *total order* is a (reflexive) partial order such that, for any x and y in the field of R, either $x = y$ or xRy or yRx. Examples: (1) the relation $<$ on the set of integers is a total order, whereas $\leqslant$ is a reflexive total order; (2) the relation $\subset$ on the set of all subsets of the set of positive integers is a partial order but not a total order, whereas the relation $\subseteq$ is a reflexive partial order but not a reflexive total order. If B is a subset of the field of a binary relation R, then an element y of B is called an *R-least element* of B if yRz for every element z of B different from y. A *well-order* (or a *well-ordering relation*) is a total order R such that every non-empty subset of the field of R has an R-least element. Examples: (1) the relation $<$ on the set of non-negative integers is a well-order; (2) the relation $<$ on the set of non-negative rational numbers is a total order but not a well-order; (3) the relation $<$ on the set of integers is a total order but not a well-order. Associated with every well-order R having field X there is a *complete induction principle*: if P is a property such that, for any u in X, whenever all z in X such that zRu have the property P, then u has the property P, then it follows that all members of X have the property P. If the set X is infinite, a proof using this principle is called a proof by *transfinite induction*. One says that a set X *can be well-ordered* if there exists a well-order whose field is X. An assumption that is useful in modern mathematics but about the validity of which there has been considerable controversy is the *well-ordering principle*: every set can be well-ordered. The well-ordering principle is equivalent (given the usual axioms of set theory) to the *axiom of choice*: for any set X of non-empty pairwise disjoint sets, there is a set Y (called a *choice set*) that contains exactly one element in common with each set in X.

Let B be a non-empty set, f a function from B into B, and g a function from B^2 into B. Write x' for $f(x)$ and $x \cap y$ for $g(x,y)$. Then $\langle B,f,g\rangle$ is called a *Boolean algebra* if B contains at least two elements and the following conditions are satisfied:

1. $x \cap y = y \cap x$ for all x and y in B
2. $(x \cap y) \cap z = x \cap (y \cap z)$ for all x, y, z in B
3. $x \cap y' = z \cap z'$ if and only if $x \cap y = x$ for all x, y, z in B.

Let $x \cup y$ stand for $(x' \cap y')'$, and write $x \leqslant y$ for $x \cap y = x$. It is easily proved that $z \cap z' = w \cap w'$ for any w and z in B; we denote the value of $z \cap z'$ by 0. Let 1 stand for $0'$. Then $z \cup z' = 1$ for all z in B. Note also that $\leqslant$ is a reflexive partial order on B, and $\langle B,f,\cup\rangle$ is a Boolean algebra. (The symbols $\cap, \cup, 0, 1$ should not be confused with the corresponding symbols used in set theory and arithmetic.) An *ideal* J in $\langle B,f,g\rangle$ is a non-empty subset of B such that (1) if $x \in J$ and $y \in J$, then $x \cup y \in J$, and (2) if $x \in J$ and $y \in B$, then $x \cap y \in J$. Clearly, $\{0\}$ and B are ideals. An ideal different from B is called a *proper* ideal. A *maximal* ideal is a proper ideal that is included in no other proper ideal. It can be shown that a proper ideal J is maximal if and only if, for any u in $B, u \in J$ or $u' \in J$. From the axiom of choice it can be proved that every Boolean algebra contains a maximal ideal, or, equivalently, that every proper ideal is included in some maximal ideal. For example, let B be the set of all subsets of a set X; for $Y \in B$, let $Y' = X - Y$, and for Y and Z in B, let $Y \cap Z$ be the ordinary set-theoretic intersection of Y and Z. Then $\langle B, ', \cap \rangle$ is a Boolean algebra. The 0 of B is the empty set $\emptyset$, and 1 is X. For each element u in X, the set J_u of all subsets of X that do not contain u is a maximal ideal. For a detailed study of Boolean algebras, see Sikorski (1960), Halmos (1963) and Mendelson (1970).

The Propositional Calculus 1

1.1 PROPOSITIONAL CONNECTIVES. TRUTH TABLES

Sentences may be combined in various ways to form more complicated sentences. We shall consider only *truth-functional* combinations, in which the truth or falsity of the new sentence is determined by the truth or falsity of its component sentences.

Negation is one of the simplest operations on sentences. Although a sentence in a natural language may be negated in many ways, we shall adopt a uniform procedure: placing a sign for negation, the symbol $\neg$, in front of the entire sentence. Thus, if A is a sentence, then $\neg A$ denotes the negation of A.

The truth-functional character of negation is made apparent in the following *truth table*:

A	$\neg A$
T	F
F	T

When A is true, $\neg A$ is false; when A is false, $\neg A$ is true. We use T and F to denote the *truth values* true and false.

Another common truth-functional operation is the *conjunction*: 'and'. The conjunction of sentences A and B will be designated by $A \wedge B$ and has the following truth table:

A	B	$A \wedge B$
T	T	T
F	T	F
T	F	F
F	F	F

$A \wedge B$ is true when and only when both A and B are true. A and B are called the *conjuncts* of $A \wedge B$. Note that there are four rows in the table, corresponding to the number of possible assignments of truth values to A and B.

In natural languages, there are two distinct uses of 'or': the inclusive and the exclusive. According to the inclusive usage, 'A or B' means 'A or B or both', whereas according to the exclusive usage, the meaning is 'A or B, but not both'. We shall introduce a special sign, $\vee$, for the inclusive connective. Its truth table is as follows:

A	B	$A \vee B$
T	T	T
F	T	T
T	F	T
F	F	F

Thus, $A \vee B$ is false when and only when both A and B are false. '$A \vee B$' is called a *disjunction*, with the *disjuncts* A and B.

Another important truth-functional operation is the *conditional*: 'if A, then B'. Ordinary usage is unclear here. Surely, 'if A, then B' is false when the *antecedent* A is true and the *consequent* B is false. However, in other cases, there is no well-defined truth value. For example, the following sentences would be considered neither true nor false:

1. If $1 + 1 = 2$, then Paris is the capital of France.
2. If $1 + 1 \neq 2$, then Paris is the capital of France.
3. If $1 + 1 \neq 2$, then Rome is the capital of France.

Their meaning is unclear, since we are accustomed to the assertion of some sort of relationship (usually causal) between the antecedent and the consequent. We shall make the convention that 'if A, then B' is false when and only when A is true and B is false. Thus, sentences 1–3 are assumed to be true. Let us denote 'if A, then B' by '$A \Rightarrow B$'. An expression '$A \Rightarrow B$' is called a *conditional*. Then $\Rightarrow$ has the following truth table:

A	B	$A \Rightarrow B$
T	T	T
F	T	T
T	F	F
F	F	T

This sharpening of the meaning of 'if A, then B' involves no conflict with ordinary usage, but rather only an extension of that usage.†

A justification of the truth table for $\Rightarrow$ is the fact that we wish 'if A and B, then B' to be true in all cases. Thus, the case in which A and B are true justifies the first line of our truth table for $\Rightarrow$, since (A and B) and B are both true. If A is

†There is a common non-truth-functional interpretation of 'if A, then B' connected with causal laws. The sentence 'if this piece of iron is placed in water at time t, then the iron will dissolve' is regarded as false even in the case that the piece of iron is not placed in water at time t – that is, even when the antecedent is false. Another non-truth-functional usage occurs in so-called counterfactual conditionals, such as 'if Sir Walter Scott had not written any novels, then there would have been no War Between the States'. (This was Mark Twain's contention in *Life on the Mississippi*: 'Sir Walter had so large a hand in making Southern character, as it existed before the war, that he is in great measure responsible for the war'.) This sentence might be asserted to be false even though the antecedent is admittedly false. However, causal laws and counterfactual conditions seem not to be needed in mathematics and logic. For a clear treatment of conditionals and other connectives, see Quine (1951). (The quotation from *Life on the Mississippi* was brought to my attention by Professor J.C. Owings, Jr.)

false and B true, then (A and B) is false while B is true. This corresponds to the second line of the truth table. Finally, if A is false and B is false, (A and B) is false and B is false. This gives the fourth line of the table. Still more support for our definition comes from the meaning of statements such as 'for every x, if x is an odd positive integer, then x^2 is an odd positive integer'. This asserts that, for every x, the statement 'if x is an odd positive integer, then x^2 is an odd positive integer' is true. Now we certainly do not want to consider cases in which x is not an odd positive integer as counterexamples to our general assertion. This supports the second and fourth lines of our truth table. In addition, any case in which x is an odd positive integer and x^2 is an odd positive integer confirms our general assertion. This corresponds to the first line of the table.

Let us denote 'A if and only if B' by '$A \Leftrightarrow B$'. Such an expression is called a *biconditional*. Clearly, $A \Leftrightarrow B$ is true when and only when A and B have the same truth value. Its truth table, therefore is:

A	B	$A \Leftrightarrow B$
T	T	T
F	T	F
T	F	F
F	F	T

The symbols $\neg, \wedge, \vee, \Rightarrow$ and $\Leftrightarrow$ will be called *propositional connectives*.[†] Any sentence built up by application of these connectives has a truth value that depends on the truth values of the constituent sentences. In order to make this dependence apparent, let us apply the name *statement form* to an expression built up from the *statement letters* A, B, C, and so on by appropriate applications of the propositional connectives.

1. All statement letters (capital italic letters) and such letters with numerical subscripts[‡] are statement forms.
2. If $\mathscr{B}$ and $\mathscr{C}$ are statement forms, then so are $(\neg\mathscr{B}), (\mathscr{B} \wedge \mathscr{C})$, $(\mathscr{B} \vee \mathscr{C}), (\mathscr{B} \Rightarrow \mathscr{C})$, and $(\mathscr{B} \Leftrightarrow \mathscr{C})$.
3. Only those expressions are statement forms that are determined to be so by means of conditions 1 and 2.[§]

Some examples of statement forms are $B, (\neg C_2), (D_3 \wedge (\neg B))$, $(((\neg B_1) \vee B_2) \Rightarrow (A_1 \wedge C_2))$, and $(((\neg A) \Leftrightarrow A) \Leftrightarrow (C \Rightarrow (B \vee C)))$.

[†]We have been avoiding and shall in the future avoid the use of quotation marks to form names whenever this is not likely to cause confusion. The given sentence should have quotation marks around each of the connectives. See Quine (1951, pp. 23–27).

[‡]For example, $A_1, A_2, A_{17}, B_{31}, C_2, \ldots$.

[§]This can be rephrased as follows: $\mathscr{C}$ is a statement form if and only if there is a finite sequence $\mathscr{B}_1, \ldots, \mathscr{B}_n (n \geq 1)$ such that $\mathscr{B}_n = \mathscr{C}$ and, if $1 \leqslant i \leqslant n$, $\mathscr{B}_i$ is either a statement letter or a negation, conjunction, disjunction, conditional or biconditional constructed from previous expressions in the sequence. Notice that we use script letters $\mathscr{A}, \mathscr{B}, \mathscr{C}, \ldots$ to stand for arbitrary expressions, whereas italic letters are used as statement letters.

For every assignment of truth values T or F to the statement letters that occur in a statement form, there corresponds, by virtue of the truth tables for the propositional connectives, a truth value for the statement form. Thus, each statement form determines a *truth function*, which can be graphically represented by a truth table for the statement form. For example, the statement form $(((\neg A) \vee B) \Rightarrow C)$ has the following truth table:

A	B	C	$(\neg A)$	$((\neg A) \vee B)$	$(((\neg A) \vee B) \Rightarrow C)$
T	T	T	F	T	T
F	T	T	T	T	T
T	F	T	F	F	T
F	F	T	T	T	T
T	T	F	F	T	F
F	T	F	T	T	F
T	F	F	F	F	T
F	F	F	T	T	F

Each row represents an assignment of truth values to the statement letters A, B and C and the corresponding truth values assumed by the statement forms that appear in the construction of $(((\neg A) \vee B) \Rightarrow C)$.

The truth table for $((A \Leftrightarrow B) \Rightarrow ((\neg A) \wedge B))$ is as follows:

A	B	$(A \Leftrightarrow B)$	$(\neg A)$	$((\neg A) \wedge B)$	$((A \Leftrightarrow B) \Rightarrow ((\neg A) \wedge B))$
T	T	T	F	F	F
F	T	F	T	T	T
T	F	F	F	F	T
F	F	T	T	F	F

If there are n distinct letters in a statement form, then there are 2^n possible assignments of truth values to the statement letters and, hence, 2^n rows in the truth table.

A truth table can be abbreviated by writing only the full statement form, putting the truth values of the statement letters underneath all occurrences of these letters, and writing, step by step, the truth values of each component statement form under the principal connective of the form†. As an example, for $((A \Leftrightarrow B) \Rightarrow ((\neg A) \wedge B))$, we obtain:

$((A$	$\Leftrightarrow$	$B)$	$\Rightarrow$	$((\neg A)$	$\wedge$	$B))$
T	T	T	F	FT	F	T
F	F	T	T	TF	T	T
T	F	F	T	FT	F	F
F	T	F	F	TF	F	F

†The *principal connective* of a statement form is the one that is applied last in constructing the form.

Exercises

1.1 Write the truth table for the exclusive usage of 'or'.
1.2 Construct truth tables for the statement forms $((A \Rightarrow B) \vee (\neg A))$ and $((A \Rightarrow (B \Rightarrow C)) \Rightarrow ((A \Rightarrow B) \Rightarrow (A \Rightarrow C)))$.
1.3 Write abbreviated truth tables for $((A \Rightarrow B) \wedge A)$ and $((A \vee (\neg C)) \Leftrightarrow B)$.
1.4 Write the following sentences as statement forms, using statement letters to stand for the *atomic sentences* – that is, those sentences that are not built up out of other sentences.

(a) If Mr Jones is happy, Mrs Jones is not happy, and if Mr Jones is not happy, Mrs Jones is not happy.
(b) Either Sam will come to the party and Max will not, or Sam will not come to the party and Max will enjoy himself.
(c) A sufficient condition for x to be odd is that x is prime.
(d) A necessary condition for a sequence s to converge is that s be bounded.
(e) A necessary and sufficient condition for the sheikh to be happy is that he has wine, women and song.
(f) Fiorello goes to the movies only if a comedy is playing.
(g) The bribe will be paid if and only if the goods are delivered.
(h) If x is positive, x^2 is positive.
(i) Karpov will win the chess tournament unless Kasparov wins today.

1.2 TAUTOLOGIES

A *truth function of n arguments* is defined to be a function of n arguments, the arguments and values of which are the truth values T or F. As we have seen, any statement form containing n distinct statement letters determines a corresponding truth function of n arguments.†

†To be precise, enumerate all statement letters as follows: $A, B, \ldots, Z; A_1, B_1, \ldots, Z_1; A_2, \ldots,$. If a statement form contains the $i_{1\text{th}}, \ldots, i_{n\text{th}}$ statement letters in this enumeration, where $i_1 < \ldots < i_n$, then the corresponding truth function is to have $x_{i_1}, \ldots, x_{i_n}$, in that order, as its arguments, where x_{i_j} corresponds to the $i_{j\text{th}}$ statement letter. For example, $(A \Rightarrow B)$ generates the truth function

x_1	x_2	$f(x_1, x_2)$
T	T	T
F	T	T
T	F	F
F	F	T

whereas $(B \Rightarrow A)$ generates the truth function

x_1	x_2	$g(x_1, x_2)$
T	T	T
F	T	F
T	F	T
F	F	T

A statement form that is always true, no matter what the truth values of its statement letters may be, is called a *tautology*. A statement form is a tautology if and only if its corresponding truth function takes only the value T, or equivalently, if, in its truth table, the column under the statement form contains only *T*s. An example of a tautology is $(A \vee (\neg A))$, the so-called *law of the excluded middle*. Other simple examples are $(\neg(A \wedge (\neg A)))$, $(A \Leftrightarrow (\neg(\neg A)))$, $((A \wedge B) \Rightarrow A)$ and $(A \Rightarrow (A \vee B))$.

$\mathscr{B}$ is said to *logically imply* $\mathscr{C}$ (or, synonymously, $\mathscr{C}$ is a *logical consequence* of $\mathscr{B}$) if and only if every truth assignment to the statement letters of $\mathscr{B}$ and $\mathscr{C}$ that makes $\mathscr{B}$ true also makes $\mathscr{C}$ true. For example, $(A \wedge B)$ logically implies A, A logically implies $(A \vee B)$, and $(A \wedge (A \Rightarrow B))$ logically implies B.

$\mathscr{B}$ and $\mathscr{C}$ are said to be *logically equivalent* if and only if $\mathscr{B}$ and $\mathscr{C}$ receive the same truth value under every assignment of truth values to the statement letters of $\mathscr{B}$ and $\mathscr{C}$. For example, A and $(\neg(\neg A))$ are logically equivalent, as are $(A \wedge B)$ and $(B \wedge A)$.

PROPOSITION 1.1

(a) $\mathscr{B}$ logically implies $\mathscr{C}$ if and only if $(\mathscr{B} \Rightarrow \mathscr{C})$ is a tautology.
(b) $\mathscr{B}$ and $\mathscr{C}$ are logically equivalent if and only if $(\mathscr{B} \Leftrightarrow \mathscr{C})$ is a tautology.

Proof

(a) (i) Assume $\mathscr{B}$ logically implies $\mathscr{C}$. Hence, every truth assignment that makes $\mathscr{B}$ true also makes $\mathscr{C}$ true. Thus, no truth asssignment makes $\mathscr{B}$ true and $\mathscr{C}$ false. Therefore, no truth assignment makes $(\mathscr{B} \Rightarrow \mathscr{C})$ false, that is, every truth assignment makes $(\mathscr{B} \Rightarrow \mathscr{C})$ true. In other words, $(\mathscr{B} \Rightarrow \mathscr{C})$ is a tautology. (ii) Assume $(\mathscr{B} \Rightarrow \mathscr{C})$ is a tautology. Then, for every truth assignment, $(\mathscr{B} \Rightarrow \mathscr{C})$ is true, and, therefore, it is not the case that $\mathscr{B}$ is true and $\mathscr{C}$ false. Hence, every truth assignment that makes $\mathscr{B}$ true makes $\mathscr{C}$ true, that is, $\mathscr{B}$ logically implies $\mathscr{C}$.
(b) $(\mathscr{B} \Leftrightarrow \mathscr{C})$ is a tautology if and only if every truth assignment makes $(\mathscr{B} \Leftrightarrow \mathscr{C})$ true, which is equivalent to saying that every truth assignment gives $\mathscr{B}$ and $\mathscr{C}$ the same truth value, that is, $\mathscr{B}$ and $\mathscr{C}$ are logically equivalent.

By means of a truth table, we have an effective procedure for determining whether a statement form is a tautology. Hence, by Proposition 1.1, we have effective procedures for determining whether a given statement form logically implies another given statement form and whether two given statement forms are logically equivalent.

To see whether a statement form is a tautology, there is another method that is often shorter than the construction of a truth table.

Examples

1. Determine whether $((A \Leftrightarrow ((\neg B) \vee C)) \Rightarrow ((\neg A) \Rightarrow B))$ is a tautology.

Assume that the statement form sometimes is F (line 1). Then $(A \Leftrightarrow ((\neg B) \vee C))$ is T and $((\neg A) \Rightarrow B)$ is F (line 2). Since $((\neg A) \Rightarrow B)$ is F, $(\neg A)$ is T and B is F (line 3). Since $(\neg A)$ is T, A is F (line 4). Since A is F and $(A \Leftrightarrow ((\neg B) \vee C))$ is T, $((\neg B) \vee C)$ is F (line 5). Since $((\neg B) \vee C)$ is F, $(\neg B)$ and C are F (line 6). Since $(\neg B)$ is F, B is T (line 7). But B is both T and F (lines 7 and 3). Hence, it is impossible for the form to be false.

$((A$	$\Leftrightarrow$	$((\neg$	$B)$	$\vee$	$C))$	$\Rightarrow$	$((\neg$	$A)$	$\Rightarrow$	$B))$	
						F					1
	T								F		2
							T			F	3
F								F			4
				F							5
		F			F						6
			T								7

2. Determine whether $((A \Rightarrow (B \vee C)) \vee (A \Rightarrow B))$ is a tautology.

Assume that the form is F (line 1). Then $(A \Rightarrow (B \vee C))$ and $(A \Rightarrow B)$ are F (line 2). Since $(A \Rightarrow B)$ is F, A is T and B is F (line 3). Since $(A \Rightarrow (B \vee C))$ is F, A is T and $(B \vee C)$ is F (line 4). Since $(B \vee C)$ is F, B and C are F (line 5). Thus, when A is T, B is F, and C is F, the form is F. Therefore, it is not a tautology.

$((A$	$\Rightarrow$	$(B$	$\vee$	$C))$	$\vee$	$(A$	$\Rightarrow$	$B))$	
					F				1
	F						F		2
						T		F	3
T			F						4
		F		F					5

Exercises

1.5 Determine whether the following are tautologies.

(a) $(((A \Rightarrow B) \Rightarrow B) \Rightarrow B)$
(b) $(((A \Rightarrow B) \Rightarrow B) \Rightarrow A)$
(c) $(((A \Rightarrow B) \Rightarrow A) \Rightarrow A)$
(d) $(((B \Rightarrow C) \Rightarrow (A \Rightarrow B)) \Rightarrow (A \Rightarrow B))$
(e) $((A \vee (\neg(B \wedge C))) \Rightarrow ((A \Leftrightarrow C) \vee B))$
(f) $(A \Rightarrow (B \Rightarrow (B \Rightarrow A)))$
(g) $((A \wedge B) \Rightarrow (A \vee C))$
(h) $((A \Leftrightarrow B) \Leftrightarrow (A \Leftrightarrow (B \Leftrightarrow A)))$
(i) $((A \Rightarrow B) \vee (B \Rightarrow A))$
(j) $((\neg(A \Rightarrow B)) \Rightarrow A)$

1.6 Determine whether the following pairs are logically equivalent.

(a) $((A \Rightarrow B) \Rightarrow A)$ and A
(b) $(A \Leftrightarrow B)$ and $((A \Rightarrow B) \wedge (B \Rightarrow A))$
(c) $((\neg A) \vee B)$ and $((\neg B) \vee A)$
(d) $(\neg(A \Leftrightarrow B))$ and $(A \Leftrightarrow (\neg B))$
(e) $(A \vee (B \Leftrightarrow C))$ and $((A \vee B) \Leftrightarrow (A \vee C))$
(f) $(A \Rightarrow (B \Leftrightarrow C))$ and $((A \Rightarrow B) \Leftrightarrow (A \Rightarrow C))$
(g) $(A \wedge (B \Leftrightarrow C))$ and $((A \wedge B) \Leftrightarrow (A \wedge C))$

1.7 Prove:

(a) $(A \Rightarrow B)$ is logically equivalent to $((\neg A) \vee B)$.
(b) $(A \Rightarrow B)$ is logically equivalent to $(\neg(A \wedge (\neg B)))$.

1.8 Prove that $\mathscr{B}$ is logically equivalent to $\mathscr{C}$ if and only if $\mathscr{B}$ logically implies $\mathscr{C}$ and $\mathscr{C}$ logically implies $\mathscr{B}$.

1.9 Show that $\mathscr{B}$ and $\mathscr{C}$ are logically equivalent if and only if, in their truth tables, the columns under $\mathscr{B}$ and $\mathscr{C}$ are the same.

1.10 Prove that $\mathscr{B}$ and $\mathscr{C}$ are logically equivalent if and only if $(\neg\mathscr{B})$ and $(\neg\mathscr{C})$ are logically equivalent.

1.11 Which of the following statement forms are logically implied by $(A \wedge B)$?

(a) A	(d) $((\neg A) \vee B)$	(g) $(A \Rightarrow B)$
(b) B	(e) $((\neg B) \Rightarrow A)$	(h) $((\neg B) \Rightarrow (\neg A))$
(c) $(A \vee B)$	(f) $(A \Leftrightarrow B)$	(i) $(A \wedge (\neg B))$

1.12 Repeat Exercise 1.11 with $(A \wedge B)$ replaced by $(A \Rightarrow B)$ and by $(\neg(A \Rightarrow B))$, respectively.

1.13 Repeat Exercise 1.11 with $(A \wedge B)$ replaced by $(A \vee B)$.

1.14 Repeat Exercise 1.11 with $(A \wedge B)$ replaced by $(A \Leftrightarrow B)$ and by $(\neg(A \Leftrightarrow B))$, respectively.

A statement form that is false for all possible truth values of its statement letters is said to be *contradictory*. Its truth table has only Fs in the column under the statement form. One example is $(A \Leftrightarrow (\neg A))$:

A	$(\neg A)$	$(A \Leftrightarrow (\neg A))$
T	F	F
F	T	F

Another is $(A \wedge (\neg A))$.

Notice that a statement form $\mathscr{B}$ is a tautology if and only if $(\neg\mathscr{B})$ is contradictory, and vice versa.

A sentence (in some natural language like English or in a formal theory)† that arises from a tautology by the substitution of sentences for all the statement letters, with occurrences of the same statement letter being replaced by the same sentence, is said to be *logically true* (according to the propositional calculus). Such a sentence may be said to be true by virtue of its truth-functional structure alone. An example is the English sentence, ‘If it is raining or it is snowing, and it is not snowing, then it is raining’, which arises by substitution from the tautology $(((A \vee B) \wedge (\neg B)) \Rightarrow A)$. A sentence that comes from a contradictory statement form by means of substitution is said to be *logically false* (according to the propositional calculus).

Now let us prove a few general facts about tautologies.

†By a formal theory we mean an artificial language in which the notions of *meaningful expressions, axioms* and *rules of inference* are precisely described (see page 34).

PROPOSITION 1.2

If $\mathscr{B}$ and $(\mathscr{B} \Rightarrow \mathscr{C})$ are tautologies, then so is $\mathscr{C}$.

Proof

Assume that $\mathscr{B}$ and $(\mathscr{B} \Rightarrow \mathscr{C})$ are tautologies. If $\mathscr{C}$ took the value F for some assignment of truth values to the statement letters of $\mathscr{B}$ and $\mathscr{C}$, then, since $\mathscr{B}$ is a tautology, $\mathscr{B}$ would take the value T and, therefore, $(\mathscr{B} \Rightarrow \mathscr{C})$ would have the value F for that assignment. This contradicts the assumption that $(\mathscr{B} \Rightarrow \mathscr{C})$ is a tautology. Hence, $\mathscr{C}$ never takes the value F.

PROPOSITION 1.3

If $\mathscr{T}$ is a tautology containing as statement letters $A_1, A_2, \ldots, A_n$, and $\mathscr{B}$ arises from $\mathscr{T}$ by substituting statement forms $\mathscr{S}_1, \mathscr{S}_2, \ldots, \mathscr{S}_n$ for $A_1, A_2, \ldots, A_n$, respectively, then $\mathscr{B}$ is a tautology; that is, substitution in a tautology yields a tautology.

Example
Let $\mathscr{T}$ be $((A_1 \wedge A_2) \Rightarrow A_1)$, let $\mathscr{S}_1$ be $(B \vee C)$ and let $\mathscr{S}_2$ be $(C \wedge D)$. Then $\mathscr{B}$ is $(((B \vee C) \wedge (C \wedge D)) \Rightarrow (B \vee C))$.

Proof

Assume that $\mathscr{T}$ is a tautology. For any assignment of truth values to the statement letters in $\mathscr{B}$, the forms $\mathscr{S}_1, \ldots, \mathscr{S}_n$ have truth values $x_1, \ldots, x_n$ (where each x_n is T or F). If we assign the values $x_1, \ldots, x_n$ to $A_1, \ldots, A_n$, respectively, then the resulting truth value of $\mathscr{T}$ is the truth value of $\mathscr{B}$ for the given assignment of truth values. Since $\mathscr{T}$ is a tautology, this truth value must be T. Thus, $\mathscr{B}$ always takes the value T.

PROPOSITION 1.4

If $\mathscr{C}_1$ arises from $\mathscr{B}_1$ by substitution of $\mathscr{C}$ for one or more occurrences of $\mathscr{B}$, then $((\mathscr{B} \Leftrightarrow \mathscr{C}) \Rightarrow (\mathscr{B}_1 \Leftrightarrow \mathscr{C}_1))$ is a tautology. Hence, if $\mathscr{B}$ and $\mathscr{C}$ are logically equivalent, then so are $\mathscr{B}_1$ and $\mathscr{C}_1$.

Example
Let $\mathscr{B}_1$ be $(C \vee D)$, let $\mathscr{B}$ be C, and let $\mathscr{C}$ be $(\neg(\neg C))$. Then $\mathscr{C}_1$ is $((\neg(\neg C)) \vee D)$. Since C and $(\neg(\neg C))$ are logically equivalent, $(C \vee D)$ and $((\neg(\neg C)) \vee D)$ are also logically equivalent.

Proof

Consider any assignment of truth values to the statement letters. If $\mathscr{B}$ and $\mathscr{C}$ have opposite truth values under this assignment, then $(\mathscr{B} \Leftrightarrow \mathscr{C})$ takes the value F, and, hence, $((\mathscr{B} \Leftrightarrow \mathscr{C}) \Rightarrow (\mathscr{B}_1 \Leftrightarrow \mathscr{C}_1))$ is T. If $\mathscr{B}$ and $\mathscr{C}$ take the same truth values, then so do $\mathscr{B}_1$ and $\mathscr{C}_1$, since $\mathscr{C}_1$ differs from $\mathscr{B}_1$ only in containing $\mathscr{C}$ in some places where $\mathscr{B}_1$ contains $\mathscr{B}$. Therefore, in this case, $(\mathscr{B} \Leftrightarrow \mathscr{C})$ is T, $(\mathscr{B}_1 \Leftrightarrow \mathscr{C}_1)$ is T, and, thus, $((\mathscr{B} \Leftrightarrow \mathscr{C}) \Rightarrow (\mathscr{B}_1 \Leftrightarrow \mathscr{C}_1))$ is T.

Parentheses

It is profitable at this point to agree on some conventions to avoid the use of so many parentheses in writing formulas. This will make the reading of complicated expressions easier.

First, we may omit the outer pair of parentheses of a statement form. (In the case of statement letters, there is no outer pair of parentheses.)

Second, we arbitrarily establish the following decreasing order of strength of the connectives: $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$. Now we shall explain a step-by-step process for restoring parentheses to an expression obtained by eliminating some or all parentheses from a statement form. Find the leftmost occurrence of the strongest connective that has not yet been processed.

(i) If the connective is $\neg$ and it precedes a statement form $\mathscr{B}$, restore left and right parentheses to obtain $(\neg\mathscr{B})$.

(ii) If the connective is a binary connective C and it is preceded by a statement form $\mathscr{B}$ and followed by a statement form $\mathscr{D}$, restore left and right parentheses to obtain $(\mathscr{B}\ C\ \mathscr{D})$.

(iii) If neither (i) nor (ii) holds, ignore the connective temporarily and find the leftmost occurrence of the strongest of the remaining unprocessed connectives and repeat (i)–(iii) for that connective.

Examples

Parentheses are restored to the expression in the first line of each of the following in the steps shown:

1. $A \Leftrightarrow (\neg B) \vee C \Rightarrow A$
   $A \Leftrightarrow ((\neg B) \vee C) \Rightarrow A$
   $A \Leftrightarrow (((\neg B) \vee C) \Rightarrow A)$
   $(A \Leftrightarrow (((\neg B) \vee C) \Rightarrow A))$
2. $A \Rightarrow \neg B \Rightarrow C$
   $A \Rightarrow (\neg B) \Rightarrow C$
   $(A \Rightarrow (\neg B)) \Rightarrow C$
   $((A \Rightarrow (\neg B)) \Rightarrow C)$
3. $B \Rightarrow \neg\neg A$
   $B \Rightarrow \neg(\neg A)$

$B \Rightarrow (\neg(\neg A))$
$(B \Rightarrow (\neg(\neg A)))$

4. $A \vee \neg(B \Rightarrow A \vee B)$
$A \vee \neg(B \Rightarrow (A \vee B))$
$A \vee (\neg(B \Rightarrow (A \vee B)))$
$(A \vee (\neg(B \Rightarrow (A \vee B))))$

Not every form can be represented without the use of parentheses. For example, parentheses cannot be further eliminated from $A \Rightarrow (B \Rightarrow C)$, since $A \Rightarrow B \Rightarrow C$ stands for $((A \Rightarrow B) \Rightarrow C)$. Likewise, the remaining parentheses cannot be removed from $\neg(A \vee B)$ or from $A \wedge (B \Rightarrow C)$.

Exercises

1.15 Eliminate as many parentheses as possible from the following forms.

(a) $((B \Rightarrow (\neg A)) \wedge C)$
(b) $(A \vee (B \vee C))$
(c) $(((A \wedge (\neg B)) \wedge C) \vee D)$
(d) $((B \vee (\neg C)) \vee (A \wedge B))$
(e) $((A \Leftrightarrow B) \Leftrightarrow (\neg(C \vee D)))$
(f) $((\neg(\neg(\neg(B \vee C)))) \Leftrightarrow (B \Leftrightarrow C))$
(g) $(\neg((\neg(\neg(B \vee C))) \Leftrightarrow (B \Leftrightarrow C)))$
(h) $((((A \Rightarrow B) \Rightarrow (C \Rightarrow D)) \wedge (\neg A)) \vee C)$

1.16 Restore parentheses to the following forms.

(a) $C \vee \neg A \wedge B$
(b) $B \Rightarrow \neg\neg\neg A \wedge C$
(c) $C \Rightarrow \neg(A \wedge B \Rightarrow C) \wedge A \Leftrightarrow B$
(d) $C \Rightarrow A \Rightarrow A \Leftrightarrow \neg A \vee B$

1.17 Determine whether the following expressions are abbreviations of statement forms and, if so, restore all parentheses.

(a) $\neg\neg A \Leftrightarrow A \Leftrightarrow B \vee C$
(b) $\neg(\neg A \Leftrightarrow A) \Leftrightarrow B \vee C$
(c) $\neg(A \Rightarrow B) \vee C \vee D \Rightarrow B$
(d) $A \Leftrightarrow (\neg A \vee B) \Rightarrow (A \wedge (B \vee C)))$
(e) $\neg A \vee B \vee C \wedge D \Leftrightarrow A \wedge \neg A$
(f) $((A \Rightarrow B \wedge (C \vee D) \wedge (A \vee D))$

1.18 If we write $\neg\mathscr{B}$ instead of $(\neg\mathscr{B})$, $\Rightarrow \mathscr{B}\mathscr{C}$ instead of $(\mathscr{B} \Rightarrow \mathscr{C})$, $\wedge\mathscr{B}\mathscr{C}$ instead of $(\mathscr{B} \wedge \mathscr{C})$, $\vee\mathscr{B}\mathscr{C}$ instead of $(\mathscr{B} \vee \mathscr{C})$, and $\Leftrightarrow \mathscr{B}\mathscr{C}$ instead of $(\mathscr{B} \Leftrightarrow \mathscr{C})$, then there is no need for parentheses. For example, $((\neg A) \wedge (B \Rightarrow (\neg D)))$, which is ordinarily abbreviated as $\neg A \wedge (B \Rightarrow \neg D)$, becomes $\wedge \neg A \Rightarrow B \neg D$. This way of writing forms is called *Polish notation*.

(a) Write $((C \Rightarrow (\neg A)) \vee B)$ and $(C \vee ((B \wedge (\neg D)) \Rightarrow C))$ in this notation.
(b) If we count $\Rightarrow, \wedge, \vee$, and $\Leftrightarrow$ each as $+1$, each statement letter as -1 and $\neg$ as 0, prove that an expression $\mathscr{B}$ in this parenthesis-free notation is a statement form if and only if (i) the sum of the symbols of $\mathscr{B}$ is -1 and (ii) the sum of the symbols in any proper initial segment of $\mathscr{B}$ is nonnegative. (If an expression $\mathscr{B}$ can be written in the form $\mathscr{C}\mathscr{D}$, where $\mathscr{C} \neq \mathscr{B}$, then $\mathscr{C}$ is called a *proper initial segment* of $\mathscr{B}$.)
(c) Write the statement forms of Exercise 1.15 in Polish notation.

(d) Determine whether the following expressions are statement forms in Polish notation. If so, write the statement forms in the standard way.

(i) $\neg \Rightarrow ABC \vee AB\neg C$ (iii) $\vee \wedge \vee \neg A \neg BC \wedge \vee AC \vee \neg C \neg A$

(ii) $\Rightarrow\Rightarrow AB \Rightarrow\Rightarrow BC \Rightarrow \neg AC$ (iv) $\vee \wedge B \wedge BBB$

1.19 Determine whether each of the following is a tautology, is contradictory, or neither.

(a) $B \Leftrightarrow (B \vee B)$
(b) $((A \Rightarrow B) \wedge B) \Rightarrow A$
(c) $(\neg A) \Rightarrow (A \wedge B)$
(d) $(A \Rightarrow B) \Rightarrow ((B \Rightarrow C) \Rightarrow (A \Rightarrow C))$
(e) $(A \Leftrightarrow \neg B) \Rightarrow A \vee B$
(f) $A \wedge (\neg(A \vee B))$
(g) $(A \Rightarrow B) \Leftrightarrow ((\neg A) \vee B)$
(h) $(A \Rightarrow B) \Leftrightarrow \neg(A \wedge (\neg B))$
(i) $(B \Leftrightarrow (B \Rightarrow A)) \Rightarrow A$
(j) $A \wedge \neg A \Rightarrow B$

1.20 If A and B are true and C is false, what are the truth values of the following statement forms?

(a) $A \vee C$
(b) $A \wedge C$
(c) $\neg A \wedge \neg C$
(d) $A \Leftrightarrow \neg B \vee C$
(e) $B \vee \neg C \Rightarrow A$
(f) $(B \vee A) \Rightarrow (B \Rightarrow \neg C)$
(g) $(B \Rightarrow \neg A) \Leftrightarrow (A \Leftrightarrow C)$
(h) $(B \Rightarrow A) \Rightarrow ((A \Rightarrow \neg C) \Rightarrow (\neg C \Rightarrow B))$

1.21 If $A \Rightarrow B$ is T, what can be deduced about the truth values of the following?

(a) $A \vee C \Rightarrow B \vee C$
(b) $A \wedge C \Rightarrow B \wedge C$
(c) $\neg A \wedge B \Leftrightarrow A \vee B$

1.22 What further truth values can be deduced from those shown?

(a) $\neg A \vee (A \underset{\mathrm{F}}{\Rightarrow} B)$
(b) $\neg(A \underset{\mathrm{T}}{\wedge} B) \Leftrightarrow \neg A \Rightarrow \neg B$
(c) $(\neg A \vee B) \underset{\mathrm{F}}{\Rightarrow} (A \Rightarrow \neg C)$
(d) $(A \underset{\mathrm{F}}{\Leftrightarrow} B) \underset{\mathrm{T}}{\Leftrightarrow} (C \Rightarrow \neg A)$

1.23 If $A \Leftrightarrow B$ is F, what can be deduced about the truth values of the following?

(a) $A \wedge B$ (b) $A \vee B$ (c) $A \Rightarrow B$ (d) $A \wedge C \Leftrightarrow B \wedge C$

1.24 Repeat Exercise 1.23, but assume that $A \Leftrightarrow B$ is T.

1.25 What further truth values can be deduced from those given?

(a) $(A \wedge B) \underset{\mathrm{F}}{\Leftrightarrow} (\underset{\mathrm{F}}{A} \vee B)$
(b) $(A \Rightarrow \neg B) \underset{\mathrm{F}}{\Rightarrow} (C \Rightarrow B)$

1.26 (a) Apply Proposition 1.3 when $\mathscr{T}$ is $A_1 \Rightarrow A_1 \vee A_2$, $\mathscr{S}_1$ is $B \wedge D$, and $\mathscr{S}_2$ is $\neg B$.

(b) Apply Proposition 1.4 when $\mathscr{B}_1$ is $(B \Rightarrow C) \wedge D$, $\mathscr{B}$ is $B \Rightarrow C$, and $\mathscr{C}$ is $\neg B \vee C$.

1.27 Show that each statement form in column I is logically equivalent to the form next to it in column II.

	I	*II*	
(a)	$A \Rightarrow (B \Rightarrow C)$	$(A \wedge B) \Rightarrow C$	
(b)	$A \wedge (B \vee C)$	$(A \wedge B) \vee (A \wedge C)$	(Distributive law)
(c)	$A \vee (B \wedge C)$	$(A \vee B) \wedge (A \vee C)$	(Distributive law)
(d)	$(A \wedge B) \vee \neg B$	$A \vee \neg B$	
(e)	$(A \vee B) \wedge \neg B$	$A \wedge \neg B$	
(f)	$A \Rightarrow B$	$\neg B \Rightarrow \neg A$	(Law of the contrapositive)
(g)	$A \Leftrightarrow B$	$B \Leftrightarrow A$	(Biconditional commutativity)
(h)	$(A \Leftrightarrow B) \Leftrightarrow C$	$A \Leftrightarrow (B \Leftrightarrow C)$	(Biconditional associativity)
(i)	$A \Leftrightarrow B$	$(A \wedge B) \vee (\neg A \wedge \neg B)$	
(j)	$\neg(A \Leftrightarrow B)$	$A \Leftrightarrow \neg B$	
(k)	$\neg(A \vee B)$	$(\neg A) \wedge (\neg B)$	(De Morgan's law)
(l)	$\neg(A \wedge B)$	$(\neg A) \vee (\neg B)$	(De Morgan's law)
(m)	$A \vee (A \wedge B)$	A	
(n)	$A \wedge (A \vee B)$	A	
(o)	$A \wedge B$	$B \wedge A$	(Commutativity of conjunction)
(p)	$A \vee B$	$B \vee A$	(Commutativity of disjunction)
(q)	$(A \wedge B) \wedge C$	$A \wedge (B \wedge C)$	(Associativity of conjunction)
(r)	$(A \vee B) \vee C$	$A \vee (B \vee C)$	(Associativity of disjunction)

1.28 Show the logical equivalence of the following pairs.

(a) $\mathscr{T} \wedge \mathscr{B}$ and $\mathscr{B}$, where $\mathscr{T}$ is a tautology.
(b) $\mathscr{T} \vee \mathscr{B}$ and $\mathscr{T}$, where $\mathscr{T}$ is a tautology.
(c) $\mathscr{F} \wedge \mathscr{B}$ and $\mathscr{F}$, where $\mathscr{F}$ is contradictory.
(d) $\mathscr{F} \vee \mathscr{B}$ and $\mathscr{B}$, where $\mathscr{F}$ is contradictory.

1.29

(a) Show the logical equivalence of $\neg(A \Rightarrow B)$ and $A \wedge \neg B$.
(b) Show the logical equivalence of $\neg(A \Leftrightarrow B)$ and $(A \wedge \neg B) \vee (\neg A \wedge B)$.
(c) For each of the following statement forms, find a statement form that is logically equivalent to its negation and in which negation signs apply only to statement letters.

(i) $A \Rightarrow (B \Leftrightarrow \neg C)$
(ii) $\neg A \vee (B \Rightarrow C)$
(iii) $A \wedge (B \vee \neg C)$

1.30 (*Duality*)

(a) If $\mathscr{B}$ is a statement form involving only $\neg$, $\wedge$, and $\vee$, and $\mathscr{B}'$ results from $\mathscr{B}$ by replacing each $\wedge$ by $\vee$ and each $\vee$ by $\wedge$, show that $\mathscr{B}$ is a tautology if and only if $\neg\mathscr{B}'$ is a tautology. Then prove that, if $\mathscr{B} \Rightarrow \mathscr{C}$ is a tautology, then so is $\mathscr{C}' \Rightarrow \mathscr{B}'$, and if $\mathscr{B} \Leftrightarrow \mathscr{C}$ is a tautology, then so is $\mathscr{B}' \Leftrightarrow \mathscr{C}'$. (Here $\mathscr{C}$ is also assumed to involve only $\neg$, $\wedge$ and $\vee$.)

(b) Among the logical equivalences in Exercise 1.27, derive (c) from (b), (e) from (d), (l) from (k), (p) from (o), and (r) from (q).

(c) If $\mathscr{B}$ is a statement form involving only $\neg$, $\wedge$ and $\vee$, and $\mathscr{B}^*$ results from $\mathscr{B}$ by interchanging $\wedge$ and $\vee$ and replacing every statement letter by its negation, show that $\mathscr{B}^*$ is logically equivalent to $\neg\mathscr{B}$. Find a statement form that is logically equivalent to the negation of $(A \vee B \vee C) \wedge (\neg A \vee \neg B \vee D)$, in which $\neg$ applies only to statement letters.

1.31

(a) Prove that a statement form that contains $\Leftrightarrow$ as its only connective is a tautology if and only if each statement letter occurs an even number of times.

(b) Prove that a statement form that contains $\neg$ and $\Leftrightarrow$ as its only connectives is a tautology if and only if $\neg$ and each statement letter occur an even number of times.

1.32 (Shannon, 1938) An electric circuit containing only on-off switches (when a switch is on, it passes current; otherwise it does not) can be represented by a diagram in which, next to each switch, we put a letter representing a necessary and sufficient condition for the switch to be on (see Figure 1.1). The condition that a current flows through this network can be given by the statement form $(A \wedge B) \vee (C \wedge \neg A)$. A statement form representating the circuit shown in Figure 1.2 is $(A \wedge B) \vee ((C \vee A) \wedge \neg B)$, which is logically equivalent to each of the following forms by virtue of the indicated logical equivalence of Exercise 1.27.

$$((A \wedge B) \vee (C \vee A)) \wedge ((A \wedge B) \vee \neg B) \qquad \text{(c)}$$
$$((A \wedge B) \vee (C \vee A)) \wedge (A \vee \neg B) \qquad \text{(d)}$$
$$((A \wedge B) \vee (A \vee C)) \wedge (A \vee \neg B) \qquad \text{(p)}$$
$$(((A \wedge B) \vee A) \vee C) \wedge (A \vee \neg B) \qquad \text{(r)}$$
$$(A \vee C) \wedge (A \vee \neg B) \qquad \text{(p), (m)}$$
$$A \vee (C \wedge \neg B) \qquad \text{(c)}$$

Hence, the given circuit is equivalent to the simpler circuit shown in Figure 1.3. (Two circuits are said to be *equivalent* if current flows through one if and only if it flows through the other, and one circuit is *simpler* if it contains fewer switches.)

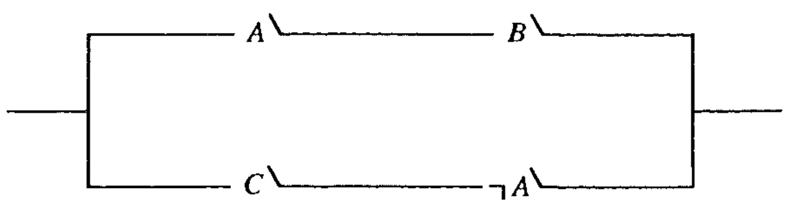

Figure. 1.1

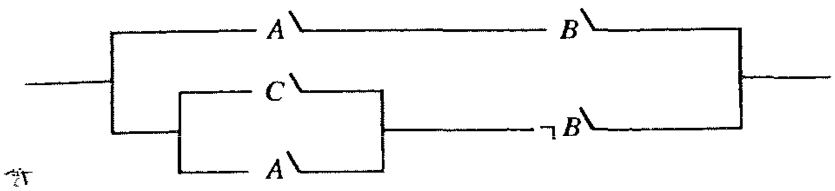

Figure. 1.2

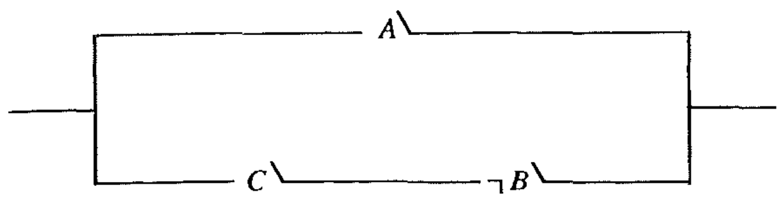

Figure. 1.3

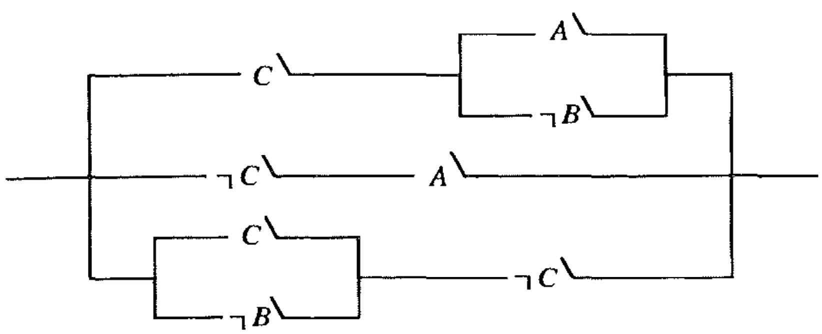

Figure. 1.4

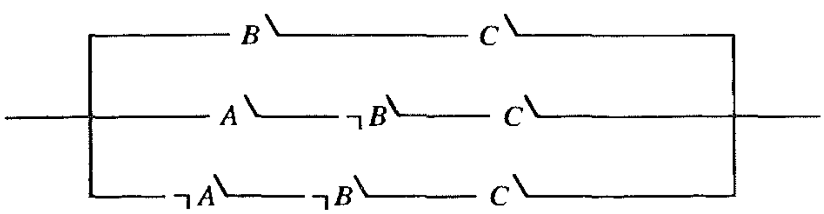

Figure. 1.5

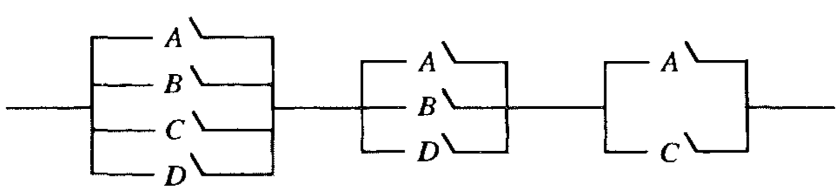

Figure. 1.6

(a) Find simpler equivalent circuits for those shown in Figures 1.4, 1.5 and 1.6.

(b) Assume that each of the three members of a committee votes *yes* on a proposal by pressing a button. Devise as simple a circuit as you can that will allow current to pass when and only when at least two of the members vote in the affirmative.

(c) We wish a light to be controlled by two different wall switches in a room in such a way that flicking either one of these switches will turn the light on if it is off and turn it off if it is on. Construct a simple circuit to do the required job.

1.33 Determine whether the following arguments are logically correct by representing each sentence as a statement form and checking whether the conclusion is logically implied by the conjunction of the assumptions. (To do this, assign T to each assumption and F to the conclusion, and determine whether a contradiction results.)

(a) If Jones is a communist, Jones is an atheist. Jones is an atheist. Therefore, Jones is a communist.
(b) If the temperature and air pressure remained constant, there was no rain. The temperature did remain constant. Therefore, if there was rain, then the air pressure did not remain constant.
(c) If Gorton wins the election, then taxes will increase if the deficit will remain high. If Gorton wins the election, the deficit will remain high. Therefore, if Gorton wins the election, taxes will increase.
(d) If the number x ends in 0, it is divisible by 5. x does not end in 0. Hence, x is not divisible by 5.
(e) If the number x ends in 0, it is divisible by 5. x is not divisible by 5. Hence, x does not end in 0.
(f) If $a = 0$ or $b = 0$, then $ab = 0$. But $ab \neq 0$. Hence, $a \neq 0$ and $b \neq 0$.
(g) A sufficient condition for f to be integrable is that g be bounded. A necessary condition for h to be continuous is that f is integrable. Hence, if g is bounded or h is continuous, then f is integrable.
(h) Smith cannot both be a running star and smoke cigarettes. Smith is not a running star. Therefore, Smith smokes cigarettes.
(i) If Jones drove the car, Smith is innocent. If Brown fired the gun, then Smith is not innocent. Hence, if Brown fired the gun, then Jones did not drive the car.

1.34 Which of the following sets of statement forms are satisfiable, in the sense that there is an assignment of truth values to the statement letters that makes all the forms in the set true?

(a) $A \Rightarrow B$
$B \Rightarrow C$
$C \vee D \Leftrightarrow \neg B$
(b) $\neg(\neg B \vee A)$
$A \vee \neg C)$
$B \Rightarrow \neg C$
(c) $D \Rightarrow B$
$A \vee \neg B$
$\neg(D \wedge A)$
D

1.35 Check each of the following sets of statements for consistency by representing the sentences as statement forms and then testing their conjunction to see whether it is contradictory.

(a) Either the witness was intimidated or, if Doherty committed suicide, a note was found. If the witness was intimidated, then Doherty did not commit suicide. If a note was found, then Doherty committed suicide.
(b) The contract is satisfied if and only if the building is completed by 30 November. The building is completed by 30 November if and only if the electrical subcontractor completes his work by 10 November. The bank loses money if and only if the contract is not satisfied. Yet the electrical subcontractor completes his work by 10 November if and only if the bank loses money.

1.3 ADEQUATE SETS OF CONNECTIVES

Every statement form containing n statement letters generates a corresponding truth function of n arguments. The arguments and values of the function are T or F. Logically equivalent forms generate the same truth function. A natural question is whether all truth functions are so generated.

PROPOSITION 1.5

Every truth function is generated by a statement form involving the connectives $\neg$, $\wedge$ and $\vee$.

Proof

(Refer to Examples 1 and 2 below for clarification.) Let $f(x_1, \ldots, x_n)$ be a truth function. Clearly f can be represented by a truth table of 2^n rows, where each row represents some assignment of truth values to the variables $x_1, \ldots, x_n$, followed by the corresponding value of $f(x_1, \ldots, x_n)$. If $1 \leqslant i \leqslant 2^n$, let C_i be the conjunction $U_1^i \wedge U_2^i \wedge \ldots \wedge U_n^i$, where U_j^i is A_j if, in the ith row of the truth table, x_j takes the value T, and U_j^i is $\neg A_j$ if x_j takes the value F in that row. Let D be the disjunction of all those C_is such that f has the value T for the ith row of the truth table. (If there are no such rows, then f always takes the value F, and we let D be $A_1 \wedge \neg A_1$, which satisfies the theorem.) Notice that D involves only $\neg$, $\wedge$ and $\vee$. To see that D has f as its corresponding truth function, let there be given an assignment of truth values to the statement letters $A_1, \ldots, A_n$, and assume that the corrreponding assignment to the variables $x_1, \ldots, x_n$ is row k of the truth table for f. Then C_k has the value T for this assignment, whereas every other C_i has the value F. If f has the value T for row k, then C_k is a disjunct of D. Hence,

D would also have the value T for this assignment. If f has the value F for row k, then C_k is not a disjunct of D and all the disjuncts take the value F for this assignment. Therefore, D would also have the value F. Thus, D generates the truth function f.

Examples

1.

x_1	x_2	$f(x_1,x_2)$
T	T	F
F	T	T
T	F	T
F	F	T

D is $(\neg A_1 \wedge A_2) \vee (A_1 \wedge \neg A_2) \vee (\neg A_1 \wedge \neg A_2)$.

2.

x_1	x_2	x_3	$g(x_1,x_2,x_3)$
T	T	T	T
F	T	T	F
T	F	T	T
F	F	T	T
T	T	F	F
F	T	F	F
T	F	F	F
F	F	F	T

$$D \text{ is } (A_1 \wedge A_2 \wedge A_3) \vee (A_1 \wedge \neg A_2 \wedge A_3) \vee (\neg A_1 \wedge \neg A_2 \wedge A_3) \\ \vee (\neg A_1 \wedge \neg A_2 \wedge \neg A_3).$$

Exercise

1.36 Find statement forms in the connectives $\neg$, $\wedge$ and $\vee$ that have the following truth functions.

x_1	x_2	x_3	$f(x_1,x_2,x_3)$	$g(x_1,x_2,x_3)$	$h(x_1,x_2,x_3)$
T	T	T	T	T	F
F	T	T	T	T	T
T	F	T	T	T	F
F	F	T	F	F	F
T	T	F	F	T	T
F	T	F	F	F	T
T	F	F	F	T	F
F	F	F	T	F	T

COROLLARY 1.6

Every truth function can be generated by a statement form containing as connectives only $\wedge$ and $\neg$, or only $\vee$ and $\neg$, or only $\Rightarrow$ and $\neg$.

Proof

Notice that $\mathscr{B} \vee \mathscr{C}$ is logically equivalent to $\neg(\neg\mathscr{B} \wedge \neg\mathscr{C})$. Hence, by the second part of Proposition 1.4, any statement form in $\wedge$, $\vee$ and $\neg$ is logically equivalent to a statement form in only $\wedge$ and $\neg$ [obtained by replacing all expressions $\mathscr{B} \vee \mathscr{C}$ by $\neg(\neg\mathscr{B} \wedge \neg\mathscr{C})$]. The other parts of the corollary are similar consequences of the following tautologies:

$$\mathscr{B} \wedge \mathscr{C} \Leftrightarrow \neg(\neg\mathscr{B} \vee \neg\mathscr{C})$$
$$\mathscr{B} \vee \mathscr{C} \Leftrightarrow (\neg\mathscr{B} \Rightarrow \mathscr{C})$$
$$\mathscr{B} \wedge \mathscr{C} \Leftrightarrow \neg(\mathscr{B} \Rightarrow \neg\mathscr{C})$$

We have just seen that there are certain pairs of connectives – for example, $\wedge$ and $\neg$ – in terms of which all truth functions are definable. It turns out that there is a single connective, $\downarrow$ (joint denial), that will do the same job. Its truth table is:

A	B	$A \downarrow B$
T	T	F
F	T	F
T	F	F
F	F	T

$A \downarrow B$ is true when and only when neither A nor B is true. Clearly, $\neg A \Leftrightarrow (A \downarrow A)$ and $(A \wedge B) \Leftrightarrow ((A \downarrow A) \downarrow (B \downarrow B))$ are tautologies. Hence, the adequacy of $\downarrow$ for the construction of all truth functions follows from Corollary 1.6.

Another connective, $|$ (alternative denial), is also adequate for this purpose. Its truth table is

A	B	$A \mid B$
T	T	F
F	T	T
T	F	T
F	F	T

$A \mid B$ is true when and only when not both A and B are true. The adequacy of $|$ follows from the tautologies $\neg A \Leftrightarrow (A \mid A)$ and $(A \vee B) \Leftrightarrow ((A \mid A) \mid (B \mid B))$.

PROPOSITION 1.7

The only binary connectives that alone are adequate for the construction of all truth functions are $\downarrow$ and $|$.

Proof

Assume that $h(A, B)$ is an adequate connective. Now, if h(T,T) were T, then any statement form built up using h alone would take the value T when all its statement letters take the value T. Hence, $\neg A$ would not be definable in terms of h. So, h(T,T) = F. Likewise, h(F,F) = T. Thus, we have the partial truth table

A	B	$h(A,B)$
T	T	F
F	T	
T	F	
F	F	T

If the second and third entries in the last column are F, F or T, T, then h is $\downarrow$ or $|$. If they are F, T, then $h(A, B) \Leftrightarrow \neg B$ is a tautology; and if they are T, F, then $h(A, B) \Leftrightarrow \neg A$ is a tautology. In both cases, h would be definable in terms of $\neg$. But $\neg$ is not adequate by itself because the only truth functions of one variable definable from it are the identity function and negation itself, whereas the truth function that is always T would not be definable.

Exercises

1.37 Prove that each of the pairs $\Rightarrow$, $\vee$ and $\neg$, $\Leftrightarrow$ is not alone adequate to express all truth functions.

1.38
(a) Prove that $A \vee B$ can be expressed in terms of $\Rightarrow$ alone.
(b) Prove that $A \wedge B$ cannot be expressed in terms of $\Rightarrow$ alone.
(c) Prove that $A \Leftrightarrow B$ cannot be expressed in terms of $\Rightarrow$ alone.

1.39 Show that any two of the connectives $\{\wedge, \Rightarrow, \Leftrightarrow\}$ serve to define the remaining one.

1.40 With one variable A, there are four truth functions:

A	$\neg A$	$A \vee \neg A$	$A \wedge \neg A$
T	F	T	F
F	T	T	F

(a) With two variable A and B, how many truth functions are there ?
(b) How many truth functions of n variables are there ?

1.41 Show that the truth function h determined by $(A \vee B) \Rightarrow \neg C$ generates all truth functions.

1.42 By a *literal* we mean a statement letter or a negation of a statement letter. A statement form is said to be in *disjunctive normal form* (dnf) if it is a disjunction consisting of one or more disjuncts, each of which is a conjunction of one or more literals – for example, $(A \wedge B) \vee (\neg A \wedge C)$, $(A \wedge B \wedge \neg A) \vee (C \wedge \neg B) \vee (A \wedge \neg C)$, $A, A \wedge B$, and $A \vee (B \vee C)$. A form is in *conjunctive normal form* (cnf) if it is a conjunction of one or more conjuncts, each of which is a disjunction of one or more literals – for example, $(B \vee C) \wedge (A \vee B)$, $(B \vee \neg C) \wedge (A \vee D)$, $A \wedge (B \vee A) \wedge (\neg B \vee A)$, $A \vee \neg B$, $A \wedge B$, A. Note that our terminology considers a literal to be a (degenerate) conjunction and a (degenerate) disjunction.

(a) The proof of Proposition 1.5 shows that every statement form $\mathscr{B}$ is logically equivalent to one in disjunctive normal form. By applying this result to $\neg\mathscr{B}$, prove that $\mathscr{B}$ is also logically equivalent to a form in conjunctive normal form.

(b) Find logically equivalent dnfs and cnfs for $\neg(A \Rightarrow B) \vee (\neg A \wedge C)$ and $A \Leftrightarrow ((B \wedge \neg A) \vee C)$. [*Hint*: Instead of relying on Proposition 1.5, it is usually easier to use Exercise 1.27(b) and (c).]

(c) A dnf (cnf) is called *full* if no disjunct (conjunct) contains two occurrences of literals with the same letter and if a letter that occurs in one disjunct (conjunct) also occurs in all the others. For example, $(A \wedge \neg A \wedge B) \vee (A \wedge B)$, $(B \wedge B \wedge C) \vee (B \wedge C)$ and $(B \wedge C) \vee B$ are not full, whereas $(A \wedge B \wedge \neg C) \vee (A \wedge B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$ and $(A \wedge \neg B)$ $\vee(B \wedge A)$ are full dnfs.

  (i) Find full dnfs and cnfs logically equivalent to $(A \wedge B) \vee \neg A$ and $\neg(A \Rightarrow B) \vee (\neg A \wedge C)$.

  (ii) Prove that every non-contradictory (non-tautologous) statement form $\mathscr{B}$ is logically equivalent to a full dnf (cnf) $\mathscr{C}$, and, if $\mathscr{C}$ contains exactly n letters, then $\mathscr{B}$ is a tautology (is contradictory) if and only if $\mathscr{C}$ has 2^n disjuncts (conjuncts).

(d) For each of the following, find a logically equivalent dnf (cnf), and then find a logically equivalent full dnf (cnf),

  (i) $(A \vee B) \wedge (\neg B \vee C)$

  (ii) $\neg A \vee (B \Rightarrow \neg C)$

  (iii) $(A \wedge \neg B) \vee (A \wedge C)$

  (iv) $(A \vee B) \Leftrightarrow \neg C$

(e) Construct statement forms in $\neg$ and $\wedge$ (respectively, in $\neg$ and $\vee$ or in $\neg$ and $\Rightarrow$) logically equivalent to the statement forms in (d).

1.43 A statement form is said to be *satisfiable* if it is true for some assignment of truth values to its statement letters. The problem of determining the satisfiability of an arbitrary cnf plays an important role in the theory of computational complexity; it is an example of a so-called *$\mathscr{N}\mathscr{P}$-complete* problem (see Garey and Johnson, 1978).

(a) Show that $\mathscr{B}$ is satisfiable if and only if $\neg\mathscr{B}$ is not a tautology.

(b) Determine whether the following are satisfiable:

  (i) $(A \vee B) \wedge (\neg A \vee B \vee C) \wedge (\neg A \vee \neg B \vee \neg C)$

  (ii) $((A \Rightarrow B) \vee C) \Leftrightarrow (\neg B \wedge (A \vee C))$

(c) Given a disjunction $\mathscr{D}$ of four or more literals: $L_1 \vee L_2 \vee \ldots \vee L_n$, let $C_1, \ldots, C_{n-2}$ be statement letters that do not occur in $\mathscr{D}$, and construct the cnf $\mathscr{E}$:

$$(L_1 \vee L_2 \vee C_1) \wedge (\neg C_1 \vee L_3 \vee C_2) \wedge (\neg C_2 \vee L_4 \vee C_3) \wedge \ldots \\ \wedge (\neg C_{n-3} \vee L_{n-1} \vee C_{n-2}) \wedge (\neg C_{n-2} \vee L_n \vee \neg C_1)$$

Show that any truth assignment satisfying $\mathscr{D}$ can be extended to a truth assignment satisfying $\mathscr{E}$ and, conversely, any truth assignment satisfying

$\mathscr{E}$ is an extension of a truth assignment satisfying $\mathscr{D}$. (This permits the reduction of the problem of satisfying cnfs to the corresponding problem for cnfs with each conjunct containing at most three literals.)

(d) For a disjunction $\mathscr{D}$ of three literals $L_1 \vee L_2 \vee L_3$, show that a form that has the properties of $\mathscr{E}$ in (c) cannot be constructed, with $\mathscr{E}$ a cnf in which each conjunct contains at most two literals (R. Cowen).

1.44 (*Resolution*) Let $\mathscr{B}$ be a cnf and let C be a statement letter. If C is a disjunct of a disjunction $\mathscr{D}_1$ in $\mathscr{B}$ and $\neg C$ is a disjunct of another disjunction $\mathscr{D}_2$ in $\mathscr{B}$, then a non-empty disjunction obtained by eliminating C from $\mathscr{D}_1$ and $\neg C$ from $\mathscr{D}_2$ and forming the disjunction of the remaining literals (dropping repetitions) is said to be obtained from $\mathscr{B}$ by *resolution on C*. For example, if $\mathscr{B}$ is

$$(A \vee \neg C \vee \neg B) \wedge (\neg A \vee D \vee \neg B) \wedge (C \vee D \vee A),$$

the first and third conjuncts yield $A \vee \neg B \vee D$ by resolution on C. In addition, the first and second conjuncts yield $\neg C \vee \neg B \vee D$ by resolution on A, and the second and third conjuncts yield $D \vee \neg B \vee C$ by resolution on A. If we conjoin to $\mathscr{B}$ any new disjunctions obtained by resolution on all variables, and if we apply the same procedure to the new cnf and keep on iterating this operation, the process must eventually stop, and the final result is denoted $\mathscr{Res}(\mathscr{B})$. In the example, $\mathscr{Res}(\mathscr{B})$ is:

$$\begin{aligned}&(A \vee \neg C \vee \neg B) \wedge (\neg A \vee D \vee \neg B) \wedge (C \vee D \vee A) \wedge (\neg C \vee \neg B \vee D)\\ &\quad \wedge (D \vee \neg B \vee C) \wedge (A \vee \neg B \vee D) \wedge (D \vee \neg B)\end{aligned}$$

(Notice that we have not been careful about specifying the order in which conjuncts or disjuncts are written, since any two arrangements will be logically equivalent.)

(a) Find $\mathscr{Res}(\mathscr{B})$ when $\mathscr{B}$ is each of the following:
   (i) $(A \vee \neg B) \wedge B$
   (ii) $(A \vee B \vee C) \wedge (A \vee \neg B \vee C)$
   (iii) $(A \vee C) \wedge (\neg A \vee B) \wedge (A \vee \neg C) \wedge (\neg A \vee \neg B)$

(b) Show that $\mathscr{B}$ logically implies $\mathscr{Res}(\mathscr{B})$.

(c) If $\mathscr{B}$ is a cnf, let $\mathscr{B}_C$ be the cnf obtained from $\mathscr{B}$ by deleting those conjuncts that contain C or $\neg C$. Let $r_C(\mathscr{B})$ be the cnf that is the conjunction of $\mathscr{B}_C$ and all those disjunctions obtained from $\mathscr{B}$ by resolution on C. For example, if $\mathscr{B}$ is the cnf in the example above, then $r_C(\mathscr{B})$ is $(\neg A \vee D \vee \neg B) \wedge (A \vee \neg B \vee D)$. Prove that, if $r_C(\mathscr{B})$ is satisfiable, then so is $\mathscr{B}$. (R. Cowen)

(d) A cnf $\mathscr{B}$ is said to be a *blatant contradiction* if it contains some letter C and its negation $\neg C$ an conjuncts. An example of a blatant contradiction is $(A \vee B) \wedge B \wedge (C \vee D) \wedge \neg B$. Prove that if $\mathscr{B}$ is unsatisfiable, then $\mathscr{Res}(\mathscr{B})$ is a blatant contradiction. [*Hint*: Use induction on the number n of letters that occur in $\mathscr{B}$. In the induction step, use (c).]

(e) Prove that $\mathscr{B}$ is unsatisfiable if and only if $\mathscr{R}e\mathscr{s}(\mathscr{B})$ is a blatant contradiction.

1.45 Let $\mathscr{B}$ and $\mathscr{D}$ be statement forms such that $\mathscr{B} \Rightarrow \mathscr{D}$ is a tautology.

(a) If $\mathscr{B}$ and $\mathscr{D}$ have no statement letters in common, show that either $\mathscr{B}$ is contradictory or $\mathscr{D}$ is a tautology.
(b) (*Craig's interpolation theorem*) If $\mathscr{B}$ and $\mathscr{D}$ have the statement letters $B_1, \ldots, B_n$ in common, prove that there is a statement form $\mathscr{C}$ having $B_1, \ldots, B_n$ as its only statement letters such that $\mathscr{B} \Rightarrow \mathscr{C}$ and $\mathscr{C} \Rightarrow \mathscr{D}$ are tautologies.
(c) Solve the special case of (b) in which $\mathscr{B}$ is $(B_1 \Rightarrow A) \wedge (A \Rightarrow B_2)$ and $\mathscr{D}$ is $(B_1 \wedge C) \Rightarrow (B_2 \wedge C)$.

1.46
(a) A certain country is inhabited only by *truth-tellers* (people who always tell the truth) and *liars* (people who always lie). Moreover, the inhabitants will respond only to *yes or no* questions. A tourist comes to a fork in a road where one branch leads to the capital and the other does not. There is no sign indicating which branch to take, but there is a native standing at the fork. What yes or no question should the tourist ask in order to determine which branch to take ? [*Hint*: Let A stand for 'You are a truth-teller' and let B stand for 'The left-hand branch leads to the capital'. Construct, by means of a suitable truth table, a statement form involving A and B such that the native's answer to the question as to whether this statement form is true will be *yes* when and only when B is true.]
(b) In a certain country, there are three kinds of people: *workers* (who always tell the truth), *businessmen* (who always lie), and *students* (who sometimes tell the truth and sometimes lie). At a fork in the road, one branch leads to the capital. A worker, a businessman and a student are standing at the side of the road but are not identifiable in any obvious way. By asking two yes or no questions, find out which fork leads to the capital (Each question may be addressed to any of the three.)

More puzzles of this kind may be found in Smullyan (1978, chap. 3; 1985, chaps 2, 4–8).

1.4 AN AXIOM SYSTEM FOR THE PROPOSITIONAL CALCULUS

Truth tables enable us to answer many of the significant questions concerning the truth-functional connectives, such as whether a given statement form is a tautology, is contradictory, or neither, and whether it logically implies or is logically equivalent to some other given statement form. The more complex parts of logic we shall treat later cannot be handled by truth tables or by any other similar effective procedure. Consequently, another approach, by means of formal axiomatic theories, will have to be tried. Although, as we have seen, the propositional calculus surrenders completely to the truth table method, it will be instructive to illustrate the axiomatic method in this simple branch of logic.

A formal theory $\mathscr{S}$ is defined when the following conditions are satisfied:

1. A countable set of symbols is given as the symbols of $\mathscr{S}$[†]. A finite sequence of symbols of $\mathscr{S}$ is called an *expression* of $\mathscr{S}$.
2. There is a subset of the set of expressions of $\mathscr{S}$ called the set of *well-formed formulas* (wfs) of $\mathscr{S}$. There is usually an effective procedure to determine whether a given expression is a wf.
3. There is a set of wfs called the set of *axioms* of $\mathscr{S}$. Most often, one can effectively decide whether a given wf is an axiom; in such a case, $\mathscr{S}$ is called an *axiomatic* theory.
4. There is a finite set $R_1, \ldots, R_n$ of relations among wfs, called *rules of inference*. For each R_i, there is a unique positive integer j such that, for every set of j wfs and each wf $\mathscr{B}$, one can effectively decide whether the given j wfs are in the relation R_i to $\mathscr{B}$, and, if so, $\mathscr{B}$ is said to *follow from* or to be a *direct consequence of* the given wfs by virtue of R_i[‡].

A *proof* in $\mathscr{S}$ is a sequence $\mathscr{B}_1, \ldots, \mathscr{B}_k$ of wfs such that, for each i, either $\mathscr{B}_i$ is an axiom of $\mathscr{S}$ or $\mathscr{B}_i$ is a direct consequence of some of the preceding wfs in the sequence by virtue of one of the rules of inference of $\mathscr{S}$.

A *theorem* of $\mathscr{S}$ is a wf $\mathscr{B}$ of $\mathscr{S}$ such that $\mathscr{B}$ is the last wf of some proof in $\mathscr{S}$. Such a proof is called a *proof of $\mathscr{B}$ in $\mathscr{S}$*.

Even if $\mathscr{S}$ is axiomatic – that is, if there is an effective procedure for checking any given wf to see whether it is an axiom – the notion of 'theorem' is not necessarily effective since, in general, there is no effective procedure for determining, given any wf $\mathscr{B}$, whether there is a proof of $\mathscr{B}$. A theory for which there is such an effective procedure is said to be *decidable*; otherwise, the theory is said to be *undecidable*.

From an intuitive standpoint, a decidable theory is one for which a machine can be devised to test wfs for theoremhood, whereas, for an undecidable theory, ingenuity is required to determine whether wfs are theorems.

A wf $\mathscr{C}$ is said to be a *consequence* in $\mathscr{S}$ of a set of Γ of wfs if and only if there is a sequence $\mathscr{B}_1, \ldots, \mathscr{B}_k$ of wfs such that $\mathscr{C}$ is $\mathscr{B}_k$ and, for each i, either $\mathscr{B}_i$ is an axiom or $\mathscr{B}_i$ is in Γ, or $\mathscr{B}_i$ is a direct consequence by some rule

[†]These 'symbols' may be thought of as arbitrary objects rather than just linguistic objects. This will become absolutely necessary when we deal with theories with uncountably many symbols in Section 2.12.

[‡]An example of a rule of inference will be the rule *modus ponens* (MP): $\mathscr{C}$ follows from $\mathscr{B}$ and $\mathscr{B} \Rightarrow \mathscr{C}$. According to our precise definition, this rule is the relation consisting of all ordered triples $\langle \mathscr{B}, \mathscr{B} \Rightarrow \mathscr{C}, \mathscr{C} \rangle$, where $\mathscr{B}$ and $\mathscr{C}$ are arbitrary wfs of the formal system.

of inference of some of the preceding wfs in the sequence. Such a sequence is called a *proof* (or *deduction*) *of* $\mathscr{C}$ *from* Γ. The members of Γ are called the *hypotheses* or *premisses* of the proof. We use $\Gamma \vdash \mathscr{C}$ as an abbreviation for '$\mathscr{C}$ is a consequence of Γ'. In order to avoid confusion when dealing with more than one theory, we write $\Gamma \vdash_{\mathscr{S}} \mathscr{C}$, adding the subscript $\mathscr{S}$ to indicate the theory in question.

If Γ is a finite set $\{\mathscr{H}_1, \ldots, \mathscr{H}_m\}$, we write $\mathscr{H}_1, \ldots, \mathscr{H}_m \vdash \mathscr{C}$ instead of $\{\mathscr{H}_1, \ldots, \mathscr{H}_m\} \vdash \mathscr{C}$. If Γ is the empty set $\emptyset$, then $\emptyset \vdash \mathscr{C}$ if and only if $\mathscr{C}$ is a theorem. It is customary to omit the sign '$\emptyset$' and simply write $\vdash \mathscr{C}$. Thus, $\vdash \mathscr{C}$ is another way of asserting that $\mathscr{C}$ is a theorem.

The following are simple properties of the notion of consequence:

1. If $\Gamma \subseteq \Delta$ and $\Gamma \vdash \mathscr{C}$, then $\Delta \vdash \mathscr{C}$.
2. $\Gamma \vdash \mathscr{C}$ if and only if there is a finite subset Δ of Γ such that $\Delta \vdash \mathscr{C}$.
3. If $\Delta \vdash \mathscr{C}$, and for each $\mathscr{B}$ in Δ, $\Gamma \vdash \mathscr{B}$, then $\Gamma \vdash \mathscr{C}$.

Assertion 1 represents the fact that if $\mathscr{C}$ is provable from a set Γ of premisses, then, if we add still more premisses, $\mathscr{C}$ is still provable. Half of 2 follows from 1. The other half is obvious when we notice that any proof of $\mathscr{C}$ from Γ uses only a finite number of premisses from Γ. Proposition 3 is also quite simple: if $\mathscr{C}$ is provable from premisses in Δ, and each premiss in Δ is provable from premisses in Γ, then $\mathscr{C}$ is provable from premisses in Γ.

We now introduce a formal axiomatic theory L for the propositional calculus.

1. The symbols of L are $\neg$, $\Rightarrow$, (,), and the letters A_i with positive integers i as subscripts: A_1, A_2, $A_3, \ldots$. The symbols $\neg$ and $\Rightarrow$ are called *primitive connectives*, and the letters A_i are called *statement letters*.
2. (a) All statement letters are wfs.
   (b) If $\mathscr{B}$ and $\mathscr{C}$ are wfs, then so are $(\neg\mathscr{B})$ and $(\mathscr{B} \Rightarrow \mathscr{C})$.†
   Thus, a wf of L is just a statement form built up from the statement letters A_i by means of the connectives $\neg$ and $\Rightarrow$.
3. If $\mathscr{B}$, $\mathscr{C}$ and $\mathscr{D}$ are wfs of L, then the following are axioms of L:
   (A1) $(\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{B}))$
   (A2) $((\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D})) \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D})))$
   (A3) $(((\neg\mathscr{C}) \Rightarrow (\neg\mathscr{B})) \Rightarrow (((\neg\mathscr{C}) \Rightarrow \mathscr{B}) \Rightarrow \mathscr{C}))$
4. The only rule of inference of L is *modus ponens*: $\mathscr{C}$ is a direct consequence of $\mathscr{B}$ and $(\mathscr{B} \Rightarrow \mathscr{C})$. We shall abbreviate applications of this rule by MP.‡

We shall use our conventions for eliminating parentheses.

†To be precise, we should add the so called extremal clause: (c) An expression is a wf if and only if it can be shown to be a wf on the basis of clauses (a) and (b). This can be made rigorous using as a model the definition in footnote § on page 13.

‡A common English synonym for modus ponens is the *detachment rule*.

Notice that the infinite set of axioms of L is given by means of three axiom schemas (A1)–(A3), with each schema standing for an infinite number of axioms. One can easily check for any given wf whether or not it is an axiom; therefore, L is axiomatic. In setting up the system L, it is our intention to obtain as theorems precisely the class of all tautologies.

We introduce other connectives by definition:

(D1) $(\mathscr{B} \wedge \mathscr{C})$ for $\neg(\mathscr{B} \Rightarrow \neg\mathscr{C})$
(D1) $(\mathscr{B} \vee \mathscr{C})$ for $(\neg\mathscr{B}) \Rightarrow \mathscr{C}$
(D3) $(\mathscr{B} \Leftrightarrow \mathscr{C})$ for $(\mathscr{B} \Rightarrow \mathscr{C}) \wedge (\mathscr{C} \Rightarrow \mathscr{B})$

The meaning of (D1), for example, is that, for any wfs $\mathscr{B}$ and $\mathscr{C}$, '$(\mathscr{B} \wedge \mathscr{C})$' is an abbreviation for '$\neg(\mathscr{B} \Rightarrow \neg\mathscr{C})$'.

LEMMA 1.8 $\vdash_L \mathscr{B} \Rightarrow \mathscr{B}$ **for all wfs** $\mathscr{B}$**.**

Proof†

We shall construct a proof in L of $\mathscr{B} \Rightarrow \mathscr{B}$.

1.	$(\mathscr{B} \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{B})) \Rightarrow$ $((\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{B})) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{B}))$	Instance of axiom schema (A2)
2.	$\mathscr{B} \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{B})$	Axiom schema (A1)
3.	$(\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{B})) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{B})$	From 1 and 2 by MP
4.	$\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{B})$	Axiom schema (A1)

†The word 'proof' is used in two distinct senses. First, it has a precise meaning defined above as a certain kind of finite sequence of wfs of L. However, in another sense, it also designates certain sequences of the English language (supplemented by various technical terms) that are supposed to serve as an argument justifying some assertion about the language L (or other formal theories). In general, the language we are studying (in this case, L) is called the *object language*, while the language in which we formulate and prove statements about the object language is called the *metalanguage*. The metalanguage might also be formalized and made the subject of study, which we would carry out in a metametalanguage, and so on. However, we shall use the English language as our (unformalized) metalanguage, although, for a substantial part of this book, we use only a mathematically weak portion of the English language. The contrast between object language and metalanguage is also present in the study of a foreign language; for example, in a Sanskrit class, Sanskrit is the object language, while the metalanguage, the language we use, is English. The distinction between *proof* and *metaproof* (i.e., a proof in the metalanguage) leads to a distinction between theorems of the object language and *metatheorems* of the metalanguage. To avoid confusion, we generally use 'proposition' instead of 'metatheorem'. The word 'metamathematics' refers to the study of logical and mathematical object languages; sometimes the word is restricted to those investigations that use what appear to the metamathematician to be constructive (or so-called finitary) methods.

5. $\mathscr{B} \Rightarrow \mathscr{B}$ From 3 and 4 by MP[†]

Exercise

1.47 Prove:

(a) $\vdash_L (\neg\mathscr{B} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{B}$

(b) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D}\ \vdash_L \mathscr{B} \Rightarrow \mathscr{D}$

(c) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D})\ \vdash_L \mathscr{C} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D})$

(d) $\vdash_L (\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$

In mathematical arguments, one often proves a statement $\mathscr{C}$ on the assumption of some other statement $\mathscr{B}$ and then concludes that 'if $\mathscr{B}$, then $\mathscr{C}$' is true. This procedure is justified for the system L by the following theorem.

PROPOSITION 1.9 (DEDUCTION THEOREM)[‡]

If Γ is a set of wfs and $\mathscr{B}$ and $\mathscr{C}$ are wfs, and $\Gamma,\ \mathscr{B} \vdash \mathscr{C}$, then $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}$. In particular, if $\mathscr{B} \vdash \mathscr{C}$, then $\vdash \mathscr{B} \Rightarrow \mathscr{C}$ (Herbrand, 1930).

Proof

Let $\mathscr{C}_1, \ldots, \mathscr{C}_n$ be a proof of $\mathscr{C}$ form $\Gamma \cup \{\mathscr{B}\}$, where $\mathscr{C}_n$ is $\mathscr{C}$. Let us prove, by induction on j, that $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}_j$ for $1 \leqslant j \leqslant n$. First of all, $\mathscr{C}_1$ must be either in Γ or an axiom of L or $\mathscr{B}$ itself. By axiom schema (A1), $\mathscr{C}_1 \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C}_1)$ is an axiom. Hence, in the first two cases, by MP, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}_1$. For the third case, when $\mathscr{C}_1$ is $\mathscr{B}$, we have $\vdash \mathscr{B} \Rightarrow \mathscr{C}_1$ by Lemma 1.8, and, therefore, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}_1$. This takes care of the case $j = 1$. Assume now that $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}_k$ for all $k < j$. Either $\mathscr{C}_j$ is an axiom, or $\mathscr{C}_j$ is in Γ, or $\mathscr{C}_j$ is $\mathscr{B}$, or $\mathscr{C}_j$ follows by modus ponens from some $\mathscr{C}_\ell$ and $\mathscr{C}_m$, where $\ell < j,\ m < j$, and $\mathscr{C}_m$ has the form $\mathscr{C}_\ell \Rightarrow \mathscr{C}_j$. In the first three cases, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}_j$ as in the case $j = 1$ above. In the last case, we have, by inductive hypothesis, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}_\ell$ and $\Gamma \vdash \mathscr{B} \Rightarrow (\mathscr{C}_\ell \Rightarrow \mathscr{C}_j)$. But, by axiom schema (A2), $\vdash (\mathscr{B} \Rightarrow (\mathscr{C}_\ell \Rightarrow \mathscr{C}_j)) \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{C}_\ell) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C}_j))$. Hence, by MP, $\Gamma \vdash (\mathscr{B} \Rightarrow \mathscr{C}_\ell) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C}_j)$, and, again by MP, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}_j$. Thus, the proof by induction is complete. The case $j = n$ is the desired result. [Notice that, given a deduction of $\mathscr{C}$ from Γ and $\mathscr{B}$, the proof just given enables us to construct a deduction of $\mathscr{B} \Rightarrow \mathscr{C}$

[†]The reader should not be discouraged by the apparently unmotivated step 1 of the proof. As in most proofs, we actually begin with the desired result, $\mathscr{B} \Rightarrow \mathscr{B}$, and then look for an appropriate axiom that may lead by MP to that result. A mixture of ingenuity and experimentation leads to a suitable instance of axiom (A2).

[‡]For the remainder of the chapter, unless something is said to the contrary, we shall omit the subscript L in $\vdash_L$. In addition, we shall use $\Gamma,\ \mathscr{B} \vdash \mathscr{C}$ to stand for $\Gamma \cup \{\mathscr{B}\} \vdash \mathscr{C}$. In general, we let $\Gamma,\ \mathscr{B}_1, \ldots, \mathscr{B}_n \vdash \mathscr{C}$ stand for $\Gamma \cup \{\mathscr{B}_1, \ldots, \mathscr{B}_n\} \vdash \mathscr{C}$.

from Γ. Also note that axiom schema (A3) was not used in proving the deduction theorem.]

COROLLARY 1.10

(a) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D} \vdash \mathscr{B} \Rightarrow \mathscr{D}$
(b) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D}),\ \mathscr{C} \vdash \mathscr{B} \Rightarrow \mathscr{D}$

Proof

For part (a):

1. $\mathscr{B} \Rightarrow \mathscr{C}$	Hyp (abbreviation for 'hypothesis')
2. $\mathscr{C} \Rightarrow \mathscr{D}$	Hyp
3. $\mathscr{B}$	Hyp
4. $\mathscr{C}$	1, 3, MP
5. $\mathscr{D}$	2, 4, MP

Thus, $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D},\ \mathscr{B} \vdash \mathscr{D}$. So, by the deduction theorem, $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D} \vdash \mathscr{B} \Rightarrow \mathscr{D}$.

To prove (b), use the deduction theorem.

LEMMA 1.11

For any wfs $\mathscr{B}$ and $\mathscr{C}$, the following wfs are theorems of L.

(a) $\neg\neg\mathscr{B} \Rightarrow \mathscr{B}$
(b) $\mathscr{B} \Rightarrow \neg\neg\mathscr{B}$
(c) $\neg\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$
(d) $(\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$
(e) $(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$
(f) $\mathscr{B} \Rightarrow (\neg\mathscr{C} \Rightarrow \neg(\mathscr{B} \Rightarrow \mathscr{C}))$
(g) $(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\neg\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow \mathscr{C})$

Proof

(a) $\vdash \neg\neg\mathscr{B} \Rightarrow \mathscr{B}$

1. $(\neg\mathscr{B} \Rightarrow \neg\neg\mathscr{B}) \Rightarrow ((\neg\mathscr{B} \Rightarrow \neg\mathscr{B}) \Rightarrow \mathscr{B})$	Axiom (A3)
2. $\neg\mathscr{B} \Rightarrow \neg\mathscr{B}$	Lemma 1.8 †
3. $(\neg\mathscr{B} \Rightarrow \neg\neg\mathscr{B}) \Rightarrow \mathscr{B}$	1, 2, Corollary 1.10(b)
4. $\neg\neg\mathscr{B} \Rightarrow (\neg\mathscr{B} \Rightarrow \neg\neg\mathscr{B})$	Axiom (A1)
5. $\neg\neg\mathscr{B} \Rightarrow \mathscr{B}$	3, 4, Corollary 1.10(a)

†Instead of writing a complete proof of $\neg\mathscr{B} \Rightarrow \neg\mathscr{B}$, we simply cite Lemma 1.8. In this way, we indicate how the proof of $\neg\neg\mathscr{B} \Rightarrow \mathscr{B}$ could be written if we wished to take the time and space to do so. This is, of course, nothing more than the ordinary application of previously proved theorems.

(b) $\vdash \mathscr{B} \Rightarrow \neg\neg\mathscr{B}$

1. $(\neg\neg\neg\mathscr{B} \Rightarrow \neg\mathscr{B}) \Rightarrow ((\neg\neg\neg\mathscr{B} \Rightarrow \mathscr{B}) \Rightarrow \neg\neg\mathscr{B})$	Axiom (A3)
2. $\neg\neg\neg\mathscr{B} \Rightarrow \neg\mathscr{B}$	Part (a)
3. $(\neg\neg\neg\mathscr{B} \Rightarrow \mathscr{B}) \Rightarrow \neg\neg\mathscr{B}$	1, 2, MP
4. $\mathscr{B} \Rightarrow (\neg\neg\neg\mathscr{B} \Rightarrow \mathscr{B})$	Axiom (A1)
5. $\mathscr{B} \Rightarrow \neg\neg\mathscr{B}$	3, 4, Corollary 1.10(a)

(c) $\vdash \neg\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$

1. $\neg\mathscr{B}$	Hyp
2. $\mathscr{B}$	Hyp
3. $\mathscr{B} \Rightarrow (\neg\mathscr{C} \Rightarrow \mathscr{B})$	Axiom (A1)
4. $\neg\mathscr{B} \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$	Axiom (A1)
5. $\neg\mathscr{C} \Rightarrow \mathscr{B}$	2, 3, MP
6. $\neg\mathscr{C} \Rightarrow \neg\mathscr{B}$	1, 4, MP
7. $(\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow ((\neg\mathscr{C} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{C})$	Axiom (A3)
8. $(\neg\mathscr{C} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{C}$	6, 7, MP
9. $\mathscr{C}$	5, 8, MP
10. $\neg\mathscr{B}, \mathscr{B} \vdash \mathscr{C}$	1–9
11. $\neg\mathscr{B} \vdash \mathscr{B} \Rightarrow \mathscr{C}$	10, Deduction theorem
12. $\vdash \neg\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$	11, Deduction theorem

(d) $\vdash (\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$

1. $\neg\mathscr{C} \Rightarrow \neg\mathscr{B}$	Hyp
2. $(\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow ((\neg\mathscr{C} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{C})$	Axiom (A3)
3. $\mathscr{B} \Rightarrow (\neg\mathscr{C} \Rightarrow \mathscr{B})$	Axiom (A1)
4. $(\neg\mathscr{C} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{C}$	1, 2, MP
5. $\mathscr{B} \Rightarrow \mathscr{C}$	3, 4, Corollary 1.10(a)
6. $\neg\mathscr{C} \Rightarrow \neg\mathscr{B} \vdash \mathscr{B} \Rightarrow \mathscr{C}$	1–5
7. $\vdash (\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$	6, deduction theorem

(e) $\vdash (\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$

1. $\mathscr{B} \Rightarrow \mathscr{C}$	Hyp
2. $\neg\neg\mathscr{B} \Rightarrow \mathscr{B}$	Part (a)
3. $\neg\neg\mathscr{B} \Rightarrow \mathscr{C}$	1, 2, Corollary 1.10(a)
4. $\mathscr{C} \Rightarrow \neg\neg\mathscr{C}$	Part (b)
5. $\neg\neg\mathscr{B} \Rightarrow \neg\neg\mathscr{C}$	3, 4, Corollary 1.10(a)
6. $(\neg\neg\mathscr{B} \Rightarrow \neg\neg\mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$	Part (d)
7. $\neg\mathscr{C} \Rightarrow \neg\mathscr{B}$	5, 6, MP
8. $\mathscr{B} \Rightarrow \mathscr{C} \vdash \neg\mathscr{C} \Rightarrow \neg\mathscr{B}$	1–7
9. $\vdash (\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$	8, deduction theorem

(f) $\vdash \mathscr{B} \Rightarrow (\neg\mathscr{C} \Rightarrow \neg(\mathscr{B} \Rightarrow \mathscr{C}))$.

Clearly, $\mathscr{B}, \mathscr{B} \Rightarrow \mathscr{C} \vdash \mathscr{C}$ by MP. Hence, $\vdash \mathscr{B} \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow \mathscr{C})$ by two uses of the deduction theorem. Now, by (e), $\vdash ((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg(\mathscr{B} \Rightarrow \mathscr{C}))$. Hence, by Corollary 1.10(a), $\vdash \mathscr{B} \Rightarrow (\neg\mathscr{C} \Rightarrow \neg(\mathscr{B} \Rightarrow \mathscr{C}))$.

(g) $\vdash (\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\neg\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow \mathscr{C})$

1. $\mathscr{B} \Rightarrow \mathscr{C}$	Hyp
2. $\neg\mathscr{B} \Rightarrow \mathscr{C}$	Hyp
3. $(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$	Part (e)
4. $\neg\mathscr{C} \Rightarrow \neg\mathscr{B}$	1, 3, MP
5. $(\neg\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\neg\mathscr{B})$	Part (e)
6. $\neg\mathscr{C} \Rightarrow \neg\neg\mathscr{B}$	2, 5, MP
7. $(\neg\mathscr{C} \Rightarrow \neg\neg\mathscr{B}) \Rightarrow ((\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow \mathscr{C})$	Axiom (A3)
8. $(\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow \mathscr{C}$	6, 7, MP
9. $\mathscr{C}$	4, 8, MP
10. $\mathscr{B} \Rightarrow \mathscr{C},\ \neg\mathscr{B} \Rightarrow \mathscr{C} \vdash \mathscr{C}$	1–9
11. $\mathscr{B} \Rightarrow \mathscr{C} \vdash (\neg\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow \mathscr{C}$	10, deduction theorem
12. $\vdash (\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\neg\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow \mathscr{C})$	11, deduction theorem

Exercises

1.48 Show that the following wfs are theorems of L.

(a) $\mathscr{B} \Rightarrow (\mathscr{B} \vee \mathscr{C})$
(b) $\mathscr{B} \Rightarrow (\mathscr{C} \vee \mathscr{B})$
(c) $\mathscr{C} \vee \mathscr{B} \Rightarrow \mathscr{B} \vee \mathscr{C}$
(d) $\mathscr{B} \wedge \mathscr{C} \Rightarrow \mathscr{B}$
(e) $\mathscr{B} \wedge \mathscr{C} \Rightarrow \mathscr{C}$
(f) $(\mathscr{B} \Rightarrow \mathscr{D}) \Rightarrow ((\mathscr{C} \Rightarrow \mathscr{D}) \Rightarrow (\mathscr{B} \vee \mathscr{C} \Rightarrow \mathscr{D}))$
(g) $((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow \mathscr{B}) \Rightarrow \mathscr{B}$
(h) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow (\mathscr{B} \wedge \mathscr{C}))$

1.49 Exhibit a complete proof in L of Lemma 1.11(c). [*Hint:* Apply the procedure used in the proof of the deduction theorem to the demonstration given earlier of Lemma 1.11(c).] Greater fondness for the deduction theorem will result if the reader tries to prove all of Lemma 1.11 without using the deduction theorem.

It is our purpose to show that a wf of L is a theorem of L if and only if it is a tautology. Half of this is very easy.

PROPOSITION 1.12

Every theorem of L is a tautology.

Proof

As an exercise, verify that all the axioms of L are tautologies. By Proposition 1.2, modus ponens leads from tautologies to other tautologies. Hence, every theorem of L is a tautology.

The following lemma is to be used in the proof that every tautology is a theorem of L.

LEMMA 1.13

Let $\mathscr{B}$ be a wf and let $B_1, \ldots, B_k$ be the statement letters that occur in $\mathscr{B}$. For a given assignment of truth values to $B_1, \ldots, B_k$, let B_j' be B_j if B_j takes the value T; and let B_j' be $\neg B_j$ if B_j takes the value F. Let $\mathscr{B}'$ be $\mathscr{B}$ if $\mathscr{B}$ takes the value T under the assignment, and let $\mathscr{B}'$ be $\neg\mathscr{B}$ if $\mathscr{B}$ takes the value F. Then $B_1', \ldots, B_k' \vdash \mathscr{B}'$.

For example, let $\mathscr{B}$ be $\neg(\neg A_2 \Rightarrow A_5)$. Then for each row of the truth table

A_2	A_5	$\neg(\neg A_2 \Rightarrow A_5)$
T	T	F
F	T	F
T	F	F
F	F	T

Lemma 1.13 asserts a corresponding deducibility relation. For instance, corresponding to the third row there is $A_2, \ \neg A_5 \vdash \neg\neg(\neg A_2 \Rightarrow A_5)$, and to the fourth row, $\neg A_2, \ \neg A_5 \vdash \neg(\neg A_2 \Rightarrow A_5)$.

Proof

The proof is by induction on the number n of occurrences of $\neg$ and $\Rightarrow$ in $\mathscr{B}$. (We assume $\mathscr{B}$ written without abbreviations.) If $n = 0$, $\mathscr{B}$ is just a statement letter B_1, and then the lemma reduces to $B_1 \vdash B_1$ and $\neg B_1 \vdash \neg B_1$. Assume now that the lemma holds for all $j < n$.

Case 1. $\mathscr{B}$ is $\neg\mathscr{C}$. Then $\mathscr{C}$ has fewer than n occurrences of $\neg$ and $\Rightarrow$.

Subcase 1a. Let $\mathscr{C}$ take the value T under the given truth value assignment. Then $\mathscr{B}$ takes the value F. So, $\mathscr{C}'$ is $\mathscr{C}$ and $\mathscr{B}'$ is $\neg\mathscr{B}$. By the inductive hypothesis applied to $\mathscr{C}$, we have $B_1', \ldots, B_k' \vdash \mathscr{C}$. Then, by Lemma 1.11(b) and MP, $B_1', \ldots, B_k' \vdash \neg\neg\mathscr{C}$. But $\neg\neg\mathscr{C}$ is $\mathscr{B}'$.

Subcase 1b. Let $\mathscr{C}$ take the value F. Then $\mathscr{B}$ takes the value T. So, $\mathscr{C}'$ is $\neg\mathscr{C}$ and $\mathscr{B}'$ is $\mathscr{B}$, By inductive hypothesis, $B_1', \ldots, B_k' \vdash \neg\mathscr{C}$. But $\neg\mathscr{C}$ is $\mathscr{B}'$.

Case 2. $\mathscr{B}$ is $\mathscr{C} \Rightarrow \mathscr{D}$. Then $\mathscr{C}$ and $\mathscr{D}$ have fewer occurrences of $\neg$ and $\Rightarrow$ than $\mathscr{B}$. So, by inductive hypothesis, $B_1', \ldots, B_k' \vdash \mathscr{C}'$ and $B_1', \ldots, B_k' \vdash \mathscr{D}'$.

Subcase 2a. $\mathscr{C}$ takes the value F. Then $\mathscr{B}$ takes the value T. So, $\mathscr{C}'$ is $\neg\mathscr{C}$ and $\mathscr{B}'$ is $\mathscr{B}$. Hence, $B_1', \ldots, B_k' \vdash \neg\mathscr{C}$. By Lemma 1.11(c) and MP, $B_1', \ldots, B_k' \vdash \mathscr{C} \Rightarrow \mathscr{D}$. But $\mathscr{C} \Rightarrow \mathscr{D}$ is $\mathscr{B}'$.

Subcase 2b. $\mathscr{D}$ takes the value T. Then $\mathscr{B}$ takes the value T. So, $\mathscr{D}'$ is $\mathscr{D}$ and $\mathscr{B}'$ is $\mathscr{B}$. Hence, $B_1', \ldots, B_k' \vdash \mathscr{D}$. Then, by axiom (A1) and MP, $B_1', \ldots, B_k' \vdash \mathscr{C} \Rightarrow \mathscr{D}$. But $\mathscr{C} \Rightarrow \mathscr{D}$ is $\mathscr{B}'$.

Subcase 2c. $\mathscr{C}$ takes the value T and $\mathscr{D}$ takes the value F. Then $\mathscr{B}$ takes the value F. So, $\mathscr{C}'$ is $\mathscr{C}$, $\mathscr{D}'$ is $\neg\mathscr{D}$, and $\mathscr{B}'$ is $\neg\mathscr{B}$. Therefore, $B_1', \ldots, B_k' \vdash \mathscr{C}$ and $B_1', \ldots, B_k' \vdash \neg\mathscr{D}$. Hence, by Lemma 1.11(f) and MP, $B_1', \ldots, B_k' \vdash \neg(\mathscr{C} \Rightarrow \mathscr{D})$. But $\neg(\mathscr{C} \Rightarrow \mathscr{D})$ is $\mathscr{B}'$.

PROPOSITION 1.14 (COMPLETENESS THEOREM)

If a wf $\mathscr{B}$ of L is a tautology, then it is a theorem of L.

Proof

(Kalmár, 1935) Assume $\mathscr{B}$ is a tautology, and let $B_1, \ldots, B_k$ be the statement letters in $\mathscr{B}$. For any truth value assignment to $B_1, \ldots, B_k$, we have, by Lemma 1.13, $B'_1, \ldots, B'_k \vdash \mathscr{B}$. ($\mathscr{B}'$ is $\mathscr{B}$ because $\mathscr{B}$ always takes the value T.) Hence, when B'_k is given the value T, we obtain $B'_1, \ldots, B'_{k-1}, B_k \vdash \mathscr{B}$, and, when B_k is given the value F, we obtain $B'_1, \ldots, B'_{k-1}, \neg B_k \vdash \mathscr{B}$. So, by the deduction theorem, $B'_1, \ldots, B'_{k-1} \vdash B_k \Rightarrow \mathscr{B}$ and $B'_1, \ldots, B'_{k-1} \vdash \neg B_k \Rightarrow \mathscr{B}$. Then by Lemma 1.11(g) and MP, $B'_1, \ldots, B'_{k-1} \vdash \mathscr{B}$. Similarly, B_{k-1} may be chosen to be T or F and, again applying the deduction theorem, Lemma 1.11(g) and MP, we can eliminate B'_{k-1} just as we eliminated B'_k. After k such steps, we finally obtain $\vdash \mathscr{B}$.

COROLLARY 1.15

If $\mathscr{C}$ is an expression involving the signs $\neg$, $\Rightarrow$, $\wedge$, $\vee$ and $\Leftrightarrow$ that is an abbreviation for a wf $\mathscr{B}$ of L, then $\mathscr{C}$ is a tautology if and only if $\mathscr{B}$ is a theorem of L.

Proof

In definitions (D1)–(D3), the abbreviating formulas replace wfs to which they are logically equivalent. Hence, by Proposition 1.4, $\mathscr{B}$ and $\mathscr{C}$ are logically equivalent, and $\mathscr{C}$ is a tautology if and only if $\mathscr{B}$ is a tautology. The corollary now follows from Propositions 1.12 and 1.14.

COROLLARY 1.16

The system L is consistent; that is, there is no wf $\mathscr{B}$ such that both $\mathscr{B}$ and $\neg\mathscr{B}$ are theorems of L.

Proof

By Proposition 1.12, every theorem of L is a tautology. The negation of a tautology cannot be a tautology and, therefore, it is impossible for both $\mathscr{B}$ and $\neg\mathscr{B}$ to be theorems of L.

Notice that L is consistent if and only if not all wfs of L are theorems. In fact, if L is consistent, then there are wfs that are not theorems (e.g., the negations of theorems). On the other hand, by Lemma 1.11(c), $\vdash_L \neg\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$, and so, if L were inconsistent, that is, if some wf $\mathscr{B}$ and its negation $\neg\mathscr{B}$ were provable, then by MP any wf $\mathscr{C}$ would be provable. (This equivalence holds for any theory that has *modus ponens* as a rule of inference and in which Lemma 1.11(c) is provable.) A theory in which not all wfs are theorems is said to be *absolutely consistent*, and this definition is applicable even to theories that do not contain a negation sign.

Exercise

1.50 Let $\mathscr{B}$ be a statement form that is not a tautology. Let L^+ be the formal theory obtained from L by adding as new axioms all wfs obtainable from $\mathscr{B}$ by substituting arbitrary statement forms for the statement letters in $\mathscr{B}$, with the same form being substituted for all occurrences of a statement letter. Show that L^+ is inconsistent.

1.5 INDEPENDENCE. MANY-VALUED LOGICS

A subset Y of the set of axioms of a theory is said to be *independent* if some wf in Y cannot be proved by means of the rules of inference from the set of those axioms not in Y.

PROPOSITION 1.17

Each of the axiom schemes (A1)–(A3) is independent.

Proof

To prove the independence of axiom schema (A1), consider the following tables:

A	$\neg A$
0	1
1	1
2	0

A	B	$A \Rightarrow B$
0	0	0
1	0	2
2	0	0
0	1	2
1	1	2
2	1	0
0	2	2
1	2	0
2	2	0

For any assignment of the values 0, 1 and 2 to the statement letters of a wf $\mathscr{B}$, these tables determine a corresponding value of $\mathscr{B}$. If $\mathscr{B}$ always takes the value 0, $\mathscr{B}$ is called *select*. Modus ponens preserves selectness, since it is easy to check that, if $\mathscr{B}$ and $\mathscr{B} \Rightarrow \mathscr{C}$ are select, so is $\mathscr{C}$. One can also verify that all instances of axiom schemas (A2) and (A3) are select. Hence, any wf derivable from (A2) and (A3) by modus ponens is select. However, $A_1 \Rightarrow (A_2 \Rightarrow A_1)$, which is an instance of (A1), is not select, since it takes the value 2 when A_1 is 1 and A_2 is 2.

To prove the independence of axiom schema (A2), consider the following tables:

A	$\neg A$
0	1
1	0
2	1

A	B	$A \Rightarrow B$
0	0	0
1	0	0
2	0	0
0	1	2
1	1	2
2	1	0
0	2	1
1	2	0
2	2	0

Let us call a wf that always takes the value 0 according to these tables *grotesque*. Modus ponens preserves grotesqueness and it is easy to verify that all instances of (A1) and (A3) are grotesque. However, the instance $(A_1 \Rightarrow (A_2 \Rightarrow A_3)) \Rightarrow ((A_1 \Rightarrow A_2) \Rightarrow (A_1 \Rightarrow A_3))$ of (A2) takes the value 2 when A_1 is 0, A_2 is 0, and A_3 is 1 and, therefore, is not grotesque.

The following argument proves the independence of (A3). Let us call a wf $\mathscr{B}$ *super* if the wf $h(\mathscr{B})$ obtained by erasing all negation signs in $\mathscr{B}$ is a tautology. Each instance of axiom schemas (A1) and (A2) is super. Also, modus ponens preserves the property of being super; for if $h(\mathscr{B} \Rightarrow \mathscr{C})$ and $h(\mathscr{B})$ are tautologies, then $h(\mathscr{C})$ is a tautology. (Just note that $h(\mathscr{B} \Rightarrow \mathscr{C})$ is $h(\mathscr{B}) \Rightarrow h(\mathscr{C})$ and use Proposition 1.2.) Hence, every wf $\mathscr{B}$ derivable from (A1) and (A2) by modus ponens is super. But $h((\neg A_1 \Rightarrow \neg A_1) \Rightarrow ((\neg A_1 \Rightarrow A_1) \Rightarrow A_1))$ is $(A_1 \Rightarrow A_1) \Rightarrow ((A_1 \Rightarrow A_1) \Rightarrow A_1)$, which is not a tautology. Therefore, $(\neg A_1 \Rightarrow \neg A_1) \Rightarrow ((\neg A_1 \Rightarrow A_1) \Rightarrow A_1)$, an instance of (A3), is not super and is thereby not derivable from (A1) and (A2) by modus ponens.

The idea used in the proof of the independence of axiom schemas (A1) and (A2) may be generalized to the notion of a *many-valued logic*. Select a positive integer n, call the numbers $0, 1, \ldots, n$ *truth values*, and choose a number m such that $0 \leqslant m < n$. The numbers $0, 1, \ldots, m$ are called *designated values*. Take a finite number of 'truth tables' representing functions from sets of the form $\{0, 1, \ldots, n\}^k$ into $\{0, 1, \ldots, n\}$. For each truth table, introduces a sign, called the corresponding *connective*. Using these connectives and statement letters, we may construct 'statement forms', and every such statement form containing j distinct letters determines a 'truth function' from $\{0, 1, \ldots, n\}^j$ into $\{0, 1, \ldots, n\}$. A statement form whose corresponding truth function takes only designated values is said to be *exceptional*. The numbers m and n and the basic truth tables are said to define a (finite) *many-valued logic* M. A formal theory involving statement letters and the connectives of M is said to be *suitable* for M if and only if the theorems of the theory coincide with the exceptional statement forms of M. All these notions obviously can be generalized to the case of an infinite number of truth values. If $n = 1$ and $m = 0$ and the truth tables are those given for $\neg$ and $\Rightarrow$ in section 1.1, then the corresponding two-valued logic is that studied in this chapter. The exceptional wfs in this case were called tautologies. The system L is suitable for this logic, as proved in Propositions 1.12 and 1.14. In the proofs of the independence of axiom schemas (A1) and (A2), two three-valued logics were used.

Exercises

1.51 Prove the independence of axiom schema (A3) by constructing appropriate 'truth tables' for $\neg$ and $\Rightarrow$.

1.52 (McKinsey and Tarski, 1948) Consider the axiomatic theory P in which there is exactly one binary connective $*$, the only rule of inference is modus ponens (that is, $\mathscr{C}$ follows from $\mathscr{B}$ and $\mathscr{B} * \mathscr{C}$), and the axioms are all wfs of the form $\mathscr{B} * \mathscr{B}$. Show that P is not suitable for any (finite) many-valued logic.

1.53 For any (finite) many-valued logic M, prove that there is an axiomatic theory suitable for M.

Further information about many-valued logics can be found in Rosser and Turquette (1952), Rescher (1969), Bolc and Borowik (1992) and Malinowski (1993).

1.6 OTHER AXIOMATIZATIONS

Although the axiom system L is quite simple, there are many other systems that would do as well. We can use, instead of $\neg$ and $\Rightarrow$, any collection of primitive connectives as long as these are adequate for the definition of all other truth-functional connectives.

Examples

L_1: $\vee$ and $\neg$ are the primitive conectives. We use $\mathscr{B} \Rightarrow \mathscr{C}$ as an abbreviation for $\neg\mathscr{B} \vee \mathscr{C}$. We have four axiom schemas: (1) $\mathscr{B} \vee \mathscr{B} \Rightarrow \mathscr{B}$; (2) $\mathscr{B} \Rightarrow \mathscr{B} \vee \mathscr{C}$; (3) $\mathscr{B} \vee \mathscr{C} \Rightarrow \mathscr{C} \vee \mathscr{B}$; and (4) $(\mathscr{C} \Rightarrow \mathscr{D}) \Rightarrow (\mathscr{B} \vee \mathscr{C} \Rightarrow \mathscr{B} \vee \mathscr{D})$. The only rule of inference is modus ponens. Here and below we use the usual rules for eliminating parentheses. This system is developed in Hilbert and Ackermann (1950).

L_2: $\wedge$ and $\neg$ are the primitive connectives. $\mathscr{B} \Rightarrow \mathscr{C}$ is an abbreviation for $\neg(\mathscr{B} \wedge \neg\mathscr{C})$. There are three axiom schemas: (1) $\mathscr{B} \Rightarrow (\mathscr{B} \wedge \mathscr{B})$;

(2) $\mathscr{B} \wedge \mathscr{C} \Rightarrow \mathscr{B}$; and (3) $(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg(\mathscr{C} \wedge \mathscr{D}) \Rightarrow \neg(\mathscr{D} \wedge \mathscr{B}))$. Modus ponens is the only rule of inference. Consult Rosser (1953) for a detailed study.

L_3: This is just like our original system L except that, instead of the axiom schemas (A1)–(A3), we have three specific axioms: (1) $A_1 \Rightarrow (A_2 \Rightarrow A_1)$; (2) $(A_1 \Rightarrow (A_2 \Rightarrow A_3)) \Rightarrow ((A_1 \Rightarrow A_2) \Rightarrow (A_1 \Rightarrow A_2))$; and (3) $(\neg A_2 \Rightarrow \neg A_1) \Rightarrow ((\neg A_2 \Rightarrow A_1) \Rightarrow A_2)$. In addition to modus ponens, we have a substitution rule: we may substitute any wf for all occurrences of a statement letter in a given wf.

L_4: The primitive connectives are $\Rightarrow$, $\wedge$, $\vee$ and $\neg$. Modus ponens is the only rule, and we have ten axiom schemas: (1) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{B})$; (2) $(\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D})) \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}))$; (3) $\mathscr{B} \wedge \mathscr{C} \Rightarrow \mathscr{B}$; (4) $\mathscr{B} \wedge \mathscr{C} \Rightarrow \mathscr{C}$; (5) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow (\mathscr{B} \wedge \mathscr{C}))$; (6) $\mathscr{B} \Rightarrow (\mathscr{B} \vee \mathscr{C})$; (7) $\mathscr{C} \Rightarrow (\mathscr{B} \vee \mathscr{C})$; (8) $(\mathscr{B} \Rightarrow \mathscr{D}) \Rightarrow ((\mathscr{C} \Rightarrow \mathscr{D}) \Rightarrow (\mathscr{B} \vee \mathscr{C} \Rightarrow \mathscr{D}))$; (9) $(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\mathscr{B} \Rightarrow \neg\mathscr{C}) \Rightarrow \neg\mathscr{B})$; and (10) $\neg\neg\mathscr{B} \Rightarrow \mathscr{B}$. This system is discussed in Kleene (1952).

Axiomatizations can be found for the propositional calculus that contain only one axiom schema. For example, if $\neg$ and $\Rightarrow$ are the primitive connectives and modus ponens the only rule of inference, then the axiom schema

$$[(((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{D} \Rightarrow \neg\mathscr{E})) \Rightarrow \mathscr{D}) \Rightarrow \mathscr{F}] \Rightarrow [(\mathscr{F} \Rightarrow \mathscr{B}) \Rightarrow (\mathscr{E} \Rightarrow \mathscr{B})]$$

is sufficient (Meredith, 1953). Another single-axiom formulation, due to Nicod (1917), uses only alternative denial $|$. Its rule of inference is: $\mathscr{D}$ follows from $\mathscr{B} \mid (\mathscr{C} \mid \mathscr{D})$ and $\mathscr{B}$, and its axiom schema is

$$(\mathscr{B}|(\mathscr{C}|\mathscr{D})) \mid \{[\mathscr{E}|(\mathscr{E}|\mathscr{E})] \mid [(\mathscr{F}|\mathscr{C})|((\mathscr{B}|\mathscr{F})|(\mathscr{B}|\mathscr{F}))]\}$$

Further information, including historical background, may be found in Church (1956) and in a paper by Lukasiewicz and Tarski in Tarski (1956, IV).

Exercises

1.54 (Hilbert and Ackermann, 1950) Prove the following results about the theory L_1.

(a) $\mathscr{B} \Rightarrow \mathscr{C} \vdash_{L_1} \mathscr{D} \vee \mathscr{B} \Rightarrow \mathscr{D} \vee \mathscr{C}$
(b) $\vdash_{L_1} (\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\mathscr{D} \Rightarrow \mathscr{B}) \Rightarrow (\mathscr{D} \Rightarrow \mathscr{C}))$
(c) $\mathscr{D} \Rightarrow \mathscr{B},\ \mathscr{B} \Rightarrow \mathscr{C} \vdash_{L_1} \mathscr{D} \Rightarrow \mathscr{C}$
(d) $\vdash_{L_1} \mathscr{B} \Rightarrow \mathscr{B}$ (i.e., $\vdash_{L_1} \neg\mathscr{B} \vee \mathscr{B}$)
(e) $\vdash_{L_1} \mathscr{B} \vee \neg\mathscr{B}$
(f) $\vdash_{L_1} \mathscr{B} \Rightarrow \neg\neg\mathscr{B}$
(g) $\vdash_{L_1} \neg\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$
(h) $\vdash_{L_1} \mathscr{B} \vee (\mathscr{C} \vee \mathscr{D}) \Rightarrow ((\mathscr{C} \vee (\mathscr{B} \vee \mathscr{D})) \vee \mathscr{B})$
(i) $\vdash_{L_1} (\mathscr{C} \vee (\mathscr{B} \vee \mathscr{D})) \vee \mathscr{B} \Rightarrow \mathscr{C} \vee (\mathscr{B} \vee \mathscr{D})$
(j) $\vdash_{L_1} \mathscr{B} \vee (\mathscr{C} \vee \mathscr{D}) \Rightarrow \mathscr{C} \vee (\mathscr{B} \vee \mathscr{D})$

(k) $\vdash_{L_1} (\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D})) \Rightarrow (\mathscr{C} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}))$
(l) $\vdash_{L_1} (\mathscr{D} \Rightarrow \mathscr{B}) \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{D} \Rightarrow \mathscr{C}))$
(m) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D}),\ \mathscr{B} \Rightarrow \mathscr{C} \vdash_{L_1} \mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D})$
(n) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D}),\ \mathscr{B} \Rightarrow \mathscr{C} \vdash_{L_1} \mathscr{B} \Rightarrow \mathscr{D}$
(o) If $\Gamma,\ \mathscr{B} \vdash_{L_1} \mathscr{C}$, then $\Gamma \vdash_{L_1} \mathscr{B} \Rightarrow \mathscr{C}$ (deduction theorem)
(p) $\mathscr{C} \Rightarrow \mathscr{B},\ \neg\mathscr{C} \Rightarrow \mathscr{B} \vdash_{L_1} \mathscr{B}$
(q) $\vdash_{L_1} \mathscr{B}$ if and only if $\mathscr{B}$ is a tautology.

1.55 (Rosser, 1953) Prove the following facts about the theory L_2.

(a) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D} \vdash_{L_2} \neg(\neg\mathscr{D} \wedge \mathscr{B})$
(b) $\vdash_{L_2} \neg(\neg\mathscr{B} \wedge \mathscr{B})$
(c) $\vdash_{L_2} \neg\neg\mathscr{B} \Rightarrow \mathscr{B}$
(d) $\vdash_{L_2} \neg(\mathscr{B} \wedge \mathscr{C}) \Rightarrow (\mathscr{C} \Rightarrow \neg\mathscr{B})$
(e) $\vdash_{L_2} \mathscr{B} \Rightarrow \neg\neg\mathscr{B}$
(f) $\vdash_{L_2} (\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$
(g) $\neg\mathscr{B} \Rightarrow \neg\mathscr{C} \vdash_{L_2} \mathscr{C} \Rightarrow \mathscr{B}$
(h) $\mathscr{B} \Rightarrow \mathscr{C} \vdash_{L_2} \mathscr{D} \wedge \mathscr{B} \Rightarrow \mathscr{C} \wedge \mathscr{D}$
(i) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D},\ \mathscr{D} \Rightarrow \mathscr{E} \vdash_{L_2} \mathscr{B} \Rightarrow \mathscr{E}$
(j) $\vdash_{L_2} \mathscr{B} \Rightarrow \mathscr{B}$
(k) $\vdash_{L_2} \mathscr{B} \wedge \mathscr{C} \Rightarrow \mathscr{C} \wedge \mathscr{B}$
(l) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D} \vdash_{L_2} \mathscr{B} \Rightarrow \mathscr{D}$
(m) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{D} \Rightarrow \mathscr{E} \vdash_{L_2} \mathscr{B} \wedge \mathscr{D} \Rightarrow \mathscr{C} \wedge \mathscr{E}$
(n) $\mathscr{C} \Rightarrow \mathscr{D} \vdash_{L_2} \mathscr{B} \wedge \mathscr{C} \Rightarrow \mathscr{B} \wedge \mathscr{D}$
(o) $\vdash_{L_2} (\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D})) \Rightarrow ((\mathscr{B} \wedge \mathscr{C}) \Rightarrow \mathscr{D})$
(p) $\vdash_{L_2} ((\mathscr{B} \wedge \mathscr{C}) \Rightarrow \mathscr{D}) \Rightarrow (\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D}))$
(q) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D}) \vdash_{L_2} \mathscr{B} \Rightarrow \mathscr{D}$
(r) $\vdash_{L_2} \mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{B} \wedge \mathscr{C})$
(s) $\vdash_{L_2} \mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{B})$
(t) If $\Gamma,\ \mathscr{B} \vdash_{L_2} \mathscr{C}$, then $\Gamma \vdash_{L_2} \mathscr{B} \Rightarrow \mathscr{C}$ (deduction theorem)
(u) $\vdash_{L_2} (\neg\mathscr{B} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{B}$
(v) $\mathscr{B} \Rightarrow \mathscr{C},\ \neg\mathscr{B} \Rightarrow \mathscr{C} \vdash_{L_2} \mathscr{C}$
(w) $\vdash_{L_2} \mathscr{B}$ if and only if $\mathscr{B}$ is a tautology.

1.56 Show that the theory L_3 has the same theorems as the theory L.

1.57 (Kleene, 1952) Derive the following facts about the theory L_4.

(a) $\vdash_{L_4} \mathscr{B} \Rightarrow \mathscr{B}$
(b) If $\Gamma,\ \mathscr{B} \vdash_{L_4} \mathscr{C}$, then $\Gamma \vdash_{L_4} \mathscr{B} \Rightarrow \mathscr{C}$ (deduction theorem)
(c) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D} \vdash_{L_4} \mathscr{B} \Rightarrow \mathscr{D}$
(d) $\vdash_{L_4} (\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$
(e) $\mathscr{B},\ \neg\mathscr{B} \vdash_{L_4} \mathscr{C}$
(f) $\vdash_{L_4} \mathscr{B} \Rightarrow \neg\neg\mathscr{B}$
(g) $\vdash_{L_4} \neg\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$
(h) $\vdash_{L_4} \mathscr{B} \Rightarrow (\neg\mathscr{C} \Rightarrow \neg(\mathscr{B} \Rightarrow \mathscr{C}))$
(i) $\vdash_{L_4} \neg\mathscr{B} \Rightarrow (\neg\mathscr{C} \Rightarrow \neg(\mathscr{B} \vee \mathscr{C}))$
(j) $\vdash_{L_4} (\neg\mathscr{C} \Rightarrow \mathscr{B}) \Rightarrow ((\mathscr{C} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{B})$
(k) $\vdash_{L_4} \mathscr{B}$ if and only if $\mathscr{B}$ is a tautology.

1.58[D] Consider the following axiomatization of the propositional calculus $\mathscr{L}$ (due to Lukasiewicz). $\mathscr{L}$ has the same wfs as our system L. Its only rule of inference is modus ponens. Its axiom schemas are:

(a) $(\neg\mathscr{B} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{B}$
(b) $\mathscr{B} \Rightarrow (\neg\mathscr{B} \Rightarrow \mathscr{C})$
(c) $(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\mathscr{C} \Rightarrow \mathscr{D}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}))$

Prove that a wf $\mathscr{B}$ of $\mathscr{L}$ is provable in $\mathscr{L}$ if and only if $\mathscr{B}$ is a tautology. [*Hint*: Show that L and $\mathscr{L}$ have the same theorems. However, remember that none of the results proved about L (such as Propositions 1.8–1.13) automatically carries over to $\mathscr{L}$. In particular, the deduction theorem is not available until it is proved for $\mathscr{L}$.]

1.59 Show that axiom schema (A3) of L can be replaced by the schema $(\neg\mathscr{B} \Rightarrow \neg\mathscr{C}) \Rightarrow (\mathscr{C} \Rightarrow \mathscr{B})$ without altering the class of theorems.

1.60 If axiom schema (10) of L_4 is replaced by the schema (10)′: $\neg\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$, then the new system L_I is called the *intuitionistic* propositional calculus.† Prove the following results about L_I.

(a) Consider an $(n+1)$-valued logic with these connectives: $\neg\mathscr{B}$ is 0 when $\mathscr{B}$ is n, and otherwise it is n; $\mathscr{B} \wedge \mathscr{C}$ has the maximum of the values of $\mathscr{B}$ and $\mathscr{C}$, whereas $\mathscr{B} \vee \mathscr{C}$ has the minimum of these values; and $\mathscr{B} \Rightarrow \mathscr{C}$ is 0 if $\mathscr{B}$ has a value not less than that of $\mathscr{C}$, and otherwise it has the same value as $\mathscr{C}$. If we take 0 as the only designated value, all theorems of L_I are exceptional.
(b) $A_1 \vee \neg A_1$ and $\neg\neg A_1 \Rightarrow A_1$ are not theorems of L_I.
(c) For any m, the wf

$$(A_1 \Leftrightarrow A_2) \vee \ldots \vee (A_1 \Leftrightarrow A_m) \vee (A_2 \Leftrightarrow A_3) \vee \ldots \vee (A_2 \Leftrightarrow A_m) \vee \ldots \vee (A_{m-1} \Leftrightarrow A_m)$$

is not a theorem of L_I

(d) (Gödel, 1933) L_I is not suitable for any finite many-valued logic.
(e) (i) If $\Gamma, \mathscr{B} \vdash_{L_I} \mathscr{C}$, then $\Gamma \vdash_{L_I} \mathscr{B} \Rightarrow \mathscr{C}$ (deduction theorem)
(ii) $\mathscr{B} \Rightarrow \mathscr{C},\ \mathscr{C} \Rightarrow \mathscr{D} \vdash_{L_I} \mathscr{B} \Rightarrow \mathscr{D}$
(iii) $\vdash_{L_I} \mathscr{B} \Rightarrow \neg\neg\mathscr{B}$
(iv) $\vdash_{L_I} (\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$
(v) $\vdash_{L_I} \mathscr{B} \Rightarrow (\neg\mathscr{B} \Rightarrow \mathscr{C})$
(vi) $\vdash_{L_I} \neg\neg(\neg\neg\mathscr{B} \Rightarrow \mathscr{B})$
(vii) $\neg\neg(\mathscr{B} \Rightarrow \mathscr{C}),\ \neg\neg\mathscr{B} \vdash_{L_I} \neg\neg\mathscr{C}$

†The principal origin of intuitionistic logic was L.E.J. Brouwer's belief that classical logic is wrong. According to Brouwer, $\mathscr{B} \vee \mathscr{C}$ is proved only when a proof of $\mathscr{B}$ or a proof of $\mathscr{C}$ has been found. As a consequence, various tautologies, such as $\mathscr{B} \vee \neg\mathscr{B}$, are not generally acceptable. For further information, consult Brouwer (1976), Heyting (1956), Kleene (1952), Troelstra (1969), and Dummett (1977). Jaśkowski (1936) showed that L_I is suitable for a many-valued logic with denumerably many values.

(viii) $\vdash_{L_1} \neg\neg\neg\mathscr{B} \Rightarrow \neg\mathscr{B}$

(b) $\vdash_{L_1} \neg\neg\mathscr{B}$ if and only if $\mathscr{B}$ is a tautology.

(c) $\vdash_{L_1} \neg\mathscr{B}$ if and only if $\neg\mathscr{B}$ is a tautology.

(d) If $\mathscr{B}$ has $\wedge$ and $\neg$ as its only connectives, then $\vdash_{L_1} \mathscr{B}$ if and only if $\mathscr{B}$ is a tautology.

1.61^A Let $\mathscr{B}$ and $\mathscr{C}$ be in the relation R if and only if $\vdash_L \mathscr{B} \Leftrightarrow \mathscr{C}$. Show that R is an equivalence relation. Given equivalence classes $[\mathscr{B}]$ and $[\mathscr{C}]$, let $[\mathscr{B}] \cup [\mathscr{C}] = [\mathscr{B} \vee \mathscr{C}]$, $[\mathscr{B}] \cap [\mathscr{C}] = [\mathscr{B} \wedge \mathscr{C}]$, and $\overline{[\mathscr{B}]} = [\neg\mathscr{B}]$. Show that the equivalence classes under R form a Boolean algebra with respect to $\cap$, $\cup$ and $^-$, called the *Lindenbaum algebra* L^* determined by L. The element 0 of L^* is the equivalence class consisting of all contradictions (i.e., negations of tautologies). The unit element 1 of L^* is the equivalence class consisting of all tautologies. Notice that $\vdash_L \mathscr{B} \Rightarrow \mathscr{C}$ if and only if $[\mathscr{B}] \leqslant [\mathscr{C}]$ in L^*, and that $\vdash_L \mathscr{B} \Leftrightarrow \mathscr{C}$ if and only if $[\mathscr{B}] = [\mathscr{C}]$. Show that a Boolean function f (built up from variables, 0, and 1, using $\cup$, $\cap$ and $^-$) is equal to the constant function 1 in all Boolean algebras if and only if $\vdash_L f^{\#}$, where $f^{\#}$ is obtained from f by changing $\cup$, $\cap$, $^-$, 0 and 1 to $\vee$, $\wedge$, $\neg$, $A_1 \wedge \neg A_1$, and $A_1 \vee \neg A_1$, respectively.

2 Quantification Theory

2.1 QUANTIFIERS

There are various kinds of logical inference that cannot be justified on the basis of the propositional calculus; for example:

1. Any friend of Martin is a friend of John.
   Peter is not John's friend.
   Hence, Peter is not Martin's friend.
2. All human beings are rational.
   Some animals are human beings.
   Hence, some animals are rational.
3. The successor of an even integer is odd.
   2 is an even integer.
   Hence, the successor of 2 is odd.

The correctness of these inferences rests not only upon the meanings of the truth-functional connectives, but also upon the meaning of such expressions as 'any', 'all' and 'some', and other linguistic constructions.

In order to make the structure of complex sentences more transparent, it is convenient to introduce special notation to represent frequently occurring expressions. If $P(x)$ asserts that x has the property P, then $(\forall x)P(x)$ means that property P holds for all x or, in other words, that everything has the property P. On the other hand, $(\exists x)P(x)$ means that some x has the property P – that is, that there is at least one object having the property P. In $(\forall x)P(x)$, '$(\forall x)$' is called a *universal quantifier*; in $(\exists x)P(x)$, '$(\exists x)$' is called an *existential quantifier*. The study of quantifiers and related concepts is the principal subject of this chapter.

Examples

1′. Inference 1 above can be represented symbolically:

$$\begin{array}{l}(\forall x)(F(x,m) \Rightarrow F(x,j)) \\ \neg F(p,j) \\ \hline \neg F(p,m)\end{array}$$

Here, $F(x,y)$ means that x is a friend of y, while m, j and p denote Martin, John, and Peter, respectively.

2′. Inference 2 becomes:

$$\begin{array}{l}(\forall x)(H(x) \Rightarrow R(x)) \\ (\exists x)(A(x) \wedge H(x)) \\ \hline (\exists x)(A(x) \wedge R(x))\end{array}$$

Here, H, R and A designate the properties of being human, rational, and an animal, respectively.

3′. Inference 3 can be symbolized as follows:

$$\begin{array}{l}(\forall x)(I(x) \wedge E(x) \Rightarrow D(s(x))) \\ I(b) \wedge E(b) \\ \hline D(s(b))\end{array}$$

Here, I, E and D designate respectively the properties of being an integer, even and odd; $s(x)$ denotes the successor of x; and b denotes the integer 2.

Notice that the validity of these inferences does not depend upon the particular meanings of $F, m, j, p, H, R, A, I, E, D, s$ and b.

Just as statement forms were used to indicate logical structure dependent upon the logical connectives, so also the form of inferences involving quantifiers, such as inferences 1–3, can be represented abstractly, as in 1′ – 3′. For this purpose, we shall use commas, parentheses, the symbols $\neg$ and $\Rightarrow$ of the propositional calculus, the universal quantifier symbol $\forall$, and the following groups of symbols:

Individual variables: $x_1, x_2, \ldots, x_n, \ldots$
Individual constants: $a_1, a_2, \ldots, a_n, \ldots$
Predicate letters: A_k^n (n and k are any positive integers)
Function letters: f_k^n (n and k are any positive integers)

The positive integer n that is a superscript of a predicate letter A_k^n or of a function letter f_k^n indicates the number of arguments, whereas the subscript k is just an indexing number to distinguish different predicate or function letters with the same number of arguments.†

In the preceding examples, x plays the role of an individual variable; m, j, p and b play the role of individual constants; F is a binary predicate letter (i.e., a predicate letter with two arguments); H, R, A, I, E and D are monadic predicate letters (i.e., predicate letters with one argument); and s is a function letter with one argument.

The function letters applied to the variables and individual constants generate the *terms*:

†For example, in arithmetic both addition and multiplication take two arguments. So, we would use one function letter, say f_1^2, for addition, and a different function letter, say f_2^2, for multiplication.

1. Variables and individual constants are terms.
2. If f_k^n is a function letter and $t_1, t_2, \ldots, t_n$ are terms, then $f_k^n(t_1, t_2, \ldots, t_n)$ is a term.
3. An expression is a term only if it can be shown to be a term on the basis of conditions 1 and 2.

Terms correspond to what in ordinary languages are nouns and noun phrases – for example, 'two', 'two plus three', and 'two plus x'.

The predicate letters applied to terms yield the *atomic formulas*; that is, if A_k^n is a predicate letter and $t_1, t_2, \ldots, t_n$ are terms, then $A_k^n(t_1, t_2, \ldots, t_n)$ is an atomic formula.

The *well-formed formulas* (wfs) of quantification theory are defined as follows:

1. Every atomic formula is a wf.
2. If $\mathscr{B}$ and $\mathscr{C}$ are wfs and y is a variable, then $(\neg\mathscr{B})$, $(\mathscr{B} \Rightarrow \mathscr{C})$, and $((\forall y)\mathscr{B})$ are wfs.
3. An expression is a wf only if it can be shown to be a wf on the basis of conditions 1 and 2.

In $((\forall y)\mathscr{B})$, '$\mathscr{B}$' is called the *scope* of the quantifier '$(\forall y)$'. Notice that $\mathscr{B}$ need not contain the variable y. In that case, we understand $((\forall y)\mathscr{B})$ to mean the same thing as $\mathscr{B}$.

The expressions $(\mathscr{B} \wedge \mathscr{C})$, $(\mathscr{B} \vee \mathscr{C})$, and $(\mathscr{B} \Leftrightarrow \mathscr{C})$ are defined as in system L (see page 36). It was unnecessary for us to use the symbol $\exists$ as a primitive symbol because we can define existential quantification as follows:

$$((\exists x)\mathscr{B}) \quad \text{stands for} \quad (\neg((\forall x)(\neg\mathscr{B})))$$

This definition is faithful to the meaning of the quantifiers: $\mathscr{B}(x)$ is true for some x if and only if it is not the case that $\mathscr{B}(x)$ is false for all x.†

Parentheses

The same conventions as made in Chapter 1 (page 20) about the omission of parentheses are made here, with the additional convention that quantifiers $(\forall y)$ and $(\exists y)$ rank in strength between $\neg, \wedge, \vee$ and $\Rightarrow, \Leftrightarrow$.

Examples

Parentheses are restored in the following steps.

1. $(\forall x_1)A_1^1(x_1) \Rightarrow A_1^2(x_2, x_1)$
   $((\forall x_1)A_1^1(x_1)) \Rightarrow A_1^2(x_2, x_1)$
   $(((\forall x_1)A_1^1(x_1)) \Rightarrow A_1^2(x_2, x_1))$

†We could have taken $\exists$ as primitive and then defined $((\forall x)\mathscr{B})$ as an abbreviation for $(\neg((\exists x)(\neg\mathscr{B})))$, since $\mathscr{B}(x)$ is true for all x if and only if it is not the case that $\mathscr{B}(x)$ is false for some x.

2. $(\forall x_1)A_1^1(x_1) \vee A_1^2(x_2, x_1)$
   $(\forall x_1)(A_1^1(x_1) \vee A_1^2(x_2, x_1))$
   $((\forall x_1)(A_1^1(x_1)) \vee A_1^2(x_2, x_1)))$
3. $(\forall x_1)(\exists x_2)A_1^2(x_1, x_2)$
   $(\forall x_1)((\exists x_2)A_1^2(x_1, x_2))$
   $((\forall x_1)((\exists x_2)A_1^2(x_1, x_2)))$

Exercises

2.1 Restore parentheses to the following.
(a) $(\forall x_1)A_1^1(x_1) \wedge \neg A_1^1(x_2)$
(b) $(\forall x_2)A_1^1(x_2) \Leftrightarrow A_1^1(x_2)$
(c) $(\forall x_2)(\exists x_1)A_1^2(x_1, x_2)$
(d) $(\forall x_1)(\forall x_3)(\forall x_4)A_1^1(x_1) \Rightarrow A_1^1(x_2) \wedge \neg A_1^1(x_1)$
(e) $(\exists x_1)(\forall x_2)(\exists x_3)A_1^1(x_1) \vee (\exists x_2)\neg(\forall x_3)A_1^2(x_3, x_2)$
(f) $(\forall x_2)\neg A_1^1(x_1) \Rightarrow A_1^3(x_1, x_1, x_2) \vee (\forall x_1)A_1^1(x_1)$
(g) $\neg(\forall x_1)A_1^1(x_1) \Rightarrow (\exists x_2)A_1^1(x_2) \Rightarrow A_1^2(x_1, x_2) \wedge A_1^1(x_2)$

2.2 Eliminate parentheses from the following wfs as far as is possible.
(a) $(((\forall x_1)(A_1^1(x_1) \Rightarrow A_1^1(x_1))) \vee ((\exists x_1)A_1^1(x_1)))$
(b) $((\neg((\exists x_2)(A_1^1(x_2) \vee A_1^1(a_1)))) \Leftrightarrow A_1^1(x_2))$
(c) $(((\forall x_1)(\neg(\neg A_1^1(a_3)))) \Rightarrow (A_1^1(x_1) \Rightarrow A_1^1(x_2)))$

An occurrence of a variable x is said to be *bound* in a wf $\mathscr{B}$ if either it is the occurrence of x in a quantifier '$(\forall x)$' in $\mathscr{B}$ or it lies within the scope of a quantifier '$(\forall x)$' in $\mathscr{B}$. Otherwise, the occurrence is said to be *free* in $\mathscr{B}$.

Examples
1. $A_1^2(x_1, x_2)$
2. $A_1^2(x_1, x_2) \Rightarrow (\forall x_1)A_1^1(x_1)$
3. $(\forall x_1)(A_1^2(x_1, x_2) \Rightarrow (\forall x_1)A_1^1(x_1))$
4. $(\exists x_1)A_1^2(x_1, x_2)$

In Example 1, the single occurrence of x_1 is free. In Example 2, the first occurrence of x_1 is free, but the second and third occurrences are bound. In Example 3, all occurrences of x_1 are bound, and in Example 4 both occurrences of x_1 are bound. (Remember that $(\exists x_1)A_1^2(x_1, x_2)$ is an abbreviation of $\neg(\forall x_1)\neg A_1^2(x_1, x_2)$.) In all four wfs, every occurrence of x_2 is free. Notice that, as in Example 2, a variable may have both free and bound occurrences in the same wf. Also observe that an occurrence of a variable may be bound in some wf $\mathscr{B}$ but free in a subformula of $\mathscr{B}$. For example, the first occurrence of x_1 is free in the wf of Example 2 but bound in the larger wf of Example 3.

A variable is said to be *free* (*bound*) in a wf $\mathscr{B}$ if it has a free (bound) occurrence in $\mathscr{B}$. Thus, a variable may be both free and bound in the same wf; for example, x_1 is free and bound in the wf of Example 2.

Exercises

2.3 Pick out the free and bound occurrences of variables in the following wfs.
(a) $(\forall x_3)(((\forall x_1)A_1^2(x_1,x_2)) \Rightarrow A_1^2(x_3,a_1))$
(b) $(\forall x_2)A_1^2(x_3,x_2) \Rightarrow (\forall x_3)A_1^2(x_3,x_2)$
(c) $((\forall x_2)(\exists x_1)A_1^3(x_1,x_2,f_1^2(x_1,x_2))) \vee \neg(\forall x_1)A_1^2(x_2,f_1^1(x_1))$
2.4 Indicate the free and bound occurrences of all variables in the wfs of Exercises 2.1 and 2.2.
2.5 Indicate the free and bound variables in the wfs of Exercises 2.1–2.3.

We shall often indicate that some of the variables $x_{i_1},\ldots,x_{i_k}$ are free variables in a wf $\mathscr{B}$ by writing $\mathscr{B}$ as $\mathscr{B}(x_{i_1},\ldots,x_{i_k})$. This does not mean that $\mathscr{B}$ contains these variables as free variables, nor does it mean that $\mathscr{B}$ does not contain other free variables. This notation is convenient because we can then agree to write as $\mathscr{B}(t_1,\ldots,t_k)$ the result of substituting in $\mathscr{B}$ the terms $t_1,\ldots,t_k$ for all free occurrences (if any) of $x_{i_1},\ldots,x_{i_k}$, respectively.

If $\mathscr{B}$ is a wf and t is a term, then t is said to be *free for x_i in $\mathscr{B}$* if no free occurrence of x_i in $\mathscr{B}$ lies within the scope of any quantifier $(\forall x_j)$, where x_j is a variable in t. This concept of t being free for x_i in a wf $\mathscr{B}(x_i)$ will have certain technical applications later on. It means that, if t is substituted for all free occurrences (if any) of x_i in $\mathscr{B}(x_i)$, no occurrence of a variable in t becomes a bound occurrence in $\mathscr{B}(t)$.

Examples
1. The term x_2 is free for x_1 in $A_1^1(x_1)$, but x_2 is not free for x_1 in $(\forall x_2)A_1^1(x_1)$.
2. The term $f_1^2(x_1,x_3)$ is free for x_1 in $(\forall x_2)A_1^2(x_1,x_2) \Rightarrow A_1^1(x_1)$ but is not free for x_1 in $(\exists x_3)(\forall x_2)A_1^2(x_1,x_2) \Rightarrow A_1^1(x_1)$.

The following facts are obvious.

1. A term that contains no variables is free for any variable in any wf.
2. A term t is free for any variable in $\mathscr{B}$ if none of the variables of t is bound in $\mathscr{B}$.
3. x_i is free for x_i in any wf.
4. Any term is free for x_i in $\mathscr{B}$ if $\mathscr{B}$ contains no free occurrences of x_i.

Exercises

2.6 Is the term $f_1^2(x_1,x_2)$ free for x_1 in the following wfs?
(a) $A_1^2(x_1,x_2) \Rightarrow (\forall x_2)A_1^1(x_2)$
(b) $((\forall x_2)A_1^2(x_2,a_1)) \vee (\exists x_2)A_1^2(x_1,x_2)$
(c) $(\forall x_1)A_1^2(x_1,x_2)$
(d) $(\forall x_2)A_1^2(x_1,x_2)$
(e) $(\forall x_2)A_1^1(x_2) \Rightarrow A_1^2(x_1,x_2)$

2.7 Justify facts 1–4 above.

When English sentences are translated into formulas, certain general guidelines will be useful:

1. A sentence of the form 'All As are Bs' becomes $(\forall x)(A(x) \Rightarrow B(x))$. For example, *Every mathematician loves music* is translated as $(\forall x)(M(x) \Rightarrow L(x))$, where $M(x)$ means *x is a mathematician* and $L(x)$ means *x loves music*.
2. A sentence of the form 'Some As are Bs' becomes $(\exists x)(A(x) \wedge B(x))$. For example, *Some New Yorkers are friendly* becomes $(\exists x)(N(x) \wedge F(x))$, where $N(x)$ means *x is a New Yorker* and $F(x)$ means *x is friendly*.
3. A sentence of the form 'No As are Bs' becomes $(\forall x)(A(x) \Rightarrow \neg B(x))$.† For example, *No philosopher understands politics* becomes $(\forall x)(P(x) \Rightarrow \neg U(x))$, where $P(x)$ means *x is a philosopher* and $U(x)$ means *x understands politics*.

Let us consider a more complicated example: *Some people respect everyone*. This can be translated as $(\exists x)(P(x) \wedge (\forall y)(P(y) \Rightarrow R(x, y)))$, where $P(x)$ means *x is a person* and $R(x, y)$ means *x respects y*.

Notice that, in informal discussions, to make formulas easier to read we may use lower-case letters u, v, x, y, z instead of our official notation x_i for individual variables, capital letters $A, B, C, \ldots$ instead of our official notation A_k^n for predicate letters, lower-case letters $f, g, h, \ldots$ instead of our official notation f_k^n for function letters, and lower-case letters $a, b, c, \ldots$ instead of our official notation a_i for individual constants.

Exercises

2.8 Translate the following sentences into wfs.

(a) Anyone who is persistent can learn logic.
(b) No politician is honest.
(c) Not all birds can fly.
(d) All birds cannot fly.
(e) x is transcendental only if it is irrational.
(f) Seniors date only juniors.
(g) If anyone can solve the problem, Hilary can.
(h) Nobody loves a loser.
(i) Nobody in the statistics class is smarter than everyone in the logic class.
(j) John hates all people who do not hate themselves.
(k) Everyone loves somebody and no one loves everybody, or somebody loves everybody and someone loves nobody.
(l) You can fool some of the people all of the time, and you can fool all the people some of the time, but you can't fool all the people all the time.

†As we shall see later, this is equivalent to $\neg(\exists x)(A(x) \wedge B(x))$.

(m) Any sets that have the same members are equal.
(n) Anyone who knows Julia loves her.
(o) There is no set belonging to precisely those sets that do not belong to themselves.
(p) There is no barber who shaves precisely those men who do not shave themselves.

2.9 Translate the following into everyday English. Note that everyday English does not use variables.

(a) $(\forall x)(M(x) \wedge (\forall y)\neg W(x,y) \Rightarrow U(x))$, where $M(x)$ means *x is a man*, $W(x,y)$ means *x is married to y*, and $U(x)$ means *x is unhappy*.
(b) $(\forall x)(V(x) \wedge P(x) \Rightarrow A(x,b))$, where $V(x)$ means *x is an even integer*, $P(x)$ means *x is a prime integer*, $A(x,y)$ means $x = y$, and b denotes 2.
(c) $\neg(\exists y)(I(y) \wedge (\forall x)(I(x) \Rightarrow L(x,y)))$, where $I(y)$ means *y is an integer* and $L(x,y)$ means $x \leqslant y$.
(d) In the following wfs, $A_1^1(x)$ means *x is a person* and $A_1^2(x,y)$ means *x hates y*.
    (i) $(\exists x)(A_1^1(x) \wedge (\forall y)(A_1^1(y) \Rightarrow A_1^2(x,y)))$
    (ii) $(\forall x)(A_1^1(x) \Rightarrow (\forall y)(A_1^1(y) \Rightarrow A_1^2(x,y)))$
    (iii) $(\exists x)(A_1^1(x) \wedge (\forall y)(A_1^1(y) \Rightarrow (A_1^2(x,y) \Leftrightarrow A_1^2(y,y))))$
(e) $(\forall x)(H(x) \Rightarrow (\exists y)(\exists z)(\neg A(y,z) \wedge (\forall u)(P(u,x) \Leftrightarrow (A(u,y) \vee A\ (u,z)))))$, where $H(x)$ means *x is a person*, $A(u,v)$ means '$u = v$', and $P(u,x)$ means *u is a parent of x*.

2.2 FIRST-ORDER LANGUAGES AND THEIR INTERPRETATIONS. SATISFIABILITY AND TRUTH. MODELS

Well-formed formulas have meaning only when an interpretation is given for the symbols. We usually are interested in interpreting wfs whose symbols come from a specific language. For that reason, we shall define the notion of a *first-order language*.†

†The adjective 'first-order' is used to distinguish the languages we shall study here from those in which there are predicates having other predicates or functions as arguments or in which predicate quantifiers or function quantifiers are permitted, or both. Most mathematical theories can be formalized within first-order languages, although there may be a loss of some of the intuitive content of those theories. Second-order languages are discussed in the appendix on second-order logic. Examples of higher-order languages are studied also in Gödel (1931), Tarski (1933), Church (1940), Hasenjaeger and Scholz (1961) and Van Bentham and Doets (1983). Differences between first-order and higher-order theories are examined in Corcoran (1980).

DEFINITION

A first-order language $\mathscr{L}$ contains the following symbols.

(a) The propositional connectives $\neg$ and $\Rightarrow$, and the universal quantifier symbol $\forall$.
(b) Punctuation marks: the left parenthesis (, the right parenthesis), and the comma.[†]
(c) Denumerably many individual variables $x_1, x_2, \ldots$.
(d) A finite or denumerable, possibly empty, set of function letters.
(e) A finite or denumerable, possibly empty, set of individual constants.
(f) A non-empty set of predicate letters.

By a *term of* $\mathscr{L}$ we mean a term whose symbols are symbols of $\mathscr{L}$.
By a *wf of* $\mathscr{L}$ we mean a wf whose symbols are symbols of $\mathscr{L}$.

Thus, in a language $\mathscr{L}$, some or all of the function letters and individual constants may be absent, and some (but not all) of the predicate letters may be absent.[‡] The individual constants, function letters and predicate letters of a language $\mathscr{L}$ are called the *non-logical constants* of $\mathscr{L}$. Languages are designed in accordance with the subject matter we wish to study. A language for arithmetic might contain function letters for addition and multiplication and a predicate letter for equality, whereas a language for geometry is likely to have predicate letters for equality and the notions of *point* and *line* but no function letters at all.

DEFINITION

Let $\mathscr{L}$ be a first-order language. An *interpretation* M of $\mathscr{L}$ consists of the following ingredients.

(a) A non-empty set D, called the *domain* of the interpretation.
(b) For each predicate letter A_j^n of $\mathscr{L}$, an assignment of an n-place relation $(A_j^n)^{\mathrm{M}}$ in D.
(c) For each function letter f_j^n of $\mathscr{L}$, an assignment of an n-place operation $(f_j^n)^{\mathrm{M}}$ in D (that is, a function from D^n into D).
(d) For each individual constant a_i of $\mathscr{L}$, an assignment of some fixed element $(a_i)^{\mathrm{M}}$ of D.

Given such an interpretation, variables are thought of as ranging over the set D, and $\neg, \Rightarrow$ and quantifiers are given their usual meaning. Remember that an n-place relation in D can be thought of as a subset of D^n, the set of all

[†]The punctuation marks are not strictly necessary; they can be avoided by redefining the notions of term and wf. However, their use makes it easier to read and comprehend formulas.

[‡]If there were no predicate letters, there would be no wfs.

n-tuples of elements of D. For example, if D is the set of human beings, then the relation 'father of' can be identified with the set of all ordered pairs $\langle x, y\rangle$ such that x is the father of y.

For a given interpretation of a language $\mathscr{L}$, a wf of $\mathscr{L}$ without free variables (called a *closed wf* or a *sentence*) represents a propostion that is true or false, whereas a wf with free variables may be satisfied (i.e., true) for some values in the domain and not satisfied (i.e., false) for the others.

Examples

Consider the following wfs:

1. $A_1^2(x_1, x_2)$
2. $(\forall x_2)A_1^2(x_1, x_2)$
3. $(\exists x_1)(\forall x_2)A_1^2(x_1, x_2)$

Let us take as domain the set of all positive integers and interpret $A_1^2(y, z)$ as $y \leqslant z$. Then wf 1 represents the expression '$x_1 \leqslant x_2$', which is satisfied by all the ordered pairs $\langle a, b\rangle$ of positive integers such that $a \leqslant b$. Wf 2 represents the expression 'For all positive integers x_2, $x_1 \leqslant x_2$,'† which is satisfied only by the integer 1. Wf 3 is a true sentence asserting that there is a smallest positive integer. If we were to take as domain the set of all integers, then wf 3 would be false.

Exercises

2.10 For the following wfs and for the given interpretations, indicate for what values the wfs are satisfied (if they contain free variables) or whether they are true or false (if they are closed wfs).

(i) $A_1^2(f_1^2(x_1, x_2), a_1)$
(ii) $A_1^2(x_1, x_2) \Rightarrow A_1^2(x_2, x_1)$
(iii) $(\forall x_1)(\forall x_2)(\forall x_3)(A_1^2(x_1, x_2) \wedge A_1^2(x_2, x_3) \Rightarrow A_1^2(x_1, x_3))$
(a) The domain is the set of positive integers, $A_1^2(y, z)$ is $y \geqslant z$, $f_1^2(y, z)$ is $y \cdot z$, and a_1 is 2.
(b) The domain is the set of integers, $A_1^2(y, z)$ is $y = z$, $f_1^2(y, z)$ is $y + z$, and a_1 is 0.
(c) The domain is the set of all sets of integers, $A_1^2(y, z)$ if $y \subseteq z$, $f_1^2(y, z)$ is $y \cap z$, and a_1 is the empty set $\emptyset$.

2.11 Describe in everyday English the assertions determined by the following wfs and interpretations.

(a) $(\forall x)(\forall y)(A_1^2(x, y) \Rightarrow (\exists z)(A_1^1(z) \wedge A_1^2(x, z) \wedge A_1^2(z, y)))$, where the domain D is the set of real numbers, $A_1^2(x, y)$ means $x < y$, and $A_1^1(z)$ means z *is a rational number*.

†In ordinary English, one would say 'x_1 is less than or equal to all positive integers'.

(b) $(\forall x)(A_1^1(x) \Rightarrow (\exists y)(A_2^1(y) \wedge A_1^2(y,x)))$, where D is the set of all days and people, $A_1^1(x)$ means *x is a day*, $A_2^1(y)$ means *y is a sucker*, and $A_1^2(y,x)$ means *y is born on day x*.
(c) $(\forall x)(\forall y)(A_1^1(x) \wedge A_1^1(y) \Rightarrow A_2^1(f_1^2(x,y)))$, where D is the set of integers, $A_1^1(x)$ means *x is odd*, $A_2^1(x)$ means *x is even*, and $f_1^2(x,y)$ denotes $x+y$.
(d) For the following wfs, D is the set of all people and $A_1^2(u,v)$ means *u loves v*.
(i) $(\exists x)(\forall y)(A_1^2(x,y)$
(ii) $(\forall y)(\exists x)A_1^2(x,y)$
(iii) $(\exists x)(\forall y)((\forall z)(A_1^2(y,z) \Rightarrow A_1^2(x,y))$
(iv) $(\exists x)(\forall y)\neg A_1^2(x,y)$

The concepts of satisfiability and truth are intuitively clear, but, following Tarski (1936), we also can provide a rigorous definition. Such a definition is necessary for carrying out precise proofs of many metamathematical results.

Satisfiability will be the fundamental notion, on the basis of which the notion of truth will be defined. Moreover, instead of talking about the n-tuples of objects that satisfy a wf that has n free variables, it is much more convenient from a technical standpoint to deal uniformly with denumerable sequences. What we have in mind is that a denumerable sequence $s = (s_1, s_2, s_3, \ldots)$ is to be thought of as satisfying a wf $\mathscr{B}$ that has $x_{j_1}, x_{j_2} \ldots, x_{j_n}$ as free variables (where $j_1 < j_2 < \ldots < j_n$) if the n-tuple $\langle s_{j_1}, s_{j_2}, \ldots, s_{j_n}\rangle$ satisfies $\mathscr{B}$ in the usual sense. For example, a denumerable sequence $(s_1, s_2, s_3, \ldots)$ of objects in the domain of an interpretation M will turn out to satisfy the wf $A_1^2(x_2, x_5)$ if and only if the ordered pair, $\langle s_2, s_5\rangle$ is in the relation $(A_1^2)^{\mathrm{M}}$ assigned to the predicate letter A_1^2 by the interpretation M.

Let M be an interpretation of a language $\mathscr{L}$ and let D be the domain of M. Let Σ be the set of all denumerable sequences of elements of D. For a wf $\mathscr{B}$ of $\mathscr{L}$, we shall define what it means for a sequence $s = (s_1, s_2, \ldots)$ in Σ to satisfy $\mathscr{B}$ in M. As a preliminary step, for a given s in Σ we shall define a function s^* that assigns to each term t of $\mathscr{L}$ an element $s^*(t)$ in D.

1. If t is a variable x_j, let $s^*(t)$ be s_j.
2. If t is an individual constant a_j, then $s^*(t)$ is the interpretation $(a_j)^{\mathrm{M}}$ of this constant.
3. If f_k^n is a function letter, $(f_k^n)^{\mathrm{M}}$ is the corresponding operation in D, and $t_1, \ldots, t_n$ are terms, then

$$s^*(f_k^n(t_1, \ldots, t_n)) = (f_k^n)^{\mathrm{M}}(s^*(t_1), \ldots, s^*(t_n))$$

Intuitively, $s^*(t)$ is the element of D obtained by substituting, for each j, a name of s_j for all occurrences of x_j in t and then performing the operations of the interpretation corresponding to the function letters of t. For instance, if t is $f_2^2(x_3, f_1^2(x_1, a_1))$ and if the interpretation has the set of integers as its domain, f_2^2 and f_1^2 are interpreted as ordinary multiplication and addition, respectively, and a_1 is interpreted as 2, then, for any sequence $s = (s_1, s_2, \ldots)$

of integers, $s^*(t)$ is the integer $s_3 \cdot (s_1 + 2)$. This is really nothing more than the ordinary way of reading mathematical expressions.

Now we proceed to the definition of satisfaction, which will be an inductive definition.

1. If $\mathscr{B}$ is an atomic wf $A_k^n(t_1, \ldots, t_n)$ and $(A_k^n)^{\mathrm{M}}$ is the corresponding n-place relation of the interpretation, then a sequence $s = (s_1, s_2, \ldots)$ satisfies $\mathscr{B}$ if and only if $(A_k^n)^{\mathrm{M}}(s^*(t_1), \ldots, s^*(t_n))$ – that is, if the n-tuple $\langle s^*(t_1), \ldots, s^*(t_n)\rangle$ is in the relation $(A_k^n)^{\mathrm{M}}$.†
2. s satisfies $\neg\mathscr{B}$ if and only if s does not satisfy $\mathscr{B}$.
3. s satisfies $\mathscr{B} \Rightarrow \mathscr{C}$ if and only if s does not satisfy $\mathscr{B}$ or s satisfies $\mathscr{C}$.
4. s satisfies $(\forall x_i)\mathscr{B}$ if and only if every sequence that differs from s in at most the ith component satisfies $\mathscr{B}$.‡

Intuitively, a sequence $s = (s_1, s_2, \ldots)$ satisfies a wf $\mathscr{B}$ if and only if, when, for each i, we replace all free occurrences of x_i (if any) in $\mathscr{B}$ by a symbol representing s_i, the resulting proposition is true under the given interpretation.

Now we can define the notions of truth and falsity of wfs for a given interpretation.

DEFINITIONS

1. A wf $\mathscr{B}$ is *true for the interpretation* M (written $\models_{\mathrm{M}} \mathscr{B}$) if and only if every sequence in Σ satisfies $\mathscr{B}$.
2. $\mathscr{B}$ is said to be *false for* M if and only if no sequence in Σ satisfies $\mathscr{B}$.
3. An interpretation M is said to be a *model* for a set Γ of wfs if and only if every wf in Γ is true for M.

The plausibility of our definition of truth will be strengthened by the fact that we can derive all of the following expected properties I–XI of the notions of truth, falsity and satisfaction. Proofs that are not explicitly given are left to the reader (or may be found in the answer to Exercise 2.12). Most

†For example, if the domain of the interpretation is the set of real numbers, the interpretation of A_1^2 is the relation $\leqslant$, and the interpretation of f_1^1 is the function e^x, then a sequence $s = (s_1, s_2, \ldots)$ of real numbers satisfies $A_1^2(f_1^1(x_2), x_5)$ if and only if $e^{s_2} \leqslant s_5$. If the domain is the set of integers, the interpretation of $A_1^4(x, y, u, v)$ is $x \cdot v = u \cdot y$, and the interpretation of a_1 is 3, then a sequence $s = (s_1, s_2, \ldots)$ of integers satisfies $A_1^4(x_3, a_1, x_1, x_3)$ if and only if $(s_3)^2 = 3s_1$.

‡In other words, a sequence $s = (s_1, s_2, \ldots, s_i, \ldots)$ satisfies $(\forall x_i)\mathscr{B}$ if and only if, for every element c of the domain, the sequence $(s_1, s_2, \ldots, c, \ldots)$ satisfies $\mathscr{B}$. Here, $(s_1, s_2, \ldots, c, \ldots)$ denotes the sequence obtained from $(s_1, s_2, \ldots, s_i, \ldots)$ by replacing the ith component s_i by c. Note also that, if s satisfies $(\forall x_i)\mathscr{B}$, then, as a special case, s satisfies $\mathscr{B}$.

of the results are also obvious if one wishes to use only the ordinary intuitive understanding of the notions of truth, falsity and satisfaction.

(I) (a) $\mathscr{B}$ is false for an interpretation M if and only if $\neg\mathscr{B}$ is true for M.
(b) $\mathscr{B}$ is true for M if and only if $\neg\mathscr{B}$ is false for M.

(II) It is not the case that both $\models_{\mathrm{M}} \mathscr{B}$ and $\models_{\mathrm{M}} \neg\mathscr{B}$; that is, no wf can be both true and false for M.

(III) If $\models_{\mathrm{M}} \mathscr{B}$ and $\models_{\mathrm{M}} \mathscr{B} \Rightarrow \mathscr{C}$, then $\models_{\mathrm{M}} \mathscr{B}$.

(IV) $\mathscr{B} \Rightarrow \mathscr{C}$ is false for M if and only if $\models_{\mathrm{M}} \mathscr{B}$ and $\models_{\mathrm{M}} \neg\mathscr{C}$.

(V)† Consider an interpretation M with domain D.
(a) A sequence s satisfies $\mathscr{B} \wedge \mathscr{C}$ if and only if s satisfies $\mathscr{B}$ and s satisfies $\mathscr{C}$.
(b) s satisfies $\mathscr{B} \vee \mathscr{C}$ if and only if s satisfies $\mathscr{B}$ or s satifies $\mathscr{C}$.
(c) s satisfies $\mathscr{B} \Leftrightarrow \mathscr{C}$ if and only if s satisfies both $\mathscr{B}$ and $\mathscr{C}$ or s satisfies neither $\mathscr{B}$ nor $\mathscr{C}$.
(d) s satisfies $(\exists x_i)\mathscr{B}$ if and only if there is a sequence s' that differs from s in at most the ith component such that s' satisfies $\mathscr{B}$. (In other words $s = (s_1, s_2, \ldots, s_i, \ldots)$ satisfies $(\exists x_i)\mathscr{B}$ if and only if there is an element c in the domain D such that the sequence $(s_1, s_2, \ldots, c, \ldots)$ satisfies $\mathscr{B}$.)

(VI) $\models_{\mathrm{M}} \mathscr{B}$ if and only if $\models_{\mathrm{M}} (\forall x_i)\mathscr{B}$. We can extend this result in the following way. By the *closure*‡ of $\mathscr{B}$ we mean the closed wf obtained from $\mathscr{B}$ by prefixing in universal quantifiers those variables, in order of descending subscripts, that are free in $\mathscr{B}$. If $\mathscr{B}$ has no free variables, the closure of $\mathscr{B}$ is defined to be $\mathscr{B}$ itself. For example, if $\mathscr{B}$ is $A_1^2(x_2, x_5) \Rightarrow \neg(\exists x_2)A_1^3(x_1, x_2, x_3)$, its closure is $(\forall x_5)(\forall x_3)(\forall x_2)(\forall x_1)\mathscr{B}$. It follows from (VI) that a wf $\mathscr{B}$ is true if and only if its closure is true.

(VII) Every instance of a tautology is true for any interpretation. (An *instance* of a statement form is a wf obtained from the statement form by substituting wfs for all statement letters, with all occurrences of the same statement letter being replaced by the same wf. Thus, an instance of $A_1 \Rightarrow \neg A_2 \vee A_1$ is $A_1^1(x_2) \Rightarrow (\neg(\forall x_1)A_1^1(x_1)) \vee A_1^1(x_2)$.) To prove (VII), show that all instances of the axioms of the system L are true and then use (III) and Proposition 1.14.

(VIII) If the free variables (if any) of a wf $\mathscr{B}$ occur in the list $x_{i_1}, \ldots, x_{i_k}$ and if the sequences s and s' have the same components in the i_1th, $\ldots$, i_kth places, then s satisfies $\mathscr{B}$ if and only if s' satisfies $\mathscr{B}$ [*Hint*: Use induction on the number of connectives and quantifiers in $\mathscr{B}$. First prove this lemma: If the variables in a term t occur in the list $x_{i_1}, \ldots, x_{i_k}$, and if s and s' have the same components in the

†Remember that $\mathscr{B} \wedge \mathscr{C}, \mathscr{B} \vee \mathscr{C}, \mathscr{B} \Leftrightarrow \mathscr{C}$ and $(\exists x_i)\mathscr{B}$ are abbreviations for $\neg(\mathscr{B} \Rightarrow \neg\mathscr{C}), \neg\mathscr{B} \Rightarrow \mathscr{C}, (\mathscr{B} \Rightarrow \mathscr{C}) \wedge (\mathscr{C} \Rightarrow \mathscr{B})$ and $\neg(\forall x_i)\neg\mathscr{B}$, respectively.

‡A better term for *closure* would be *universal closure*.

i_1th, ..., i_kth places, then $s^*(t) = (s')^*(t)$. In particular, if t contains no variables at all, $s^*(t) = (s')^*(t)$ for any sequences s and s'.]

Although, by (VIII), a particular wf $\mathscr{B}$ with k free variables is essentially satisfied or not only by k-tuples, rather than by denumerable sequences, it is more convenient for a general treatment of satisfaction to deal with infinite rather than finite sequences. If we were to define satisfaction using finite sequences, conditions 3 and 4 of the definition of satisfaction would become much more complicated.

Let $x_{i_1}, \ldots, x_{i_k}$ be k distinct variables in order of increasing subscripts. Let $\mathscr{B}(x_{i_1}, \ldots, x_{i_k})$ be a wf that has $x_{i_1}, \ldots, x_{i_k}$ as its only free variables. The set of k-tuples $\langle b_1, \cdots, b_k\rangle$ of elements of the domain D such that any sequence with $b_1, \cdots, b_k$ in its i_1th, ..., i_kth places, respectively, satisfies $\mathscr{B}(x_{i_1}, \ldots, x_{i_k})$ is called the *relation* (or *property*†) *of the interpretation defined by* $\mathscr{B}$. Extending our terminology, we shall say that every k-tuple $\langle b_1, \ldots, b_k\rangle$ in this relation *satisfies* $\mathscr{B}(x_{i_1}, \ldots, x_{i_k})$ in the interpretation M; this will be written $\models_{\mathrm{M}} \mathscr{B}[b_1, \ldots, b_k]$. This extended notion of satisfaction corresponds to the original intuitive notion.

Examples

1. If the domain D of M is the set of human beings, $A_1^2(x, y)$ is interpreted as x *is a brother of* y, and $A_2^2(x, y)$ is interpreted as x *is a parent of* y, then the binary relation on D corresponding to the wf $\mathscr{B}(x_1, x_2)$: $(\exists x_3)(A_1^2(x_1, x_3) \wedge A_2^2(x_3, x_2))$ is the relation of unclehood. $\models_{\mathrm{M}} \mathscr{B}[b, c]$ when and only when b is an uncle of c.
2. If the domain is the set of positive integers, A_1^2 is interpreted as $=$, f_1^2 is interpreted as multiplication, and a_1 is interpreted as 1, then the wf $\mathscr{B}(x_1)$:

$$\neg A_1^2(x_1, a_1) \wedge (\forall x_2)((\exists x_3)A_1^2(x_1, f_1^2(x_2, x_3)) \Rightarrow A_1^2(x_2, x_1) \vee A_1^2(x_2, a_1))$$

determines the property of being a prime number. Thus $\models_{\mathrm{M}} \mathscr{B}[k]$ if and only if k is a prime number.

(IX) If $\mathscr{B}$ is a closed wf of a language $\mathscr{L}$, then, for any interpretation M, either $\models_{\mathrm{M}} \mathscr{B}$ or $\models_{\mathrm{M}} \neg\mathscr{B}$ – that is, either $\mathscr{B}$ is true for M or $\mathscr{B}$ is false for M. [*Hint:* Use (VIII).] Of course, $\mathscr{B}$ may be true for some interpretations and false for others. (As an example, consider $A_1^1(a_1)$. If M is an interpretation whose domain is the set of positive integers, A_1^1 is interpreted as the property of being a prime, and the interpretation of a_1 is 2, then $A_1^1(a_1)$ is true. If we change the interpretation by interpreting a_1 as 4, then $A_1^1(a_1)$ becomes false.)

If $\mathscr{B}$ is not closed – that is, if $\mathscr{B}$ contains free variables – $\mathscr{B}$ may be neither true nor false for some interpretation. For example, if $\mathscr{B}$ is $A_1^2(x_1, x_2)$ and we consider an interpretation in which the domain is the set of integers and

†A property is defined when $k = 1$.

$A_1^2(y, z)$ is interpreted as $y < z$, then $\mathscr{B}$ is satisfied by only those sequences $s = (s_1, s_2, \ldots)$ of integers in which $s_1 < s_2$. Hence, $\mathscr{B}$ is neither true nor false for this interpretation. On the other hand, there are wfs that are not closed but that nevertheless are true or false for every interpretation. A simple example is the wf $A_1^1(x_1) \vee \neg A_1^1(x_1)$, which is true for every interpretation.

(X) Assume t is free for x_i in $\mathscr{B}(x_i)$. Then $(\forall x_i)\mathscr{B}(x_i) \Rightarrow \mathscr{B}(t)$ is true for all interpretations.

The proof of (X) is based upon the following lemmas.

LEMMA 1

If t and u are terms, s is a sequence in Σ, t' results from t by replacing all occurrences of x_i by u, and s' results from s by replacing the ith component of s by $s^*(u)$, then $s^*(t') = (s')^*(t)$. [*Hint:* Use induction on the length of t.†]

LEMMA 2

Let t be free for x_i in $\mathscr{B}(x_i)$. Then:

(a) A sequence $s = (s_1, s_2, \ldots)$ satisfies $\mathscr{B}(t)$ if and only if the sequence s', obtained from s by substituting $s^*(t)$ for s_i in the ith place, satisfies $\mathscr{B}(x_i)$. [*Hint:* Use induction on the number of occurrences of connectives and quantifiers in $\mathscr{B}(x_i)$, applying Lemma 1.]
(b) If $(\forall x_i)\mathscr{B}(x_i)$ is satisfied by the sequence s, then $\mathscr{B}(t)$ also is satisfied by s.

(XI) If $\mathscr{B}$ does not contain x_i free, then $(\forall x_i)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow (\forall x_i)\mathscr{C})$ is true for all interpretations.

Proof

Assume (XI) is not correct. Then $(\forall x_i)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow (\forall x_i)\mathscr{C})$ is not true for some interpretation. By condition 3 of the definition of satisfication, there is a sequence s such that s satisfies $(\forall x_i)(\mathscr{B} \Rightarrow \mathscr{C})$ and s does not satisfy $\mathscr{B} \Rightarrow (\forall x_i)\mathscr{C}$. From the latter and condition 3, s satisfies $\mathscr{B}$ and s does not satisfy $(\forall x_i)\mathscr{C}$. Hence, by condition 4, there is a sequence s', differing from s in at most the ith place, such that s' does not satisfy $\mathscr{C}$. Since x_i is free in neither $(\forall x_i)(\mathscr{B} \Rightarrow \mathscr{C})$ nor $\mathscr{B}$, and since s satisfies both of these wfs, it follows by (VIII) that s' also satisfies both $(\forall x_i)(\mathscr{B} \Rightarrow \mathscr{C})$ and $\mathscr{B}$. Since s' satisfies

†The *length* of an expression is the number of occurrences of symbols in the expression.

$(\forall x_i)(\mathscr{B} \Rightarrow \mathscr{C})$, it follows by condition 4 that s' satisfies $\mathscr{B} \Rightarrow \mathscr{C}$. Since s' satisfies $\mathscr{B} \Rightarrow \mathscr{C}$ and $\mathscr{B}$, condition 3 implies that s' satisfies $\mathscr{C}$, which contradicts the fact that s' does not satisfy $\mathscr{C}$. Hence, (XI) is established.

Exercises

2.12 Verify (I)–(X).

2.13 Prove that a closed wf $\mathscr{B}$ is true for M if and only if $\mathscr{B}$ is satisfied by *some* sequence s in Σ. (Remember that Σ is the set of denumerable sequences of elements in the domain of M.)

2.14 Find the properties or relations determined by the following wfs and interpretations.

(a) $[(\exists u)A_1^2(f_1^2(x,u),y)] \wedge [(\exists v)A_1^2(f_1^2(x,v),z)]$, where the domain D is the set of integers, A_1^2 is $=$, and f_1^2 is multiplication.

(b) Here, D is the set of non-negative integers, A_1^2 is $=$, a_1 denotes 0, f_1^2 is addition, and f_2^2 is multiplication.
   (i) $[(\exists z)(\neg A_1^2(z,a_1) \wedge A_1^2(f_1^2(x,z),y))]$
   (ii) $(\exists y)A_1^2(x, f_2^2(y,y))$

(c) $(\exists x_3)A_1^2(f_1^2(x_1,x_3),x_2)$, where D is the set of positive integers, A_1^2 is $=$, and f_1^2 is multiplication,

(d) $A_1^1(x_1) \wedge (\forall x_2)\neg A_1^2(x_1,x_2)$, where D is the set of all living people, $A_1^1(x)$ means x *is a man* and $A_1^2(x,y)$ means x *is married to* y.

(e) (i) $(\exists x_1)(\exists x_2)(A_1^2(x_1,x_3) \wedge A_1^2(x_2,x_4) \wedge A_2^2(x_1,x_2))$
   (ii) $(\exists x_3)(A_1^2(x_1,x_3) \wedge A_1^2(x_3,x_2))$
   where D is the set of all people, $A_1^2(x,y)$ means x *is a parent of* y, and $A_2^2(x,y)$ means x *and* y *are siblings*.

(f) $(\forall x_3)((\exists x_4)(A_1^2(f_1^2(x_4,x_3),x_1) \wedge (\exists x_4)(A_1^2(f_1^2(x_4,x_3),x_2)) \Rightarrow A_1^2(x_3,a_1))$, where D is the set of positive integers, A_1^2 is $=$, f_1^2 is multiplication, and a_1 denotes 1.

(g) $\neg A_1^2(x_2,x_1) \wedge (\exists y)(A_1^2(y,x_1) \wedge A_2^2(x_2,y))$, where D is the set of all people, $A_1^2(u,v)$ means u *is a parent of* v, and $A_2^2(u,v)$ means u *is a wife of* v.

2.15 For each of the following sentences and interpretations, write a translation into ordinary English and determine its truth or falsity.

(a) The domain D is the set of non-negative integers, A_1^2 is $=$, f_1^2 is addition, f_2^2 is multiplication, a_1 denotes 0 and a_2 denotes 1.
   (i) $(\forall x)(\exists y)(A_1^2(x, f_1^2(y,y)) \vee A_1^2(x, f_1^2(f_1^2(y,y),a_2)))$
   (ii) $(\forall x)(\forall y)(A_1^2(f_2^2(x,y),a_1) \Rightarrow A_1^2(x,a_1) \vee A_1^2(y,a_1))$
   (iii) $(\exists y)A_1^2(f_1^2(y,y),a_2)$

(b) Here, D is the set of integers, A_1^2 is $=$, and f_1^2 is addition.
   (i) $(\forall x_1)(\forall x_2)A_1^2(f_1^2(x_1,x_2), f_1^2(x_2,x_1))$
   (ii) $(\forall x_1)(\forall x_2)(\forall x_3)A_1^2(f_1^2(x_1, f_1^2(x_2,x_3)), f_1^2(f_1^2(x_1,x_2),x_3))$
   (iii) $(\forall x_1)(\forall x_2)(\exists x_3)A_1^2(f_1^2(x_1,x_3),x_2)$

(c) The wfs are the same as in part (b), but the domain is the set of positive integers, A_1^2 is $=$, and $f_1^2(x,y)$ is x^y.

(d) The domain is the set of rational numbers, A_1^2 is $=$, A_2^2 is $<$, f_1^2 is multiplication, $f_1^1(x)$ is $x+1$, and a_1 denotes 0.

(i) $(\exists x)A_1^2(f_1^2(x,x), f_1^1(f_1^1(a_1)))$

(ii) $(\forall x)(\forall y)(A_2^2(x,y) \Rightarrow (\exists z)(A_2^2(x,z) \wedge A_2^2(z,y)))$

(iii) $(\forall x)(\neg A_1^2(x,a_1) \Rightarrow (\exists y)A_1^2(f_1^2(x,y), f_1^1(a_1)))$

(e) The domain is the set of non-negative integers, $A_1^2(u,v)$ means $u \leqslant v$, and $A_1^3(u,v,w)$ means $u+v=w$.

(i) $(\forall x)(\forall y)(\forall z)(A_1^3(x,y,z) \Rightarrow A_1^3(y,x,z))$

(ii) $(\forall x)(\forall y)(A_1^3(x,x,y) \Rightarrow A_1^2(x,y))$

(iii) $(\forall x)(\forall y)(A_1^2(x,y) \Rightarrow A_1^3(x,x,y))$

(iv) $(\exists x)(\forall y)A_1^3(x,y,y)$

(v) $(\exists y)(\forall x)A_1^2(x,y)$

(vi) $(\forall x)(\forall y)(A_1^2(x,y) \Leftrightarrow (\exists z)A_1^3(x,z,y))$

(f) The domain is the set of natural numbers, $A_1^2(u,v)$ means $u=v$, $f_1^2(u,v) = u+v$, and $f_2^2(u,v) = u \cdot v$

$(\forall x)(\exists y)(\exists z)A_1^2(x, f_1^2(f_2^2(y,y), f_2^2(z,z)))$

DEFINITIONS

A wf $\mathscr{B}$ is said to be *logically valid* if and only if $\mathscr{B}$ is true for every interpretation.†

$\mathscr{B}$ is said to be *satisfiable* if and only if there is an interpretation for which $\mathscr{B}$ is satisfied by at least one sequence.

It is obvious that $\mathscr{B}$ is logically valid if and only if $\neg\mathscr{B}$ is not satisfiable, and $\mathscr{B}$ is satisfiable if and only if $\neg\mathscr{B}$ is not logically valid.

If $\mathscr{B}$ is a closed wf, then we know that $\mathscr{B}$ is either true or false for any given interpretation; that is, $\mathscr{B}$ is satisfied by all sequences or by none. Therefore, if $\mathscr{B}$ is closed, then $\mathscr{B}$ is satisfiable if and only if $\mathscr{B}$ is true for some interpretation.

A set Γ of wfs is said to be *satisfiable* if and only if there is an interpretation in which there is a sequence that satisfies every wf of Γ.

It is impossible for both a wf $\mathscr{B}$ and its negation $\neg\mathscr{B}$ to be logically valid. For if $\mathscr{B}$ is true for an interpretation, then $\neg\mathscr{B}$ is false for that interpretation.

We say that $\mathscr{B}$ is *contradictory* if and only if $\mathscr{B}$ is false for every interpretation, or, equivalently, if and only if $\neg\mathscr{B}$ is logically valid.

$\mathscr{B}$ is said to *logically imply* $\mathscr{C}$ if and only if, in every interpretation, every sequence that satisfies $\mathscr{B}$ also satisfies $\mathscr{C}$. More generally, $\mathscr{C}$ is said to be a

†The mathematician and philosopher G.W. Leibniz (1646–1716) gave a similar definition: $\mathscr{B}$ is logically valid if and only if $\mathscr{B}$ is true in all 'possible worlds'.

logical consequence of a set Γ of wfs if and only if, in every interpretation, every sequence that satisfies every wf in Γ also satisfies $\mathscr{C}$.

$\mathscr{B}$ and $\mathscr{C}$ are said to be *logically equivalent* if and only if they logically imply each other.

The following assertions are easy consequences of these definitions.

1. $\mathscr{B}$ logically implies $\mathscr{C}$ if and only if $\mathscr{B} \Rightarrow \mathscr{C}$ is logically valid.
2. $\mathscr{B}$ and $\mathscr{C}$ are logically equivalent if and only if $\mathscr{B} \Leftrightarrow \mathscr{C}$ is logically valid.
3. If $\mathscr{B}$ logically implies $\mathscr{C}$ and $\mathscr{B}$ is true in a given interpretation, then so is $\mathscr{C}$.
4. If $\mathscr{C}$ is a logical consequence of a set Γ of wfs and all wfs in Γ are true in a given interpretation, then so is $\mathscr{C}$.

Exercise 2.16

Prove assertions 1–4.

Examples

1. Every instance of a tautology is logically valid (VII).
2. If t is free for x in $\mathscr{B}(x)$, then $(\forall x)\mathscr{B}(x) \Rightarrow \mathscr{B}(t)$ is logically valid (X).
3. If $\mathscr{B}$ does not contain x free, then $(\forall x)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow (\forall x)\mathscr{C})$ is logically valid (XI).
4. $\mathscr{B}$ is logically valid if and only if $(\forall y_1)\dots(\forall y_n)\mathscr{B}$ is logically valid (VI).
5. The wf $(\forall x_2)(\exists x_1)A_1^2(x_1,x_2) \Rightarrow (\exists x_1)(\forall x_2)A_1^2(x_1,x_2)$ is not logically valid. As a counterexample, let the domain D be the set of integers and let $A_1^2(y,z)$ mean $y<z$. Then $(\forall x_2)(\exists x_1)A_1^2(x_1,x_2)$ is true but $(\exists x_1)(\forall x_2)$ $A_1^2(x_1,x_2)$ is false.

Exercises

2.17 Show that the following wfs are not logically valid.

(a) $[(\forall x_1)A_1^1(x_1) \Rightarrow (\forall x_1)A_2^1(x_1)] \Rightarrow [(\forall x_1)(A_1^1(x_1) \Rightarrow A_2^1(x_1))]$

(b) $[(\forall x_1)(A_1^1(x_1) \vee A_2^1(x_1))] \Rightarrow [((\forall x_1)A_1^1(x_1)) \vee (\forall x_1)A_2^1(x_1)]$

2.18 Show that the following wfs are logically valid.†

(a) $\mathscr{B}(t) \Rightarrow (\exists x_i)\mathscr{B}(x_i)$ if t is free for x_i in $\mathscr{B}(x_i)$

(b) $(\forall x_i)\mathscr{B} \Rightarrow (\exists x_i)\mathscr{B}$

(c) $(\forall x_i)(\forall x_j)\mathscr{B} \Rightarrow (\forall x_j)(\forall x_i)\mathscr{B}$

(d) $(\forall x_i)\mathscr{B} \Leftrightarrow \neg(\exists x_i)\neg\mathscr{B}$

(e) $(\forall x_i)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\forall x_i)\mathscr{B} \Rightarrow (\forall x_i)\mathscr{C})$

(f) $((\forall x_i)\mathscr{B}) \wedge (\forall x_i)\mathscr{C} \Leftrightarrow (\forall x_i)(\mathscr{B} \wedge \mathscr{C})$

(g) $((\forall x_i)\mathscr{B}) \vee (\forall x_i)\mathscr{C} \Rightarrow (\forall x_i)(\mathscr{B} \vee \mathscr{C})$

(h) $(\exists x_i)(\exists x_j)\mathscr{B} \Leftrightarrow (\exists x_j)(\exists x_i)\mathscr{B}$

†At this point, one can use intuitive arguments or one can use the rigorous definitions of satisfaction and truth, as in the argument above for (XI). Later on, we shall discover another method for showing logical validity.

(i) $(\exists x_i)(\forall x_j)\mathscr{B} \Rightarrow (\forall x_j)(\exists x_i)\mathscr{B}$

2.19 (a) If $\mathscr{B}$ is a closed wf, show that $\mathscr{B}$ logically implies $\mathscr{C}$ if and only if $\mathscr{C}$ is true for every interpretation for which $\mathscr{B}$ is true.

(b) Although, by (VI), $(\forall x_1)A_1^1(x_1)$ is true whenever $A_1^1(x_1)$ is true, find an interpretation for which $A_1^1(x_1) \Rightarrow (\forall x_1)A_1^1(x_1)$ is not true. (Hence, the hypothesis that $\mathscr{B}$ is a closed wf is essential in (a).)

2.20 Prove that, if the free variables of $\mathscr{B}$ are $y_1, \ldots, y_n$, then $\mathscr{B}$ is satisfiable if and only if $(\exists y_1)\ldots(\exists y_n)\mathscr{B}$ is satisfiable.

2.21 Produce counterexamples to show that the following wfs are not logically valid (that is, in each case, find an interpretation for which the wf is not true).

(a) $$[(\forall x)(\forall y)(\forall z)(A_1^2(x,y) \wedge A_1^2(y,z) \Rightarrow A_1^2(x,z)) \wedge (\forall x)\neg A_1^2(x,x)] \Rightarrow (\exists x)(\forall y)\neg A_1^2(x,y)$$

(b) $(\forall x)(\exists y)A_1^2(x,y) \Rightarrow (\exists y)A_1^2(y,y)$

(c) $(\exists x)(\exists y)A_1^2(x,y) \Rightarrow (\exists y)A_1^2(y,y)$

(d) $[(\exists x)A_1^1(x) \Leftrightarrow (\exists x)A_2^1(x)] \Rightarrow (\forall x)(A_1^1(x) \Leftrightarrow A_2^1(x))$

(e) $(\exists x)(A_1^1(x) \Rightarrow A_2^1(x)) \Rightarrow ((\exists x)A_1^1(x) \Rightarrow (\exists x)A_2^1(x))$

(f) $$[(\forall x)(\forall y)(A_1^2(x,y) \Rightarrow A_1^2(y,x)) \wedge (\forall x)(\forall y)(\forall z)(A_1^2(x,y) \wedge A_1^2(y,z) \Rightarrow A_1^2(x,z))] \Rightarrow (\forall x)A_1^2(x,x)$$

(g)^D $(\exists x)(\forall y)(A_1^2(x,y) \wedge \neg A_1^2(y,x) \Rightarrow [A_1^2(x,x) \Leftrightarrow A_1^2(y,y)])$

(h) $$(\forall x)(\forall y)(\forall z)(A_1^2(x,x) \wedge (A_1^2(x,z) \Rightarrow A_1^2(x,y) \vee A_1^2(y,z))) \Rightarrow (\exists y)(\forall z)A_1^2(y,z)$$

(i) $(\exists x)(\forall y)(\exists z)((A_1^2(y,z) \Rightarrow A_1^2(x,z)) \Rightarrow (A_1^2(x,x) \Rightarrow A_1^2(y,x)))$

2.22 By introducing appropriate notation, write the sentences of each of the following arguments as wfs and determine whether the argument is correct, that is, determine whether the conclusion is logically implied by the conjunction of the premisses

(a) All scientists are neurotic. No vegetarians are neurotic. Therefore, no vegetarians are scientists.

(b) All men are animals. Some animals are carnivorous. Therefore, some men are carnivorous.

(c) Some geniuses are celibate. Some students are not celibate. Therefore, some students are not geniuses.

(d) Any barber in Jonesville shaves exactly those men in Jonesville who do not shave themselves. Hence, there is no barber in Jonesville.

(e) For any numbers x, y, z, if $x > y$ and $y > z$, then $x > z$. $x > x$ is false for all numbers x. Therefore, for any numbers x and y, if $x > y$, then it is not the case that $y > x$.

(f) No student in the statistics class is smarter than every student in the logic class. Hence, some student in the logic class is smarter than every student in the statistics class.

(g) Everyone who is sane can understand mathematics. None of Hegel's sons can understand mathematics. No madmen are fit to vote. Hence, none of Hegel's sons is fit to vote.

(h) For every set x, there is a set y such that the cardinality of y is greater than the cardinality of x. If x is included in y, the cardinality of x is not greater than the cardinality of y. Every set is included in V. Hence, V is not a set.

(i) For all positive integers $x, x \leqslant x$. For all positive integers x, y, z, if $x \leqslant y$ and $y \leqslant z$, then $x \leqslant z$. For all positive integers x and y, $x \leqslant y$ or $y \leqslant x$. Therefore, there is a positive integer y such that, for all positive integers $x, y \leqslant x$.

(j) For any integers x, y, z, if $x > y$ and $y > z$, then $x > z$. $x > x$ is false for all integers x. Therefore, for any integers x and y, if $x > y$, then it is not the case that $y > x$.

2.23 Determine whether the following sets of wfs are compatible – that is, whether their conjunction is satisfiable.

(a) $(\exists x)(\exists y)A_1^2(x,y)$
$(\forall x)(\forall y)(\exists z)(A_1^2(x,z) \wedge A_1^2(z,y))$

(b) $(\forall x)(\exists y)A_1^2(y,x)$
$(\forall x)(\forall y)(A_1^2(x,y) \Rightarrow \neg A_1^2(y,x))$
$(\forall x)(\forall y)(\forall z)(A_1^2(x,y) \wedge A_1^2(y,z) \Rightarrow A_1^2(x,z))$

(c) All unicorns are animals.
No unicorns are animals.

2.24 Determine whether the following wfs are logically valid.

(a) $\neg(\exists y)(\forall x)(A_1^2(x,y) \Leftrightarrow \neg A_1^2(x,x))$
(b) $[(\exists x)A_1^1(x) \Rightarrow (\exists x)A_2^1(x)] \Rightarrow (\exists x)(A_1^1(x) \Rightarrow A_2^1(x))$
(c) $(\exists x)(A_1^1(x) \Rightarrow (\forall y)A_1^1(y))$
(d) $(\forall x)(A_1^1(x) \vee A_2^1(x)) \Rightarrow (((\forall x)A_1^1(x)) \vee (\exists x)A_2^1(x))$
(e) $(\exists x)(\exists y)(A_1^2(x,y) \Rightarrow (\forall z)A_1^2(z,y))$
(f) $(\exists x)(\exists y)(A_1^1(x) \Rightarrow A_2^1(y)) \Rightarrow (\exists x)(A_1^1(x) \Rightarrow A_2^1(x))$
(g) $(\forall x)(A_1^1(x) \Rightarrow A_2^1(x)) \Rightarrow \neg(\forall x)(A_1^1(x) \Rightarrow \neg A_2^1(x))$
(h) $(\exists x)A_1^2(x,x) \Rightarrow (\exists x)(\exists y)A_1^2(x,y)$

2.25 Exhibit a logically valid wf that is not an instance of a tautology. However, show that any logically valid *open* wf (that is, a wf without quantifiers) must be an instance of a tautology.

2.26 (a) Find a satisfiable closed wf that is not true in any interpretation whose domain has only one member.

(b) Find a satisfiable closed wf that is not true in any interpretation whose domain has fewer than three members.

2.3 FIRST-ORDER THEORIES

In the case of the propositional calculus, the method of truth tables provides an effective test as to whether any given statement form is a tautology. However, there does not seem to be any effective process for determining whether a given wf is logically valid, since, in general, one has to check the truth of a wf for interpretations with arbitrarily large finite or infinite domains. In fact, we shall see later that, according to a plausible definition of 'effective', it may actually be proved that there is no effective way to test for logical validity. The axiomatic method, which was a luxury in the study of the propositional calculus, thus appears to be a necessity in the study of wfs involving quantifiers,† and we therefore turn now to the consideration of first-order theories.

Let $\mathscr{L}$ be a first-order language. A *first-order theory* in the language $\mathscr{L}$ will be a formal theory K whose symbols and wfs are the symbols and wfs of $\mathscr{L}$ and whose axioms and rules of inference are specified in the following way.‡

The axioms of K are divided into two classes: the logical axioms and the proper (or non-logical) axioms.

LOGICAL AXIOMS

If $\mathscr{B}, \mathscr{C}$ and $\mathscr{D}$ are wfs of $\mathscr{L}$, then the following are logical axioms of K:

(A1) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{B})$

(A2) $(\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D})) \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}))$

(A3) $(\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow ((\neg\mathscr{C} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{C})$

(A4) $(\forall x_i)\mathscr{B}(x_i) \Rightarrow \mathscr{B}(t)$ if $\mathscr{B}(x_i)$ is a wf of $\mathscr{L}$ and t is a term of $\mathscr{L}$ that is free for x_i in $\mathscr{B}(x_i)$. Note here that t may be identical with x_i so that all wfs $(\forall x_i)\mathscr{B} \Rightarrow \mathscr{B}$ are axioms by virtue of axiom (A4).

(A5) $(\forall x_i)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow (\forall x_i)\mathscr{C})$ if $\mathscr{B}$ contains no free occurrences of x_i.

†There is still another reason for a formal axiomatic approach. Concepts and propositions that involve the notion of interpretation and related ideas such as truth and model are often called *semantical* to distinguish them from *syntactical* concepts, which refer to simple relations among symbols and expressions of precise formal languages. Since semantical notions are set-theoretic in character, and since set theory, because of the paradoxes, is considered a rather shaky foundation for the study of mathematical logic, many logicians consider a syntactical approach, consisting of a study of formal axiomatic theories using only rather weak number-theoretic methods, to be much safer. For further discussions, see the pioneering study on semantics by Tarski (1936), as well as Kleene (1952), Church (1956) and Hilbert and Bernays (1934).

‡The reader might wish to review the definition of *formal theory* in Section 1.4. We shall use the terminology (proof, theorem, consequence, axiomatic, $\vdash \mathscr{B}$ etc.) and notation ($\Gamma \vdash \mathscr{B}, \vdash \mathscr{B}$) introduced there.

PROPER AXIOMS

These cannot be specified, since they vary from theory to theory. A first-order theory in which there are no proper axioms is called a first-order *predicate calculus.*

RULES OF INFERENCE

The rules of inference of any first-order theory are:

1. Modus ponens: $\mathscr{C}$ follows from $\mathscr{B}$ and $\mathscr{B} \Rightarrow \mathscr{C}$.
2. Generalization: $(\forall x_i)\mathscr{B}$ follows from $\mathscr{B}$.

We shall use the abbreviations MP and Gen, respectively, to indicate applications of these rules.

DEFINITION

Let K be a first-order theory in the language $\mathscr{L}$. By a *model* of K we mean an interpretation of $\mathscr{L}$ for which all the axioms of K are true.

By (III) and (VI) on page 61, if the rules of modus ponens and generalization are applied to wfs that are true for a given interpretation, then the results of these applications are also true. Hence *every theorem of* K *is true in every model of* K.

As we shall see, the logical axioms are so designed that the logical consequences (in the sense defined on pages 65–6) of the closures of the axioms of K are precisely the theorems of K. In particular, if K is a first-order predicate calculus, it turns out that the theorems of K are just those wfs of K that are logically valid.

Some explanation is needed for the restrictions in axiom schemas (A4) and (A5). In the case of (A4), if t were not free for x_i in $\mathscr{B}(x_i)$, the following unpleasant result would arise: let $\mathscr{B}(x_1)$ be $\neg(\forall x_2)A_1^2(x_1, x_2)$ and let t be x_2. Notice that t is not free for x_1 in $\mathscr{B}(x_1)$. Consider the following pseudo-instance of axiom (A4):

$$(\nabla) \quad (\forall x_1)(\neg(\forall x_2)A_1^2(x_1, x_2)) \Rightarrow \neg(\forall x_2)A_1^2(x_2, x_2)$$

Now take as interpretation any domain with at least two members and let A_1^2 stand for the identity relation. Then the antecedent of (∇) is true and the consequent false. Thus, (∇) is false for this interpretation.

In the case of axiom (A5), relaxation of the restriction that x_i not be free in $\mathscr{B}$ would lead to the following disaster. Let $\mathscr{B}$ and $\mathscr{C}$ both be $A_1^1(x_1)$. Thus, x_1 is free in $\mathscr{B}$. Consider the following pseudo-instance of axiom (A5):

$$(\nabla\nabla) \quad (\forall x_1)(A_1^1(x_1) \Rightarrow A_1^1(x_1)) \Rightarrow (A_1^1(x_1) \Rightarrow (\forall x_1)A_1^1(x_1))$$

The antecedent of $(\nabla\nabla)$ is logically valid. Now take as domain the set of integers and let $A_1^1(x)$ mean that x *is even*. Then $(\forall x_1)A_1^1(x_1)$ is false. So, any sequence $s = (s_1, s_2, \ldots)$ for which s_1 is even does not satisfy the consequent of $(\nabla\nabla)$.† Hence, $(\nabla\nabla)$ is not true for this interpretation.

Examples of first-order theories

1. *Partial order*. Let the language $\mathscr{L}$ have a single predicate letter A_2^2 and no function letters and individual constants. We shall write $x_i < x_j$ instead of $A_2^2(x_i, x_j)$. The theory K has two proper axioms.
   (a) $(\forall x_1)(\neg\, x_1 < x_1)$ (irreflexivity)
   (b) $(\forall x_1)(\forall x_2)(\forall x_3)(x_1 < x_2 \wedge x_2 < x_3 \Rightarrow x_1 < x_3)$ (transitivity)

   A model of the theory is called a *partially ordered structure*.

2. *Group theory*. Let the language $\mathscr{L}$ have one predicate letter A_1^2, one function letter f_1^2, and one individual constant a_1. To conform with ordinary notation, we shall write $t = s$ instead of $A_1^2(t, s), t + s$ instead of $f_1^2(t, s)$, and 0 instead of a_1. The proper axioms of K are:

   (a) $(\forall x_1)(\forall x_2)(\forall x_3)(x_1 + (x_2 + x_3) = (x_1 + x_2) + x_3)$ (associativity)
   (b) $(\forall x_1)(0 + x_1 = x_1)$ (identity)
   (c) $(\forall x_1)(\exists x_2)(x_2 + x_1 = 0)$ (inverse)
   (d) $(\forall x_1)(x_1 = x_1)$ (reflexivity of $=$)
   (e) $(\forall x_1)(\forall x_2)(x_1 = x_2 \Rightarrow x_2 = x_1)$ (symmetry of $=$)
   (f) $(\forall x_1)(\forall x_2)(\forall x_3)(x_1 = x_2 \wedge x_2 = x_3 \Rightarrow x_1 = x_3)$ (transitivity of $=$)
   (g) $(\forall x_1)(\forall x_2)(\forall x_3)(x_2 = x_3 \Rightarrow$
   $x_1 + x_2 = x_1 + x_3 \wedge x_2 + x_1 = x_3 + x_1)$ (substitutivity of $=$)

A model for this theory, in which the interpretation of $=$ is the identity relation, is called a *group*. A group is said to be *abelian* if, in addition, the wf $(\forall x_1)(\forall x_2)(x_1 + x_2 = x_2 + x_1)$ is true.

The theories of partial order and of groups are both axiomatic. In general, any theory with a finite number of proper axioms is axiomatic, since it is obvious that one can effectively decide whether any given wf is a logical axiom.

2.4 PROPERTIES OF FIRST-ORDER THEORIES

All the results in this section refer to an arbitrary first-order theory K. Instead of writing $\vdash_K \mathscr{B}$, we shall sometimes simply write $\vdash \mathscr{B}$. Moreover, we shall refer to first-order theories simply as *theories*, unless something is said to the contrary.

†Such a sequence would satisfy $A_1^1(x_1)$, since s_1 is even, but would not satisfy $(\forall x_1)A_1^1(x_1)$, since no sequence satisfies $(\forall x_1)A_1^1(x_1)$.

PROPOSITION 2.1

Every wf $\mathscr{B}$ of K that is an instance of a tautology is a theorem of K, and it may be proved using only axioms (A1)–(A3) and MP.

Proof

$\mathscr{B}$ arises from a tautology $\mathscr{T}$ by substitution. By Proposition 1.14, there is a proof of $\mathscr{T}$ in L. In such a proof, make the same substitution of wfs of K for statement letters as were used in obtaining $\mathscr{B}$ from $\mathscr{T}$, and, for all statement letters in the proof that do not occur in $\mathscr{T}$, substitute an arbitrary wf of K. Then the resulting sequence of wfs is a proof of $\mathscr{B}$, and this proof uses only axiom schemes (A1)–(A3) and MP.

The application of Proposition 2.1 in a proof will be indicated by writing 'Tautology'.

PROPOSITION 2.2

Every theorem of a first-order predicate calculus is logically valid.

Proof

Axioms (A1)–(A3) are logically valid by property (VII) of the notion of truth (see page 61), and axioms (A4) and (A5) are logically valid by properties (X) and (XI). By properties (III) and (VI), the rules of inference MP and Gen preserve logical validity. Hence, every theorem of a predicate calculus is logically valid.

Example
The wf $(\forall x_2)(\exists x_1)A_1^2(x_1,x_2) \Rightarrow (\exists x_1)(\forall x_2)A_1^2(x_1,x_2)$ is not a theorem of any first-order predicate calculus, since it is not logically valid (by Example 5, p. 66).

DEFINITION

A theory K is *consistent* if no wf $\mathscr{B}$ and its negation $\neg\mathscr{B}$ are both provable in K. A theory is *inconsistent* if it is not consistent.

COROLLARY 2.3

Any first-order predicate calculus is consistent.

Proof

If a wf $\mathscr{B}$ and its negation $\neg\mathscr{B}$ were both theorems of a first-order predicate calculus, then, by Proposition 2.2, both $\mathscr{B}$ and $\neg\mathscr{B}$ would be logically valid, which is impossible.

Notice that, in an inconsistent theory K, every wf $\mathscr{C}$ of K is provable in K. In fact, assume that $\mathscr{B}$ and $\neg\mathscr{B}$ are both provable in K. Since the wf $\mathscr{B} \Rightarrow (\neg\mathscr{B} \Rightarrow \mathscr{C})$ is an instance of a tautology, that wf is, by Proposition 2.1, provable in K. Then two applications of MP would yield $\vdash \mathscr{C}$.

It follows from this remark that, if some wf of a theory K is not a theorem of K, then K is consistent.

The deduction theorem (Proposition 1.9) for the propositional calculus cannot be carried over without modification to first-order theories. For example, for any wf $\mathscr{B}$, $\mathscr{B} \vdash_K (\forall x_i)\mathscr{B}$, but it is not always the case that $\vdash_K \mathscr{B} \Rightarrow (\forall x_i)\mathscr{B}$. Consider a domain containing at least two elements c and d. Let K be a predicate calculus and let $\mathscr{B}$ be $A_1^1(x_1)$. Interpret A_1^1 as a property that holds only for c. Then $A_1^1(x_1)$ is satisfied by any sequence $s = (s_1, s_2, \ldots)$ in which $s_1 = c$, but $(\forall x_1)A_1^1(x_1)$ is satisfied by no sequence at all. Hence, $A_1^1(x_1) \Rightarrow (\forall x_1)A_1^1(x_1)$ is not true in this interpretation, and so it is not logically valid. Therefore, by Proposition 2.2, $A_1^1(x_1) \Rightarrow (\forall x_1)A_1^1(x_1)$ is not a theorem of K.

A modified, but still useful, form of the deduction theorem may be derived, however. Let $\mathscr{B}$ be a wf in a set Γ of wfs and assume that we are given a deduction $\mathscr{D}_1, \ldots, \mathscr{D}_n$ from Γ, together with justification for each step in the deduction. We shall say that $\mathscr{D}_i$ *depends upon* $\mathscr{B}$ in this proof if and only if:

(1) $\mathscr{D}_i$ is $\mathscr{B}$ and the justification for $\mathscr{D}_i$ is that it belongs to Γ, or
(2) $\mathscr{D}_i$ is justified as a direct consequence by MP or Gen of some preceding wfs of the sequence, where at least one of these preceding wfs depends upon $\mathscr{B}$.

Example

$\mathscr{B}, (\forall x_1)\mathscr{B} \Rightarrow \mathscr{C} \vdash (\forall x_1)\mathscr{C}$

$(\mathscr{D}_1)$	$\mathscr{B}$	Hyp
$(\mathscr{D}_2)$	$(\forall x_1)\mathscr{B}$	$(\mathscr{D}_1)$, Gen
$(\mathscr{D}_3)$	$(\forall x_1)\mathscr{B} \Rightarrow \mathscr{C}$	Hyp
$(\mathscr{D}_4)$	$\mathscr{C}$	$(\mathscr{D}_2), (\mathscr{D}_3)$, MP
$(\mathscr{D}_5)$	$(\forall x_1)\mathscr{C}$	$(\mathscr{D}_4)$, Gen

Here, $(\mathscr{D}_1)$ depends upon $\mathscr{B}$, $(\mathscr{D}_2)$ depends upon $\mathscr{B}$, $(\mathscr{D}_3)$ depends upon $(\forall x_1)\mathscr{B} \Rightarrow \mathscr{C}$, $(\mathscr{D}_4)$ depends upon $\mathscr{B}$ and $(\forall x_1)\mathscr{B} \Rightarrow \mathscr{C}$, and $(\mathscr{D}_5)$ depends upon $\mathscr{B}$ and $(\forall x_1)\mathscr{B} \Rightarrow \mathscr{C}$.

PROPOSITION 2.4

If $\mathscr{C}$ does not depend upon $\mathscr{B}$ in a deduction showing that $\Gamma, \mathscr{B} \vdash \mathscr{C}$, then $\Gamma \vdash \mathscr{C}$.

Proof

Let $\mathscr{D}_1 \cdots, \mathscr{D}_n$ be a deduction of $\mathscr{C}$ from Γ and $\mathscr{B}$, in which $\mathscr{C}$ does not depend upon $\mathscr{B}$. (In this deduction, $\mathscr{D}_n$ is $\mathscr{C}$.) As an inductive hypothesis, let us assume that the proposition is true for all deductions of length less than n. If $\mathscr{C}$ belongs to Γ or is an axiom, then $\Gamma \vdash \mathscr{C}$. If $\mathscr{C}$ is a direct consequence of one or two preceding wfs by Gen or MP, then, since $\mathscr{C}$ does not depend upon $\mathscr{B}$, neither do these preceding wfs. By the inductive hypothesis, these preceding wfs are deducible from Γ alone. Consequently, so is $\mathscr{C}$.

PROPOSITION 2.5 (DEDUCTION THEOREM)

Assume that, in some deduction showing that Γ, $\mathscr{B} \vdash \mathscr{C}$, no application of Gen to a wf that depends upon $\mathscr{B}$ has as its quantified variable a free variable of $\mathscr{B}$. The $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}$.

Proof

Let $\mathscr{D}_1, \ldots, \mathscr{D}_n$ be a deduction of $\mathscr{C}$ from Γ and $\mathscr{B}$, satisfying the assumption of our proposition. (In this deduction, $\mathscr{D}_n$ is $\mathscr{C}$.) Let us show by induction that $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_i$ for each $i \leqslant n$. If $\mathscr{D}_i$ is an axiom or belongs to Γ, then $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_i$, since $\mathscr{D}_i \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}_i)$ is an axiom. If $\mathscr{D}_i$ is $\mathscr{B}$, then $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_i$, since, by Proposition 2.1, $\vdash \mathscr{B} \Rightarrow \mathscr{B}$. If there exist j and k less than i such that $\mathscr{D}_k$ is $\mathscr{D}_j \Rightarrow \mathscr{D}_i$, then, by inductive hypothesis, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_j$ and $\Gamma \vdash \mathscr{B} \Rightarrow (\mathscr{D}_j \Rightarrow \mathscr{D}_i)$. Now, by axiom (A2), $\vdash (\mathscr{B} \Rightarrow (\mathscr{D}_j \Rightarrow \mathscr{D}_i)) \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{D}_j) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}_i))$. Hence, by MP twice, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_i$. Finally, suppose that there is some $j < i$ such that $\mathscr{D}_i$ is $(\forall x_k)\mathscr{D}_j$. By the inductive hypothesis, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_j$, and, by the hypothesis of the theorem, either $\mathscr{D}_j$ does not depend upon $\mathscr{B}$ or x_k is not a free variable of $\mathscr{B}$. If $\mathscr{D}_j$ does not depend upon $\mathscr{B}$, then, by Proposition 2.4, $\Gamma \vdash \mathscr{D}_j$ and, consequently, by Gen, $\Gamma \vdash (\forall x_k)\mathscr{D}_j$. Thus, $\Gamma \vdash \mathscr{D}_i$. Now, by axiom (A1), $\vdash \mathscr{D}_i \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}_i)$. So, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_i$ by MP. If, on the other hand, x_k is not a free variable of $\mathscr{B}$, then, by axiom (A5), $\vdash (\forall x_k)(\mathscr{B} \Rightarrow \mathscr{D}_j) \Rightarrow (\mathscr{B} \Rightarrow (\forall x_k)\mathscr{D}_j)$. Since $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_j$, we have, by Gen, $\Gamma \vdash (\forall x_k)(\mathscr{B} \Rightarrow \mathscr{D}_j)$, and so, by MP, $\Gamma \vdash \mathscr{B} \Rightarrow (\forall x_k)\mathscr{D}_j$; that is, $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{D}_i$. This completes the induction, and our proposition is just the special case $i = n$.

The hypothesis of Proposition 2.5 is rather cumbersome; the following weaker corollaries often prove to be more useful.

COROLLARY 2.6

If a deduction showing that $\Gamma, \mathscr{B} \vdash \mathscr{C}$ involves no application of Gen of which the quantified variables is free in $\mathscr{B}$, then $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}$.

COROLLARY 2.7

If $\mathscr{B}$ is a closed wf and $\Gamma, \mathscr{B} \vdash \mathscr{C}$, then $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}$.

EXTENSION OF PROPOSITIONS 2.4–2.7

In Propositions 2.4–2.7, the following additional conclusion can be drawn from the proofs. The new proof of $\Gamma \vdash \mathscr{B} \Rightarrow \mathscr{C}$ (in Proposition 2.4, of $\Gamma \vdash \mathscr{C}$) involves an application of Gen to a wf depending upon a wf $\mathscr{E}$ of Γ only if there is an application of Gen in the given proof of $\Gamma, \mathscr{B} \vdash \mathscr{C}$ that involves the same quantified variable and is applied to a wf that depends upon $\mathscr{E}$. (In the proof of Proposition 2.5, one should observe that $\mathscr{D}_j$ depends upon a premiss $\mathscr{E}$ of Γ in the original proof if and only if $\mathscr{B} \Rightarrow \mathscr{D}_j$ depends upon $\mathscr{E}$ in the new proof.)

This supplementary conclusion will be useful when we wish to apply the deduction theorem several times in a row to a given deduction – for example, to obtain $\Gamma \vdash \mathscr{D} \Rightarrow (\mathscr{B} \Rightarrow \mathscr{C})$ from $\Gamma, \mathscr{D}, \mathscr{B} \vdash \mathscr{C}$; from now on, it is to be considered an integral part of the statements of Propositions 2.4–2.7.

Example
$\vdash (\forall x_1)(\forall x_2)\mathscr{B} \Rightarrow (\forall x_2)(\forall x_1)\mathscr{B}$

Proof

1. $(\forall x_1)(\forall x_2)\mathscr{B}$	Hyp
2. $(\forall x_1)(\forall x_2)\mathscr{B} \Rightarrow (\forall x_2)\mathscr{B}$	(A4)
3. $(\forall x_2)\mathscr{B}$	1, 2, MP
4. $(\forall x_2)\mathscr{B} \Rightarrow \mathscr{B}$	(A4)
5. $\mathscr{B}$	3, 4, MP
6. $(\forall x_1)\mathscr{B}$	5, Gen
7. $(\forall x_2)(\forall x_1)\mathscr{B}$	6, Gen

Thus, by 1–7, we have $(\forall x_1)(\forall x_2)\mathscr{B} \vdash (\forall x_2)(\forall x_1)\mathscr{B}$, where, in the deduction, no application of Gen has as a quantified variable a free variable of $(\forall x_1)(\forall x_2)\mathscr{B}$. Hence, by Corollary 2.6, $\vdash (\forall x_1)(\forall x_2)\mathscr{B} \Rightarrow (\forall x_2)(\forall x_1)\mathscr{B}$.

Exercises

2.27 Derive the following theorems.
(a) $\vdash (\forall x)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\forall x)\mathscr{B} \Rightarrow (\forall x)\mathscr{C})$

(b) $\vdash (\forall x)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\exists x)\mathscr{B} \Rightarrow (\exists x)\mathscr{C})$
(c) $\vdash (\forall x)(\mathscr{B} \wedge \mathscr{C}) \Leftrightarrow ((\forall x)\mathscr{B}) \wedge (\forall x)\mathscr{C}$
(d) $\vdash (\forall y_1) \ldots (\forall y_n)\mathscr{B} \Rightarrow \mathscr{B}$
(e) $\vdash \neg(\forall x)\mathscr{B} \Rightarrow (\exists x)\neg\mathscr{B}$

2.28$^{\text{D}}$ Let K be a first-order theory and let $K^{\#}$ be an axiomatic theory having the following axioms:

(a) $(\forall y_1) \ldots (\forall y_n)\mathscr{B}$, where $\mathscr{B}$ is any axiom of K and $y_1, \ldots, y_n (n \geqslant 0)$ are any variables (none at all when $n = 0$);
(b) $(\forall y_1) \ldots (\forall y_n)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow [(\forall y_1) \ldots (\forall y_n)\mathscr{B} \Rightarrow (\forall y_1) \ldots (\forall y_n)\mathscr{C}]$ where $\mathscr{B}$ and $\mathscr{C}$ are any wfs and $y_1 \ldots, y_n$ are any variables.

Moreover, $K^{\#}$ has modus ponens as its only rule of inference. Show that $K^{\#}$ has the same theorems as K. Thus, at the expense of adding more axioms, the generalization rule can be dispensed with.

2.29 Carry out the proof of the Extension of Propositions 2.4–2.7 above.

2.5 ADDITIONAL METATHEOREMS AND DERIVED RULES

For the sake of smoothness in working with particular theories later, we shall introduce various techniques for constructing proofs. In this section it is assumed that we are dealing with an arbitray theory K.

Often one wants to obtain $\mathscr{B}(t)$ from $(\forall x)\mathscr{B}(x)$, where t is a term free for x in $\mathscr{B}(x)$. This is allowed by the following *derived rule*.

PARTICULARIZATION RULE A4

If t is free for x in $\mathscr{B}(x)$, then $(\forall x)\mathscr{B}(x) \vdash \mathscr{B}(t)$.†

Proof

From $(\forall x)\mathscr{B}(x)$ and the instance $(\forall x)\mathscr{B}(x) \Rightarrow \mathscr{B}(t)$ of axiom (A4), we obtain $\mathscr{B}(t)$ by modus ponens.

Since x is free for x in $\mathscr{B}(x)$, a special case of rule A4 is: $(\forall x)\mathscr{B} \vdash \mathscr{B}$.

There is another very useful derived rule, which is essentially the contrapositive of rule A4.

†From a strict point of view, $(\forall x)\mathscr{B}(x) \vdash \mathscr{B}(t)$ states a fact about derivability. Rule A4 should be taken to mean that, if $(\forall x)\mathscr{B}(x)$ occurs as a step in a proof, we may write $\mathscr{B}(t)$ as a later step (if t is free for x in $\mathscr{B}(x)$). As in this case, we shall often state a derived rule in the form of the corresponding derivability result that justifies the rule.

EXISTENTIAL RULE E4

Let t be a term that is free for x in a wf $\mathscr{B}(x,t)$, and let $\mathscr{B}(t,t)$ arise from $\mathscr{B}(x,t)$ by replacing all free occurrences of x by t. ($\mathscr{B}(x,t)$ may or may not contain occurrences of t.) Then, $\mathscr{B}(t,t) \vdash (\exists x)\mathscr{B}(x,t)$.

Proof

It suffices to show that $\vdash \mathscr{B}(t,t) \Rightarrow (\exists x)\mathscr{B}(x,t)$. But, by axiom (A4), $\vdash (\forall x)\neg\mathscr{B}(x,t) \Rightarrow \neg\mathscr{B}(t,t)$. Hence, by the tautology $(A \Rightarrow \neg B) \Rightarrow (B \Rightarrow \neg A)$ and MP, $\vdash \mathscr{B}(t,t) \Rightarrow \neg(\forall x)\neg\mathscr{B}(x,t)$, which, in abbreviated form, is $\vdash \mathscr{B}(t,t) \Rightarrow (\exists x)\mathscr{B}(x,t)$.

A special case of rule E4 is $\mathscr{B}(t) \vdash (\exists x)\mathscr{B}(x)$, whenever t is free for x in $\mathscr{B}(x)$. In particular, when t is x itself, $\mathscr{B}(x) \vdash (\exists x)\mathscr{B}(x)$.

Example

$\vdash (\forall x)\mathscr{B} \Rightarrow (\exists x)\mathscr{B}$

1. $(\forall x)\mathscr{B}$	Hyp
2. $\mathscr{B}$	1, rule A4
3. $(\exists x)\mathscr{B}$	2, rule E4
4. $(\forall x)\mathscr{B} \vdash (\exists x)\mathscr{B}$	1–3
5. $\vdash (\forall x)\mathscr{B} \Rightarrow (\exists x)\mathscr{B}$	1–4, Corollary 2.6

The following derived rules are extremely useful.

Negation elimination: $\neg\neg\mathscr{B} \vdash \mathscr{B}$
Negation introduction: $\mathscr{B} \vdash \neg\neg\mathscr{B}$
Conjunction elimination: $\mathscr{B} \wedge \mathscr{C} \vdash \mathscr{B}$
$\mathscr{B} \wedge \mathscr{C} \vdash \mathscr{C}$
$\neg(\mathscr{B} \wedge \mathscr{C}) \vdash \neg\mathscr{B} \vee \neg\mathscr{C}$
Conjunction introduction: $\mathscr{B}, \mathscr{C} \vdash \mathscr{B} \wedge \mathscr{C}$
Disjunction elimination: $\mathscr{B} \vee \mathscr{C}, \neg\mathscr{B} \vdash \mathscr{C}$
$\mathscr{B} \vee \mathscr{C}, \neg\mathscr{C} \vdash \mathscr{B}$
$\neg(\mathscr{B} \vee \mathscr{C}) \vdash \neg\mathscr{B} \wedge \neg\mathscr{C})$
$\mathscr{B} \Rightarrow \mathscr{D}, \mathscr{C} \Rightarrow \mathscr{D}, \mathscr{B} \vee \mathscr{C} \vdash \mathscr{D}$
Disjunction introduction: $\mathscr{B} \vdash \mathscr{B} \vee \mathscr{C}$
$\mathscr{C} \vdash \mathscr{B} \vee \mathscr{C}$
Conditional elimination: $\mathscr{B} \Rightarrow \mathscr{C}, \neg\mathscr{C} \vdash \neg\mathscr{B}$
$\mathscr{B} \Rightarrow \neg\mathscr{C}, \mathscr{C} \vdash \neg\mathscr{B}$
$\neg\mathscr{B} \Rightarrow \mathscr{C}, \neg\mathscr{C} \vdash \mathscr{B}$
$\neg\mathscr{B} \Rightarrow \neg\mathscr{C}, \mathscr{C} \vdash \mathscr{B}$
$\neg(\mathscr{B} \Rightarrow \mathscr{C}) \vdash \mathscr{B}$
$\neg(\mathscr{B} \Rightarrow \mathscr{C}) \vdash \neg\mathscr{C}$
Conditional introduction: $\mathscr{B}, \neg\mathscr{C} \vdash \neg(\mathscr{B} \Rightarrow \mathscr{C})$
Conditional contrapositive: $\mathscr{B} \Rightarrow \mathscr{C} \vdash \neg\mathscr{C} \Rightarrow \neg\mathscr{B}$

$$\neg\mathscr{C} \Rightarrow \neg\mathscr{B} \vdash \mathscr{B} \Rightarrow \mathscr{C}$$

Biconditional elimination: $\mathscr{B} \Leftrightarrow \mathscr{C}, \mathscr{B} \vdash \mathscr{C}$ $\quad \mathscr{B} \Leftrightarrow \mathscr{C}, \neg\mathscr{B} \vdash \neg\mathscr{C}$

$\mathscr{B} \Leftrightarrow \mathscr{C}, \mathscr{C} \vdash \mathscr{B}$ $\quad \mathscr{B} \Leftrightarrow \mathscr{C}, \neg\mathscr{C} \vdash \neg\mathscr{B}$

$\mathscr{B} \Leftrightarrow \mathscr{C} \vdash \mathscr{B} \Rightarrow \mathscr{C}$ $\quad \mathscr{B} \Leftrightarrow \mathscr{C} \vdash \mathscr{C} \Rightarrow \mathscr{B}$

Biconditional introduction: $\mathscr{B} \Rightarrow \mathscr{C}, \mathscr{C} \Rightarrow \mathscr{B} \vdash \mathscr{B} \Leftrightarrow \mathscr{C}$

Biconditional negation: $\mathscr{B} \Leftrightarrow \mathscr{C} \vdash \neg\mathscr{B} \Leftrightarrow \neg\mathscr{C}$

$\neg\mathscr{B} \Leftrightarrow \neg\mathscr{C} \vdash \mathscr{B} \Leftrightarrow \mathscr{C}$

Proof by contradiction: If a proof of $\Gamma, \neg\mathscr{B} \vdash \mathscr{C} \wedge \neg\mathscr{C}$ involves no application of Gen using a variable free in $\mathscr{B}$, then $\Gamma \vdash \mathscr{B}$. (Similarly, one obtains $\Gamma \vdash \neg\mathscr{B}$ from $\Gamma, \mathscr{B} \vdash \mathscr{C} \wedge \neg\mathscr{C}$.)

Exercises

2.30 Justify the derived rules listed above.

2.31 Prove the following.

(a) $\vdash (\forall x)(\forall y)A_1^2(x,y) \Rightarrow (\forall x)A_1^2(x,x)$
(b) $\vdash [(\forall x)\mathscr{B}] \vee [(\forall x)\mathscr{C}] \Rightarrow (\forall x)(\mathscr{B} \vee \mathscr{C})$
(c) $\vdash \neg(\exists x)\mathscr{B} \Rightarrow (\forall x)\neg\mathscr{B}$
(d) $\vdash (\forall x)\mathscr{B} \Rightarrow (\forall x)(\mathscr{B} \vee \mathscr{C})$
(e) $\vdash (\forall x)(\forall y)(A_1^2(x,y) \Rightarrow \neg A_1^2(y,x)) \Rightarrow (\forall x)\neg A_1^2(x,x)$
(f) $\vdash [(\exists x)\mathscr{B} \Rightarrow (\forall x)\mathscr{C}] \Rightarrow (\forall x)(\mathscr{B} \Rightarrow \mathscr{C})$
(g) $\vdash (\forall x)(\mathscr{B} \vee \mathscr{C}) \Rightarrow [(\forall x)\mathscr{B}] \vee (\exists x)\mathscr{C}$
(h) $\vdash (\forall x)(A_1^2(x,x) \Rightarrow (\exists y)A_1^2(x,y))$
(i) $\vdash (\forall x)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow [(\forall x)\neg\mathscr{C} \Rightarrow (\forall x)\neg\mathscr{B}]$
(j) $\vdash (\exists y)[A_1^1(y) \Rightarrow (\forall y)A_1^1(y)]$
(k) $\vdash (\forall x)\mathscr{B} \Rightarrow (\forall x)(\mathscr{B} \vee \mathscr{C})$
(l) $\vdash [(\forall x)(\forall y)(\mathscr{B}(x,y) \Rightarrow \mathscr{B}(y,x)) \wedge (\forall x)(\forall y)(\forall z)(\mathscr{B}(x,y) \wedge \mathscr{B}(y,z) \Rightarrow \mathscr{B}(x,z))] \Rightarrow (\forall x)(\forall y)(\mathscr{B}(x,y) \Rightarrow \mathscr{B}(x,x))$.

2.32 Assume that $\mathscr{B}$ and $\mathscr{C}$ are wfs and that x is not free in $\mathscr{B}$. Prove the following.

(a) $\vdash \mathscr{B} \Rightarrow (\forall x)\mathscr{B}$
(b) $\vdash \mathscr{B} \Rightarrow (\exists x)\mathscr{B}$
(c) $\vdash (\mathscr{B} \Rightarrow (\forall x)\mathscr{C}) \Leftrightarrow (\forall x)(\mathscr{B} \Rightarrow \mathscr{C})$
(d) $\vdash ((\exists x)\mathscr{C} \Rightarrow \mathscr{B}) \Leftrightarrow (\forall x)(\mathscr{C} \Rightarrow \mathscr{B})$

We need a derived rule that will allow us to replace a part $\mathscr{C}$ of a wf $\mathscr{B}$ by a wf that is provably equivalent to $\mathscr{C}$. For this purpose, we first must prove the following auxiliary result.

LEMMA 2.8

For any wfs $\mathscr{B}$ and $\mathscr{C}$, $\vdash (\forall x)(\mathscr{B} \Leftrightarrow \mathscr{C}) \Rightarrow ((\forall x)\mathscr{B} \Leftrightarrow (\forall x)\mathscr{C})$.

Proof

1. $(\forall x)(\mathscr{B} \Leftrightarrow \mathscr{C})$	Hyp
2. $(\forall x)\mathscr{B}$	Hyp
3. $\mathscr{B} \Leftrightarrow \mathscr{C}$	1, rule A4
4. $\mathscr{B}$	2, rule A4
5. $\mathscr{C}$	3, 4, biconditional elimination
6. $(\forall x)\mathscr{C}$	5, Gen
7. $(\forall x)(\mathscr{B} \Leftrightarrow \mathscr{C}), (\forall x)\mathscr{B} \vdash (\forall x)\mathscr{C}$	1–6
8. $(\forall x)(\mathscr{B} \Leftrightarrow \mathscr{C}) \vdash (\forall x)\mathscr{B} \Rightarrow (\forall x)\mathscr{C}$	1–7, Corollary 2.6
9. $(\forall x)(\mathscr{B} \Leftrightarrow \mathscr{C}) \vdash (\forall x)\mathscr{C} \Rightarrow (\forall x)\mathscr{B}$	Proof like that of 8
10. $(\forall x)(\mathscr{B} \Leftrightarrow \mathscr{C}) \vdash (\forall x)\mathscr{B} \Leftrightarrow (\forall x)\mathscr{C}$	8, 9, Bioconditional introduction
11. $\vdash (\forall x)(\mathscr{B} \Leftrightarrow \mathscr{C}) \Rightarrow ((\forall x)\mathscr{B} \Leftrightarrow (\forall x)\mathscr{C})$	1–10, Corollary 2.6

PROPOSITION 2.9

If $\mathscr{C}$ is a subformula of $\mathscr{B}$, $\mathscr{B}'$ is the result of replacing zero or more occurrences of $\mathscr{C}$ in $\mathscr{B}$ by a wf $\mathscr{D}$, and every free variable of $\mathscr{C}$ or $\mathscr{D}$ that is also a bound variable of $\mathscr{B}$ occurs in the list $y_1, \ldots, y_k$, then:

(a) $\vdash [(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{B} \Leftrightarrow \mathscr{B}')$ (Equivalence theorem)
(b) If $\vdash \mathscr{C} \Leftrightarrow \mathscr{D}$, then $\vdash \mathscr{B} \Leftrightarrow \mathscr{B}'$ (Replacement theorem)
(c) If $\vdash \mathscr{C} \Leftrightarrow \mathscr{D}$ and $\vdash \mathscr{B}$, then $\vdash \mathscr{B}'$

Example
(a) $\vdash (\forall x)(A_1^1(x) \Leftrightarrow (A_2^1(x)) \Rightarrow [(\exists x)A_1^1(x) \Leftrightarrow (\exists x)\ A_2^1(x)]$

Proof

(a) We use induction on the number of connectives and quantifiers in $\mathscr{B}$. Note that, if zero occurrences are replaced, $\mathscr{B}'$ is $\mathscr{B}$ and the wf to be proved is an instance of the tautology $A \Rightarrow (B \Leftrightarrow B)$. Note also that, if $\mathscr{C}$ is identical with $\mathscr{B}$ and this occurrence of $\mathscr{C}$ is replaced by $\mathscr{D}$, the wf to be proved, $[(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{B} \Leftrightarrow \mathscr{B}')$, is derivable by Exercise 2.27(d). Thus, we may assume that $\mathscr{C}$ is a proper part of $\mathscr{B}$ and that at least one occurrence of $\mathscr{C}$ is replaced. Our inductive hypothesis is that the result holds for all wfs with fewer connectives and quantifiers than $\mathscr{B}$.

Case 1. $\mathscr{B}$ is an atomic wf. Then $\mathscr{C}$ cannot be a proper part of $\mathscr{B}$.

Case 2. $\mathscr{B}$ is $\neg\mathscr{E}$. Let $\mathscr{B}'$ be $\neg\mathscr{E}'$. By inductive hypothesis, $\vdash [(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{E} \Leftrightarrow \mathscr{E}')$. Hence, by a suitable instance of the tautology $(C \Rightarrow (A \Leftrightarrow B)) \Rightarrow (C \Rightarrow (\neg A \Leftrightarrow \neg B))$ and MP, we obtain $\vdash [(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{B} \Leftrightarrow \mathscr{B}')$.

Case 3. $\mathscr{B}$ is $\mathscr{E} \Rightarrow \mathscr{F}$. Let $\mathscr{B}'$ be $\mathscr{E}' \Rightarrow \mathscr{F}'$. By inductive hypothesis, $\vdash [(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{E} \Leftrightarrow \mathscr{E}')$ and $\vdash [(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{F} \Leftrightarrow \mathscr{F}')$. Using a suitable instance of the tautology

$$(A \Rightarrow (B \Leftrightarrow C)) \wedge (A \Rightarrow (D \Leftrightarrow E)) \Rightarrow (A \Rightarrow [(B \Rightarrow D) \Leftrightarrow (C \Rightarrow E)])$$

we obtain $\vdash [(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{B} \Leftrightarrow \mathscr{B}')$.

Case 4. $\mathscr{B}$ is $(\forall x)\mathscr{E}$. Let $\mathscr{B}'$ be $(\forall x)\mathscr{E}'$. By inductive hypothesis, $\vdash [(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{E} \Leftrightarrow \mathscr{E}')$. Now, x does not occur free in $(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})$ because, if it did, it would be free in $\mathscr{C}$ or $\mathscr{D}$ and, since it is bound in $\mathscr{B}$, it would be one of $y_1, \ldots, y_k$ and it would not be free in $(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})$. Hence, using axiom (A5), we obtain $\vdash (\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D}) \Rightarrow (\forall x)(\mathscr{E} \Leftrightarrow \mathscr{E}')$. However, by Lemma 2.8, $\vdash (\forall x)(\mathscr{E} \Leftrightarrow \mathscr{E}') \Rightarrow ((\forall x)\mathscr{E} \Leftrightarrow (\forall x)\mathscr{E}')$. Then, by a suitable tautology and MP, $\vdash [(\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})] \Rightarrow (\mathscr{B} \Leftrightarrow \mathscr{B}')$.

(b) From $\vdash \mathscr{C} \Leftrightarrow \mathscr{D}$, by several applications of Gen, we obtain $\vdash (\forall y_1) \ldots (\forall y_k)(\mathscr{C} \Leftrightarrow \mathscr{D})$. Then, by (a) and MP, $\vdash \mathscr{B} \Leftrightarrow \mathscr{B}'$.

(c) Use part (b) and biconditional elimination.

Exercises

2.33 Prove the following:

(a) $\vdash (\exists x)\neg\mathscr{B} \Leftrightarrow \neg(\forall x)\mathscr{B}$

(b) $\vdash (\forall x)\mathscr{B} \Leftrightarrow \neg(\exists x)\neg\mathscr{B}$

(c) $\vdash (\exists x)(\mathscr{B} \Rightarrow \neg(\mathscr{C} \vee \mathscr{D})) \Rightarrow (\exists x)(\mathscr{B} \Rightarrow \neg\mathscr{C} \wedge \neg\mathscr{D})$

(d) $\vdash (\forall x)(\exists y)(\mathscr{B} \Rightarrow \mathscr{C}) \Leftrightarrow (\forall x)(\exists y)(\neg\mathscr{B} \vee \mathscr{C})$

(e) $\vdash (\forall x)(\mathscr{B} \Rightarrow \neg\mathscr{C}) \Leftrightarrow \neg(\exists x)(\mathscr{B} \wedge \mathscr{C})$

2.34 Show by a counterexample that we cannot omit the quantifiers $(\forall y_1) \ldots (\forall y_k)$ in Proposition 2.9(a).

2.35 If $\mathscr{C}$ is obtained from $\mathscr{B}$ by erasing all quantifiers $(\forall x)$ or $(\exists x)$ whose scope does not contain x free, prove that $\vdash \mathscr{B} \Leftrightarrow \mathscr{C}$.

2.36 For each wf $\mathscr{B}$ below, find a wf $\mathscr{C}$ such that $\vdash \mathscr{C} \Leftrightarrow \neg\mathscr{B}$ and negation signs in $\mathscr{C}$ apply only to atomic wfs.

(a) $(\forall x)(\forall y)(\exists z)A_1^3(x, y, z)$

(b) $(\forall \varepsilon)(\varepsilon > 0 \Rightarrow (\exists \delta)(\delta > 0 \wedge (\forall x)(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon))$

(c) $(\forall \varepsilon)(\varepsilon > 0 \Rightarrow (\exists n)(\forall m)(m > n \Rightarrow |a_m - b| < \varepsilon))$

2.37 Let $\mathscr{B}$ be a wf that does not contain $\Rightarrow$ and $\Leftrightarrow$. Exchange universal and existential quantifiers and exchange $\wedge$ and $\vee$. The result $\mathscr{B}^*$ is called the *dual* of $\mathscr{B}$.

(a) In any predicate calculus, prove the following.

(i) $\vdash \mathscr{B}$ if and only if $\vdash \neg\mathscr{B}^*$

(ii) $\vdash \mathscr{B} \Rightarrow \mathscr{C}$ if and only if $\vdash \mathscr{C}^* \Rightarrow \mathscr{B}^*$.

(iii) $\vdash \mathscr{B} \Leftrightarrow \mathscr{C}$ if and only if $\vdash \mathscr{B}^* \Leftrightarrow \mathscr{C}^*$.

(iv) $\vdash (\exists x)(\mathscr{B} \vee \mathscr{C}) \Leftrightarrow [((\exists x)\mathscr{B}) \vee (\exists x)\mathscr{C}]$. [*Hint*: Use Exercise 2.27(c).]

(b) Show that the duality results of part (a), (i)–(iii), do not hold for arbitrary theories.

2.6 RULE C

It is very common in mathematics to reason in the following way. Assume that we have proved a wf of the form $(\exists x)\mathscr{B}(x)$. Then we say, let b be an object such that $\mathscr{B}(b)$. We continue the proof, finally arriving at a formula that does not involve the arbitrarily chosen element b.

For example, let us say that we wish to show that $(\exists x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x)), (\forall x)\mathscr{B}(x) \vdash (\exists x)\mathscr{C}(x)$.

1. $(\exists x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x))$	Hyp
2. $(\forall x)\mathscr{B}(x)$	Hyp
3. $\mathscr{B}(b) \Rightarrow \mathscr{C}(b)$ for some b	1
4. $\mathscr{B}(b)$	2, rule A4
5. $\mathscr{C}(b)$	3, 4, MP
6. $(\exists x)\mathscr{C}(x)$	5, rule E4

Such a proof seems to be perfectly legitimate on an intuitive basis. In fact, we can achieve the same result without making an arbitrary choice of an element b as in step 3. This can be done as follows:

1. $(\forall x)\mathscr{B}(x)$	Hyp
2. $(\forall x)\neg\mathscr{C}(x)$	Hyp
3. $\mathscr{B}(x)$	1, rule A4
4. $\neg\mathscr{C}(x)$	2, rule A4
5. $\neg(\mathscr{B}(x) \Rightarrow \mathscr{C}(x))$	3, 4, conditional introduction
6. $(\forall x)\neg(\mathscr{B}(x) \Rightarrow \mathscr{C}(x))$	5, Gen
7. $(\forall x)\mathscr{B}(x), (\forall x)\neg\mathscr{C}(x) \vdash$ $(\forall x)\neg(\mathscr{B}(x) \Rightarrow \mathscr{C}(x))$	1–6
8. $(\forall x)\mathscr{B}(x) \vdash (\forall x)\neg\mathscr{C}(x)$ $\Rightarrow (\forall x)\neg(\mathscr{B}(x) \Rightarrow \mathscr{C}(x))$	1–7, corollary 2.6
9. $(\forall x)\mathscr{B}(x) \vdash \neg(\forall x)\neg(\mathscr{B}(x)$ $\Rightarrow \mathscr{C}(x)) \Rightarrow \neg(\forall x)\neg\mathscr{C}(x)$	8, contrapositive
10. $(\forall x)\mathscr{B}(x) \vdash (\exists x)(\mathscr{B}(x) \Rightarrow$ $\mathscr{C}(x)) \Rightarrow (\exists x)\mathscr{C}(x)$	Abbreviation of 9
11. $(\exists x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x)),$ $(\forall x)\mathscr{B}(x) \vdash (\exists x)\mathscr{C}(x)$	10, MP

In general, any wf that can be proved using a finite number of arbitrary choices can also be proved without such acts of choice. We shall call the rule that permits us to go from $(\exists x)\mathscr{B}(x)$ to $\mathscr{B}(b)$, *rule C* ('C' for 'choice'). More precisely, a rule C deduction in a first-order theory K is defined in the following manner: $\Gamma \vdash_C \mathscr{B}$ if and only if there is a sequence of wfs $\mathscr{D}_1, \cdots, \mathscr{D}_n$ such that $\mathscr{D}_n$ is $\mathscr{B}$ and the following four conditions hold:

1. For each $i < n$, either
   (a) $\mathscr{D}_i$ is an axiom of K, or
   (b) $\mathscr{D}_i$ is in Γ, or
   (c) $\mathscr{D}_i$ follows by MP or Gen from preceding wfs in the sequence, or
   (d) there is a preceding wf $(\exists x)\mathscr{C}(x)$ such that $\mathscr{D}_i$ is $\mathscr{C}(d)$, where d is a new individual constant (rule C).
2. As axioms in condition 1(a), we also can use all logical axioms that involve the new individual constants already introduced in the sequence by applications of rule C.
3. No application of Gen is made using a variable that is free in some $(\exists x)\mathscr{C}(x)$ to which rule C has been previously applied.
4. $\mathscr{B}$ contains none of the new individual constants introduced in the sequence in any application of rule C.

A word should be said about the reason for including condition 3. If an application of rule C to a wf $(\exists x)\mathscr{C}(x)$ yields $\mathscr{C}(d)$, then the object referred to by d may depend on the values of the free variables in $(\exists x)\mathscr{C}(x)$. So that one object may not satisfy $\mathscr{C}(x)$ for *all* values of the free variables in $(\exists x)\mathscr{C}(x)$. For example, without clause 3, we could proceed as follows:

1. $(\forall x)(\exists y)A_1^2(x,y)$	Hyp
2. $(\exists y)A_1^2(x,y)$	1, rule A4
3. $A_1^2(x,d)$	2, rule C
4. $(\forall x)A_1^2(x,d)$	3, Gen
5. $(\exists y)(\forall x)A_1^2(x,y)$	4, rule E4

However, there is an interpretation for which $(\forall x)(\exists y)A_1^2(x,y)$ is true but $(\exists y)(\forall x)A_1^2(x,y)$ is false. Take the domain to be the set of integers and let $A_1^2(x,y)$ mean that $x < y$.

PROPOSITION 2.10

If $\Gamma \vdash_C \mathscr{B}$, then $\Gamma \vdash \mathscr{B}$. Moreover, from the following proof it is easy to verify that, if there is an application of Gen in the new proof of $\mathscr{B}$ from Γ using a certain variable and applied to a wf depending upon a certain wf of Γ, then there was such an application of Gen in the original proof.[†]

[†]The first formulation of a version of rule C similar to that given here seems to be due to Rosser (1953).

Proof

Let $(\exists y_1)\mathscr{C}_1(y_1), \dots, (\exists y_k)\mathscr{C}_k(y_k)$ be the wfs in order of occurrence to which rule C is applied in the proof of $\Gamma \vdash_C \mathscr{B}$, and let $d_1, \dots, d_k$ be the corresponding new individual constants. Then Γ, $\mathscr{C}_1(d_1), \dots, \mathscr{C}_k(d_k) \vdash \mathscr{B}$. Now, by condition 3 of the definition above, Corollary 2.6 is applicable, yielding $\Gamma, \mathscr{C}_1(d_1), \dots, \mathscr{C}_{k-1}(d_{k-1}) \vdash \mathscr{C}_k(d_k) \Rightarrow \mathscr{B}$. We replace d_k everywhere by a variable z that does not occur in the proof.
Then

$$\Gamma, \mathscr{C}_1(d_1), \dots, \mathscr{C}_{k-1}(d_{k-1}) \vdash \mathscr{C}_k(z) \Rightarrow \mathscr{B}$$

and, by Gen,

$$\Gamma, \mathscr{C}_1(d_1), \dots, \mathscr{C}_{k-1}(d_{k-1}) \vdash (\forall z)(\mathscr{C}_k(z) \Rightarrow \mathscr{B})$$

Hence, by Exercise 2.32(d),

$$\Gamma, \mathscr{C}_1(d_1), \dots, \mathscr{C}_{k-1}(d_{k-1}) \vdash (\exists y_k)\mathscr{C}_k(y_k) \Rightarrow \mathscr{B}$$

But,

$$\Gamma, \mathscr{C}_1(d_1), \dots, \mathscr{C}_{k-1}(d_{k-1}) \vdash (\exists y_k)\mathscr{C}_k(y_k)$$

Hence, by MP,

$$\Gamma, \mathscr{C}_1(d_1), \dots, \mathscr{C}_{k-1}(d_{k-1}) \vdash \mathscr{B}$$

Repeating this argument, we can eliminate $\mathscr{C}_{k-1}(d_{k-1}), \dots, \mathscr{C}_1(d_1)$ one after the other, finally obtaining $\Gamma \vdash \mathscr{B}$.

Example

$\vdash (\forall x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x)) \Rightarrow ((\exists x)\mathscr{B}(x) \Rightarrow (\exists x)\mathscr{C}(x))$

1. $(\forall x)(\mathscr{B}x) \Rightarrow \mathscr{C}(x))$	Hyp
2. $(\exists x)\mathscr{B}(x)$	Hyp
3. $\mathscr{B}(d)$	2, rule C
4. $\mathscr{B}(d) \Rightarrow \mathscr{C}(d)$	1, rule A4
5. $\mathscr{C}(d)$	3, 4, MP
6. $(\exists x)\mathscr{C}(x)$	5, rule E4
7. $(\forall x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x)), (\exists x)\mathscr{B}(x) \vdash_C (\exists x)\mathscr{C}(x)$	$1-6$
8. $(\forall x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x)), (\exists x)\mathscr{B}(x) \vdash (\exists x)\mathscr{C}(x)$	7, Proposition 2.10
9. $(\forall x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x)) \vdash (\exists x)\mathscr{B}(x) \Rightarrow (\exists x)\mathscr{C}(x)$	$1-8$, corollary 2.6
10. $\vdash (\forall x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x)) \Rightarrow ((\exists x)\mathscr{B}(x) \Rightarrow (\exists x)\mathscr{C}(x))$	$1-9$, corollary 2.6

Exercises

Use rule C and Proposition 2.10 to prove Exercises 2.38–2.45.

2.38 $\vdash (\exists x)\mathscr{B}(x) \Rightarrow \mathscr{C}(x)) \Rightarrow ((\forall x)\mathscr{B}(x) \Rightarrow (\exists x)\mathscr{C}(x))$

2.39 $\vdash \neg(\exists y)(\forall x)(A_1^2(x, y) \Leftrightarrow \neg A_1^2(x, x))$

2.40 $\vdash [(\forall x)(A_1^1(x) \Rightarrow A_2^1(x) \vee A_3^1(x)) \wedge \neg(\forall x)(A_1^1(x) \Rightarrow A_2^1(x))]$
$\Rightarrow (\exists x)(A_1^1(x) \wedge A_3^1(x))$

2.41 $\vdash [(\exists x)\mathscr{B}(x)] \wedge [(\forall x)\mathscr{C}(x)] \Rightarrow (\exists x)(\mathscr{B}(x) \wedge \mathscr{C}(x))$

2.42 $\vdash (\exists x)\mathscr{C}(x) \Rightarrow (\exists x)(\mathscr{B}(x) \vee \mathscr{C}(x))$

2.43 $\vdash (\exists x)(\exists y)\mathscr{B}(x,y) \Leftrightarrow (\exists y)(\exists x)\mathscr{B}(x,y)$

2.44 $\vdash (\exists x)(\forall y)\mathscr{B}(x,y) \Rightarrow (\forall y)(\exists x)\mathscr{B}(x,y)$

2.45 $\vdash (\exists x)(\mathscr{B}(x) \wedge \mathscr{C}(x)) \Rightarrow ((\exists x)\mathscr{B}(x)) \wedge (\exists x)\mathscr{C}(x)$

2.46 What is wrong with the following alleged derivations?

(a)

1. $(\exists x)\mathscr{B}(x)$	Hyp
2. $\mathscr{B}(d)$	1, rule C
3. $(\exists x)\mathscr{C}(x)$	Hyp
4. $\mathscr{C}(d)$	3, rule C
5. $\mathscr{B}(d) \wedge \mathscr{C}(d)$	2, 4, conjunction introduction
6. $(\exists x)(\mathscr{B}(x) \wedge \mathscr{C}(x))$	5, rule E4
7. $(\exists x)\mathscr{B}(x), (\exists x)\mathscr{C}(x) \vdash (\exists x)(\mathscr{B}(x) \wedge \mathscr{C}(x))$	1–6, Proposition 2.10

(b)

1. $(\exists x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x))$	Hyp
2. $(\exists x)\mathscr{B}(x)$	Hyp
3. $\mathscr{B}(d) \Rightarrow \mathscr{C}(d)$	1, rule C
4. $\mathscr{B}(d)$	2, rule C
5. $\mathscr{C}(d)$	3, 4, MP
6. $(\exists x)\mathscr{C}(x)$	5, rule E4
7. $(\exists x)(\mathscr{B}(x) \Rightarrow \mathscr{C}(x)), (\exists x)\mathscr{B}(x) \vdash (\exists x)\mathscr{C}(x)$	1–6, Proposition 2.10

2.7 COMPLETENESS THEOREMS

We intend to show that the theorems of a first-order predicate calculus K are precisely the same as the logically valid wfs of K. Half of this result was proved in Proposition 2.2. The other half will follow from a much more general proposition established later. First we must prove a few preliminary lemmas.

If x_i and x_j are distinct, then $\mathscr{B}(x_i)$ and $\mathscr{B}(x_j)$ are said to be *similar* if and only if x_j is free for x_i in $\mathscr{B}(x_i)$ and $\mathscr{B}(x_i)$ has no free occurrences of x_j. It is assumed here that $\mathscr{B}(x_j)$ arises from $\mathscr{B}(x_i)$ by substituting x_j for all free occurrences of x_i. It is easy to see that, if $\mathscr{B}(x_i)$ and $\mathscr{B}(x_j)$ are similar, then x_i is free for x_j in $\mathscr{B}(x_j)$ and $\mathscr{B}(x_j)$ has no free occurrences of x_i. Thus, if $\mathscr{B}(x_i)$ and $\mathscr{B}(x_j)$ are similar, then $\mathscr{B}(x_j)$ and $\mathscr{B}(x_i)$ are similar. Intuitively, $\mathscr{B}(x_i)$ and $\mathscr{B}(x_j)$ are similar if and only if $\mathscr{B}(x_i)$ and $\mathscr{B}(x_j)$ are the same except that $\mathscr{B}(x_i)$ has free occurrences of x_i in exactly those places where $\mathscr{B}(x_j)$ has free occurrences of x_j.

Example

$(\forall x_3)[A_1^2(x_1,x_3) \vee A_1^1(x_1)]$ and $(\forall x_3)[A_1^2(x_2,x_3) \vee A_1^1(x_2)]$ are similar.

LEMMA 2.11

If $\mathscr{B}(x_i)$ and $\mathscr{B}(x_j)$ are similar, then $\vdash (\forall x_i)\mathscr{B}(x_i) \Leftrightarrow (\forall x_j)\mathscr{B}(x_j)$.

Proof

$\vdash (\forall x_i)\mathscr{B}(x_i) \Rightarrow \mathscr{B}(x_j)$ by axiom (A4). Then, by Gen, $\vdash (\forall x_j)((\forall x_i)\mathscr{B}(x_i) \Rightarrow \mathscr{B}(x_j))$, and so, by axiom (A5) and MP, $\vdash (\forall x_i)\mathscr{B}(x_i) \Rightarrow (\forall x_j)\mathscr{B}(x_j)$. Similarly, $\vdash (\forall x_j)\mathscr{B}(x_j) \Rightarrow (\forall x_i)\mathscr{B}(x_i)$. Hence, by biconditional introduction, $\vdash (\forall x_i)\mathscr{B}(x_i) \Leftrightarrow (\forall x_j)\mathscr{B}(x_j)$.

Exercises

2.47 If $\mathscr{B}(x_i)$ and $\mathscr{B}(x_j)$ are similar, prove that $\vdash (\exists x_i)\mathscr{B}(x_i) \Leftrightarrow (\exists x_j)\mathscr{B}(x_j)$.
2.48 *Change of bound variables.* If $\mathscr{B}(x)$ is similar to $\mathscr{B}(y)$, $(\forall x)\mathscr{B}(x)$ is a subformula of $\mathscr{C}$, and $\mathscr{C}'$ is the result of replacing one or more occurrences of $(\forall x)\mathscr{B}(x)$ in $\mathscr{C}$ by $(\forall y)\mathscr{B}(y)$, prove that $\vdash \mathscr{C} \Leftrightarrow \mathscr{C}'$.

LEMMA 2.12

If a closed wf $\neg\mathscr{B}$ of a theory K is not provable in K, and if K′ is the theory obtained from K by adding $\mathscr{B}$ as a new axiom, then K′ is consistent.

Proof

Assume K′ inconsistent. Then, for some wf $\mathscr{C}$, $\vdash_{K'} \mathscr{C}$ and $\vdash_{K'} \neg\mathscr{C}$. Now, $\vdash_{K'} \mathscr{C} \Rightarrow (\neg\mathscr{C} \Rightarrow \neg\mathscr{B})$ by Proposition 2.1. So, by two applications of MP, $\vdash_{K'} \neg\mathscr{B}$. Now, any use of $\mathscr{B}$ as an axiom in a proof in K′ can be regarded as a hypothesis in a proof in K. Hence, $\mathscr{B} \vdash_K \neg\mathscr{B}$. Since $\mathscr{B}$ is closed, we have $\vdash_K \mathscr{B} \Rightarrow \neg\mathscr{B}$ by Corollary 2.7. However, by Proposition 2.1, $\vdash_K (\mathscr{B} \Rightarrow \neg\mathscr{B}) \Rightarrow \neg\mathscr{B}$. Therefore, by MP, $\vdash_K \neg\mathscr{B}$, contradicting our hypothesis.

Exercise

2.49 If a closed wf $\mathscr{B}$ of a theory K is not provable in K, and if K′ is the theory obtained from K by adding $\neg\mathscr{B}$ as a new axiom, then K′ is consistent.

LEMMA 2.13

The set of expressions of a language $\mathscr{L}$ is denumerable. Hence, the same is true of the set of terms, the set of wfs and the set of closed wfs.

Proof

First assign a distinct positive integer $g(u)$ to each symbol u as follows: $g(()=3$, $g())=5$, $g(,)=7$, $g(\neg)=9$, $g(\Rightarrow)=11$, $g(\forall)=13$, $g(x_k)=13+8k$, $g(a_k)=7+8k$, $g(f_k^n)=1+8(2^n3^k)$, and $g(A_k^n)=3+8(2^n3^k)$. Then, to an expression $u_0u_1\ldots u_r$ associate the number $2^{g(u_0)}3^{g(u_1)}\ldots p_r^{g(u_r)}$, where p_j is the jth prime number, starting with $p_0=2$. (Example: the number of $A_1^1(x_2)$ is $2^{51}3^35^{29}7^5$.) We can enumerate all expressions in the order of their associated numbers; so, the set of expressions is denumerable.

If we can effectively tell whether any given symbol is a symbol of $\mathscr{L}$, then this enumeration can be effectively carried out, and, in addition, we can effectively decide whether any given number is the number of an expression of $\mathscr{L}$. The same holds true for terms, wfs and closed wfs. If a theory K in the language $\mathscr{L}$ is axiomatic, that is, if we can effectively decide whether any given wf is an axiom of K, then we can effectively enumerate the theorems of K in the following manner. Starting with a list consisting of the first axiom of K in the enumeration just specified, add to the list all the direct consequences of this axiom by MP and by Gen used only once and with x_1 as quantified variable. Add the second axiom to this new list and write all new direct consequences by MP and Gen of the wfs in this augmented list, with Gen used only once and with x_1 and x_2 as quantified variables. If at the kth step we add the kth axiom and apply MP and Gen to the wfs in the new list (with Gen applied only once for each of the variables $x_1,\ldots,x_k$), we eventually obtain in this manner all theorems of K. However, in contradistinction to the case of expressions, terms, wfs and closed wfs, it turns out that there are axiomatic theories K for which we cannot tell in advance whether any given wf of K will eventually appear in the list of theorems.

DEFINITIONS

(i) A theory K is said to be *complete* if, for every closed wf $\mathscr{B}$ of K, either $\vdash_K \mathscr{B}$ or $\vdash_K \neg\mathscr{B}$.

(ii) A theory K′ is said to be an *extension* of a theory K if every theorem of K is a theorem of K′. (We also say in such a case that K is a *subtheory* of K′.)

PROPOSITION 2.14 (LINDENBAUM'S LEMMA)

If K is a consistent theory, then there is a consistent, complete extension of K.

Proof

Let $\mathscr{B}_1, \mathscr{B}_2, \ldots$ be an enumeration of all closed wfs of the language of K, by Lemma 2.13. Define a sequence $J_0, J_1, J_2, \ldots$ of theories in the following way. J_0 is K. Assume J_n is defined, with $n \geqslant 0$. If it is not the case that $\vdash_{J_n} \neg\mathscr{B}_{n+1}$, then let J_{n+1} be obtained from J_n by adding $\mathscr{B}_{n+1}$ as an additional axiom. On the other hand, if $\vdash_{J_n} \neg\mathscr{B}_{n+1}$, let $J_{n+1} = J_n$. Let J be the theory obtained by taking as axioms all the axioms of all the J_is. Clearly, J_{i+1} is an extension of J_i, and J is an extension of all the J_is including $J_0 = K$. To show that J is consistent, it suffices to prove that every J_i is consistent because a proof of a contradiction in J, involving as it does only a finite number of axioms, is also a proof of a contradition in some J_i. We prove the consistency of the J_is, by induction. By hypothesis, $J_0 = K$ is consistent. Assume that J_i is consistent. If $J_{i+1} = J_i$, then J_{i+1} is consistent. If $J_i \neq J_{i+1}$, and therefore, by the definition of $J_{i+1}, \neg\mathscr{B}_{i+1}$ is not provable in J_i, then, by Lemma 2.12, J_{i+1} is also consistent. So, we have proved that all the J_is are consistent and, therefore, that J is consistent. To prove the completeness of J, let $\mathscr{C}$ be any closed wf of K. Then $\mathscr{C} = \mathscr{B}_{j+1}$ for some $j \geqslant 0$. Now, either $\vdash_{J_j} \neg\mathscr{B}_{j+1}$ or $\vdash_{J_{j+1}} \mathscr{B}_{j+1}$, since, if it is not the case that $\vdash_{J_j} \neg\mathscr{B}_{j+1}$, then $\mathscr{B}_{j+1}$ is added as an axiom in J_{j+1}. Therefore, either $\vdash_J \neg\mathscr{B}_{j+1}$ or $\vdash_J \mathscr{B}_{j+1}$. Thus, J is complete.

Note that even if one can effectively determine whether any wf is an axiom of K, it may not be possible to do the same with (or even to enumerate effectively) the axioms of J; that is, J may not be axiomatic even if K is. This is due to the possibility of not being able to determine, at each step, whether or not $\neg\mathscr{B}_{n+1}$ is provable in J_n.

Exercises

2.49 Show that a theory K is complete if and only if, for any closed wfs $\mathscr{B}$ and $\mathscr{C}$ of K, if $\vdash_K \mathscr{B} \vee \mathscr{C}$, then $\vdash_K \mathscr{B}$ or $\vdash_K \mathscr{C}$.
2.50^D Prove that every consistent decidable theory has a consistent, decidable, complete extension.

DEFINITIONS

1. A *closed term* is a term without variables.
2. A theory K is a *scapegoat theory* if, for any wf $\mathscr{B}(x)$ that has x as its only free variable, there is a closed term t such that

$$\vdash_K (\exists x) \neg\mathscr{B}(x) \Rightarrow \neg\mathscr{B}(t)$$

LEMMA 2.15

Every consistent theory K has a consistent extension K′ such that K′ is a scapegoat theory and K′ contains denumerably many closed terms.

Proof

Add to the symbols of K a denumerable set $\{b_1, b_2, \ldots\}$ of new individual constants. Call this new theory K_0. Its axioms are those of K plus those logical axioms that involve the symbols of K and the new constants. K_0 is consistent. For, if not, there is a proof in K_0 of a wf $\mathscr{B} \wedge \neg\mathscr{B}$. Replace each b_i appearing in this proof by a variable that does not appear in the proof. This transforms axioms into axioms and preserves the correctness of the applications of the rules of inference. The final wf in the proof is still a contradiction, but now the proof does not involve any of the b_is and therefore is a proof in K. This contradicts the consistency of K. Hence, K_0 is consistent.

By Lemma 2.13, let $F_1(x_{i_1}), F_2(x_{i_2}), \ldots, F_k(x_{i_k}), \ldots$ be an enumeration of all wfs of K_0 that have one free variable. Choose a sequence $b_{j_1}, b_{j_2}, \ldots$ of some of the new individual constants such that each b_{j_k} is not contained in any of the wfs $F_1(x_{i_1}), \ldots, F_k(x_{i_k})$ and such that b_{j_k} is different from each of $b_{j_1}, \ldots, b_{j_{k-1}}$. Consider the wf

$$(S_k) \quad (\exists x_{i_k})\neg F_k(x_{i_k}) \Rightarrow \neg F_k(b_{j_k})$$

Let K_n be the theory obtained by adding $(S_1), \ldots, (S_n)$ to the axioms of K_0, and let K_∞ be the theory obtained by adding all the (S_i)s as axioms to K_0. Any proof in K_∞ contains only a finite number of the (S_i)s and, therefore, will also be a proof in some K_n. Hence, if all the K_ns are consistent, so is K_∞. To demonstrate that all the K_ns are consistent, proceed by induction. We know that K_0 is consistent. Assume that K_{n-1} is consistent but that K_n is inconsistent $(n \geqslant 1)$. Then, as we know, any wf is provable in K_n (by the tautology $\neg A \Rightarrow (A \Rightarrow B)$, Proposition 2.1 and MP). In particular, $\vdash_{K_n} \neg(S_n)$. Hence, $(S_n) \vdash_{K_{n-1}} \neg(S_n)$. Since (S_n) is closed, we have, by Corollary 2.7, $\vdash_{K_{n-1}} (S_n) \Rightarrow \neg(S_n)$. But, by the tautology $(A \Rightarrow \neg A) \Rightarrow \neg A$, Proposition 2.1 and MP, we then have $\vdash_{K_{n-1}} \neg(S_n)$; that is, $\vdash_{K_{n-1}} \neg[(\exists x_{i_n})\neg F_n(x_{i_n}) \Rightarrow \neg F_n(b_{j_n})]$. Now, by conditional elimination, we obtain $\vdash_{K_{n-1}} (\exists x_{i_n})\neg F_n(x_{i_n})$ and $\vdash_{K_{n-1}} \neg\neg F_n(b_{j_n})$, and then, by negation elimination, $\vdash K_1 F_n(b_{j_n})$. From the latter and the fact that b_{j_n} does not occur in $(S_0), \ldots, (S_{n-1})$, we conclude $\vdash_{K_{n-1}} F_n(x_r)$, where x_r is a variable that does not occur in the proof of $F_n(b_{j_n})$. (Simply replace in the proof all occurrences of b_{j_n} by x_r.) By Gen, $\vdash_{K_{n-1}} (\forall x_r)\, F_n(x_r)$, and then, by Lemma 2.11 and biconditional elimination, $\vdash_{K_{n-1}} (\forall x_{i_n})F_n(x_{i_n})$. (We use the fact that $F_n(x_r)$ and $F_n(x_{i_n})$ are similar.) But we already have $\vdash_{K_{n-1}} (\exists x_{i_n})\neg F_n(x_{i_n})$, which is an abbreviation of $\vdash_{K_{n-1}} \neg(\forall x_{i_n})\neg\neg F_n(x_{i_n})$, whence, by the replacement theorem, $\vdash_{K_{n-1}} \neg(\forall x_{i_n})F_n(x_{i_n})$, contradicting the hypothesis that K_{n-1} is consistent. Hence, K_n must also be consistent. Thus K_∞ is consistent, it is an extension of K, and it is clearly a scapegoat theory.

LEMMA 2.16

Let J be a consistent, complete scapegoat theory. Then J has a model M whose domain is the set D of closed terms of J.

Proof

For any individual constant a_i of J, let $(a_i)^{\mathrm{M}} = a_i$. For any function letter f_k^n of J and for any closed terms $t_1, \ldots t_n$ of J, let $(f_k^n)^{\mathrm{M}}(t_1, \ldots, t_n) = f_k^n(t_1, \ldots, t_n)$. (Notice that $f_k^n(t_1, \ldots, t_n)$ is a closed term. Hence, $(f_k^n)^{\mathrm{M}}$ is an n-ary operation on D.) For any predicate letter A_k^n of J, let $(A_k^n)^{\mathrm{M}}$ consist of all n-tuples $\langle t_1, \ldots, t_n \rangle$ of closed terms $t_1, \ldots, t_n$ of J such that $\vdash_{\mathrm{J}} A_k^n(t_1, \ldots, t_n)$. It now suffices to show that, for any closed wf $\mathscr{C}$ of J:

$$(\Box) \qquad \models_{\mathrm{M}} \mathscr{C} \quad \text{if and only if} \quad \vdash_{\mathrm{J}} \mathscr{C}$$

(If this is established and $\mathscr{B}$ is any axiom of J, let $\mathscr{C}$ be the closure of $\mathscr{B}$. By Gen, $\vdash_{\mathrm{J}} \mathscr{C}$. By ($\Box$), $\models_{\mathrm{M}} \mathscr{C}$. By (VI) on page 61, $\models_{\mathrm{M}} \mathscr{B}$. Hence, M would be a model of J.) The proof of ($\Box$) is by induction on the number r of connectives and quantifiers in $\mathscr{C}$. Assume that ($\Box$) holds for all closed wfs with fewer than r connectives and quantifiers.

Case 1. $\mathscr{C}$ is a closed atomic wf $A_k^n(t_1, \ldots, t_n)$. Then ($\Box$) is a direct consequence of the definition of $(A_k^n)^{\mathrm{M}}$.

Case 2. $\mathscr{C}$ is $\neg\mathscr{D}$. If $\mathscr{C}$ is true for M, then $\mathscr{D}$ is false for M and so, by inductive hypothesis, not-$\vdash_{\mathrm{J}} \mathscr{D}$. Since J is complete and $\mathscr{D}$ is closed, $\vdash_{\mathrm{J}} \neg\mathscr{D}$ – that is, $\vdash_{\mathrm{J}} \mathscr{C}$. Conversely, if $\mathscr{C}$ is not true for M, then $\mathscr{D}$ is true for M. Hence, $\vdash_{\mathrm{J}} \mathscr{D}$. Since J is consistent, not-$\vdash_{\mathrm{J}} \neg\mathscr{D}$, that is, not-$\vdash_{\mathrm{J}} \mathscr{C}$.

Case 3. $\mathscr{C}$ is $\mathscr{D} \Rightarrow \mathscr{E}$. Since $\mathscr{C}$ is closed, so are $\mathscr{D}$ and $\mathscr{E}$. If $\mathscr{C}$ is false for M, then $\mathscr{D}$ is true and $\mathscr{E}$ is false. Hence, by inductive hypothesis, $\vdash_{\mathrm{J}} \mathscr{D}$ and not-$\vdash_{\mathrm{J}} \mathscr{E}$. By the completeness of J, $\vdash_{\mathrm{J}} \neg\mathscr{E}$. Therefore, by an instance of the tautology $D \Rightarrow (\neg E \Rightarrow \neg(D \Rightarrow E))$ and two applications of MP, $\vdash_{\mathrm{J}} \neg(\mathscr{D} \Rightarrow \mathscr{E})$, that is, $\vdash_{\mathrm{J}} \neg\mathscr{C}$, and so, by the consistency of J, not-$\vdash_{\mathrm{J}} \mathscr{C}$. Conversely, if not-$\vdash_{\mathrm{J}} \mathscr{C}$, then, by the completeness of J, $\vdash_{\mathrm{J}} \neg\mathscr{C}$, that is, $\vdash_{\mathrm{J}} \neg(\mathscr{D} \Rightarrow \mathscr{E})$. By conditional elimination, $\vdash_{\mathrm{J}} \mathscr{D}$ and $\vdash_{\mathrm{J}} \neg\mathscr{E}$. Hence, by ($\Box$) for $\mathscr{D}$, $\mathscr{D}$ is true for M. By the consistency of J, not-$\vdash_{\mathrm{J}} \mathscr{E}$ and, therefore, by ($\Box$) for $\mathscr{E}$, $\mathscr{E}$ is false for M. Thus, since $\mathscr{D}$ is true for M and $\mathscr{E}$ is false for M, $\mathscr{C}$ is false for M.

Case 4. $\mathscr{C}$ is $(\forall x_m)\mathscr{D}$.

Case 4a. $\mathscr{D}$ is a closed wf. By inductive hypothesis, $\models_{\mathrm{M}} \mathscr{D}$ if and only if $\vdash_{\mathrm{J}} \mathscr{D}$. By Exercise 2.32(a), $\vdash_{\mathrm{J}} \mathscr{D} \Leftrightarrow (\forall x_m)\mathscr{D}$. So, $\vdash_{\mathrm{J}} \mathscr{D}$ if and only if $\vdash_{\mathrm{J}} (\forall x_m)\mathscr{D}$, by biconditional elimination. Moreover, $\models_{\mathrm{M}} \mathscr{D}$ if and only if $\models_{\mathrm{M}} (\forall x_m)\mathscr{D}$ by property (VI) on page 61. Hence, $\models_{\mathrm{M}} \mathscr{C}$ if and only if $\vdash_{\mathrm{J}} \mathscr{C}$.

Case 4b. $\mathscr{D}$ is not a closed wf. Since $\mathscr{C}$ is closed, $\mathscr{D}$ has x_m as its only free variable, say $\mathscr{D}$ is $F(x_m)$. Then $\mathscr{C}$ is $(\forall x_m)F(x_m)$.

(i) Assume $\models_M \mathscr{C}$ and not-$\vdash_J \mathscr{C}$. By the completeness of J, $\vdash_J \neg\mathscr{C}$, that is, $\vdash_J \neg(\forall x_m)F(x_m)$. Then, by Exercise 2.33(a) and biconditional elimination, $\vdash_J (\exists x_m)\neg F(x_m)$. Since J is a scapegoat theory, $\vdash_J \neg F(t)$ for some closed term t of J. But $\models_M \mathscr{C}$, that is, $\models_M (\forall x_m)F(x_m)$. Since $(\forall x_m)F(x_m) \Rightarrow F(t)$ is true for M by property (X) on page 63, $\models_M F(t)$. Hence, by ($\square$) for $F(t)$, $\vdash_J F(t)$. This contradicts the consistency of J. Thus, if $\models_M \mathscr{C}$, then, $\vdash_J \mathscr{C}$.

(ii) Assume $\vdash_J \mathscr{C}$ and not-$\models_M \mathscr{C}$. Thus,

$$(\#) \quad \vdash_J (\forall x_m)F(x_m)(\#\#) \quad \text{not}- \models_M (\forall x_m)F(x_m).$$

By (##), some sequence of elements of the domain D does not satisfy $(\forall x_m)F(x_m)$. Hence, some sequence s does not satisfy $F(x_m)$. Let t be the ith component of s. Notice that $s^*(u) = u$ for all closed terms u of J (by the definition of $(a_i)^M$ and $(f_k^n)^M$). Observe also that $F(t)$ has fewer connectives and quantifiers than $\mathscr{C}$ and, therefore, the inductive hypothesis applies to $F(t)$, that is, ($\square$) holds for $F(t)$. Hence, by Lemma 2(a) on page 63, s does not satisfy $F(t)$. So, $F(t)$ is false for M. But, by (#) and rule A4, $\vdash_J F(t)$, and so, by ($\square$) for $F(t)$, $\models_M F(t)$. This contradiction shows that, if $\vdash_J \mathscr{C}$, then $\models_M \mathscr{C}$.

Now we can prove the fundamental theorem of quantification theory. By a *denumerable model* we mean a model in which the domain is denumerable.

PROPOSITION 2.17†

Every consistent theory K has a denumerable model.

Proof

By Lemma 2.15, K has a consistent extension K′ such that K′ is a scapegoat theory and has denumerably many closed terms. By Lindenbaum's lemma, K′ has a consistent, complete extension J that has the same symbols at K′. Hence, J is also a scapegoat theory. By Lemma 2.16, J has a model M whose domain is the denumerable set of closed terms of J. Since J is an extension of K, M is a denumerable model of K.

†The proof given here is essentially due to Henkin (1949), as simplified by Hasenjaeger (1953). The result was originally proved by Gödel (1930). Other proofs have been published by Rasiowa and Sikorski (1951; 1952) and Beth (1951), using (Boolean) algebraic and topological methods, respectively. Still other proofs may be found in Hintikka (1955a, b) and in Beth (1959).

COROLLARY 2.18

Any logically valid wf $\mathscr{B}$ of a theory K is a theorem of K.

Proof

We need consider only closed wfs $\mathscr{B}$, since a wf $\mathscr{D}$ is logically valid if and only if its closure is logically valid, and $\mathscr{D}$ is provable in K if and only if its closure is provable in K. So, let $\mathscr{B}$ be a logically valid closed wf of K. Assume that not-$\vdash_K \mathscr{B}$. By Lemma 2.12, if we add $\neg\mathscr{B}$ as a new axiom to K, the new theory K′ is consistent. Hence, by Proposition 2.17, K′ has a model M. Since $\neg\mathscr{B}$ is an axiom of K′, $\neg\mathscr{B}$ is true for M. But, since $\mathscr{B}$ is logically valid, $\mathscr{B}$ is true for M. Hence, $\mathscr{B}$ is both true and false for M, which is impossible (by (II) on page 61). Thus, $\mathscr{B}$ must be a theorem of K.

COROLLARY 2.19. (GÖDEL'S COMPLETENESS THEOREM, 1930)

In any predicate calculus, the theorems are precisely the logically valid wfs.

Proof

This follows from Proposition 2.2 and Corollary 2.18. (Gödel's original proof runs along quite different lines. For other proofs, see Beth (1951), Dreben (1952), Hintikka (1955a, b) and Rasiowa and Sikorski (1951; 1952).)

COROLLARY 2.20

Let K be any theory.

(a) A wf $\mathscr{B}$ is true in every denumerable model of K if and only if $\vdash_K \mathscr{B}$.
(b) If, in every model of K, every sequence that satisfies all wfs in a set Γ of wfs also satisfies a wf $\mathscr{B}$, then $\Gamma \vdash_K \mathscr{B}$.
(c) If a wf $\mathscr{B}$ of K is a logical consequence of a set Γ of wfs of K, then $\Gamma \vdash_K \mathscr{B}$.
(d) If a wf $\mathscr{B}$ of K is a logical consequence of a wf $\mathscr{C}$ of K, then $\mathscr{C} \vdash_K \mathscr{B}$.

Proof

(a) We may assume $\mathscr{B}$ is closed. If not-$\vdash_K \mathscr{B}$, then the theory $K' = K + \{\neg\mathscr{B}\}$ is consistent.† Hence, by Proposition 2.17, K′ has a denumerable model M. However, $\neg\mathscr{B}$, being an axiom of K′, is true for

†If K is a theory and Δ is a set of wfs of K, then K + Δ denotes the theory obtained from K by adding the wfs of Δ as axioms.

M. By hypothesis, since M is a denumerable model of K, $\mathscr{B}$ is true for M. Therefore, $\mathscr{B}$ is true and false for M, which is impossible.

(b) Consider the theory $K + \Gamma$. By the hypothesis, $\mathscr{B}$ is true for every model of this theory. Hence, by (a), $\vdash_{K+\Gamma} \mathscr{B}$. So, $\Gamma \vdash_K \mathscr{B}$.

Part (c) is a consequence of (b), and part (d) is a special case of (c).

Corollaries 2.18–2.20 show that the 'syntactical' approach to quantification theory by means of first-order theories is equivalent to the 'semantical' approach through the notions of interpretations, models, logical validity, and so on. For the propositional calculus, Corollary 1.15 demonstrated the analogous equivalence between the semantical notion (tautology) and the syntactical notion (theorem of L). Notice also that, in the propositional calculus, the completeness of the system L (see Proposition 1.14) led to a solution of the decision problem. However, for first-order theories, we cannot obtain a decision procedure for logical validity or, equivalently, for provability in first-order predicate calculi. We shall prove this and related results in Section 3.6.

COROLLARY 2.21. (SKOLEM–LÖWENHEIM THEOREM, 1920, 1915)

Any theory that has a model has a denumerable model.

Proof

If K has a model, then K is consistent, since no wf can be both true and false for the same model M. Hence, by Proposition 2.17, K has a denumerable model.

The following stronger consequence of Proposition 2.17 is derivable.

COROLLARY 2.22^{A}

For any cardinal number $\mathfrak{m} \geqslant \aleph_0$, any consistent theory K has a model of cardinality $\mathfrak{m}$.

Proof

By Proposition 2.17, we know that K has a denumerable model. Therefore, it suffices to prove the following lemma.

LEMMA

If $\mathfrak{m}$ and $\mathfrak{n}$ are two cardinal numbers such that $\mathfrak{m} \leqslant \mathfrak{n}$ and if K has a model of cardinality $\mathfrak{m}$, then K has a model of cardinality $\mathfrak{n}$.

Proof

Let M be a model of K with domain D of cardinality $\mathfrak{m}$. Let D' be a set of cardinality $\mathfrak{n}$ that contains D. Extend the model M to an interpretation M′ that has D' as domain in the following way. Let c be a fixed element of D. We stipulate that the elements of $D' - D$ behave like c. For example, if B_j^n is the interpretation in M of the predicate letter A_j^n and $(B_j^n)'$ is the new interpretation in M′, then for any $d_1, \ldots, d_n$ in D′, $(B_j^n)'$ holds for $(d_1, \ldots, d_n)$ if and only if B_j^n holds for $(u_1, \ldots, u_n)$, where $u_i = d_i$ if $d_i \in$ D and $u_i = c$ if $d_i \in D' - D$. The interpretation of the function letters is extended in an analogous way, and the individual constants have the same interpretations as in M. It is an easy exercise to show, by induction on the number of connectives and quantifiers in a wf $\mathscr{B}$, that $\mathscr{B}$ is true for M′ if and only if it is true for M. Hence, M′ is a model of K of cardinality $\mathfrak{n}$.

Exercises

2.51 For any theory K, if $\Gamma \vdash_K \mathscr{B}$ and each wf in Γ is true for a model M of K, show that $\mathscr{B}$ is true for M.

2.52 If a wf $\mathscr{B}$ without quantifiers is provable in a predicate calculus, prove that $\mathscr{B}$ is an instance of a tautology and, hence, by Proposition 2.1, has a proof without quantifiers using only axioms (A1)–(A3) and MP. [*Hint*: if $\mathscr{B}$ were not a tautology, one could construct an interpretation, having the set of terms that occur in $\mathscr{B}$ as its domain, for which $\mathscr{B}$ is not true, contradicting Proposition 2.2.] Note that this implies the consistency of the predicate calculus and also provides a decision procedure for the provability of wfs without quantifiers.

2.53 Show that $\vdash_K \mathscr{B}$ if and only if there is a wf $\mathscr{C}$ that is the closure of the conjunction of some axioms of K such that $\mathscr{C} \Rightarrow \mathscr{B}$ is logically valid.

2.54 *Compactness*. If all finite subsets of the set of axioms of a theory K have models, prove that K has a model.

2.55 (a) For any wf $\mathscr{B}$, prove that there is only a finite number of interpretations of $\mathscr{B}$ on a given domain of finite cardinality k.

(b) For any wf $\mathscr{B}$, prove that there is an effective way of determining whether $\mathscr{B}$ is true for all interpretations with domain of some fixed cardinality k.

(c) Let a wf $\mathscr{B}$ be called *k-valid* if it is true for all interpretations that have a domain of k elements. Call $\mathscr{B}$ *precisely k-valid* if it is k-valid but not $(k+1)$-valid. Show that $(k+1)$-validity implies k-validity and give an example of a wf that is precisely k-valid. (See Hilbert and Bernays (1934, § 4–5) and Wajsberg (1933).)

2.56 Show that the following wf is true for all finite domains but is false for some infinite domain.

$$(\forall x)(\forall y)(\forall z)[A_1^2(x,x) \wedge (A_1^2(x,y) \wedge A_1^2(y,z) \Rightarrow A_1^2(x,z)) \wedge (A_1^2(x,y) \vee A_1^2(y,x))] \Rightarrow (\exists y)(\forall x)A_1^2(y,x)$$

2.57 Prove that there is no theory K whose models are exactly the interpretations with finite domains.

2.58 Let $\mathscr{B}$ be any wf that contains no quantifiers, function letters, or individual constants.

(a) Show that a closed *prenex* wf $(\forall x_1)\dots(\forall x_n)(\exists y_1)\dots(\exists y_m)\mathscr{B}$, with $m \geqslant 0$ and $n \geqslant 1$, is logically valid if and only if it is true for every interpretation with a domain of n objects.

(b) Prove that a closed prenex wf $(\exists y_1)\dots(\exists y_m)\mathscr{B}$ is logically valid if and only if it is true for all interpretations with a domain of one element.

(c) Show that there is an effective procedure to determine the logical validity of all wfs of the forms given in (a) and (b).

2.59 Let K_1 and K_2 be theories in the same language $\mathscr{L}$. Assume that any interpretation M of $\mathscr{L}$ is a model of K_1 if and only if M is not a model of K_2. Prove that K_1 and K_2 are finitely axiomatizable, that is, there are finite sets of sentences Γ and Δ such that, for any sentence $\mathscr{B}$, $\vdash_{K_1} \mathscr{B}$ if and only if $\Gamma \vdash \mathscr{B}$, and $\vdash_{K_2} \mathscr{B}$ if and only if $\Delta \vdash \mathscr{B}$.†

2.60 A set Γ of sentences is called an *independent axiomatization* of a theory K if (a) all sentences in Γ are theorems of K, (b) $\Gamma \vdash \mathscr{B}$ for every theorem $\mathscr{B}$ of K, and (c) for every sentence $\mathscr{C}$ of Γ, it is not the case that $\Gamma - \{\mathscr{C}\} \vdash \mathscr{C}$ †. Prove that every theory K has an independent axiomatization.

2.61[A] If, for some cardinal $\mathfrak{m} \geqslant \aleph_0$, a wf $\mathscr{B}$ is true for every interpretation of cardinality $\mathfrak{m}$, prove that $\mathscr{B}$ is logically valid.

2.62[A] If a wf $\mathscr{B}$ is true for all interpretations of cardinality $\mathfrak{m}$ prove that $\mathscr{B}$ is true for all interpretations of cardinality less than or equal to $\mathfrak{m}$.

2.63 (a) Prove that a theory K is a scapegoat theory if and only if, for any wf $\mathscr{B}(x)$ with x as its only free variable, there is a closed term t such that $\vdash_K (\exists x)\mathscr{B}(x) \Rightarrow \mathscr{B}(t)$.

(b) Prove that a theory K is a scapegoat theory if and only if, for any wf $\mathscr{B}(x)$ with x as its only free variable such that $\vdash_K (\exists x)\mathscr{B}(x)$, there is a closed term t such that $\vdash_K \mathscr{B}(t)$.

(c) Prove that no predicate calculus is a scapegoat theory.

2.8 FIRST-ORDER THEORIES WITH EQUALITY

Let K be a theory that has as one of its predicate letters A_1^2. Let us write $t = s$ as an abbreviation for $A_1^2(t,s)$, and $t \neq s$ as an abbreviation for $\neg A_1^2(t,s)$.

†Here, an expression $\Gamma \vdash \mathscr{B}$, without any subscript attached to $\vdash$, means that $\mathscr{B}$ is derivable from Γ using only logical axioms, that is within the predicate calculus.

Then K is called a *first-order theory with equality* (or simply a *theory with equality*) if the following are theorems of K:

(A6) $(\forall x_1)x_1 = x_1$ (reflexivity of equality)
(A7) $x = y \Rightarrow (\mathscr{B}(x,x) \Rightarrow \mathscr{B}(x,y))$ (substitutivity of equality)

where x and y are any variables, $\mathscr{B}(x,x)$ is any wf, and $\mathscr{B}(x,y)$ arises from $\mathscr{B}(x,x)$ by replacing some, but not necessarily all, free occurrences of x by y, with the proviso that y is free for x in $\mathscr{B}(x,x)$. Thus, $\mathscr{B}(x,y)$ may or may not contain free occurrences of x.

The numbering (A6) and (A7) is a continuation of the numbering of the logical axioms.

PROPOSITION 2.23

In any theory with equality,

(a) $\vdash t = t$ for any term t;
(b) $\vdash t = s \Rightarrow s = t$ for any terms t and s;
(c) $\vdash t = s \Rightarrow (s = r \Rightarrow t = r)$ for any terms t, s and r.

Proof

(a) By (A6), $\vdash (\forall x_1)\, x_1 = x_1$. Hence, by rule A4, $\vdash t = t$.
(b) Let x and y be variables not occurring in t or s. Letting $\mathscr{B}(x,x)$ be $x = x$ and $\mathscr{B}(x,y)$ be $y = x$ in schema (A7), $\vdash x = y \Rightarrow (x = x \Rightarrow y = x)$. But, by (a), $\vdash x = x$. So, by an instance of the tautology $(A \Rightarrow (B \Rightarrow C)) \Rightarrow (B \Rightarrow (A \Rightarrow C))$ and two applications of MP, we have $\vdash x = y \Rightarrow y = x$. Two applications of Gen yield $\vdash (\forall x)(\forall y)(x = y \Rightarrow y = x)$, and then two applications of rule A4 give $\vdash t = s \Rightarrow s = t$.
(c) Let x, y and z be three variables not occurring in t, s, or r. Letting $\mathscr{B}(y,y)$ be $y = z$ and $\mathscr{B}(y,x)$ be $x = z$ in (A7), with x and y interchanged, we obtain $\vdash y = x \Rightarrow (y = z \Rightarrow x = z)$. But, by (b), $\vdash x = y \Rightarrow y = x$. Hence, using an instance of the tautology $(A \Rightarrow B) \Rightarrow ((B \Rightarrow C) \Rightarrow (A \Rightarrow C))$ and two applications of MP, we obtain $\vdash x = y \Rightarrow (y = z \Rightarrow x = z)$. By three applications of Gen, $\vdash (\forall x)(\forall y)(\forall z)(x = y \Rightarrow (y = z \Rightarrow x = z))$, and then, by three uses of rule A4, $\vdash t = s \Rightarrow (s = r \Rightarrow t = r)$.

Exercises

2.64 Show that (A6) and (A7) are true for any interpretation M in which $(A_1^2)^M$ is the identity relation on the domain of the interpretation.
2.65 Prove the following in any theory with equality.

(a) $\vdash (\forall x)(\mathscr{B}(x) \Leftrightarrow (\exists y)(x = y \wedge \mathscr{B}(y)))$ if y does not occur in $\mathscr{B}(x)$
(b) $\vdash (\forall x)(\mathscr{B}(x) \Leftrightarrow (\forall y)(x = y \Rightarrow \mathscr{B}(y)))$ if y does not occur in $\mathscr{B}(x)$
(c) $\vdash (\forall x)(\exists y)\ x = y$
(d) $\vdash x = y \Rightarrow f(x) = f(y)$, where f is any function letter of one argument
(e) $\vdash \mathscr{B}(x) \wedge x = y \Rightarrow \mathscr{B}(y)$, if y is free for x in $\mathscr{B}(x)$
(f) $\vdash \mathscr{B}(x) \wedge \neg\mathscr{B}(y) \Rightarrow x \neq y$, if y is free for x in $\mathscr{B}(x)$

We can reduce schema (A7) to a few simpler cases.

PROPOSITION 2.24

Let K be a theory for which (A6) holds and (A7) holds for all atomic wfs $\mathscr{B}(x,x)$ in which there are no individual constants. Then K is a theory with equality, that is, (A7) holds for all wfs $\mathscr{B}(x,x)$.

Proof

We must prove (A7) for all wfs $\mathscr{B}(x,x)$. It holds for atomic wfs by assumption. Note that we have the results of Proposition 2.23, since its proof used (A7) only with atomic wfs without individual constants. Note also that we have (A7) for all atomic wfs $\mathscr{B}(x,x)$. For if $\mathscr{B}(x,x)$ contains individual constants, we can replace those individual constants by new variables, obtaining a wf $\mathscr{B}^*(x,x)$ without individual constants. By hypothesis, the corresponding instance of (A7) with $\mathscr{B}^*(x,x)$ is a theorem; we can then apply Gen with respect to the new variables, and finally apply rule A4 one or more times to obtain (A7) with respect to $\mathscr{B}(x,x)$.

Proceeding by induction on the number n of connectives and quantifiers in $\mathscr{B}(x,x)$, we assume that (A7) holds for all $k < n$.

Case 1. $\mathscr{B}(x,x)$ is $\neg\mathscr{C}(x,x)$. By inductive hypothesis, we have $\vdash y = x \Rightarrow (\mathscr{C}(x,y) \Rightarrow \mathscr{C}(x,x))$, since $\mathscr{C}(x,x)$ arises from $\mathscr{C}(x,y)$ by replacing some occurrences of y by x. Hence, by Proposition 2.23(b), instances of the tautologies $(A \Rightarrow B) \Rightarrow (\neg\mathscr{B} \Rightarrow \neg A)$ and $(A \Rightarrow B) \Rightarrow ((B \Rightarrow C) \Rightarrow (A \Rightarrow C))$ and MP, we obtain $\vdash x = y \Rightarrow (\mathscr{B}(x,x) \Rightarrow \mathscr{B}(x,y))$.

Case 2. $\mathscr{B}(x,x)$ is $\mathscr{C}(x,x) \Rightarrow \mathscr{D}(x,x)$. By inductive hypothesis and Proposition 2.23(b), $\vdash x = y \Rightarrow (\mathscr{C}(x,y) \Rightarrow \mathscr{C}(x,x))$ and $\vdash x = y \Rightarrow (\mathscr{D}(x,x) \Rightarrow \mathscr{D}(x,y))$. Hence, by the tautology $(A \Rightarrow (C_1 \Rightarrow C)) \Rightarrow [(A \Rightarrow (D \Rightarrow D_1)) \Rightarrow (A \Rightarrow ((C \Rightarrow D) \Rightarrow (C_1 \Rightarrow D_1)))]$, we have $\vdash x = y \Rightarrow (\mathscr{B}(x,x) \Rightarrow \mathscr{B}(x,y))$.

Case 3. $\mathscr{B}(x,x)$ is $(\forall z)\mathscr{C}(x,x,z)$. By inductive hypothesis, $\vdash x = y \Rightarrow (\mathscr{C}(x,x,z) \Rightarrow \mathscr{C}(x,y,z))$. Now, by Gen and axiom (A5), $\vdash x = y \Rightarrow (\forall z)(\mathscr{C}(x,x,z) \Rightarrow \mathscr{C}(x,y,z))$. By Exercise 2.27(a), $\vdash (\forall z)\ (\mathscr{C}(x,x,z) \Rightarrow \mathscr{C}(x,y,z)) \Rightarrow [(\forall z)\mathscr{C}(x,x,z) \Rightarrow (\forall z)\mathscr{C}(x,y,z)]$, and so, by the tautology $(A \Rightarrow B) \Rightarrow ((B \Rightarrow C) \Rightarrow (A \Rightarrow C))$, $\vdash x = y \Rightarrow (\mathscr{B}(x,x) \Rightarrow \mathscr{B}(x,y))$.

The instances of (A7) can be still further reduced.

PROPOSITION 2.25

Let K be a theory in which (A6) holds and the following are true.

(a) Schema (A7) holds for all atomic wfs $\mathscr{B}(x,x)$ such that no function letters or individual constants occur in $\mathscr{B}(x,x)$ and $\mathscr{B}(x,y)$ comes from $\mathscr{B}(x,x)$ by replacing exactly one occurrence of x by y.

(b) $\vdash x = y \Rightarrow f_j^n(z_1,\ldots,z_n) = f_j^n(w_1,\ldots,w_n)$, where f_j^n is any function letter of K, $z_1,\ldots,z_n$ are variables, and $f_j^n(w_1,\ldots,w_n)$ arises from $f_j^n(z_1,\ldots,z_n)$ by replacing exactly one occurrence of x by y.

Then K is a theory with equality.

Proof

By repeated application, our assumptions can be extended to replacements of more than one occurrence of x by y. Also, Proposition 2.23 is still derivable. By Proposition 2.24, it suffices to prove (A7) for only atomic wfs without individual constants. But, hypothesis (a) enables us easily to prove

$$\vdash (y_1 = z_1 \wedge \ldots \wedge y_n = z_n) \Rightarrow (\mathscr{B}(y_1,\ldots,y_n) \Rightarrow \mathscr{B}(z_1,\ldots,z_n))$$

for all variables $y_1,\ldots,y_n$, $z_1,\ldots,z_n$ and any atomic wf $\mathscr{B}(y_1,\ldots,y_n)$ without function letters or individual constants. Hence, it suffices to show:

(*) If $t(x,x)$ is a term without individual constants and $t(x,y)$ comes from $t(x,x)$ by replacing some occurrences of x by y, then $\vdash x = y \Rightarrow t(x,x) = t(x,y)$.†

But (*) can be proved, using hypothesis (b), by induction on the number of function letters in $t(x,x)$, and we leave this as an exercise.

It is easy to see from Proposition 2.25 that, when the language of K has only finitely many predicate and function letters, it is only necessary to verify (A7) for a finite list of special cases (in fact, n wfs for each A_j^n and n wfs for each f_j^n).

Exercises

2.66 Let K_1 be a theory whose language has only $=$ as a predicate letter and no function letters or individual constants. Let its proper axioms be $(\forall x_1)\, x_1 = x_1$, $(\forall x_1)(\forall x_2)(x_1 = x_2 \Rightarrow x_2 = x_1)$ and $(\forall x_1)(\forall x_2)(\forall x_3)(x_1 = x_2 \Rightarrow (x_2 = x_3 \Rightarrow x_1 = x_3))$. Show that K_1 is a theory with equality. [*Hint*: It

†The reader can clarify how (*) is applied by using it to prove the following instance of (A7): $\vdash x = y \Rightarrow (A_1^1(f_1^1(x)) \Rightarrow A_1^1(f_1^1(y)))$. Let $t(x,x)$ be $f_1^1(x)$ and let $t(x,y)$ be $f_1^1(y)$.

suffices to prove that $\vdash x_1 = x_3 \Rightarrow (x_1 = x_2 \Rightarrow x_3 = x_2)$ and $\vdash x_2 = x_3 \Rightarrow (x_1 = x_2 \Rightarrow x_1 = x_3)$.] K_1 is called the *pure first-order theory of equality.*

2.67 Let K_2 be a theory whose language has only $=$ and $<$ as predicate letters and no function letters or individual constants. Let K_2 have the following proper axioms.

(a) $(\forall x_1)\, x_1 = x_1$
(b) $(\forall x_1)(\forall x_2)(x_1 = x_2 \Rightarrow x_2 = x_1)$
(c) $(\forall x_1)(\forall x_2)(\forall x_3)(x_1 = x_2 \Rightarrow (x_2 = x_3 \Rightarrow x_1 = x_3))$
(d) $(\forall x_1)(\exists x_2)(\exists x_3)(x_1 < x_2 \wedge x_3 < x_1)$
(e) $(\forall x_1)(\forall x_2)(\forall x_3)(x_1 < x_2 \wedge x_2 < x_3 \Rightarrow x_1 < x_3)$
(f) $(\forall x_1)(\forall x_2)(x_1 = x_2 \Rightarrow \neg\, x_1 < x_2)$
(g) $(\forall x_1)(\forall x_2)(x_1 < x_2 \vee x_1 = x_2 \vee x_2 < x_1)$
(h) $(\forall x_1)(\forall x_2)(x_1 < x_2 \Rightarrow (\exists x_3)(x_1 < x_3 \wedge x_3 < x_2))$

Using Proposition 2.25, show that K_2 is a theory with equality. K_2 is called the *theory of densely ordered sets with neither first nor last element.*

2.68 Let K be any theory with equality. Prove the following.

(a) $\vdash x_1 = y_1 \wedge \ldots \wedge x_n = y_n \Rightarrow t(x_1, \ldots, x_n) = t(y_1, \ldots, y_n)$, where $t(y_1, \ldots, y_n)$ arises from the term $t(x_1, \ldots, x_n)$ by substitution of $y_1, \ldots, y_n$ for $x_1, \ldots, x_n$, respectively.
(b) $\vdash x_1 = y_1 \wedge \ldots \wedge x_n = y_n \Rightarrow (\mathscr{B}(x_1, \ldots, x_n) \Leftrightarrow \mathscr{B}(y_1, \ldots, y_n))$, where $\mathscr{B}(y_1, \ldots, y_n)$ is obtained by substituting $y_1, \ldots, y_n$ for one or more occurrences of $x_1, \ldots, x_n$, respectively, in the wf $\mathscr{B}(x_1, \ldots, x_n)$, and $y_1, \ldots, y_n$ are free for $x_1, \ldots, x_n$, respectively, in the wf $\mathscr{B}(x_1, \ldots, x_n)$.

Examples.
(In the literature, 'elementary' is sometimes used instead of 'first-order'.)

1. *Elementary theory* G *of groups*: predicate letter $=$, function letter f_1^2, and individual constant a_1. We abbreviate $f_1^2\,(t,s)$ by $t + s$ and a_1 by 0. The proper axioms are the following.
   (a) $x_1 + (x_2 + x_3) = (x_1 + x_2) + x_3$
   (b) $x_1 + 0 = x_1$
   (c) $(\forall x_1)(\exists x_2) x_1 + x_2 = 0$
   (d) $x_1 = x_1$
   (e) $x_1 = x_2 \Rightarrow x_2 = x_1$
   (f) $x_1 = x_2 \Rightarrow (x_2 = x_3 \Rightarrow x_1 = x_3)$
   (g) $x_1 = x_2 \Rightarrow (x_1 + x_3 = x_2 + x_3 \wedge x_3 + x_1 = x_3 + x_2)$
   That G is a theory with equality follows easily from Proposition 2.25. If one adds to the axioms the following wf:
   (h) $x_1 + x_2 = x_2 + x_1$
the new theory is called the *elementary theory of abelian groups.*
2. *Elementary theory* F *of fields*: predicate letter $=$, function letters f_1^2 and f_2^2, and individual constants a_1 and a_2. Abbreviate $f_1^2(t,s)$ by

$t+s$, $f_2^2(t,s)$ by $t \cdot s$, and a_1 and a_2 by 0 and 1. As proper axioms, take (a)–(h) of Example 1 plus the following.

(i) $x_1 = x_2 \Rightarrow (x_1 \cdot x_3 = x_2 \cdot x_3 \wedge x_3 \cdot x_1 = x_3 \cdot x_2)$
(j) $x_1 \cdot (x_2 \cdot x_3) = (x_1 \cdot x_2) \cdot x_3$
(k) $x_1 \cdot (x_2 + x_3) = (x_1 \cdot x_2) + (x_1 \cdot x_3)$
(l) $x_1 \cdot x_2 = x_2 \cdot x_1$
(m) $x_1 \cdot 1 = x_1$
(n) $x_1 \neq 0 \Rightarrow (\exists x_2)\ x_1 \cdot x_2 = 1$
(o) $0 \neq 1$

F is a theory with equality. Axioms (a)–(m) define the elementary theory $\mathrm{R_C}$ of commutative rings with unit. If we add to F the predicate letter A_2^2, abbreviate $A_2^2(t,s)$ by $t < s$, and add axioms (e), (f) and (g) of Exercise 2.67, as well as $x_1 < x_2 \Rightarrow x_1 + x_3 < x_2 + x_3$ and $x_1 < x_2 \wedge 0 < x_3 \Rightarrow x_1 \cdot x_3 < x_2 \cdot x_3$, then the new theory $\mathrm{F}_<$ is called the *elementary theory of ordered fields.*

Exercise

2.69 (a) What formulas must be derived in order to use Proposition 2.25 to conclude that the theory G of Example 1 is a theory with equality?
(b) Show that the axioms (d)–(f) of equality mentioned in Example 1 can be replaced by (d) and
(f′): $x_1 = x_2 \Rightarrow (x_3 = x_2 \Rightarrow x_1 = x_3)$.

One often encounters theories K in which = may be defined; that is, there is a wf $\mathscr{E}(x,y)$ with two free variables x and y, such that, if we abbreviate $\mathscr{E}(t,s)$ by $t = s$, then axioms (A6) and (A7) are provable in K. We make the convention that, if t and s are terms that are not free for x and y, respectively, in $\mathscr{E}(x,y)$, then, by suitable changes of bound variables (see Exercise 2.48), we replace $\mathscr{E}(x,y)$ by a logically equivalent wf $\mathscr{E}^*(x,y)$ such that t and s are free for x and y, respectively, in $\mathscr{E}^*(x,y)$; then $t = s$ is to be the abbreviation of $\mathscr{E}^*(t,s)$. Proposition 2.23 and analogues of Propositions 2.24 and 2.25 hold for such theories. There is no harm in extending the term *theory with equality* to cover such theories.

In theories with equality it is possible to define in the following way phrases that use the expression 'There exists one and only one x such that...'.

DEFINITION

$(\exists_1 x)\mathscr{B}(x)$ for $(\exists x)\mathscr{B}(x) \wedge (\forall x)(\forall y)(\mathscr{B}(x) \wedge \mathscr{B}(y) \Rightarrow x = y)$

In this definition, the new variable y is assumed to be the first variable that does not occur in $\mathscr{B}(x)$. A similar convention is to be made in all other definitions where new variables are introduced.

Exercise

2.70 In any theory with equality, prove the following.

(a) $\vdash (\forall x)(\exists_1 y)\ x = y$

(b) $\vdash (\exists_1 x)\mathscr{B}(x) \Leftrightarrow (\exists x)(\forall y)(x = y \Leftrightarrow \mathscr{B}(y))$

(c) $\vdash (\forall x)(\mathscr{B}(x) \Leftrightarrow \mathscr{C}(x)) \Rightarrow [(\exists_1 x)\mathscr{B}(x) \Leftrightarrow (\exists_1 x)\mathscr{C}(x)]$

(d) $\vdash (\exists_1 x)(\mathscr{B} \vee \mathscr{C}) \Rightarrow ((\exists_1 x)\mathscr{B}) \vee (\exists_1 x)\mathscr{C}$

(e) $\vdash (\exists_1 x)\mathscr{B}(x) \Leftrightarrow (\exists x)(\mathscr{B}(x) \wedge (\forall y)(\mathscr{B}(y) \Rightarrow y = x))$

In any model for a theory K with equality, the relation E in the model corresponding to the predicate letter = is an equivalence relation (by Proposition 2.23). If this relation E is the identity relation in the domain of the model, then the model is said to be *normal.*

Any model M for K can be *contracted* to a normal model M* for K by taking the domain D^* of M* to be the set of equivalence classes determined by the relation E in the domain D of M. For a predicate letter A_j^n and for any equivalence classes $[b_1], \ldots, [b_n]$ in D^* determined by elements $b_1, \ldots, b_n$ in D, we let $(A_j^n)^{M^*}$ hold for $([b_1], \ldots, [b_n])$ if and only if $(A_j^n)^{M}$ holds for $(b_1, \ldots, b_n)$. Notice that it makes no difference which representatives $b_1, \ldots, b_n$ we select in the given equivalence classes because, from (A7), $\vdash x_1 = y_1 \wedge \ldots \wedge\ x_n = y_n \Rightarrow (A_j^n(x_1, \ldots, x_n) \Leftrightarrow A_j^n(y_1, \ldots, y_n))$. Likewise, for any function letter f_j^n and any equivalence classes $[b_1], \ldots, [b_n]$ in D^*, let $(f_j^n)^{M^*}([b_1], \ldots, [b_n]) = [(f_j^n)^{M}(b_1, \ldots, b_n)]$. Again note that this is independent of the choice of the representatives $b_1, \ldots, b_n$, since, from (A7), we can prove $\vdash x_1 = y_1 \wedge \ldots \wedge x_n = y_n \Rightarrow f_j^n(x_1, \ldots, x_n) = f_j^n(y_1, \ldots, y_n)$. For any individual constant a_i let $(a_i)^{M^*} = [(a_i)^{M}]$. The relation E* corresponding to = in the model M* is the identity relation in D^*: $E^*([b_1], [b_2])$ if and only if $E(b_1, b_2)$, that is, if and only if $[b_1] = [b_2]$. Now one can easily prove by induction the following lemma: If $s = (b_1, b_2, \ldots)$ is a denumerable sequence of elements of D, and $s' = ([b_1], [b_2], \ldots)$ is the corresponding sequence of equivalence classes, then a wf $\mathscr{B}$ is satisfied by s in M if and only if $\mathscr{B}$ is satisfied by s' in M*. It follows that, for any wf $\mathscr{B}$, $\mathscr{B}$ is true for M if and only if $\mathscr{B}$ is true for M*. Hence, because M is a model of K, M* is a normal model of K.

PROPOSITION 2.26 (EXTENSION OF PROPOSITION 2.17)

(Gödel, 1930) Any consistent theory with equality K has a finite or denumerable normal model.

Proof

By Proposition 2.17, K has a denumerable model M. Hence, the contraction of M to a normal model yields a finite or denumerable normal model M* because the set of equivalence classes in a denumerable set D is either finite or denumerable.

COROLLARY 2.27 (EXTENSION OF THE SKOLEM–LÖWENHEIM THEOREM)

Any theory with equality K that has an infinite normal model M has a denumerable normal model.

Proof

Add to K the denumerably many new individual constants $b_1, b_2, \ldots$ together with the axioms $b_i \neq b_j$ for $i \neq j$. Then the new theory K′ is consistent. If K′ were inconsistent, there would be a proof in K′ of a contradiction $\mathscr{C} \wedge \neg\mathscr{C}$, where we may assume that $\mathscr{C}$ is a wf of K. But this proof uses only a finite number of the new axioms: $b_{i_1} \neq b_{j_1}, \ldots, b_{i_n} \neq b_{j_n}$. Now, M can be extended to a model $\mathrm{M}^{\#}$ of K plus the axioms $b_{i_1} \neq b_{j_1}, \ldots, b_{i_n} \neq b_{j_n}$; in fact, since M is an infinite normal model, we can choose interpretations of $b_{i_1}, b_{j_1}, \ldots, b_{i_n}, b_{j_n}$, so that the wfs $b_{i_1} \neq b_{j_1}, \ldots, b_{i_n} \neq b_{j_n}$ are true. But, since $\mathscr{C} \wedge \neg\mathscr{C}$ is derivable from these wfs and the axioms of K, it would follow that $\mathscr{C} \wedge \neg\mathscr{C}$ is true for $\mathrm{M}^{\#}$, which is impossible. Hence, K′ must be consistent. Now, by Proposition 2.26, K′ has a finite or denumerable normal model N. But, since, for $i \neq j$, the wfs $b_i \neq b_j$ are axioms of K′, they are true for N. Thus, the elements in the domain of N that are the interpretations of $b_1, b_2, \ldots$ must be distinct, which implies that the domain of N is infinite and, therefore, denumerable.

Exercises

2.71 We define $(\exists_n x)\mathscr{B}(x)$ by induction on $n \geqslant 1$. The case $n = 1$ has already been taken care of. Let $(\exists_{n+1} x)\mathscr{B}(x)$ stand for $(\exists y)(\mathscr{B}(y) \wedge (\exists_n x)(x \neq y \wedge \mathscr{B}(x)))$.

(a) Show that $(\exists_n x)\mathscr{B}(x)$ asserts that there are exactly n objects for which $\mathscr{B}$ holds, in the sense that in any normal model for $(\exists_n x)\mathscr{B}(x)$ there are exactly n objects for which the property corresponding to $\mathscr{B}(x)$ holds.

(b) (i) For each positive integer n, write a closed wf $\mathscr{B}_n$ such that $\mathscr{B}_n$ is true in a normal model when and only when that model contains at least n elements.

(ii) Prove that the theory K, whose axioms are those of the pure theory of equality K_1 (see Exercise 2.66), plus the axioms $\mathscr{B}_1, \mathscr{B}_2, \ldots$, is not finitely axiomatizable, that is, there is no theory K′ with a finite number of axioms such that K and K′ have the same theorems.

(iii) For a normal model, state in ordinary English the meaning of $\neg\mathscr{B}_{n+1}$.

(c) Let n be a positive integer and consider the wf $(\mathscr{E}_n)$ $(\exists_n x)x = x$. Let L_n be the theory $K_1 + \{\mathscr{E}_n\}$, where K_1 is the pure theory of equality.

(i) Show that a normal model M is a model of L_n if and only if there are exactly n elements in the domain of M.

(ii) Define a procedure for determining whether any given sentence is a theorem of L_n and show that L_n is a complete theory.

2.72 (a) Prove that, if a theory with equality K has arbitrarily large finite normal models, then it has a denumerable normal model.

(b) Prove that there is no theory with equality whose normal models are precisely all finite normal interpretations.

2.73 Prove that any predicate calculus with equality is consistent. (A predicate calculus with equality is assumed to have (A1)–(A7) as its only axioms.)

2.74[D] Prove the independence of axioms (A1)–(A7) in any predicate calculus with equality.

2.75 If $\mathscr{B}$ is a wf that does not contain the = symbol and $\mathscr{B}$ is provable in a predicate calculus with equality K, show that $\mathscr{B}$ is provable in K without using (A6) or (A7).

2.76[D] Show that = can be defined in any theory whose language has only a finite number of predicate letters and no function letters.

2.77 (a)[A] Find a non-normal model of the elementary theory of groups G.

(b) Show that any model M of a theory with equality K can be extended to a non-normal model of K. [*Hint*: Use the argument in the proof of the lemma within the proof of Corollary 2.22.]

2.78 Let $\mathscr{B}$ be a wf of a theory with equality. Show that $\mathscr{B}$ is true in every normal model of K if and only if $\vdash_K \mathscr{B}$.

2.79 Write the following as wfs of a theory with equality.

(a) There are at least three moons of Jupiter.

(b) At most two people know everyone in the class.

2.80 If $P(u)$ means *u is a person*, $G(u, v)$ means *u is a grandparent of v*, and $u = v$ means that u and v are identical, translate the following wf into ordinary English:

$$(\forall x)(P(x) \Rightarrow (\exists x_1)(\exists x_2)(\exists x_3)(\exists x_4)(x_1 \neq x_2 \wedge x_1 \neq x_3 \wedge x_1 \neq x_4 \wedge \\ x_2 \neq x_3 \wedge x_2 \neq x_4 \wedge x_3 \neq x_4 \wedge G(x_1, x) \wedge G(x_2, x) \wedge G(x_3, x) \wedge \\ G(x_4, x) \wedge (\forall y)(G(y, x) \Rightarrow y = x_1 \vee y = x_2 \vee y = x_3 \vee y = x_4)))$$

2.81 Consider the wf

$$(*) \quad (\forall x)(\forall y)(\exists z)(z \neq x \wedge z \neq y \wedge A(z)).$$

Show that $(*)$ is true in a normal model M of a theory with equality if and only if there exist in the domain of M at least three things having property $A(z)$.

2.82 Let the language $\mathscr{L}$ have the four predicate letters $=$, P, S and L. Read $u = v$ as *u and v are identical*, $P(u)$ as *u is a point*, $S(u)$ as *u is a line*, and $L(u, v)$ as *u lies on v*. Let the theory of equality $\mathbb{G}$ of *planar incidence geometry* have, in addition to axioms (A1)–(A7), the following non-logical axioms.

(1) $P(x) \Rightarrow \neg S(x)$
(2) $L(x, y) \Rightarrow P(x) \wedge S(y)$
(3) $S(x) \Rightarrow (\exists y)(\exists z)(y \neq z \wedge L(y, x) \wedge L(z, x))$
(4) $P(x) \wedge P(y) \wedge x \neq y \Rightarrow (\exists_1 z)(S(z) \wedge L(x, z) \wedge L(y, z))$
(5) $(\exists x)(\exists y)(\exists z)(P(x) \wedge P(y) \wedge P(z) \wedge \neg \mathscr{C}(x, y, z))$

where $\mathscr{C}(x, y, z)$ is the wf $(\exists u)(S(u) \wedge L(x, u) \wedge L(y, u) \wedge L(z, u))$, which is read as x, y, z *are collinear*.

(a) Translate (1)–(5) into ordinary geometric language.
(b) Prove $\vdash_{\mathbb{G}} (\forall u)(\forall v)\ (S(u) \wedge S(v) \wedge u \neq v \Rightarrow (\forall x)(\forall y)\ (L(x, u) \wedge L(x, v) \wedge L(y, u) \wedge L(y, v) \Rightarrow x = y))$, and translate this theorem into ordinary geometric language.
(c) Let $R(u, v)$ stand for $S(u) \wedge S(v) \wedge \neg(\exists w)(L(w, u) \wedge L(w, v))$. Read $R(u, v)$ as *u and v are distinct parallel lines*.
  (i) Prove: $\vdash_{\mathbb{G}} R(u, v) \Rightarrow u \neq v$
  (ii) Show that there exists a normal model of $\mathbb{G}$ with a finite domain in which the following sentence is true:

$$(\forall x)(\forall y)(S(x) \wedge P(y) \wedge \neg L(y, x) \Rightarrow (\exists_1 z)(L(y, z) \wedge R(z, x)))$$

(d) Show that there exists a model of $\mathbb{G}$ in which the following sentence is true:

$$(\forall x)(\forall y)(S(x) \wedge S(y) \wedge x \neq y \Rightarrow \neg R(x, y))$$

2.9 DEFINITIONS OF NEW FUNCTION LETTERS AND INDIVIDUAL CONSTANTS

In mathematics, once we have proved, for any $y_1, \ldots, y_n$, the existence of a unique object u that has a property $\mathscr{B}(u, y_1, \ldots, y_n)$, we often introduce a new function letter $f(y_1, \ldots, y_n)$ such that $\mathscr{B}(f(y_1, \ldots, y_n), y_1, \ldots, y_n)$ holds for all $y_1, \ldots, y_n$. In cases where we have proved the existence of a unique object u that satisfies a wf $\mathscr{B}(u)$ and $\mathscr{B}(u)$ contains u as its only free variable, then we introduce a new individual constant b such that $\mathscr{B}(b)$ holds. It is generally acknowledged that such definitions, though convenient, add nothing really new to the theory. This can be made precise in the following manner.

PROPOSITION 2.28

Let K be a theory with equality. Assume that $\vdash_K (\exists_1 u)\mathscr{B}(u, y_1, \ldots, y_n)$. Let $K^{\#}$ be the theory with equality obtained by adding to K a new function letter f of n arguments and the proper axiom $\mathscr{B}(f(y_1, \ldots, y_n), y_1, \ldots, y_n)$,[†] as well as all instances of axioms (A1)–(A7) that involve f. Then there is an effective transformation mapping each wf $\mathscr{C}$ of $K^{\#}$ into a wf $\mathscr{C}^{\#}$ of K such that:

(a) If f does not occur in $\mathscr{C}$, then $\mathscr{C}^{\#}$ is $\mathscr{C}$.
(b) $(\neg\mathscr{C})^{\#}$ is $\neg(\mathscr{C}^{\#})$.
(c) $(\mathscr{C} \Rightarrow \mathscr{D})^{\#}$ is $\mathscr{C}^{\#} \Rightarrow \mathscr{D}^{\#}$.
(d) $((\forall x)\mathscr{C})^{\#}$ is $(\forall x)(\mathscr{C}^{\#})$.
(e) $\vdash_{K^{\#}} (\mathscr{C} \Leftrightarrow \mathscr{C}^{\#})$.
(f) If $\vdash_{K^{\#}} \mathscr{C}$, then $\vdash_K \mathscr{C}^{\#}$.

Hence, if $\mathscr{C}$ does not contain f and $\vdash_{K^{\#}} \mathscr{C}$, then $\vdash_K \mathscr{C}$.

Proof

By a *simple f-term* we mean an expression $f(t_1, \ldots, t_n)$ in which $t_1, \ldots, t_n$ are terms that do not contain f. Given an atomic wf $\mathscr{C}$ of $K^{\#}$, let $\mathscr{C}^*$ be the result of replacing the leftmost occurrence of a simple term $f(t_1, \ldots, t_n)$ in $\mathscr{C}$ by the first variable v not in $\mathscr{C}$ or $\mathscr{B}$. Call the wf $(\exists v)(\mathscr{B}(v, t_1, \ldots, t_n) \wedge \mathscr{C}^*)$ the *f-transform* of $\mathscr{C}$. If $\mathscr{C}$ does not contain f, then let $\mathscr{C}$ be its own f-transform. Clearly, $\vdash_{K^{\#}} (\exists v)(\mathscr{B}(v, t_1, \ldots, t_n) \wedge \mathscr{C}^*) \Leftrightarrow \mathscr{C}$. (Here, we use $\vdash_K (\exists_1 u)\mathscr{B}(u, y_1, \ldots, y_n)$ and the axiom $\mathscr{B}(f(y_1, \ldots, y_n), y_1, \ldots, y_n)$ of $K^{\#}$.) Since the f-transform $\mathscr{C}'$ of $\mathscr{C}$ contains one less f than $\mathscr{C}$ and $\vdash_{K^{\#}} \mathscr{C}' \Leftrightarrow \mathscr{C}$, if we take successive f-transforms, eventually we obtain a wf $\mathscr{C}^{\#}$ that does not contain f and such that $\vdash_{K^{\#}} \mathscr{C}^{\#} \Leftrightarrow \mathscr{C}$. Call $\mathscr{C}^{\#}$ the *f-less transform* of $\mathscr{C}$. Extend the definition to all wfs of $K^{\#}$ by letting $(\neg\mathscr{D})^{\#}$ be $\neg(\mathscr{D}^{\#})$, $(\mathscr{D} \Rightarrow \mathscr{E})^{\#}$ be $\mathscr{D}^{\#} \Rightarrow \mathscr{E}^{\#}$, and $((\forall x)\mathscr{D})^{\#}$ be $(\forall x)\mathscr{D}^{\#}$. Properties (a)–(e) of Proposition 2.28 are then obvious. To prove property (f), it suffices, by property (e), to show that, if $\mathscr{C}$ does not contain f and $\vdash_{K^{\#}} \mathscr{C}$, then $\vdash_{K^{\#}} \mathscr{C}$. We may assume that $\mathscr{C}$ is a closed wf, since a wf and its closure are deducible from each other.

Assume that M is a model of K. Let M_1 be the normal model obtained by contracting M. We know that a wf is true for M if and only if it is true for M_1. Since $\vdash_K (\exists_1 u)\mathscr{B}(u, y_1, \ldots, y_n)$, then, for any $b_1, \ldots, b_n$ in the domain of M_1, there is a unique c in the domain of M_1 such that $\models_{M_1} \mathscr{B}[c, b_1, \ldots, b_n]$. If we define $f_1(b_1, \ldots, b_n)$ to be c, then, taking f_1 to be the interpretation of the function letter f, we obtain from M_1 a model $M^{\#}$ of $K^{\#}$. For the logical axioms of $K^{\#}$ (including the equality axioms of $K^{\#}$) are true in any normal

[†]It is better to take this axiom in the form $(\forall u)(u = f(y_1, \ldots, y_n) \Rightarrow \mathscr{B}(u, y_1, \ldots, y_n))$, since $f(y_1, \ldots, y_n)$ might not be free for u in $\mathscr{B}(u, y_1, \ldots, y_n)$.

interpretation, and the axiom $\mathscr{B}(f(y_1,\dots,y_n),y_1,\dots,y_n)$ also holds in $\mathrm{M}^\#$ by virtue of the definition of f_1. Since the other proper axioms of $\mathrm{K}^\#$ do not contain f and since they are true for M_1, they are also true for $\mathrm{M}^\#$. But $\vdash_{\mathrm{K}^\#} \mathscr{C}$. Therefore, $\mathscr{C}$ is true for $\mathrm{M}^\#$, but since $\mathscr{C}$ does not contain f, $\mathscr{C}$ is true for M_1 and hence also for M. Thus, $\mathscr{C}$ is true for every model of K. Therefore, by Corollary 2.20(a), $\vdash_{\mathrm{K}} \mathscr{C}$. (In the case where $\vdash_{\mathrm{K}} (\exists_1 u)\mathscr{B}(u)$ and $\mathscr{B}(u)$ contains only u as a free variable, we form $\mathrm{K}^\#$ by adding a new individual constant b and the axiom $\mathscr{B}(b)$. Then the analogue of Proposition 2.28 follows from practically the same proof as the one just given.)

Exercise

2.83 Find the f-less transforms of the following wfs.

(a) $(\forall x)(\exists y)(A_1^3(x,y,f(x,y_1,\dots,y_n)) \Rightarrow f(y,x,\dots,x)=x)$
(b) $A_1^1(f(y_1,\dots,y_{n-1},f(y_1,\dots,y_n))) \wedge (\exists x)A_1^2(x,f(y_1,\dots,y_n))$

Note that Proposition 2.28 also applies when we have introduced several new symbols $f_1,\dots,f_m$ because we can assume that we have added each f_i to the theory already obtained by the addition of $f_1,\dots,f_{i-1}$; then m successive applications of Proposition 2.28 are necessary. The resulting wf $\mathscr{C}^\#$ of K can be considered an $(f_1,\dots,f_m)$-free transform of $\mathscr{C}$ into the language of K.

Examples

1. In the elementary theory G of groups, one can prove $(\exists_1 y)\ x+y=0$. Then introduce a new function f of one argument, abbreviate $f(t)$ by $(-t)$, and add the new axiom $x+(-x)=0$. By Proposition 2.28, we now are not able to prove any wf of G that we could not prove before. Thus, the definition of $(-t)$ adds no really new power to the original theory.
2. In the elementary theory F of fields, one can prove that $(\exists_1 y)((x \neq 0 \wedge x\cdot y=1) \vee (x=0 \text{ and } y=0))$. We then introduce a new function letter g of one argument, abbreviate $g(t)$ by t^{-1}, and add the axiom $(x\neq 0 \wedge x\cdot x^{-1}=1)\vee(x=0 \text{ and } x^{-1}=0)$, from which one can prove $x\neq 0 \Rightarrow x\cdot x^{-1}=1$.

From Proposition 2.28 we can see that, in theories with equality, only predicate letters are needed; function letters and individual constants are dispensable. If f_j^n is a function letter, we can replace it by a new predicate letter A_k^{n+1} if we add the axiom $(\exists_1 u)A_k^{n+1}(u,y_1,\dots,y_n)$. An individual constant is to be replaced by a new predicate letter A_k^1 if we add the axiom $(\exists_1 u)A_k^1(u)$.

Example

In the elementary theory G of groups, we can replace $+$ and 0 by predicate letters A_1^3 and A_1^1 if we add the axioms $(\forall x_1)(\forall x_2)\ (\exists_1 x_3)A_1^3(x_1,x_2,x_3)$ and $(\exists_1 x_1)A_1^1(x_1)$, and if we replace axioms (a), (b), (c) and (g) by the following:

(a′) $A_1^3(x_2,x_3,u) \wedge A_1^3(x_1,u,v) \wedge A_1^3(x_1,x_2,w) \wedge A_1^3(w,x_3,y) \Rightarrow v = y$
(b′) $A_1^1(y) \wedge A_1^3(x,y,z) \Rightarrow z = x$
(c′) $(\exists y)(\forall u)(\forall v)(A_1^1(u) \wedge A_1^3(x,y,v) \Rightarrow v = u)$
(g′) $[x_1 = x_2 \wedge A_1^3(x_1,y,z) \wedge A_1^3(x_2,y,u) \wedge A_1^3(y,x_1,v) \wedge A_1^3(y,x_2,w)]$
$\Rightarrow z = u \wedge v = w$

Notice that the proof of Proposition 2.28 is highly non-constructive, since it uses semantical notions (model, truth) and is based upon Corollary 2.20(a), which was proved in a non-constructive way. Constructive syntactical proofs have been given for Proposition 2.28 (see Kleene, 1952, § 74), but, in general, they are quite complex.

Descriptive phrases of the kind 'the u such that $\mathscr{B}(u,y_1,\ldots,y_n)$' are very common in ordinary language and in mathematics. Such phrases are called *definite descriptions*. We let $\imath u(\mathscr{B}(u,y_1,\ldots,y_n))$ denote the unique object u such that $\mathscr{B}(u,y_1,\ldots,y_n)$ if there is such a unique object. If there is no such unique object, either we may let $\imath u(\mathscr{B}(u,y_1,\ldots,y_n))$ stand for some fixed object, or we may consider it meaningless. (For example, we may say that the phrases 'the present king of France' and 'the smallest integer' are meaningless or we may arbitrarily make the convention that they denote 0.) There are various ways of incorporating these $\imath$-terms in formalized theories, but since in most cases the same results are obtained by using new function letters or individual constants as above, and since they all lead to theorems similar to Proposition 2.28, we shall not discuss them any further here. For details, see Hilbert and Bernays (1934) and Rosser (1939; 1953).

2.10 PRENEX NORMAL FORMS

A wf $(Q_1y_1)\ldots(Q_ny_n)\mathscr{B}$, where each (Q_i,y_i) is either $(\forall y_i)$ or $(\exists y_i)$, y_i is different from y_j for $i \neq j$, and $\mathscr{B}$ contains no quantifiers, is said to be in *prenex normal form*. (We include the case $n = 0$, when there are no quantifiers at all.) We shall prove that, for every wf, we can construct an equivalent prenex normal form.

LEMMA 2.29

In any theory, if y is not free in $\mathscr{D}$, and $\mathscr{C}(x)$ and $\mathscr{C}(y)$ are similar, then the following hold.

(a) $\vdash ((\forall x)\mathscr{C}(x) \Rightarrow \mathscr{D}) \Leftrightarrow (\exists y)(\mathscr{C}(y) \Rightarrow \mathscr{D})$
(b) $\vdash ((\exists x)\mathscr{C}(x) \Rightarrow \mathscr{D}) \Leftrightarrow (\forall y)(\mathscr{C}(y) \Rightarrow \mathscr{D})$
(c) $\vdash (\mathscr{D} \Rightarrow (\forall x)\mathscr{C}(x)) \Leftrightarrow (\forall y)(\mathscr{D} \Rightarrow \mathscr{C}(y))$
(d) $\vdash (\mathscr{D} \Rightarrow (\exists x)\mathscr{C}(x)) \Leftrightarrow (\exists y)(\mathscr{D} \Rightarrow \mathscr{C}(y))$
(e) $\vdash \neg(\forall x)\mathscr{C} \Leftrightarrow (\exists x)\neg\mathscr{C}$
(f) $\vdash \neg(\exists x)\mathscr{C} \Leftrightarrow (\forall x)\neg\mathscr{C}$

Proof

For part (a):

1. $(\forall x)\mathscr{C}(x) \Rightarrow \mathscr{D}$	Hyp
2. $\neg(\exists y)(\mathscr{C}(y) \Rightarrow \mathscr{D})$	Hyp
3. $\neg\neg(\forall y)\neg(\mathscr{C}(y) \Rightarrow \mathscr{D})$	2, abbreviation
4. $(\forall y)\neg(\mathscr{C}(y) \Rightarrow \mathscr{D})$	3, negation elimination
5. $(\forall y)(\mathscr{C}(y) \wedge \neg\mathscr{D})$	4, tautology, Proposition 2.9(c)
6. $\mathscr{C}(y) \wedge \neg\mathscr{D}$	5, rule A4
7. $\mathscr{C}(y)$	6, conjunction elimination
8. $(\forall y)\mathscr{C}(y)$	7, Gen
9. $(\forall x)\mathscr{C}(x)$	8, Lemma 2.11, biconditional elimination
10. $\mathscr{D}$	1, 9, MP
11. $\neg\mathscr{D}$	6, conjunction elimination
12. $\mathscr{D} \wedge \neg\mathscr{D}$	10, 11, conjunction introduction
13. $(\forall x)\mathscr{C}(x) \Rightarrow \mathscr{D}$, $\neg(\exists y)(\mathscr{C}(y) \Rightarrow \mathscr{D}) \vdash \mathscr{D} \wedge \neg\mathscr{D}$	1–12
14. $(\forall x)\mathscr{C}(x) \Rightarrow \mathscr{D} \vdash (\exists y)(\mathscr{C}(y) \Rightarrow \mathscr{D})$	1–13, proof by contradiction
15. $\vdash (\forall x)\mathscr{C}(x) \Rightarrow \mathscr{D} \Rightarrow (\exists y)(\mathscr{C}(y) \Rightarrow \mathscr{D})$	1–14, Corollary 2.6

The converse is proven in the following manner.

1. $(\exists y)(\mathscr{C}(y) \Rightarrow \mathscr{D})$	Hyp
2. $(\forall x)\mathscr{C}(x)$	Hyp
3. $\mathscr{C}(b) \Rightarrow \mathscr{D}$	1, rule C
4. $\mathscr{C}(b)$	2, rule A4
5. $\mathscr{D}$	3, 4, MP
6. $(\exists y(\mathscr{C}(y) \Rightarrow \mathscr{D})$, $(\forall x)\mathscr{C}(x) \vdash_C \mathscr{D}$	1–5
7. $(\exists y)(\mathscr{C}(y) \Rightarrow \mathscr{D})$, $(\forall x)\mathscr{C}(x) \vdash \mathscr{D}$	6, Proposition 2.10
8. $\vdash (\exists y)(\mathscr{C}(y) \Rightarrow \mathscr{D}) \Rightarrow ((\forall x)\mathscr{C}(x) \Rightarrow \mathscr{D})$	1–7, Corollary 2.6 twice

Part (a) follows from the two proofs above by biconditional introduction. Parts (b)–(f) are proved easily and left as an exercise. (Part (f) is trivial, and (e) appeared as Exercise 2.33(a); (c) and (d) follow easily from (b) and (a), respectively.)

Lemma 2.29 allows us to move interior quantifiers to the front of a wf. This is the essential process in the proof of the following proposition.

PROPOSITION 2.30

There is an effective procedure for transforming any wf $\mathscr{B}$ into a wf $\mathscr{C}$ in prenex normal form such that $\vdash \mathscr{B} \Leftrightarrow \mathscr{C}$.

Proof

We describe the procedure by induction on the number k of occurrences of connectives and quantifiers in $\mathscr{B}$. (By Exercise 2.32(a, b), we may assume that the quantified variables in the prefix that we shall obtain are distinct.) If $k = 0$, then let $\mathscr{C}$ be $\mathscr{B}$ itself. Assume that we can find a corresponding $\mathscr{C}$ for all wfs with $k < n$, and assume that $\mathscr{B}$ has n occurrences of connectives and quantifiers.

Case 1. If $\mathscr{B}$ is $\neg\mathscr{D}$, then, by inductive hypothesis, we can construct a wf $\mathscr{E}$ in prenex normal form such that $\vdash \mathscr{D} \Leftrightarrow \mathscr{E}$. Hence, $\vdash \neg\mathscr{D} \Leftrightarrow \neg\mathscr{E}$ by biconditional negation. Thus, $\vdash \mathscr{B} \Leftrightarrow \neg\mathscr{E}$, and, by applying parts (e) and (f) of Lemma 2.29 and the replacement theorem (Proposition 2.9(b)), we can find a wf $\mathscr{C}$ in prenex normal form such that $\vdash \neg\mathscr{E} \Leftrightarrow \mathscr{C}$. Hence, $\vdash \mathscr{B} \Leftrightarrow \mathscr{C}$.

Case 2. If $\mathscr{B}$ is $\mathscr{D} \Rightarrow \mathscr{E}$, then, by inductive hypothesis, we can find wfs $\mathscr{D}_1$ and $\mathscr{E}_1$ in prenex normal form such that $\vdash \mathscr{D} \Leftrightarrow \mathscr{D}_1$ and $\vdash \mathscr{E} \Leftrightarrow \mathscr{E}_1$. Hence, by a suitable tautology and MP, $\vdash (\mathscr{D} \Rightarrow \mathscr{E}) \Leftrightarrow (\mathscr{D}_1 \Rightarrow \mathscr{E}_1)$, that is, $\vdash \mathscr{B} \Leftrightarrow (\mathscr{D}_1 \Rightarrow \mathscr{E}_1)$. Now, applying parts (a)–(d) of Lemma 2.29 and the replacement theorem, we can move the quantifiers in the prefixes of $\mathscr{D}_1$ and $\mathscr{E}_1$ to the front, obtaining a wf $\mathscr{C}$ in prenex normal form such that $\vdash \mathscr{B} \Leftrightarrow \mathscr{C}$.

Case 3. If $\mathscr{B}$ is $(\forall x)\mathscr{D}$, then, by inductive hypothesis, there is a wf $\mathscr{D}_1$ in prenex normal form such that $\vdash \mathscr{D} \Leftrightarrow \mathscr{D}_1$; hence, $\vdash \mathscr{B} \Leftrightarrow (\forall x)\mathscr{D}_1$ by Gen, Lemma 2.8, and MP. But $(\forall x)\mathscr{D}_1$ is in prenex normal form.

Examples

1. Let $\mathscr{B}$ be $(\forall x)(A_1^1(x) \Rightarrow (\forall y)(A_2^2(x,y) \Rightarrow \neg(\forall z)A_3^2(y,z)))$. By part (e) of Lemma 2.29: $(\forall x)(A_1^1(x) \Rightarrow (\forall y)[A_2^2(x,y) \Rightarrow (\exists z)\neg A_3^2(y,z)])$.
   By part (d): $(\forall x)(A_1^1(x) \Rightarrow (\forall y)(\exists u)[A_2^2(x,y) \Rightarrow \neg A_3^2(y,u)])$.
   By part (c): $(\forall x)(\forall v)(A_1^1(x) \Rightarrow (\exists u)[A_2^2(x,v) \Rightarrow \neg A_3^2(v,u)])$.
   By part (d): $(\forall x)(\forall v)(\exists w)(A_1^1(x) \Rightarrow (A_2^2(x,v) \Rightarrow \neg A_3^2(v,w)))$.
   Changing bound variables: $(\forall x)(\forall y)(\exists z)(A_1^1(x) \Rightarrow (A_2^2(x,y) \Rightarrow \neg A_3^2(y,z)))$.
2. Let $\mathscr{B}$ be $A_1^2(x,y) \Rightarrow (\exists y)[A_1^1(y) \Rightarrow ([(\exists x)A_1^1(x)] \Rightarrow A_2^1(y))]$.
   By part (b): $A_1^2(x,y) \Rightarrow (\exists y)(A_1^1(y) \Rightarrow (\forall u)[A_1^1(u) \Rightarrow A_2^1(y)])$.
   By part (c): $A_1^2(x,y) \Rightarrow (\exists y)(\forall v)(A_1^1(y) \Rightarrow [A_1^1(v) \Rightarrow A_2^1(y)])$.
   By part (d): $(\exists w)(A_1^2(x,y) \Rightarrow (\forall v)[A_1^1(w) \Rightarrow (A_1^1(v) \Rightarrow A_2^1(w))])$.
   By part (c): $(\exists w)(\forall z)(A_1^2(x,y) \Rightarrow [A_1^1(w) \Rightarrow (A_1^1(z) \Rightarrow A_2^1(w))])$.

Exercise

2.84 Find prenex normal forms equivalent to the following wfs.

(a) $[(\forall x)(A_1^1(x) \Rightarrow A_1^2(x, y))] \Rightarrow ([(\exists y)A_1^1(y)] \Rightarrow (\exists z)A_1^2(y, z))$

(b) $(\exists x)A_1^2(x, y) \Rightarrow (A_1^1(x) \Rightarrow \neg(\exists u)A_1^2(x, u))$

A predicate calculus in which there are no function letters or individual constants and in which, for any positive integer n, there are infiitely many predicate letters with n arguments, will be called a *pure predicate calculus*. For pure predicate calculi we can find a very simple prenex normal form theorem. A wf in prenex normal form such that all existential quantifiers (if any) precede all universal quantifiers (if any) is said to be in *Skolem normal form*.

PROPOSITION 2.31

In a pure predicate calculus, there is an effective procedure assigning to each wf $\mathscr{B}$ another wf $\mathscr{S}$ in Skolem normal form such that $\vdash \mathscr{B}$ if and only if $\vdash \mathscr{S}$ (or, equivalently, by Gödel's completeness theorem, such that $\mathscr{B}$ is logically valid if and only if $\mathscr{S}$ is logically valid).

Proof

First we may assume that $\mathscr{B}$ is a closed wf, since a wf is provable if and only if its closure is provable. By Proposition 2.30 we may also assume that $\mathscr{B}$ is in prenex normal form. Let the *rank* r of $\mathscr{B}$ be the number of universal quantifiers in $\mathscr{B}$ that precede existential quantifiers. By induction on the rank, we shall describe the process for finding Skolem normal forms. Clearly, when the rank is 0, we already have the Skolem normal form. Let us assume that we can construct Skolem normal forms when the rank is less than r, and let r be the rank of $\mathscr{B}$. $\mathscr{B}$ can be written as follows: $(\exists y_1)\ldots(\exists y_n)(\forall u)\mathscr{C}(y_1, \ldots, y_n, u)$, where $\mathscr{C}(y_1, \ldots, y_n, u)$ has only $y_1, \ldots, y_n, u$ as its free variables. Let A_j^{n+1} be the first predicate letter of $n+1$ arguments that does not occur in $\mathscr{B}$. Construct the wf

$$(\mathscr{B}_1) \qquad (\exists y_1)\ldots(\exists y_n)([(\forall u)(\mathscr{C}(y_1, \ldots, y_n, u) \Rightarrow A_j^{n+1}(y_1, \ldots, y_n, u))] \Rightarrow (\forall u)A_j^{n+1}(y_1, \ldots, y_n, u))$$

Let us show that $\vdash \mathscr{B}$ if and only if $\vdash \mathscr{B}_1$. Assume $\vdash \mathscr{B}_1$. In the proof of $\mathscr{B}_1$, replace all occurrences of $A_j^{n+1}(z_1, \ldots, z_n, w)$ by $\mathscr{C}^*(z_1, \ldots, z_n, w))$, where $\mathscr{C}^*$ is obtained from $\mathscr{C}$ by replacing all bound variables having free occurrences in the proof by new variables not occurring in the proof. The result is a proof of

$$(\exists y_1)\dots(\exists y_n)(((\forall u)(\mathscr{C}(y_1,\dots,y_n,u) \Rightarrow \mathscr{C}^*(y_1,\dots y_n,u))) \\ \Rightarrow (\forall u)\mathscr{C}^*(y_1,\dots,y_n,u))$$

($\mathscr{C}^*$ was used instead of $\mathscr{C}$ so that applications of axiom (A4) would remain applications of the same axiom.) Now, by changing the bound variables back again, we see that

$$\vdash (\exists y_1)\dots(\exists y_n)[(\forall u)(\mathscr{C}(y_1,\dots,y_n,u) \Rightarrow \mathscr{C}(y_1,\dots,y_n,u)) \\ \Rightarrow (\forall u)\mathscr{C}(y_1,\dots,y_n,u)]$$

Since $\vdash (\forall u)(\mathscr{C}(y_1,\dots,y_n,u) \Rightarrow \mathscr{C}(y_1,\dots,y_n,u))$, we obtain, by the replacement theorem, $\vdash (\exists y_1)\dots(\exists y_n)(\forall u)\mathscr{C}(y_1,\dots,y_n,u)$, that is, $\vdash \mathscr{B}$. Conversely, assume that $\vdash \mathscr{B}$. By rule C, we obtain $(\forall u)\mathscr{C}(b_1,\dots,b_n,u)$. But, $\vdash (\forall u)\mathscr{D} \Rightarrow ((\forall u)(\mathscr{D} \Rightarrow \mathscr{E}) \Rightarrow (\forall u)\mathscr{E})$ (see Exercise 2.27 (a)) for any wfs $\mathscr{D}$ and $\mathscr{E}$. Hence, $\vdash_C (\forall u)(\mathscr{C}(b_1,\dots,b_n,u) \Rightarrow A_j^{n+1}(b_1,\dots,b_n,u)) \Rightarrow (\forall u)A_j^{n+1}(b_1,\dots,b_n,u)$. So, by rule E4, $\vdash_C (\exists_{y_1})\dots(\exists y_n)$ $([(\forall u)(\mathscr{C}(b_1, \dots b_n,u) \Rightarrow A_j^{n+1}(y_1,\dots,y_n,u))] \Rightarrow (\forall u)A_j^{n+1}(y_1, \dots,y_n,u))$, that is, $\vdash_C \mathscr{B}_1$. By Proposition 2.10, $\vdash \mathscr{B}_1$. A prenex normal form of $\mathscr{B}_1$ has the form $\mathscr{B}_2$: $(\exists y_1)\dots(\exists y_n)(\exists u)(Q_1z_1)\dots(Q_sz_s)(\forall v)\mathscr{G}$, where $\mathscr{G}$ has no quantifiers and $(Q_1,z_1)\dots(Q_sz_s)$ is the prefix of $\mathscr{C}$. [In deriving the prenex normal form, first, by Lemma 2.29(a), we pull out the first $(\forall u)$, which changes to $(\exists u)$; then we pull out of the first conditional the quantifiers in the prefix of $\mathscr{C}$. By Lemma 2.29(a, b), this exchanges existential and universal quantifiers, but then we again pull these out of the second conditional of $\mathscr{B}_1$, which brings the prefix back to its original form. Finally, by Lemma 2.29(c), we bring the second $(\forall u)$ out to the prefix, changing it to a new quantifier $(\forall v)$.] Clearly, $\mathscr{B}_2$ has rank one less than the rank of $\mathscr{B}$ and, by Proposition 2.30, $\vdash \mathscr{B}_1 \Leftrightarrow \mathscr{B}_2$. But, $\vdash \mathscr{B}$ if and only if $\vdash \mathscr{B}_1$. Hence, $\vdash \mathscr{B}$ if and only if $\vdash \mathscr{B}_2$. By inductive hypothesis, we can find a Skolem normal form for $\mathscr{B}_2$, which is also a Skolem normal form for $\mathscr{B}$.

Example

$\mathscr{B}:(\forall x)(\forall y)(\exists z)\mathscr{C}(x,y,z)$, where $\mathscr{C}$ contains no quantifiers

$\mathscr{B}_1:(\forall x)((\forall y)(\exists z)\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)) \Rightarrow (\forall x)A_j^1(x)$, where A_j^1 is not in $\mathscr{C}$.

We obtain the prenex normal form of $\mathscr{B}_1$:

$(\exists x)([(\forall y)(\exists z)\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)] \Rightarrow (\forall x)A_j^1(x))$ 2.29(a)

$(\exists x)((\exists y)[(\exists z)\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)] \Rightarrow (\forall x)A_j^1(x))$ 2.29(a)

$(\exists x)((\exists y)(\forall z)[\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)] \Rightarrow (\forall x)A_j^1(x))$ 2.29(b)

$(\exists x)(\forall y)[(\forall z)(\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)) \Rightarrow (\forall x)A_j^1(x)]$ 2.29(b)

$(\exists x)(\forall y)(\exists z)([\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)] \Rightarrow (\forall x)A_j^1(x))$ 2.29(a)

$(\exists x)(\forall y)(\exists z)(\forall v)[(\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)) \Rightarrow A_j^1(v)]$ 2.29(c)

We repeat this process again: Let $\mathscr{D}(x,y,z,v)$ be $(\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)) \Rightarrow A_j^1(v)$. Let A_k^2 not occur in $\mathscr{D}$. Form:

$$(\exists x)([(\forall y)[(\exists z)(\forall v)(\mathscr{D}(x,y,z,v)) \Rightarrow A_k^2(x,y)]] \Rightarrow (\forall y)A_k^2(x,y))$$

$$(\exists x)(\exists y)[[(\exists z)(\forall v)(\mathscr{D}(x,y,z,v)) \Rightarrow A_k^2(x,y)] \Rightarrow (\forall y)A_k^2(x,y)] \qquad 2.29(a)$$

$$(\exists x)(\exists y)(\exists z)(\forall v)([\mathscr{D}(x,y,z,v) \Rightarrow A_k^2(x,y)] \Rightarrow (\forall y)A_k^2(x,y)) \qquad 2.29(a,b)$$

$$(\exists x)(\exists y)(\exists z)(\forall v)(\forall w)([\mathscr{D}(x,y,z,v) \Rightarrow A_k^2(x,y)] \Rightarrow A_k^2(x,w)) \qquad 2.29(c)$$

Thus, a Skolem normal form of $\mathscr{B}$ is:

$$(\exists x)(\exists y)(\exists z)(\forall v)(\forall w)([((\mathscr{C}(x,y,z) \Rightarrow A_j^1(x)) \Rightarrow A_j^1(v)) \Rightarrow A_k^2(x,y)] \Rightarrow A_k^2(x,w))$$

Exercises

2.85 Find Skolem normal forms for the following wfs.
(a) $\neg(\exists x)A_1^1(x) \Rightarrow (\forall u)(\exists y)(\forall x)A_1^3(u,x,y)$
(b) $(\forall x)(\exists y)(\forall u)(\exists v)A_1^4(x,y,u,v)$

2.86 Show that there is an effective procedure that gives, for each wf $\mathscr{B}$ of a pure predicate calculus, another wf $\mathscr{D}$ of this calculus of the form $(\forall y_1)\ldots(\forall y_n)(\exists z_1)\ldots(\exists z_m)\mathscr{C}$, such that $\mathscr{C}$ is quantifier-free, $n, m \geqslant 0$, and $\mathscr{B}$ is satisfiable if and only if $\mathscr{D}$ is satisfiable. [*Hint:* Apply Proposition 2.31 to $\neg\mathscr{B}$.]

2.87 Find a Skolem normal form $\mathscr{S}$ for $(\forall x)(\exists y)A_1^2(x,y)$ and show that it is not the case that $\vdash \mathscr{S} \Leftrightarrow (\forall x)(\exists y)A_1^2(x,y)$. Hence, a Skolem normal form for a wf $\mathscr{B}$ is not necessarily logically equivalent to $\mathscr{B}$, in contradistinction to the prenex normal form given by Proposition 2.30.

2.11 ISOMORPHISM OF INTERPRETATIONS. CATEGORICITY OF THEORIES

We shall say that an interpretation M of some language $\mathscr{L}$ is *isomorphic* with an interpretation M* of $\mathscr{L}$ if and only if there is a one-one correspondence g (called an isomorphism) of the domain D of M with the domain D^* of M* such that:

1. For any predicate letter A_j^n of $\mathscr{L}$ and for any $b_1, \ldots, b_n$ in D, $\models_{\mathrm{M}} A_j^n[b_1, \ldots, b_n]$ if and only if $\models_{\mathrm{M}^*} A_j^n[g(b_1), \ldots, g(b_n)]$.
2. For any function letter f_j^n of $\mathscr{L}$ and for any $b_1, \ldots, b_n$ in D, $g((f_j^n)^{\mathrm{M}}(b_1, \ldots, b_n)) = (f_j^n)^{\mathrm{M}^*}(g(b_1), \ldots, g(b_n))$.
3. For any individual constant a_j of $\mathscr{L}$, $g((a_j)^{\mathrm{M}}) = (a_j)^{\mathrm{M}^*}$.

   The notation $\mathrm{M} \approx \mathrm{M}^*$ will be used to indicate that M is isomorphic with M*. Notice that, if $\mathrm{M} \approx \mathrm{M}^*$, then the domains of M and M* must be of the same cardinality.

PROPOSITION 2.32

If g is an isomorphism of M with M*, then:

(a) for any wf $\mathscr{B}$ of $\mathscr{L}$, any sequence $s = (b_1, b_2, \ldots)$ of elements of the domain D of M, and the corresponding sequence $g(s) = (g(b_1), g(b_2), \ldots)$, s satisfies $\mathscr{B}$ in M if and only if $g(s)$ satisfies $\mathscr{B}$ in M*;
(b) hence, $\models_{\mathrm{M}} \mathscr{B}$ if and only if $\models_{\mathrm{M}^*} \mathscr{B}$.

Proof

Part (b) follows directly from part (a). The proof of part (a) is by induction on the number of connectives and quantifiers in $\mathscr{B}$ and is left as an exercise.

From the definition of isomorphic interpretations and Proposition 2.32 we see that isomorphic interpretations have the same 'structure' and, thus, differ in no essential way.

Exercises

2.88 Prove that, if M is an interpretation with domain D and D^* is a set that has the same cardinality as D, then one can define an interpretation M* with domain D^* such that M is isomorphic with M*.
2.89 Prove the following: (a) M is isomorphic with M. (b) If M_1 is isomorphic with M_2, then M_2 is isomorphic with M_1. (c) If M_1 is isomorphic with M_2 and M_2 is isomorphic with M_3, then M_1 is isomorphic with M_3.

A theory with equality K is said to be $\mathfrak{m}-$*categorical*, where $\mathfrak{m}$ is a cardinal number, if and only if: any two normal models of K of cardinality $\mathfrak{m}$ are isomorphic; and K has at least one normal model of cardinality $\mathfrak{m}$ (see Łoś, 1954c).

Examples

1. Let K^2 be the pure theory of equality K_1 (see page 98) to which has been added axiom (E2): $(\exists x_1)(\exists x_2)(x_1 \neq x_2 \wedge (\forall x_3)\ (x_3 = x_1 \vee x_3 = x_2))$. Then K^2 is 2-categorical. Every normal model of K^2 has exactly two elements. More generally, define (En) to be:

$$(\exists x_1) \ldots (\exists x_n)\left(\bigwedge_{1 \leqslant i < j \leqslant n} x_i \neq x_j \wedge (\forall y)(y = x_1 \vee \ldots \vee y = x_n) \right)$$

   where $\bigwedge_{1 \leqslant i < j \leqslant n} x_i \neq x_j$ is the conjunction of all wfs $x_i \neq x_j$ with $1 \leqslant i < j \leqslant n$. Then, if K^n is obtained from K_1 by adding (En) as an axiom, K^n is n-categorical, and every normal model of K^n has exactly n elements.
2. The theory K_2 (see page 98) of densely ordered sets with neither first nor last element is $\aleph_0-$categorical (see Kamke, 1950, p. 71: every denumer-

able normal model of K_2 is isomorphic with the model consisting of the set of rational numbers under their natural ordering). But one can prove that K_2 is not $\mathfrak{m}$–categorical for any $\mathfrak{m}$ different from $\aleph_0$.

Exercises

2.90[A] Find a theory with equality that is not $\aleph_0$–categorical but is $\mathfrak{m}$–categorical for all $\mathfrak{m} > \aleph_0$. [*Hint:* Consider the theory G_C of abelian groups (see page 98). For each integer n, let ny stand for the term $(y + y) + \ldots + y$ consisting of the sum of n ys. Add to G_C the axioms $(\mathscr{B}_n)$: $(\forall x)(\exists_1 y)(ny = x)$ for all $n \geqslant 2$. The new theory is the theory of uniquely divisible abelian groups. Its normal models are essentially vector spaces over the field of rational numbers. However, any two vector spaces over the rational numbers of the same non-denumerable cardinality are isomorphic, and there are denumerable vector spaces over the rational numbers that are not isomorphic (see Bourbaki, 1947).]

2.91[A] Find a theory with equality that is $\mathfrak{m}$–categorical for all infinite cardinals $\mathfrak{m}$. [*Hint:* Add to the theory G_C of abelian groups the axiom $(\forall x_1)(2x_1 = 0)$. The normal models of this theory are just the vector spaces over the field of integers modulo 2. Any two such vector spaces of the same cardinality are isomorphic (see Bourbaki, 1947).]

2.92 Show that the theorems of the theory K^n in Example 1 above are precisely the set of all wfs of K^n that are true in all normal models of cardinality n.

2.93[A] Find two non-isomorphic densely ordered sets of cardinality $2^{\aleph_0}$ with neither first nor last element. (This shows that the theory K_2 of Example 2 is not $2^{\aleph_0}$–categorical.)

Is there a theory with equality that is $\mathfrak{m}$–categorical for some non-countable cardinal $\mathfrak{m}$ but not $\mathfrak{n}$–categorical for some other non-countable cardinal $\mathfrak{n}$? In Example 2 we found a theory that is only $\aleph_0$-categorical; in Exercise 2.90 we found a theory that is $\mathfrak{m}$–categorical for all infinite $\mathfrak{m} > \aleph_0$ but not $\aleph_0$–categorical, and in Exercise 2.91, a theory that is $\mathfrak{m}$–categorical for all infinite $\mathfrak{m}$. The elementary theory G of groups is not $\mathfrak{m}$–categorical for any infinite $\mathfrak{m}$. The problem is whether these four cases exhaust all the possibilities. That this is so was proved by Morley (1965).

2.12 GENERALIZED FIRST-ORDER THEORIES. COMPLETENESS AND DECIDABILITY†

If, in the definition of the notion of first-order language, we allow a non-countable number of predicate letters, function letters, and individual

†Presupposed in parts of this section is a slender acquaintance with ordinal and cardinal numbers (see Chapter 4; or Kamke, 1950; or Sierpinski, 1958).

constants, we arrive at the notion of a *generalized first-order language*. The notions of *interpretation* and *model* extend in an obvious way to a generalized first-order language. A *generalized first-order theory* in such a language is obtained by taking as proper axioms any set of wfs of the language. Ordinary first-order theories are special cases of generalized first-order theories. The reader may easily check that all the results for first-order theories, through Lemma 2.12, hold also for generalized first-order theories without any changes in the proofs. Lemma 2.13 becomes Lemma 2.13′: if the set of symbols of a generalized theory K has cardinality $\aleph_\alpha$, then the set of expressions of K also can be well-ordered and has cardinality $\aleph_\alpha$. (First, fix a well-ordering of the symbols of K. Second, order the expressions by their length, which is some positive integer, and then stipulate that if e_1 and e_2 are two distinct expressions of the same length k, amd j is the first place in which they differ, then e_1 *precedes* e_2 if the jth symbol of e_1 precedes the jth symbol of e_2 according to the given well-ordering of the symbols of K.) Now, under the same assumption as for Lemma 2.13′, Lindenbaum's Lemma 2.14′ can be proved for generalized theories much as before, except that all the enumerations (of the wfs $\mathscr{B}_i$ and of the theories J_i) are transfinite, and the proof that J is consistent and complete uses transfinite induction. The analogue of Henkin's Proposition 2.17 runs as follows.

PROPOSITION 2.33

If the set of symbols of a consistent generalized theory K has cardinality $\aleph_\alpha$, then K has a model of cardinality $\aleph_\alpha$.

Proof

The original proof of Lemma 2.15 is modified in the following way. Add $\aleph_\alpha$ new individual constants $b_1, b_2, \ldots, b_\lambda, \ldots$. As before, the new theory K_0 is consistent. Let $F_1(x_{i_1}), \ldots, F_\lambda(x_{i_\lambda}), \ldots$ $(\lambda < \omega_\alpha)$ be a sequence consisting of all wfs of K_0 with exactly one free variable. Let (S_λ) be the sentence $(\exists x_{i_\lambda})\neg F_\lambda(x_{i_\lambda}) \Rightarrow \neg F_\lambda(b_{j_\lambda})$, where the sequence $b_{j_1}, b_{j_2}, \ldots b_{j_\lambda}, \ldots$ of distinct individual constants is chosen so that b_{j_λ} does not occur in $F_\beta(x_{i_\beta})$ for $\beta \leqslant \lambda$. The new theory K_∞, obtained by adding all the wfs (S_λ) as axioms, is proved to be consistent by a transfinite induction analogous to the inductive proof in Lemma 2.15. K_∞ is a scapegoat theory that is an extension of K and contains $\aleph_\alpha$ closed terms. By the extended Lindenbaum Lemma 2.14′, K_∞ can be extended to a consistent, complete scapegoat theory J with $\aleph_\alpha$ closed terms. The same proof as in Lemma 2.16 provides a model M of J of cardinality $\aleph_\alpha$.

COROLLARY 2.34

(a) If the set of symbols of a consistent generalized theory with equality K has cardinality $\aleph_\alpha$, then K has a normal model of cardinality less than or equal to $\aleph_\alpha$.

(b) If, in addition, K has an infinite normal model (or if K has arbitrarily large finite normal models), then K has a normal model of any cardinality $\aleph_\beta \geqslant \aleph_\alpha$.

(c) In particular, if K is an ordinary theory with equality (i.e., $\aleph_\alpha = \aleph_0$) and K has an infinite normal model (or if K has arbitrarily large finite normal models), then K has a normal model of any cardinality $\aleph_\beta (\beta \geqslant 0)$.

Proof

(a) The model guaranteed by Proposition 2.33 can be contracted to a normal model consisting of equivalence classes in a set of cardinality $\aleph_\alpha$. Such a set of equivalence classes has cardinality less than or equal to $\aleph_\alpha$.

(b) Assume $\aleph_\beta \geqslant \aleph_\alpha$. Let $b_1, b_2, \ldots$ be a set of new individual constants of cardinality $\aleph_\beta$, and add the axioms $b_\lambda \neq b_\mu$ for $\lambda \neq \mu$. As in the proof of Corollary 2.27, this new theory is consistent and so, by (a), has a normal model of cardinality less than or equal to $\aleph_\beta$ (since the new theory has $\aleph_\beta$ new symbols). But, because of the axioms $b_\lambda \neq b_\mu$, the normal model has exactly $\aleph_\beta$ elements.

(c) This is a special case of (b).

Exercise

2.94 If the set of symbols of a predicate calculus with equality K has cardinality $\aleph_\alpha$, prove that there is an extension K$'$ of K (with the same symbols as K) such that K$'$ has normal model of cardinality $\aleph_\alpha$, but K$'$ has no normal model of cardinality less than $\aleph_\alpha$.

From Lemma 2.12 and Corollary 2.34(a, b), it follows easily that, if a generalized theory with equality K has $\aleph_\alpha$ symbols, is $\aleph_\beta$-categorical for some $\beta \geqslant \alpha$, and has no finite models, then K is complete, in the sense that, for any closed wf $\mathscr{B}$, either $\vdash_K \mathscr{B}$ or $\vdash_K \neg\mathscr{B}$ (Vaught, 1954). If not -$\vdash_K \mathscr{B}$ and not-$\vdash_K \neg\mathscr{B}$, then the theories K$' = K + \{\neg\mathscr{B}\}$ and K$'' =$ K $+ \{\mathscr{B}\}$ are consistent by Lemma 2.12, and so, by Corollary 2.34(a), there are normal models M$'$ and M$''$ of K$'$ and K$''$, respectively, of cardinality less than or equal to $\aleph_\alpha$. Since K has no finite models, M$'$ and M$''$ are infinite. Hence, by Corollary 2.34(b), there are normal models N$'$ and N$''$ of K$'$ and K$''$, respectively, of cardinality $\aleph_\beta$. By the $\aleph_\beta$-categoricity of K, N$'$ and N$''$ must be isomorphic. But, since $\neg\mathscr{B}$ is true in N′ and $\mathscr{B}$ is true in N″, this is impossible by Proposition 2.32(b). Therefore, either $\vdash_K \mathscr{B}$ or $\vdash_K \neg\mathscr{B}$.

In particular, if K is an ordinary theory with equality that has no finite models and is $\aleph_\beta$-categorical for some $\beta \geqslant 0$, then K is complete. As an example, consider the theory K_2 of densely ordered sets with neither first nor last element (see page 98). K_2 has no finite models and is $\aleph_0$-cateogrical.

If an ordinary theory K is axiomatic (i.e., one can effectively decide whether any wf is an axiom) and complete, then K is decidable, that is, there is an effective procedure to determine whether any given wf is a theorem. To see this, remember (see page 86) that if a theory is axiomatic, one can effectively enumerate the theorems. Any wf $\mathscr{B}$ is provable if and only if its closure is provable. Hence, we may confine our attention to closed wfs $\mathscr{B}$. Since K is complete, either $\mathscr{B}$ is a theorem or $\neg\mathscr{B}$ is a theorem, and, therefore, one or the other will eventually turn up in our enumeration of theorems. This provides an effective test for theoremhood. Notice, that if K is inconsistent, then every wf is a theorem and there is an obvious decision procedure; if K is consistent, then not both $\mathscr{B}$ and $\neg\mathscr{B}$ can show up as theorems and we need only wait until one or the other appears.

If an ordinary axiomatic theory with equality K has no finite models and is $\aleph_\beta$-categorical for some $\beta \geqslant 0$, then, by what we have proved, K is decidable. In particular, the theory K_2 discussed above is decidable.

In certain cases, there is a more direct method of proving completeness or decidability. Let us take as an example the theory K_2 of densely ordered sets with neither first nor last element. Langford (1927) has given the following procedure for K_2. Consider any closed wf $\mathscr{B}$. By proposition 2.30, we can assume that $\mathscr{B}$ is in prenex normal form $(Q_1 y_1) \ldots (Q_n y_n)\mathscr{C}$, where $\mathscr{C}$ contains no quantifiers. If $(Q_n y_n)$ is $(\forall y_n)$, replace $(\forall y_n)\mathscr{C}$ by $\neg(\exists y_n)\neg\mathscr{C}$. In all cases, then, we have, at the right side of the wf, $(\exists y_n)\mathscr{D}$, where $\mathscr{D}$ has no quantifiers. Any negation $x \neq y$ can be replaced by $x < y \vee y < x$, and $\neg(x < y)$ can be replaced by $x = y \vee y < x$. Hence, all negation signs can be eliminated from $\mathscr{D}$. We can now put $\mathscr{D}$ into disjunctive normal form, that is, a disjunction of conjunctions of atomic wfs (see Exercise 1.42). Now $(\exists y_n)(\mathscr{D}_1 \vee \mathscr{D}_2 \vee \ldots \vee \mathscr{D}_k)$ is equivalent to $(\exists y_n)\mathscr{D}_1 \vee (\exists y_n)\mathscr{D}_2 \vee \ldots \vee (\exists y_n)\mathscr{D}_k$. Consider each $(\exists y_n)\mathscr{D}_i$ separately. $\mathscr{D}_i$ is a conjunction of atomic wfs of the form $t < s$ and $t = s$. If $\mathscr{D}_i$ does not contain y_n, just erase $(\exists y_n)$. Note that, if a wf $\mathscr{E}$ does not contain y_n, then $(\exists y_n)(\mathscr{E} \wedge \mathscr{F})$ may be replaced by $\mathscr{E} \wedge (\exists y_n)\mathscr{F}$. Hence, we are reduced to the consideration of $(\exists y_n)\mathscr{F}$, where $\mathscr{F}$ is a conjunction of atomic wfs of the form $t < s$ or $t = s$, each of which contains y_n. Now, if one of the conjuncts is $y_n = z$ for some z different from y_n, then replace in $\mathscr{F}$ all occurrences of y_n by z and erase $(\exists y_n)$. If we have $y_n = y_n$ alone, then just erase $(\exists y_n)$. If we have $y_n = y_n$ as one conjunct among others, then erase $y_n = y_n$. If $\mathscr{F}$ has a conjunct $y_n < y_n$, then replace all of $(\exists y_n)\mathscr{F}$ by $y_n < y_n$. If $\mathscr{F}$ consists of $y_n < z_n \wedge \ldots \wedge y_n < z_j \wedge u_1 < y_n \wedge \ldots \wedge u_m < y_n$, then replace $(\exists y_n)\mathscr{F}$ by the conjunction of all the wfs $u_i < z_p$ for $1 \leqslant i \leqslant m$ and $1 \leqslant p \leqslant j$. If all the us or all the z_ps are missing, replace $(\exists y_n)\mathscr{F}$ by $y_n = y_n$. This exhausts all possibilities and, in every case, we have replaced $(\exists y_n)\mathscr{F}$ by a wf containing no quantifiers, that is, we have eliminated the quantifier $(\exists y_n)$. We are left with $(Q_1 y_1)\dots(Q_{n-1}y_{n-1})\mathscr{G}$, where $\mathscr{G}$ contains no quantifiers. Now we apply the same procedure successively to $(Q_{n-1}y_{n-1}), \dots, (Q_1 y_1)$. Finally we are left with a wf without quantifiers, built up of wfs of the form $x = x$ and $x < x$. If we replace $x = x$ by $x = x \Rightarrow x = x$ and $x < x$ by $\neg(x = x \Rightarrow x = x)$, the result is either an instance of a tautology or the negation of such an instance. Hence, by Proposition 2.1, either the result or its negation is provable. Now, one can easily check that all the replacements we have made in this whole reduction procedure applied to $\mathscr{B}$ have been replacements of wfs $\mathscr{H}$ by other wfs $\mathscr{U}$ such that $\vdash_K \mathscr{H} \Leftrightarrow \mathscr{U}$. Hence, by the replacement theorem, if our final result $\mathscr{R}$ is provable, then so is the original wf $\mathscr{B}$, and, if $\neg\mathscr{R}$ is provable, then so is $\neg\mathscr{B}$. Thus, K_2 is complete and decidable.

The method used in this proof, the successive elimination of existential quantifiers, has been applied to other theories. It yields a decision procedure (see Hilbert and Bernays, 1934, § 5) for the pure theory of equality K_1 (see page 98). It has been applied by Tarski (1951) to prove the completeness and decidability of elementary algebra (i.e., of the theory of real-closed fields; see van der Waerden, 1949) and by Szmielew (1955) to prove the decidability of the theory G_C of abelian groups.

Exercises

2.95 (Henkin, 1955) If an ordinary theory with equality K is finitely axiomatizable and $\aleph_\alpha$-categorical for some α, prove that K is decidable.

2.96 (a) Prove the decidability of the pure theory K_1 of equality.
(b) Give an example of a theory with equality that is $\aleph_\alpha$-categorical for some α, but is incomplete.

Mathematical applications

1. Let F be the elementary theory of fields (see page 98). We let n stand for the term $1 + 1 + \dots + 1$, consisting of the sum of n 1s. Then the assertion that a field has characteristic p can be expressed by the wf $\mathscr{C}_p$: $p = 0$. A field has characteristic 0 if and only if it does not have characteristic p for any prime p. Then for any closed wf $\mathscr{B}$ of F that is true for all fields of characteristic 0, there is a prime number q such that $\mathscr{B}$ is true for all fields of characteristic greater than or equal to q. To see this, notice that, if F_0 is obtained from F by adding as axioms $\neg\mathscr{C}_2, \neg\mathscr{C}_3, \dots, \neg\mathscr{C}_p, \dots$ (for all primes p), the normal models of F_0 are the fields of characteristic 0. Hence, by Exercise 2.77, $\vdash_{F_0} \mathscr{B}$. But then, for some finite set of new axioms

$\neg\mathscr{C}_{q_1}, \neg\mathscr{C}_{q_2}, \ldots, \neg\mathscr{C}_{q_n}$, we have $\neg\mathscr{C}_{q_1}, \neg\mathscr{C}_{q_2}, \ldots, \neg\mathscr{C}_{q_n} \vdash_{\mathrm{F}} \mathscr{B}$. Let q be a prime greater than all $q_1, \ldots, q_n$, In every field of characteristic greater than or equal to q, the wfs $\neg\mathscr{C}_{q_1}, \neg\mathscr{C}_{q_2}, \ldots, \neg\mathscr{C}_{q_n}$ are true; hence, $\mathscr{B}$ is also true. (Other applications in algebra may be found in A. Robinson (1951) and Cherlin (1976).)

2. A *graph* may be considered as a set with a symmetric binary relation R (i.e., the relation that holds between two vertices if and only if they are connected by an edge). Call a graph k-colourable if and only if the graph can be divided into k disjoint (possibly empty) sets such that no two elements in the same set are in the relation R. (Intuitively, these sets correspond to k colours, each colour being painted on the points in the corresponding set, with the proviso that two points connected by an edge are painted different colours.) Notice that any subgraph of a k-colourable graph is k-colourable. Now we can show that, if every finite subgraph of a graph $\mathscr{G}$ is k-colourable, and if $\mathscr{G}$ can be well-ordered, then the whole graph $\mathscr{G}$ is k-colourable. To prove this, construct the following generalized theory with equality K (Beth, 1953). There are two binary predicate letters, $A_1^2(=)$ and A_2^2 (corresponding to the relation R on $\mathscr{G}$); there are k monadic predicate letters $A_1^1, \ldots, A_k^1$ (corresponding to the k subsets into which we hope to divide the graph); and there are individual constants a_c, one for each element c of the graph $\mathscr{G}$. As proper axioms, in addition to the usual assumptions (A6) and (A7), we have the following wfs:

(I)	$\neg A_2^2(x,x)$	(irreflexivity of R)
(II)	$A_2^2(x,y) \Rightarrow A_2^2(y,x)$	(symmetry of R)
(III)	$(\forall x)(A_1^1(x) \vee A_2^1(x) \vee \ldots \vee A_k^1(x))$	(division into k classes)
(IV)	$(\forall x)\neg(A_i^1(x) \wedge A_j^1(x))$, for $1 \leqslant i < j \leqslant k$	(disjointness of the k classes)

(V) $(\forall x)(\forall y)(A_i^1(x) \wedge A_i^1(y) \Rightarrow \neg A_2^2(x,y))$ for $1 \leqslant i \leqslant k$ (two elements of the same class are not in the relation R)

(VI) $a_b \neq a_c$, for any two distinct elements b and c of $\mathscr{G}$

(VII) $A_2^2(a_b, a_c)$, if $R(b,c)$ holds in $\mathscr{G}$

Now, any finite set of these axioms involves only a finite number of the individual constants $a_{c_1}, \ldots, a_{c_n}$, and since the corresponding subgraph $\{c_1, \ldots, c_n\}$ is, by assumption, k-colourable, the given finite set of axioms has a model and is, therefore, consistent. Since any finite set of axioms is consistent, K is consistent. By Corollary 2.34(a), K has a normal model of cardinality less than or equal to the cardinality of $\mathscr{G}$. This model is a k-colourable graph and, by (VI)–(VII), has $\mathscr{G}$ as a subgraph. Hence $\mathscr{G}$ is also k-colourable. (Compare this proof with a standard mathematical proof of the same result by Bruijn and Erdös (1951). Generally, use of the method above replaces complicated applications of Tychonoff's theorem or König's Unendlichkeits lemma.)

Exercises

2.97[A] (Łoś, 1954b) A group B is said to be *orderable* if there exists a binary relation R on B that totally orders B such that, if xRy, then $(x+z)R(y+z)$ and $(z+x)R(z+y)$. Show, by a method similar to that used in Example 2 above, that a group B is orderable if and only if every finitely generated subgroup is orderable (if we assume that the set B can be well-ordered).

2.98[A] Set up a theory for algebraically closed fields of characteristic p ($\geqslant 0$) by adding to the theory F of fields the new axioms P_n, where P_n states that every non-constant polynomial of degree n has a root, as well as axioms that determine the characteristic. Show that every wf of F that holds for one algebraically closed field of characteristic 0 holds for all of them. [*Hint*: This theory is $\aleph_\beta$-categorical for $\beta > 0$, is axiomatizable, and has no finite models. See A. Robinson (1952).]

2.99 By ordinary mathematical reasoning, solve the *finite marriage problem*. Given a finite set M of m men and a set N of women such that each man knows only a finite number of women and, for $1 \leqslant k \leqslant m$, any subset of M having k elements knows at least k women of N (i.e., there are at least k women in N who know at least one of the k given men), then it is possible to marry (monogamously) all the men of M to women in N so that every man is married to a women whom he knows. [*Hint* (Halmos and Vaughn, 1950): $m = 1$ is trival. For $m > 1$, use induction, considering the cases: (I) for all k with $1 \leqslant k < m$, every set of k men knows at least $k+1$ women; and (II) for some k with $1 \leqslant k < m$, there is a set of k men knowing exactly k women.] Extend this result to the infinite case, that is, when M is infinite and well-orderable and the assumptions above hold for all finite k. [*Hint*: Construct an appropriate generalized theory with equality, analogous to that in Example 2 above, and use Corollary 2.34(a).]

2.100 Prove that there is no generalized theory with equality K, having one predicate letter $<$ in addition to $=$, such that the normal models of K are exactly those normal interpretations in which the interpretation of $<$ is a well-ordering of the domain of the interpretation.

Let $\mathscr{B}$ be a wf in prenex normal form. If $\mathscr{B}$ is not closed, form its closure instead. Suppose, for example, $\mathscr{B}$ is $(\exists y_1)(\forall y_2)(\forall y_3)(\exists y_4)(\exists y_5)(\forall y_6)$ $\mathscr{C}(y_1, y_2, y_3, y_4, y_5, y_6)$, where $\mathscr{C}$ contains no quantifiers. Erase $(\exists y_1)$ and replace y_1 in $\mathscr{C}$ by a new individual constant b_1: $(\forall y_2)(\forall y_3)(\exists y_4)(\exists y_5)(\forall y_6)$ $\mathscr{C}(b_1, y_2, y_3, y_4, y_5, y_6)$. Erase $(\forall y_2)$ and $(\forall y_3)$, obtaining $(\exists y_4)(\exists y_5)(\forall y_6)$ $\mathscr{C}(b_1, y_2, y_3, y_4, y_5, y_6)$. Now erase $(\exists y_4)$ and replace y_4 in $\mathscr{C}$ by $g(y_2, y_3)$, where g is a new function letter: $(\exists y_5)(\forall y_6)\mathscr{C}(b_1, y_2, y_3, g(y_2, y_3), y_5, y_6)$. Erase $(\exists y_5)$ and replace y_5 by $h(y_2, y_3)$, where h is another new function letter: $(\forall y_6)$ $\mathscr{C}(b_1, y_2, y_3, g(y_2, y_3), h(y_2, y_3), y_6)$. Finally, erase $(\forall y_6)$. The resulting wf $\mathscr{C}(b_1, y_2, y_3, g(y_2, y_3), h(y_2, y_3), y_6)$ contains no quantifiers and will be denoted by $\mathscr{B}^*$. Thus, by introducing new function letters and individual constants, we can eliminate the quantifiers from a wf.

Examples

1. If $\mathscr{B}$ is $(\forall y_1)(\exists y_2)(\forall y_3)(\forall y_4)(\exists y_5)\mathscr{C}(y_1, y_2, y_3, y_4, y_5)$, where $\mathscr{C}$ is quantifier-free, then $\mathscr{B}^*$ is of the form $\mathscr{C}(y_1, g(y_1), y_3, y_4, h(y_1, y_3, y_4))$.
2. If $\mathscr{B}$ is $(\exists y_1)(\exists y_2)(\forall y_3)(\forall y_4)(\exists y_5)\mathscr{C}(y_1, y_2, y_3, y_4, y_5)$, where $\mathscr{C}$ is quantifier-free, then $\mathscr{B}^*$ is of the form $\mathscr{C}(b, c, y_3, y_4, g(y_3, y_4))$.

Notice that $\mathscr{B}^* \vdash \mathscr{B}$, since we can put the quantifiers back by applications of Gen and rule E4. (To be more precise, in the process of obtaining $\mathscr{B}^*$, we drop all quantifiers and, for each existentially quantified variable y_i, we substitute a term $g(z_1, \ldots, z_k)$, where g is a new function letter and $z_1, \ldots, z_k$ are the variables that were universally quantified in the prefix preceding $(\exists y_i)$. If there are no such variables $z_1, \ldots, z_k$, we replace y_i by a new individual constant.)

PROPOSITION 2.35 (SECOND ε-THEOREM)

(Rasiowa, 1956; Hilbert and Bernays, 1939) Let K be a generalized theory. Replace each axiom $\mathscr{B}$ of K by $\mathscr{B}^*$. (The new function letters and individual constants introduced for one axiom are to be different from those introduced for another axiom.) Let K* be the generalized theory with the proper axioms $\mathscr{B}^*$. Then:

(a) If $\mathscr{D}$ is a wf of K and $\vdash_{K^*} \mathscr{D}$, then $\vdash_K \mathscr{D}$.
(b) K is consistent if and only if K* is consistent.

Proof

(a) Let $\mathscr{D}$ be a wf of K such that $\vdash_{K^*} \mathscr{D}$. Consider the ordinary theory K° whose axioms $\mathscr{B}_1, \ldots, \mathscr{B}_n$ are such that $\mathscr{B}_1^*, \ldots, \mathscr{B}_n^*$ are the axioms used in the proof of $\mathscr{D}$. Let K°* be the theory whose axioms are $\mathscr{B}_1^*, \ldots, \mathscr{B}_n^*$. Hence $\vdash_{K^{\circ *}} \mathscr{D}$. Assume that M is a denumerable model of K°. We may assume that the domain of M is the set P of positive integers (see Exercise 2.88). Let $\mathscr{B}$ be any axiom of K°. For example, suppose that $\mathscr{B}$ has the form $(\exists y_1)(\forall y_2)(\forall y_3)(\exists y_4)\mathscr{C}(y_1, y_2, y_3, y_4)$, where $\mathscr{C}$ is quantifier-free. $\mathscr{B}^*$ has the form $\mathscr{C}(b, y_2, y_3, g(y_2, y_3))$. Extend the model M step by step in the following way (noting that the domain always remains P); since $\mathscr{B}$ is true for M, $(\exists y_1)(\forall y_2)(\forall y_3)(\exists y_4)\mathscr{C}(y_1, y_2, y_3, y_4)$ is true for M. Let the interpretation b^* of b be the least positive integer y_1 such that $(\forall y_2)(\forall y_3)(\exists y_4)\mathscr{C}(y_1, y_2, y_3, y_4)$ is true for M. Hence, $(\exists y_4)\mathscr{C}(b, y_2, y_3, y_4)$ is true in this extended model. For any positive integers y_2 and y_3, let the interpretation of $g(y_2, y_3)$ be the least positive integer y_4 such that $\mathscr{C}(b, y_2, y_3, y_4)$ is true in the extended model. Hence, $\mathscr{C}(b, y_2, y_3, g(y_2, y_3))$ is true in the extended model. If we do this for all the axioms $\mathscr{B}$ of K°, we obtain a model M* of K°*. Since $\vdash_{K^{\circ *}} \mathscr{D}$, $\mathscr{D}$ is true for M*. Since M* differs from M only in having interpretations of the new individual constants and function letters, and since $\mathscr{D}$ does not contain any of those symbols, $\mathscr{D}$ is true for M. Thus, $\mathscr{D}$ is true in every denumerable model of K°. Hence, $\vdash_{K^\circ} \mathscr{D}$, by Corollary 2.20(a). Since the axioms of K° are axioms of K, we have $\vdash_K \mathscr{D}$. (For a constructive proof of an equivalent result, see Hilbert and Bernays (1939).)

b) Clearly, K* is an extension of K, since $\mathscr{B}^* \vdash \mathscr{B}$. Hence, if K* is consistent, so is K. Conversely, assume K is consistent. Let $\mathscr{D}$ be any wf of K. If K* is inconsistent, $\vdash_{K^*} \mathscr{D} \wedge \neg\mathscr{D}$. By (a), $\vdash_K \mathscr{D} \wedge \neg\mathscr{D}$, contradicting the consistency of K.

Let us use the term *generalized completeness theorem* for the proposition that every consistent generalized theory has a model. If we assume that every set can be well-ordered (or, equivalently, the axiom of choice), then the generalized completeness theorem is a consequence of Proposition 2.33.

By the *maximal ideal theorem* (MI) we mean the proposition that every proper ideal of a Boolean algebra can be extended to a maximal ideal.† This is equivalent to the Boolean representation theorem, which states that every Boolean algebra is isomorphic to a Boolean algebra of sets. (Compare Stone (1936). For the theory of Boolean algebras, see Sikorski (1960) or Mendelson (1970).) The usual proofs of the MI theorem use the axiom of choice, but it is a remarkable fact that the MI theorem is equivalent to the generalized completeness theorem, and this equivalence can be proved without using the axiom of choice.

PROPOSITION 2.36

(Loś, 1954a; Rasiowa and Sikorski, 1951; 1952) The generalized completeness theorem is equivalent to the maximal ideal theorem.

Proof

(a) Assume the generalized completeness theorem. Let B be a Boolean algebra. Construct a generalized theory with equality K having the binary function letters $\cup$ and $\cap$, the singulary function letter f_1^1 [we denote $f_1^1(t)$ by $\bar{t}$], predicate letters $=$ and A_1^1, and, for each element b in B, an individual constant a_b. By the complete description of B, we mean the following sentences: (i) $a_b \neq a_c$ if b and c are distinct elements of B; (ii) $a_b \cup a_c = a_d$ if b, c, d are elements of B such that $b \cup c = d$ in B; (iii) $a_b \cap a_c = a_e$ if b, c, e are elements of b such that $b \cap c = e$ in B; and (iv) $\bar{a}_b = a_c$ if b and c are elements of B such that $\bar{b} = c$ in B, where $\bar{b}$ denotes the complement of b. As

†Since {0} is a proper ideal of a Boolean algebra, this implies (and is implied by) the proposition that every Boolean algebra has a maximal ideal.

axioms of K we take a set of axioms for a Boolean algebra, axioms (A6) and (A7) for equality, the complete description of B, and axioms asserting that A_1^1 determines a maximal ideal (i.e., $A_1^1(x \cap \bar{x})$, $A_1^1(x) \wedge A_1^1(y) \Rightarrow A_1^1(x \cup y)$, $A_1^1(x) \Rightarrow A_1^1(x \cap y)$, $A_1^1(x) \vee A_1^1(\bar{x})$, and $\neg A_1^1(x \cup \bar{x})$). Now K is consistent, for, if there were a proof in K of a contradiction, this proof would contain only a finite number of the symbols $a_b, a_c, \ldots$—say, $a_{b_1}, \ldots, a_{b_n}$. The elements $b_1, \ldots, b_n$ generate a finite subalgebra B′ of B. Every finite Boolean algebra clearly has a maximal ideal. Hence, B′ is a model for the wfs that occur in the proof of the contradiction, and therefore the contradiction is true in B′, which is impossible. Thus, K is consistent and, by the generalized completeness theorem, K has a model. That model can be contracted to a normal model of K, which is a Boolean algebra A with a maximal ideal I. Since the complete description of B is included in the axioms of K, B is a subalgebra of A, and then $I \cap B$ is a maximal ideal in B.

(b) Assume the maximal ideal theorem. Let K be a consistent generalized theory. For each axiom $\mathscr{B}$ of *K*, form the wf $\mathscr{B}^*$ obtained by constructing a prenex normal form for $\mathscr{B}$ and then eliminating the quantifiers through the addition of new individual constants and function letters (see the example preceding the proof of Proposition 2.35). Let $K^\#$ be a new theory having the wfs $\mathscr{B}^*$, plus all instances of tautologies, as its axioms, such that its wfs contain no quantifiers and its rules of inference are modus ponens and a rule of substitution for variables (namely, substitution of terms for variables). Now, $K^\#$ is consistent, since the theorems of $K^\#$ are also theorems of the consistent K^* of Proposition 2.35. Let B be the Lindenbaum algebra determined by $K^\#$ (i.e., for any wfs $\mathscr{C}$ and $\mathscr{D}$, let $\mathscr{C}$ Eq $\mathscr{D}$ mean that $\vdash_{K^\#} \mathscr{C} \Leftrightarrow \mathscr{D}$; Eq is an equivalence relation; let $[\mathscr{C}]$ be the equivalence class of $\mathscr{C}$; define $[\mathscr{C}] \cup [\mathscr{D}] = [\mathscr{C} \vee \mathscr{D}]$, $[\mathscr{C}] \cap [\mathscr{D}] = [\mathscr{C} \wedge \mathscr{D}]$, $\overline{[\mathscr{C}]} = [\neg\mathscr{C}]$; under these operations, the set of equivalence classes is a Boolean algebra, called the Lindenbaum algebra of $K^\#$). By the maximal ideal theorem, let I be a maximal ideal in B. Define a model M of $K^\#$ having the set of terms of $K^\#$ as its domain; the individual constants and function letters are their own interpretations, and, for any predicate letter A_j^n, we say that $A_j^n(t_1, \ldots, t_n)$ is true in M if and only if $[A_j^n(t_1, \ldots, t_n)]$ is not in I. One can show easily that a wf $\mathscr{C}$ of $K^\#$ is true in M if and only if $[\mathscr{C}]$ is not in I. But, for any theorem $\mathscr{D}$ of $K^\#$, $[\mathscr{D}] = 1$, which is not in I. Hence, M is a model for $K^\#$. For any axiom $\mathscr{B}$ of K, every substitution instance of $\mathscr{B}^*(y_1, \ldots, y_n)$ is a theorem in $K^\#$; therefore, $\mathscr{B}^*(y_1, \ldots, y_n)$ is true for all $y_1, \ldots, y_n$ in the model. It follows easily, by reversing the process through which $\mathscr{B}^*$ arose from $\mathscr{B}$, that $\mathscr{B}$ is true in the model. Hence, M is a model for K.

The maximal ideal theorem (and, therefore, also the generalized completeness theorem) turns out to be strictly weaker than the axiom of choice (see Halpern, 1964).

Exercise

2.101 Show that the generalized completeness theorem implies that every set can be totally ordered (and, therefore, that the axiom of choice holds for any set of non-empty disjoint finite sets).

The natural algebraic structures corresponding to the propositional calculus are Boolean algebras (see Exercise 1.60, and Rosenbloom, 1950, chaps 1 and 2). For first-order theories, the presence of quantifiers introduces more algebraic structure. For example, if K is a first-order theory, then, in the corresponding Lindenbaum algebra B, $[(\exists x)\mathscr{B}(x)] = \Sigma_t[\mathscr{B}(t)]$, where Σ_t indicates the least upper bound in B, and t ranges over all terms of K that are free for x in $\mathscr{B}(x)$. Two types of algebraic structure have been proposed to serve as algebraic counterparts of quantification theory. The first, cylindrical algebras, have been studied extensively by Tarski, Thompson, Henkin, Monk and others (see Henkin, Monk and Tarski, 1971). The other approach is the theory of polyadic algebras, invented and developed by Halmos (1962).

2.13 ELEMENTARY EQUIVALENCE. ELEMENTARY EXTENSIONS

Two interpretations M_1 and M_2 of a generalized first-order language $\mathscr{L}$ are said to be *elementarily equivalent* (written $M_1 \equiv M_2$) if the sentences of $\mathscr{L}$ true for M_1 are the same as the sentences true for M_2. Intuitively, $M_1 \equiv M_2$ if and only if M_1 and M_2 cannot be distinguished by means of the language $\mathscr{L}$. Of course, since $\mathscr{L}$ is a generalized first-order language, $\mathscr{L}$ may have non-denumerably many symbols.

Clearly, (1) $M \equiv M$; (2) if $M_1 \equiv M_2$, then $M_2 \equiv M_1$; (3) if $M_1 \equiv M_2$ and $M_2 \equiv M_3$, then $M_1 \equiv M_3$.

Two models of a complete theory K must be elementarily equivalent, since the sentences true in these models are precisely the sentences provable in K. This applies, for example, to any two densely ordered sets without first or last elements (see page 116).

We already know, by Proposition 2.32(b), that isomorphic models are elementarily equivalent. The converse, however, is not true. Consider, for example, any complete theory K that has an infinite normal model. By Corollary 2.34(b), K has normal models of any infinite cardinality $\aleph_\alpha$. If we take two normal models of K of different cardinality, they are elementarily equivalent but not isomorphic. A concrete example is the complete theory K_2 of densely ordered sets that have neither first nor last element. The rational numbers and the real numbers, under their natural orderings, are elementarily equivalent non-isomorphic models of K_2.

Exercises

2.102 Let K_∞, the theory of infinite sets, consist of the pure theory K_1 of equality plus the axioms $\mathscr{B}_n$, where $\mathscr{B}_n$ asserts that there are at least n elements. Show that any two models of K_∞ are elementarily equivalent (see Exercises 2.66 and 2.96(a)).

2.103^D If M_1 and M_2 are elementarily equivalent normal models and M_1 is finite, prove that M_1 and M_2 are isomorphic.

2.104 Let K be a theory with equality having $\aleph_\alpha$ symbols.

(a) Prove that there are at most $2^{\aleph_\alpha}$ models of K, no two of which are elementarily equivalent.

(b) Prove that there are at most $2^{\aleph_\gamma}$ mutually non-isomorphic models of K of cardinality $\aleph_\beta$, where γ is the maximum of α and β.

2.105 Let M be any infinite normal model of a theory with equality K having $\aleph_\alpha$ symbols. Prove that, for any cardinal $\aleph_\gamma \geqslant \aleph_\alpha$, there is a normal model M* of K of cardinality $\aleph_\alpha$ such that $M \equiv M^*$.

A model M_2 of a language $\mathscr{L}$ is said to be an *extension* of a model M_1 of $\mathscr{L}$ (written $M_1 \subseteq M_2$)† if the following conditions hold:

1. The domain D_1 of M_1 is a subset of the domain D_2 of M_2.
2. For any individual constant c of $\mathscr{L}$, $c^{M_2} = c^{M_1}$, where c^{M_2} and c^{M_1} are the interpretations of c in M_2 and M_1.
3. For any function letter f_j^n of $\mathscr{L}$ and any $b_1, \ldots, b_n$ in D, $(f_j^n)^{M_2}(b_1, \ldots, b_n) = (f_j^n)^{M_1}(b_1, \ldots, b_n)$.
4. For any predicate letter A_j^n of $\mathscr{L}$ and any $b_1, \ldots, b_n$ in D, $\models M_1 A_j^n[b_1, \ldots, b_n]$ if and only if $\models_{M_2} A_j^n[b_1, \ldots, b_n]$.

When $M_1 \subseteq M_2$, one also says that M_1 is a *substructure* (or *submodel*) of M_2.

Examples

1. If $\mathscr{L}$ contains only the predicate letters $=$ and $<$, then the set of rational numbers under its natural ordering is an extension of the set of integers under its natural ordering.
2. If $\mathscr{L}$ is the language of field theory (with the predicate letter $=$, function letters $+$ and $\times$, and individual constants 0 and 1), then the field of real numbers is an extension of the field of rational numbers, the field of rational numbers is an extension of the ring of integers, and the ring of integers is an extension of the 'semiring' of non-negative integers. For any fields F_1 and F_2, $F_1 \subseteq F_2$ if and only if F_1 is a subfield of F_2 in the usual algebraic sense.

†The reader will have no occasion to confuse this use of $\subseteq$ with that for the inclusion relation.

Exercises

2.106 Prove:
(a) $M \subseteq M$;
(b) if $M_1 \subseteq M_2$ and $M_2 \subseteq M_3$, then $M_1 \subseteq M_3$;
(c) If $M_1 \subseteq M_2$ and $M_2 \subseteq M_1$, then $M_1 = M_2$.

2.107 Assume $M_1 \subseteq M_2$.
(a) Let $\mathscr{B}(x_1, \ldots, x_n)$ be a wf of the form $(\forall y_1) \ldots (\forall y_m)\mathscr{C}(x_1, \ldots x_n, y_1 \ldots, y_m)$, where $\mathscr{C}$ is quantifier-free. Show that, for any $b_1, \ldots, b_n$ in the domain of M_1, if $\models_{M_2} \mathscr{B}[b_1, \ldots, b_n]$, then $\models_{M_1} \mathscr{B}[b_1, \ldots, b_n]$. In particular, any sentence $(\forall y_1) \ldots (\forall y_m)\mathscr{C}(y_1 \ldots, y_m)$, where $\mathscr{C}$ is quantifier-free, is true in M_1 if it is true in M_2.
(b) Let $\mathscr{B}(x_1, \ldots, x_n)$ be a wf of the form $(\exists y_1) \ldots (\exists y_m)\mathscr{C}(x_1, \ldots x_n, y_1, \ldots, y_m)$, where $\mathscr{C}$ is quantifier-free. Show that, for any $b_1, \ldots, b_n$ in the domain of M_1, if $\models_{M_1} \mathscr{B}[b_1, \ldots, b_n]$,then $\models_{M_2} \mathscr{B}[b_1, \ldots, b_n]$. In particular, any sentence $(\exists y_1) \ldots (\exists y_m)\, \mathscr{C}(y_1 \ldots, y_m)$, where $\mathscr{C}$ is quantifier-free, is true in M_2 if it is true in M_1.

2.108 (a) Let K be the predicate calculus of the language of field theory. Find a model M of K and a non-empty subset X of the domain D of M such that there is no substructure of M having domain X.
(b) If K is a predicate calculus with no individual constants or function letters, show that, if M is a model of K and X is a subset of the domain D of M, then there is one and only one substructure of M having domain X.
(c) Let K be any predicate calculus. Let M be any model of K and let X be any subset of the domain D of M. Let Y be the intersection of the domains of all submodels M^* of M such that X is a subset of the domain D_{M^*} of M^*. Show that there is one and only one submodel of M having domain Y. (This submodel is called the *submodel generated* by X.)

A somewhat stronger relation between interpretations than 'extension' is useful in model theory. Let M_1 and M_2 be models of some language $\mathscr{L}$. We say that M_2 is an *elementary extension* of M_1 (written $M_1 \leqslant_e M_2$) if (1) $M_1 \subseteq M_2$ and (2) for any wf $\mathscr{B}(y_1 \ldots, y_n)$ of $\mathscr{L}$ and for any $b_1, \ldots, b_n$ in the domain D_1 of M_1, $\models_{M_1} \mathscr{B}[b_1, \ldots, b_n]$ if and only if $\models_{M_2} \mathscr{B}[b_1, \ldots, b_n]$. (In particular, for any sentence $\mathscr{B}$ of $\mathscr{L}$, $\mathscr{B}$ is true for M_1 if and only if $\mathscr{B}$ is true for M_2.) When $M_1 \leqslant_e M_2$, we shall also say that M_1 is an *elementary substructure* (or *elementary submodel*) of M_2.

It is obvious that, if $M_1 \leqslant_e M_2$, then $M_1 \subseteq M_2$ and $M_1 \equiv M_2$. The converse is not true, as the following example shows. Let G be the elementary theory of groups (see page 98). G has the predicate letter $=$, function letter $+$, and individual constant 0. Let I be the group of integers and E the group of even integers. Then $E \subseteq I$ and $I \cong E$. (The function g

such that $g(x) = 2x$ for all x in I is an isomorphism of I with E.) Consider the wf $\mathscr{B}(y)$: $(\exists x)(x + x = y)$. Then $\models_I \mathscr{B}[2]$, but not-$\models_E \mathscr{B}[2]$. Thus, I is not an elementary extension of E. (This example shows the stronger result that even assuming $M_1 \subseteq M_2$ and $M_1 \cong M_2$ does not imply $M_1 \leqslant_e M_2$.)

The following theorem provides an easy method for showing that $M_1 \leqslant_e M_2$.

PROPOSITION 2.37 (Tarski and Vaught, 1957)

Let $M_1 \subseteq M_2$. Assume the following condition:

($) For every wf $\mathscr{B}(x_1, \ldots x_k)$ of the form $(\exists y)\mathscr{C}(x_1, \ldots x_k, y)$ and for all $b_1, \ldots, b_k$ in the domain D_1 of M_1, if $\models_{M_2} \mathscr{B}[b_1, \ldots, b_k]$, then there is some d in D_1 such that $\models_{M_2} \mathscr{C}[b_1, \ldots, b_k, d]$.

Then $M_1 \leqslant_e M_2$.

Proof

Let us prove:

(*) $\models_{M_1} \mathscr{D}[b_1, \ldots, b_k]$ if and only if $\models_{M_2} \mathscr{D}[b_1, \ldots, b_k]$ for any wf $\mathscr{D}(x_1, \ldots, x_k)$ and any $b_1, \ldots, b_k$ in D_1.

The proof is by induction on the number m of connectives and quantifiers in $\mathscr{D}$. If $m = 0$, then (*) follows from clause 4 of the definition of $M_1 \subseteq M_2$. Now assume that (*) holds true for all wfs having fewer than m connectives and quantifiers.

Case 1. $\mathscr{D}$ is $\neg\mathscr{E}$. By inductive hypothesis, $\models_{M_1} \mathscr{E}[b_1, \ldots, b_k]$ if and only if $\models_{M_2} \mathscr{E}[b_1, \ldots, b_k]$. Using the fact that not - $\models_{M_1} \mathscr{E}[b_1, \ldots, b_k]$ if and only if $\models_{M_1} \neg\mathscr{E}[b_1, \ldots, b_k]$, and similarly for M_2, we obtain (*).

Case 2. $\mathscr{D}$ is $\mathscr{E} \Rightarrow \mathscr{F}$. By inductive hypothesis, $\models_{M_1} \mathscr{E}[b_1, \ldots, b_k]$ if and only if $\models_{M_2} \mathscr{E}[b_1, \ldots, b_k]$ and similarly for $\mathscr{F}$. (*) then follows easily.

Case 3. $\mathscr{D}$ is $(\exists y)\mathscr{E}(x_1, \ldots x_n, y)$. By inductive hypothesis,

(**) $\models_{M_1} \mathscr{E}[b_1, \ldots, b_k, d]$ if and only if $\models_{M_2} \mathscr{E}[b_1, \ldots, b_k, d]$, for any $b_1, \ldots, b_k, d$ in D_1.

Case 3a. Assume $\models_{M_1} (\exists y)\mathscr{E}(x_1, \ldots x_k, y)[b_1, \ldots, b_k]$ for some $b_1, \ldots, b_k$ in D_1. Then $\models_{M_1} \mathscr{E}[b_1, \ldots, b_k, d]$ for some d in D_1. So, by (**), $\models_{M_2} \mathscr{E}[b_1, \ldots, b_k, d]$. Hence, $\models_{M_2} (\exists y)\mathscr{E}(x_1, \ldots x_k, y)[b_1, \ldots, b_k]$.

Case 3b. Assume $\models_{M_2} (\exists y)\mathscr{E}(x_1, \ldots x_k, y)[b_1, \ldots, b_k]$ for some $b_1, \ldots, b_k$ in D_1. By assumption ($), there exists d in D_1 such that $\models_{M_2} \mathscr{E}[b_1, \ldots, b_k, d]$. Hence, by (**), $\models_{M_1} \mathscr{E}[b_1, \ldots, b_k, d]$ and therefore $\models_{M_1} (\exists y)\mathscr{E}(x_1, \ldots x_k, y)$ $[b_1, \ldots, b_k]$.

This completes the induction proof, since any wf is logically equivalent to a wf that can be built up from atomic wfs by forming negations, conditionals and existential quantifications.

Exercises

2.109 Prove:

(a) $M \leqslant_e M$;
(b) if $M_1 \leqslant_e M_2$ and $M_2 \leqslant_e M_3$, then $M_1 \leqslant_e M_3$;
(c) if $M_1 \leqslant_e M$ and $M_2 \leqslant_e M$ and $M_1 \subseteq M_2$, then $M_1 \leqslant_e M_2$.

2.110 Let K be the theory of totally ordered sets with equality (axioms (a)–(c) and (e)–(g) of Exercise 2.67). Let M_1 and M_2 be the models for K with domains the set of positive integers and the set of non-negative integers, respectively (under their natural orderings in both cases). Prove that $M_1 \subseteq M_2$ and $M_1 \simeq M_2$, but $M_1 \not\leqslant_e M_2$.

Let M be an interpretation of a language $\mathscr{L}$. Extend $\mathscr{L}$ to a language $\mathscr{L}^*$ by adding a new individual constant a_d for every member d of the domain of M. We can extend M to an interpretation of $\mathscr{L}^*$ by taking d as the interpretation of a_d. By the *diagram* of M we mean the set of all true sentences of M of the forms $A_j^n(a_{d_1}, \ldots, a_{d_n})$, $\neg A_j^n(a_{d_1}, \ldots, a_{d_n})$, and $f_j^n(a_{d_1}, \ldots, a_{d_n}) = a_{d_m}$. In particular, $a_{d_1} \neq a_{d_2}$ belongs to the diagram if $d_1 \neq d_2$. By the *complete diagram* of M we mean the set of all sentences of $\mathscr{L}^*$ that are true for M.

Clearly, any model $M^{\#}$ of the complete diagram of M determines an elementary extension $M^{\#\#}$ of M,† and vice versa.

Exercise

2.111 (a) Let M_1 be a denumerable normal model of an ordinary theory K with equality such that every element of the domain of M_1 is the interpretation of some closed term of K.
(i) Show that, if $M_1 \subseteq M_2$ and $M_1 \equiv M_2$, then $M_1 \leqslant_e M_2$.
(ii) Prove that there is a denumerable normal elementary extension M_3 of M_1 such that M_1 and M_3 are not isomorphic.
(b) Let K be a predicate calculus with equality having two function letters $+$ and $\times$ and two individual constants 0 and 1. Let M be the standard model of arithmetic with domain the set of natural numbers, and $+, \times, 0$ and 1 having their ordinary meaning. Prove that M has a denumerable normal elementary extension that is not isomorphic to M, that is, there is a denumerable nonstandard model of arithmetic.

†The elementary extension $M^{\#\#}$ of M is obtained from $M^{\#}$ by forgetting about the interpretations of the a_ds.

PROPOSITION 2.38 (UPWARD SKOLEM–LÖWENHEIM–TARSKI THEOREM)

Let K be a theory with equality having $\aleph_\alpha$ symbols, and let M be a normal model of K with domain of cardinality $\aleph_\beta$. Let γ be the maximum of α and β. Then, for any $\delta \geqslant \gamma$, there is a model M* of cardinality $\aleph_\delta$ such that $\mathrm{M} \neq \mathrm{M}^*$ and $\mathrm{M} \leqslant_e \mathrm{M}^*$.

Proof

Add to the complete diagram of M a set of cardinality $\aleph_\delta$ of new individual constants b_τ, together with axioms $b_\tau \neq b_\rho$ for distinct τ and ρ and axioms $b_\tau \neq a_d$ for all individual constants a_d corresponding to members d of the domain of M. This new theory $\mathrm{K}^\#$ is consistent, since M can be used as a model for any finite number of axioms of $\mathrm{K}^\#$. (If $b_{\tau_1}, \ldots, b_{\tau_k}, a_{d_1}, \ldots, a_{d_m}$ are the new individual constants in these axioms, interpret $b_{\tau_1}, \ldots, b_{\tau_k}$ as distinct elements of the domain of M different from $d_1, \ldots, d_m$.) Hence, by Corollary 2.34 (a), $\mathrm{K}^\#$ has a normal model $\mathrm{M}^\#$ of cardinality $\aleph_\delta$ such that $\mathrm{M} \subseteq \mathrm{M}^\#$, $\mathrm{M} \neq \mathrm{M}^\#$, and $\mathrm{M} \leqslant_e \mathrm{M}^\#$.

PROPOSITION 2.39 (DOWNWARD SKOLEM–LÖWENHEIM–TARSKI THEOREM)

Let K be a theory having $\aleph_\alpha$ symbols, and let M be a model of K with domain of cardinality $\aleph_\gamma \geqslant \aleph_\alpha$. Assume A is a subset of the domain D of M having cardinality $\mathfrak{n}$, and assume $\aleph_\beta$ is such that $\aleph_\gamma \geqslant \aleph_\beta \geqslant \max(\aleph_\alpha, \mathfrak{n})$. Then there is an elementary submodel M* of M of cardinality $\aleph_\beta$ and with domain D^* including A.

Proof

Since $\mathfrak{n} \leqslant \aleph_\beta \leqslant \aleph_\gamma$, we can add $\aleph_\beta$ elements of D to A to obtain a larger set B of cardinality $\aleph_\beta$. Consider any subset C of D having cardinality $\aleph_\beta$. For every wf $\mathscr{B}(y_1, \ldots, y_n, z)$ of K, and any $c_1, \ldots, c_n$ in C such that $\models_{\mathrm{M}} (\exists z)\mathscr{B}(y_1, \ldots, y_n, z)[c_1, \ldots, c_n]$, add to C the first element d of D (with respect to some fixed well-ordering of D) such that $\models_{\mathrm{M}} (\exists z)\mathscr{B}[c_1, \ldots, c_n, d]$. Denote the so-enlarged set by $C^\#$. Since K has $\aleph_\alpha$ symbols, there are $\aleph_\alpha$ wfs. Since $\aleph_\alpha \leqslant \aleph_\beta$, there are at most $\aleph_\beta$ new elements in $C^\#$ and, therefore, the cardinality of $C^\#$ is $\aleph_\beta$. Form by induction a sequence of sets $C_0, C_1, \ldots$ by setting $C_0 = B$ and $C_{n+1} = C_n^\#$. Let $D^* = \bigcup_{n \in \omega} C_n$. Then the cardinality of D^* is $\aleph_\beta$. In addition, D^* is closed under all the functions $(f_j^n)^{\mathrm{M}}$. (Assume $d_1, \ldots, d_n$ in D^*. We may assume $d_1, \ldots, d_n$ in C_k for some k. Now $\models_{\mathrm{M}} (\exists z)(f_j^n(x_1, \ldots, x_n) = z)[d_1, \ldots, d_n]$. Hence, $(f_j^n)^{\mathrm{M}}(d_1, \ldots, d_n)$, being the first and only member d of D such that $\models_M (f_j^n(x_1, \ldots, x_n) = z)[d_1, \ldots, d_n, d]$, must belong to $C_k^{\#} = C_{k+1} \subseteq D^*$.) Similarly, all interpretations $(a_j)^M$ of individual constants are in D^*. Hence, D^* determines a substructure M^* of M. To show that $M^* \leqslant_e M$, consider any wf $\mathscr{B}(y_1, \ldots, y_n, z)$ and any $d_1, \ldots, d_n$ in D^* such that $\models_M (\exists z)\mathscr{B}(y_1, \ldots, y_n, z)[d_1, \ldots, d_n]$. There exists C_k such that $d_1, \ldots, d_n$ are in C_k. Let d be the first element of D such that $\models_M \mathscr{B}[d_1, \ldots, d_n, d]$. Then $d \in C_k^{\#} = C_{k+1} \subseteq D^*$. So, by the Tarski–Vaught theorem (Proposition 2.37) $M^* \leqslant_e M$.

2.14 ULTRAPOWERS. NON-STANDARD ANALYSIS

By a *filter*† on a non-empty set A we mean a set $\mathscr{F}$ of subsets of A such that:

1. $A \in \mathscr{F}$
2. $B \in \mathscr{F} \wedge C \in \mathscr{F} \Rightarrow B \cap C \in \mathscr{F}$
3. $B \in \mathscr{F} \wedge B \subseteq C \Rightarrow C \in \mathscr{F}$

Examples

Let $B \subseteq A$. The set $\mathscr{F}_B = \{C \mid B \subseteq C \subseteq A\}$ is a filter on A. $\mathscr{F}_B$ consists of all subsets of A that include B. Any filter of the form $\mathscr{F}_B$ is called a *principal* filter. In particular, $\mathscr{F}_A = \{A\}$ and $\mathscr{F}_\emptyset = \mathscr{P}(A)$ are principal filters. The filter $\mathscr{P}(A)$ is said to be *improper* and every other filter is said to be *proper*.

Exercises

2.112 Show that a filter $\mathscr{F}$ on A is proper if and only if $\emptyset \notin \mathscr{F}$.

2.113 Show that a filter $\mathscr{F}$ on A is a principal filter if and only if the intersection of all sets in $\mathscr{F}$ is a member of $\mathscr{F}$.

2.114 Prove that every finite filter is a principal filter. In particular, any filter on a finite set A is a principal filter.

2.115 Let A be infinite and let $\mathscr{F}$ be the set of all subsets of A that are complements of finite sets: $\mathscr{F} = \{C | (\exists W)(C = A - W \wedge \text{Fin}(W)\}$, where $\text{Fin}(W)$ means that W is finite. Show that $\mathscr{F}$ is a non-principal filter on A.

2.116 Assume A has cardinality $\aleph_\beta$. Let $\aleph_\alpha \leqslant \aleph_\beta$. Let $\mathscr{F}$ be the set of all subsets of A whose complements have cardinality $< \aleph_\alpha$. Show that $\mathscr{F}$ is a non-principal filter on A.

2.117 A collection $\mathscr{G}$ of sets is said to have the *finite intersection property* if $B_1 \cap B_2 \cap \ldots \cap B_k \neq \emptyset$ for any sets $B_1, B_2, \ldots, B_k$ in $\mathscr{G}$. If $\mathscr{G}$ is a collection of

†The notion of a filter is related to that of an ideal. A subset $\mathscr{F}$ of $\mathscr{P}(A)$ is a filter on A if and only if the set $\mathscr{G} = \{A - B \mid B \in \mathscr{F}\}$ of complements of sets in $\mathscr{F}$ is an ideal in the Boolean algebra $\mathscr{P}(A)$. Remember that $\mathscr{P}(A)$ denotes the set of all subsets of A.

subsets of A having the finite intersection property and $\mathscr{H}$ is the set of all finite intersections $B_1 \cap B_2 \cap \ldots \cap B_k$ of sets in $\mathscr{G}$, show that $\mathscr{F} = \{D|(\exists C)(B \in \mathscr{H} \wedge C \subseteq D \subseteq A)\}$ is a proper filter on A.

DEFINITION

A filter $\mathscr{F}$ on a set A is called an *ultrafilter* on A if $\mathscr{F}$ is a maximal proper filter on A, that is, $\mathscr{F}$ is a proper filter on A and there is no proper filter $\mathscr{G}$ on A such that $\mathscr{F} \subset \mathscr{G}$.

Example
Let $d \in A$. The principal filter $\mathscr{F}_d = \{B|d \in B \wedge B \subseteq A\}$ is an ultrafilter on A. Assume that $\mathscr{G}$ is a filter on A such that $\mathscr{F}_d \subset \mathscr{G}$. Let $C \in \mathscr{G} - \mathscr{F}_d$. Then $C \subseteq A$ and $d \notin C$. Hence, $d \in A - C$. Thus, $A - C \in \mathscr{F}_d \subset \mathscr{G}$. Since $\mathscr{G}$ is a filter and C and $A - C$ are both in $\mathscr{G}$, then $\emptyset = C \cap (A - C) \in \mathscr{G}$. Hence, $\mathscr{G}$ is not a proper filter.

Exercises

2.118 Let $\mathscr{F}$ be a proper filter on A and assume that $B \subseteq A$ and $A - B \notin \mathscr{F}$. Prove that there is a proper filter $\mathscr{F}' \supseteq \mathscr{F}$ such that $B \in \mathscr{F}'$.
2.119 Let $\mathscr{F}$ be a proper filter on A. Prove that $\mathscr{F}$ is an ultrafilter on A if and only if, for every $B \subseteq A$, either $B \in \mathscr{F}$ or $A - B \in \mathscr{F}$.
2.120 Let $\mathscr{F}$ be a proper filter on A. Show that $\mathscr{F}$ is an ultrafilter on A if and only if, for all B and C in $\mathscr{P}(A)$, if $B \notin \mathscr{F}$ and $C \notin \mathscr{F}$, then $B \cup C \notin \mathscr{F}$.
2.121 (a) Show that every principal ultrafilter on A is of the form $\mathscr{F}_d = \{B|d \in B \wedge B \subseteq A\}$ for some d in A.
(b) Show that a non-principal ultrafilter on A contains no finite sets.
2.122 Let $\mathscr{F}$ be a filter on A and let $\mathscr{I}$ be the corresponding ideal: $B \in \mathscr{I}$ if and only if $A - B \in \mathscr{F}$. Prove that $\mathscr{F}$ is an ultrafilter on A if and only if $\mathscr{I}$ is a maximal ideal.
2.123 Let X be a chain of proper filters on A, that is, for any B and C in X, eithr $B \subseteq C$ or $C \subseteq B$. Prove that the union $\bigcup X = \{a|(\exists B)(B \in X \wedge a \in B)\}$ is a proper filter on A, and $B \subseteq \bigcup X$ for all B in X.

PROPOSITION 2.40 (ULTRAFILTER THEOREM)

Every proper filter $\mathscr{F}$ on a set A can be extended to an ultrafilter on A[†]

[†]We assume the generalized completeness theorem.

Proof

Let $\mathscr{F}$ be a proper filter on A. Let $\mathscr{I}$ be the corresponding proper ideal: $B \in \mathscr{I}$ if and only if $A - B \in \mathscr{F}$. By Proposition 2.36, every ideal can be extended to a maximal ideal. In particular, $\mathscr{I}$ can be extended to a maximal ideal $\mathscr{H}$. If we let $\mathscr{U} = \{B | A - B \in \mathscr{H}\}$, then $\mathscr{U}$ is easily seen to be an ultrafilter and $\mathscr{F} \subseteq \mathscr{U}$.

Alternatively, the existence of an ultrafilter including $\mathscr{F}$ can be proved easily on the basis of Zorn's lemma. (In fact, consider the set X of all proper filters $\mathscr{F}'$ such that $\mathscr{F} \subseteq \mathscr{F}'$. X is partially ordered by $\subset$, and any $\subset$-chain in X has an upper bound in X, namely, by Exercise 2.123, the union of all filters in the chain. Hence, by Zorn's lemma, there is a maximal element $\mathscr{F}^*$ in X, which is the required ultrafilter.) However, Zorn's lemma is equivalent to the axiom of choice, which is a stronger assumption than the generalized completeness theorem.

COROLLARY 2.41

If A is an infinite set, there exists a non-principal ultrafilter on A.

Proof

Let $\mathscr{F}$ be the filter on A consisting of all complements $A - B$ of finite subsets B of A (see Exercise 2.115). By Proposition 2.40, there is an ultrafilter $\mathscr{U} \supseteq \mathscr{F}$. Assume $\mathscr{U}$ is a principal ultrafilter. By Exercise 2.121(a), $\mathscr{U} = \mathscr{F}_d$ for some $d \in A$. Then $A - \{d\} \in \mathscr{F} \subseteq \mathscr{U}$. Also, $\{d\} \in \mathscr{U}$. Hence, $\emptyset = \{d\} \cap (A - \{d\}) \in \mathscr{U}$, contradicting the fact that an ultrafilter is proper.

Reduced direct products

We shall now study an important way of constructing models. Let K be any predicate calculus with equality. Let J be a non-empty set and, for each j in J, let M_j be some normal model of K. In other words, consider a function F assigning to each j in J some normal model. We denote $F(j)$ by M_j.

Let $\mathscr{F}$ be a filter on J. For each j in J, let D_j denote the domain of the model M_j. By the Cartesian product $\Pi_{j\in J}D_j$ we mean the set of all functions f with domain J such that $f(j) \in D_j$ for all j in J. If $f \in \Pi_{j\in J}D_j$, we shall refer to $f(j)$ as the jth *component* of f. Let us define a binary relation $=_{\mathscr{F}}$ in $\Pi_{j\in J}D_j$ as follows:

$$f =_{\mathscr{F}} g \text{ if and only if } \{j | f(j) = g(j)\} \in \mathscr{F}$$

If we think of the sets in $\mathscr{F}$ as being 'large' sets, then, borrowing a phrase from measure theory, we read $f =_{\mathscr{F}} g$ as '$f(j) = g(j)$ almost everywhere'.

It is easy to see that $=_{\mathscr{F}}$ is an equivalence relation: (1) $f =_{\mathscr{F}} f$; (2) if $f =_{\mathscr{F}} g$ then $g =_{\mathscr{F}} f$; (3) if $f =_{\mathscr{F}} g$ and $g =_{\mathscr{F}} h$, then $f =_{\mathscr{F}} h$. For the proof of (3), observe that $\{j|f(j) = g(j)\} \cap \{j|g(j) = h(j)\} \subseteq \{j|f(j) = h(j)\}$. If $\{j|f(j) = g(j)\}$ and $\{j|g(j) = h(j)\}$ are in $\mathscr{F}$, then so is their intersection and, therefore, also $\{j|f(j) = h(j)\}$.

On the basis of the equivalence relation $=_{\mathscr{F}}$, we can divide $\Pi_{j\in J}D_j$ into equivalence classes: for any f in $\Pi_{j\in J}D_j$, we define its equivalence class $f_{\mathscr{F}}$ as $\{g|f =_{\mathscr{F}} g\}$. Clearly, (1) $f \in f_{\mathscr{F}}$; (2) $f_{\mathscr{F}} = h_{\mathscr{F}}$ if and only if $f =_{\mathscr{F}} h$; and (3) if $f_{\mathscr{F}} \neq h_{\mathscr{F}}$, then $f_{\mathscr{F}} \cap h_{\mathscr{F}} = \emptyset$. We denote the set of equivalence classes $f_{\mathscr{F}}$ by $\Pi_{j\in J}D_j/\mathscr{F}$. Intuitively, $\Pi_{j\in J}D_j/\mathscr{F}$ is obtained from $\Pi_{j\in J}D_j$ by identifying (or merging) elements of $\Pi_{j\in J}D_j$ that are equal almost everywhere.

Now we shall define a model M of K with domain $\Pi_{j\in J}D_j/\mathscr{F}$.

1. Let c be any individual constant of K and let c_j be the interpretation of c in M_j. Then the interpretation of c in M will be $f_{\mathscr{F}}$, where f is the function such that $f(j) = c_j$ for all j in J. We denote f by $\{c_j\}_{j\in J}$.
2. Let f_k^n be any function letter of K and let A_k^n be any predicate letter of K. Their interpretations $(f_k^n)^{\mathrm{M}}$ and $(A_k^n)^{\mathrm{M}}$ are defined in the following manner. Let $(g_1)_{\mathscr{F}}, \dots, (g_n)_{\mathscr{F}}$ be any members of $\Pi_{j\in J}D_j/\mathscr{F}$.

(a) $(f_k^n)^{\mathrm{M}}((g_1)_{\mathscr{F}}, \dots, (g_n)_{\mathscr{F}}) = h_{\mathscr{F}}$, where $h(j) = (f_k^n)^{\mathrm{M}_j}(g_1(j), \dots, g_n(j))$ for all j in J.

(b) $(A_k^n)^{\mathrm{M}}((g_1)_{\mathscr{F}}, \dots, (g_n)_{\mathscr{F}})$ holds if and only if $\{j| \models_{\mathrm{M}_j} A_k^n[g_1(j), \dots, g_n(j)]\} \in \mathscr{F}$.

Intuitively, $(f_k^n)^{\mathrm{M}}$ is calculated componentwise, and $(A_k^n)^{\mathrm{M}}$ holds if and only if A_k^n holds in almost all components. Definitions (a) and (b) have to be shown to be independent of the choice of the representatives $g_1, \dots, g_n$ in the equivalence classes $(g_1)_{\mathscr{F}}, \dots, (g_n)_{\mathscr{F}}$: if $g_1 =_{\mathscr{F}} g_1^*, \dots, g_n =_{\mathscr{F}} g_n^*$, and $h^*(j) = (f_k^n)^{\mathrm{M}_j}(g_1^*(j), \dots, g_n^*(j))$, then (i) $h_{\mathscr{F}} =_{\mathscr{F}} h^*_{\mathscr{F}}$ and (ii) $\{j| \models_{\mathrm{M}_j} A_k^n[g_1(j), \dots, g_n(j)]\} \in \mathscr{F}$ if and only if $\{j| \models_{\mathrm{M}_j} A_k^n[g_1^*(j), \dots, g_n^*(j)]\} \in \mathscr{F}$.

Part (i) follows from the inclusion

$$\begin{aligned}&\{j|g_1(j) = g_1^*(j)\} \cap \dots \cap \{j|g_n(j) = g_n^*(j)\} \subseteq \\ &\qquad \{j|(f_k^n)^{\mathrm{M}_j}(g_1(j), \dots, g_n(j)) = (f_k^n)^{\mathrm{M}_j}(g_1^*(j), \dots, g_n^*(j))\}\end{aligned}$$

Part (ii) follows from the inclusions:

$$\begin{aligned}&\{j|g_1(j) = g_1^*(j)\} \cap \dots \cap \{j|g_n(j) = g_n^*(j)\} \subseteq \\ &\quad \{j| \models_{\mathrm{M}_j} A_k^n[g_1(j), \dots, g_n(j)] \text{ if and only if } \models_{\mathrm{M}_j} A_k^n[g_1^*(j), \dots, g_n^*(j)]\}\end{aligned}$$

and

$$\begin{aligned}&\{j| \models_{\mathrm{M}_j} A_k^n[g_1(j), \dots, g_n(j)]\} \cap \{j| \models_{\mathrm{M}_j} A_k^n[g_1(j), \dots, g_n(j)] \text{ if and} \\ &\quad \text{only if } \models_{\mathrm{M}_j} A_k^n[g_1^*(j), \dots, g_n^*(j)]\} \subseteq \{j| \models_{\mathrm{M}_j} A_k^n[g_1^*(j), \dots, g_n^*(j)]\}\end{aligned}$$

In the case of the equality relation $=$, which is an abbreviation for A_1^2,

$$(A_1^2)^{\mathrm{M}}(g_{\mathscr{F}}, h_{\mathscr{F}}) \text{ if and only if } \{j \mid \models_{\mathrm{M}_j} A_1^2[g(j), h(j)]\} \in \mathscr{F}$$
$$\text{if and only if } \{j \mid g(j) = h(j)\} \in \mathscr{F}$$
$$\text{if and only if } g =_{\mathscr{F}} h$$

that is, if and only if $g_{\mathscr{F}} = h_{\mathscr{F}}$. Hence, the interpretation $(A_1^2)^{\mathrm{M}}$ is the identity relation and the model M is normal.

The model M just defined will be denoted $\Pi_{j \in J}\mathrm{M}_j/\mathscr{F}$ and will be called a *reduced direct product*. When $\mathscr{F}$ is an ultrafilter, $\Pi_{j \in J}\mathrm{M}_j/\mathscr{F}$ is called an *ultraproduct*. When $\mathscr{F}$ is an ultrafilter and all the M_js are the same model N, then $\Pi_{j \in J}\mathrm{M}_j/\mathscr{F}$ is denoted $\mathrm{N}^J/\mathscr{F}$ and is called an *ultrapower*.

Examples

1. Choose a fixed element r of the index set J, and let $\mathscr{F}$ be the principal ultrafilter $\mathscr{F}_r = \{B \mid r \in B \wedge B \subseteq J\}$. Then, for any f, g in $\Pi_{j \in J}D_j$, $f =_{\mathscr{F}} g$ if and only if $\{j \mid f(j) = g(j)\} \in \mathscr{F}$, that is, if and only if $f(r) = g(r)$. Hence, a member of $\Pi_{j \in J}D_j/\mathscr{F}$ consists of all f in $\Pi_{j \in J}D_j$ that have the same rth component. For any predicate letter A_k^n of K and any $g_1, \ldots, g_n$ in $\Pi_{j \in J}D_j$, $\models_{\mathrm{M}} A_k^n[(g_1)_{\mathscr{F}}, \ldots, (g_n)_{\mathscr{F}}]$ if and only if $\{j \mid \models_{\mathrm{M}_j} A_k^n[g_1(j), \ldots, g_n(j)]\} \in \mathscr{F}$, that is, if and only if $\models_{\mathrm{M}_r} A_k^n[g_1(j), \ldots, g_n(j)]$. Hence, it is easy to verify that the function $\varphi : \Pi_{j \in J}D_j/\mathscr{F} \to D_r$, defined by $\varphi(g_{\mathscr{F}}) = g(r)$ is an isomorphism of $\Pi_{j \in J}\mathrm{M}_j/\mathscr{F}$ with M_r. Thus, when $\mathscr{F}$ is a principal ultrafilter, the ultraproduct $\Pi_{j \in J}\mathrm{M}_j/\mathscr{F}$ is essentially the same as one of its components and yields nothing new.
2. Let $\mathscr{F}$ be the filter $\{J\}$. Then, for any f, g in $\Pi_{j \in J}D_j$, $f =_{\mathscr{F}} g$ if and only if $\{j \mid f(j) = g(j)\} \in \mathscr{F}$, that is, if and only if $f(j) = g(j)$ for all j in J, or if and only if $f = g$. Thus, every member of $\Pi_{j \in J}D_j/\mathscr{F}$ is a singleton $\{g\}$ for some g in $\Pi_{j \in J}D_j$. Moreover, $(f_k^n)^M((g_1)_{\mathscr{F}}, \ldots, (g_n)_{\mathscr{F}}) = \{g\}$, where g is such that $g(j) = (f_k^n)^{\mathrm{M}_j}(g_1(j), \ldots, g_n(j))$ for all j in J. Also, $\models_{\mathrm{M}} A_k^n[(g_1)_{\mathscr{F}}, \ldots, (g_n)_{\mathscr{F}}]$ if and only if $\models_{\mathrm{M}_j} A_k^n[g_1(j), \ldots, g_n(j)]$ for all j in J. Hence, $\Pi_{j \in J}\mathrm{M}_j/_{\mathscr{F}}$ is, in this case, essentially the same as the ordinary 'direct product' $\Pi_{j \in J}\mathrm{M}_j$, in which the operations and relations are defined componentwise.
3. Let $\mathscr{F}$ be the improper filter $\mathscr{P}(J)$. Then, for any f, g in $\Pi_{j \in J}D_j$, $f =_{\mathscr{F}} g$ if and only if $\{j \mid f(j) = g(j)\} \in \mathscr{F}$, that is, if and only if $\{j \mid f(j) = g(j)\} \in \mathscr{P}(J)$. Thus, $f =_{\mathscr{F}} g$ for all f and g, and $\Pi_{j \in J}D_j/\mathscr{F}$ consists of only one element. For any predicate letter A_k^n, $\models_{\mathrm{M}} A_k^n[f_{\mathscr{F}}, \ldots, f_{\mathscr{F}}]$ if and only if $\{j \models_{\mathrm{M}_j} A_k^n[f(j), \ldots, f(j)])\} \in \mathscr{P}(J)$; that is, every atomic wf is true.

The basic theorem on ultraproducts is due to Loś (1955b).

PROPOSITION 2.42 (LOŚ'S THEOREM)

Let $\mathscr{F}$ be an ultrafilter on a set J and let $\mathrm{M} = \Pi_{j \in J}\mathrm{M}_j/\mathscr{F}$ be an ultraproduct.

(a) Let $s = ((g_1)_{\mathscr{F}}, (g_2)_{\mathscr{F}}, \ldots)$ be a denumerable sequence of elements of $\Pi_{j\in J} D_j/\mathscr{F}$. For each j in J, let s_j be the denumerable sequence $(g_1(j), g_2(j), \ldots)$ in D_j. Then, for any wf $\mathscr{B}$ of K, s satisfies $\mathscr{B}$ in M if and only if $\{j|s_j$ satisfies $\mathscr{B}$ in $\mathrm{M}_j\} \in \mathscr{F}$.

(b) For any sentence $\mathscr{B}$ of K, $\mathscr{B}$ is true in $\Pi_{j\in J}\mathrm{M}_j/\mathscr{F}$ if and only if $\{j| \models_{\mathrm{M}_j} \mathscr{B}\} \in \mathscr{F}$. (Thus, (b) asserts that a sentence $\mathscr{B}$ is true in an ultraproduct if and only if it is true in almost all components.)

Proof

(a) We shall use induction on the number m of connectives and quantifiers in $\mathscr{B}$. We can reduce the case $m = 0$ to the following subcases:† (i) $A^n_k(x_{i_1}, \ldots, x_{i_n})$; (ii) $x_\ell = f^n_k(x_{i_1}, \ldots, x_{i_n})$; and (iii) $x_\ell = a_k$. For subcase (i), s satisfies $A^n_k(x_{i_1}, \ldots, x_{i_n})$ if and only if $\models_{\mathrm{M}} A^n_k[(g_{i_1})_{\mathscr{F}}, \ldots, (g_{i_n})_{\mathscr{F}}]$, which is equivalent to $\{j| \models_{\mathrm{M}_j} A^n_k[g_{i_1}(j), \ldots, g_{i_n}(j)]\} \in \mathscr{F}$; that is $\{j|\ s_j$ satisfies $A^n_k\ (x_{i_1}, \ldots, x_{i_n})$ in $\mathrm{M}_j\} \in \mathscr{F}$. Subcases (ii) and (iii) are handled in similar fashion.

Now, let us assume the result holds for all wfs that have fewer than m connectives and quantifiers.

Case 1. $\mathscr{B}$ is $\neg\mathscr{C}$. By inductive hypothesis, s satisfies $\mathscr{C}$ in M if and only if $\{j|s_j$ satisfies $\mathscr{C}$ in $\mathrm{M}_j\} \in \mathscr{F}$. s satisfies $\neg\mathscr{C}$ in M if and only if $\{j|s_j$ satisfies $\mathscr{C}$ in $\mathrm{M}_j\} \notin \mathscr{F}$. But, since $\mathscr{F}$ is an ultrafilter, the last condition is equivalent, by exercise 2.119, to $\{j|s_j$ satisfies $\neg\mathscr{C}$ in $\mathrm{M}_j\} \in \mathscr{F}$.

Case 2. $\mathscr{B}$ is $\mathscr{C} \wedge \mathscr{D}$. By inductive hypothesis, s satisfies $\mathscr{C}$ in M if and only if $\{j|s_j$ satisfies $\mathscr{C}$ in $M_j\} \in \mathscr{F}$, and s satisfies $\mathscr{D}$ in M if and only if $\{j|s_j$ satisfies $\mathscr{D}$ in $\mathrm{M}_j\} \in \mathscr{F}$. Therefore, s satisfies $\mathscr{C} \wedge \mathscr{D}$ if and only if both of the indicated sets belong to $\mathscr{F}$. But, this is equivalent to their intersection belonging to $\mathscr{F}$, which, in turn, is equivalent to $\{j|s_j$ satisfies $\mathscr{C} \wedge \mathscr{D}$ in $\mathrm{M}_j\} \in \mathscr{F}$.

Case 3. $\mathscr{B}$ is $(\exists x_i)\mathscr{C}$. Assume s satisfies $(\exists x_i)\mathscr{C}$. Then there exists h in $\Pi_{j\in J} D_j$ such that s' satisfies $\mathscr{C}$ in M, where s' is the same as s except that $h_{\mathscr{F}}$ is the ith component of s'. By inductive hypothesis, s' satisfies $\mathscr{C}$ in M if and only if $\{j|s'_j$ satisfies $\mathscr{C}$ in $\mathrm{M}_j\} \in \mathscr{F}$. Hence, $\{j|s_j$ satisfies $(\exists x_i)\mathscr{C}$ in $\mathrm{M}_j\} \in \mathscr{F}$, since, if s'_j satisfies $\mathscr{C}$ in M_j then s_j satisfies $(\exists x_i)\mathscr{C}$ in M_j.

Conversely, assume $W = \{j|s_j$ satisfies$(\exists x_i)\mathscr{C}$ in $\mathrm{M}_j\} \in \mathscr{F}$. For each j in W, choose some s'_j such that s'_j is the same as s_j except in at most the ith component and s'_j satisfies $\mathscr{C}$. Now define h in $\Pi_{j\in J} D_j$ as follows: for j in W, let $h(j)$ be the ith component of s'_j, and , for $j \notin W$, choose $h(j)$ to be an

†A wf $A^n_k(t_1, \ldots, t_n)$ can be replaced by $(\forall u_1)\ldots(\forall u_n)(u_1 = t_1 \wedge \ldots \wedge u_n = t_n \Rightarrow A^n_k(u_1, \ldots, u_n))$, and a wf $x = f^n_k(t_1, \ldots, t_n)$ can be replaced by $(\forall z_1)\ldots(\forall z_n)(z_1 = t_1 \wedge \ldots \wedge z_n = t_n \Rightarrow x = f^n_k(z_1, \ldots, z_n))$. In this way, every wf is equivalent to a wf built up from wfs of the forms (i)–(iii) by applying connectives and quantifiers.

arbitrary element of D_j. Let s'' be the same as s except that its ith component is $h_{\mathscr{F}}$. Then $W \subseteq \{j | s''_j$ satisfies $\mathscr{C}$ in $M_j\} \in \mathscr{F}$. Hence, by the inductive hypothesis, s'' satisfies $\mathscr{C}$ in M. Therefore, s satisfies $(\exists x_i)\mathscr{C}$ in M.

(b) This follows from part (a) by noting that a sentence $\mathscr{B}$ is true in a model if and only if some sequence satisfies $\mathscr{B}$.

COROLLARY 2.43

If M is a model and $\mathscr{F}$ is an ultrafilter on J, and if M* is the ultrapower $M^J/\mathscr{F}$, then $M^* \equiv M$.

Proof

Let $\mathscr{B}$ be any sentence. Then, by Proposition 2.42(b), $\mathscr{B}$ is true in M* if and only if $\{j | \mathscr{B}$ is true in M $\} \in \mathscr{F}$. If $\mathscr{B}$ is true in M, $\{j | \mathscr{B}$ is true in M $\} = J \in \mathscr{F}$. If $\mathscr{B}$ is false in M, $\{j | \mathscr{B}$ is true in M $\} = \emptyset \notin \mathscr{F}$.

Corollary 2.43 can be strengthened considerably. For each c in the domain D of M, let $c^{\#}$ stand for the constant function such that $c^{\#}(j) = c$ for all j in J. Define the function ψ such that, for each c in D, $\psi(c) = (c^{\#})_{\mathscr{F}} \in D^J/\mathscr{F}$, and denote the range of ψ by $M^{\#}$. $M^{\#}$ obviously contains the interpretations in M* of the individual constants. Moreover, $M^{\#}$ is closed under the operations $(f_k^n)^{M^*}$; for $(f_k^n)^{M^*}((c_1^{\#})_{\mathscr{F}}, \ldots, (c_n^{\#})_{\mathscr{F}})$ is $h_{\mathscr{F}}$, where $h(j) = (f_k^n)^M$ $(c_1, \ldots, c_n)$ for all j in J, and $(f_k^n)^M((c_1, \ldots, c_n)$ is a fixed element b of D. So, $h_{\mathscr{F}} = (b^{\#})_{\mathscr{F}} \in M^{\#}$. Thus, $M^{\#}$ is a substructure of M*.

COROLLARY 2.44

ψ is an isomorphism of M with $M^{\#}$, and $M^{\#} \leqslant_e M^*$.

Proof

(a) By definition of $M^{\#}$, the range of ψ is $M^{\#}$.

(b) ψ is one–one. (For any c, d in D, $(c^{\#})_{\mathscr{F}} = (d^{\#})_{\mathscr{F}}$ if and only if $c^{\#} =_{\mathscr{F}} d^{\#}$, which is equivalent to $\{j | c^{\#}(j) = d^{\#}(j)\} \in \mathscr{F}$; that is, $\{j | c = d\} \in \mathscr{F}$. If $c \neq d$, $\{j | c = d\} = \emptyset \notin \mathscr{F}$, and, therefore, $\psi(c) \neq \psi(d)$.

(c) For any $c_1, \ldots, c_n$ in D, $(f_k^n)^{M^*}(\psi(c_1) \ldots, \psi(c_n)) = (f_k^n)^{M^*}((c_1^{\#})_{\mathscr{F}}, \ldots, (c_n^{\#})_{\mathscr{F}}) = h_{\mathscr{F}}$, where $h(j) = (f_k^n)^M$ $(c_1^{\#}(j), \ldots, c_n^{\#}(j)) = (f_k^n)^M(c_1, \ldots, c_n)$. Thus, $h_{\mathscr{F}} = ((f_k^n)^M(c_1, \ldots, c_n))^{\#}/\mathscr{F} = \psi((f_k^n)^M$ $(c_1, \ldots, c_n))$.

(d) $\models_{M^*} A_k^n[\psi(c_1) \ldots, \psi(c_n)]$ if and only if $\{j | \models_M A_k^n(\psi(c_1)(j), \ldots, \psi(c_n)(j))\} \in \mathscr{F}$, which is equivalent to $\{j | \models_M A_k^n(c_1, \ldots, c_n)\} \in \mathscr{F}$, that is, $\models_M A_k^n[c_1, \ldots, c_n]$. Thus, ψ is an isomorphism of M with $M^{\#}$.

To see that $M^{\#} \leqslant_e M^*$, let $\mathscr{B}$ be any wf and $(c_1^{\#})_{\mathscr{F}}, \ldots, (c_n^{\#})_{\mathscr{F}} \in M^{\#}$. Then, by proposition 2.42(a), $\models_{M^*} \mathscr{B}[(c_1^{\#})_{\mathscr{F}}, \ldots (c_n^{\#})_{\mathscr{F}}]$ if and only if $\{j \mid \models_M \mathscr{B}[c_1^{\#}(j), \ldots, c_n^{\#}(j)]\} \in \mathscr{F}$, which is equivalent to $\{j \mid \models_M \mathscr{B}[c_1, \ldots, c_n]\} \in \mathscr{F}$, which, in turn, is equivalent to $\models_M \mathscr{B}[c_1, \ldots, c_n]$, that is, to $\models_{M^{\#}} \mathscr{B}[(c_1^{\#})_{\mathscr{F}}, \ldots, (c_n^{\#})_{\mathscr{F}}]$, since ψ is an isomorphism of M with $M^{\#}$.

Exercises

2.124 (The compactness theorem again; see Exercise 2.54) If all finite subsets of a set of sentences Γ have a model, then Γ has a model.

2.125

(a) A class $\mathscr{W}$ of interpretations of a language $\mathscr{L}$ is called *elementary* if there is a set Γ of sentences of $\mathscr{L}$ such that $\mathscr{W}$ is the class of all models of Γ. Prove that $\mathscr{W}$ is elementary if and only if $\mathscr{W}$ is closed under elementary equivalence and the formation of ultraproducts.

(b) A class $\mathscr{W}$ of interpretations of a language $\mathscr{L}$ will be called *sentential* if there is a sentence $\mathscr{B}$ of $\mathscr{L}$ such that $\mathscr{W}$ is the class of all models of $\mathscr{B}$. Prove that a class $\mathscr{W}$ is sentential if and only if both $\mathscr{W}$ and its complement $\overline{\mathscr{W}}$ (all interpretations of $\mathscr{L}$ not in $\mathscr{W}$) are closed with respect to elementary equivalence and ultraproducts.

(c) Prove that theory K of fields of characteristic 0 (see page 117) is axiomatizable but not finitely axiomatizable.

Non-standard analysis

From the invention of the calculus until relatively recent times the idea of *infinitesimals* has been an intuitively meaningful tool for finding new results in analysis. The fact that there was no rigourous foundation for infinitesimals was a source of embarrassment and led mathematicians to discard them in favour of the rigorous limit ideas of Cauchy and Weierstrass. However, about forty years ago, Abraham Robinson discovered that it was possible to resurrect infinitesimals in an entirely legitimate and precise way. This can be done by constructing models that are elementarily equivalent to, but not isomorphic to, the ordered field of real numbers. Such models can be produced either by using Proposition 2.33 or as ultrapowers. We shall sketch here the method based on ultrapowers.

Let R be the set of real numbers. Let K be a generalized predicate calculus with equality having the following symbols:

1. For each real number r, there is an individual constant a_r.
2. For every n-ary operation φ on R, there is a function letter f_φ.
3. For every n-ary relation Φ on R, there is a predicate letter A_Φ.

We can think of R as forming the domain of a model $\mathscr{R}$ for K; we simply let $(a_r)^{\mathscr{R}} = r$, $(f_\varphi)^{\mathscr{R}} = \varphi$, and $(A_\Phi)^{\mathscr{R}} = \Phi$.

Let $\mathscr{F}$ be a non-principal ultrafilter on the set ω of natural numbers. We can then form the ultrapower $\mathscr{R}^* = \mathscr{R}^\omega/\mathscr{F}$. We denote the domain $R^\omega/\mathscr{F}$ of $\mathscr{R}^*$ by R^*. By Corollary 2.43, $\mathscr{R}^* \equiv \mathscr{R}$ and, therefore, $\mathscr{R}^*$ has all the properties formalizable in K that $\mathscr{R}$ possesses. Moreover, by Corollary 2.44, $\mathscr{R}^*$ has an elementary submodel $\mathscr{R}^\#$ that is an isomorphic image of $\mathscr{R}$. The domain $R^\#$ of $\mathscr{R}^\#$ consists of all elements $(c^\#)_\mathscr{F}$ corresponding to the constant functions $c^\#(i) = c$ for all i in ω. We shall sometimes refer to the members of $R^\#$ also as real numbers; the elements of $R^* - R^\#$ will be called *non-standard reals*.

That there exist non-standard reals can be shown by explicitly exhibiting one. Let $\imath(j) = j$ for all j in ω. Then $\imath_\mathscr{F} \in R^*$. However, $(c^\#)_\mathscr{F} < \imath_\mathscr{F}$ for all c in R, by virtue of Loś's theorem and the fact that $\{j|c^\#(j) < \imath(j)\} = \{j|c < j\}$, being the set of all natural numbers greater than a fixed real number, is the complement of a finite set and is, therefore, in the non-principal ultrafilter $\mathscr{F}$. $\imath_\mathscr{F}$ is an 'infinitely large' non-standard real. (The relation $<$ used in the assertion $(c^\#)_\mathscr{F} < \imath_\mathscr{F}$ is the relation on the ultrapower $\mathscr{R}^*$ corresponding to the predicate letter $<$ of K. We use the symbol $<$ instead of $(<)^{\mathscr{R}^*}$ in order to avoid excessive notation, and we shall often do the same with other relations and functions, such as $u + v$, $u \times v$, and $|u|$.)

Since $\mathscr{R}^*$ possesses all the properties of $\mathscr{R}$ formalizable in K, $\mathscr{R}^*$ is an ordered field having the real number field $\mathscr{R}^\#$ as a proper subfield. ($\mathscr{R}^*$ is non-Archimedean: the element $\imath_\mathscr{F}$ defined above is greater than all the natural numbers $(n^\#)_\mathscr{F}$ of $\mathscr{R}^*$.) Let R_1, the set of 'finite' elements of R^*, contain those elements z such that $|z| < u$ for some real number u in $R^\#$. (R_1 is easily seen to form a subring of R^*.) Let R_0 consist of 0 and the 'infinitesimals' of R^*, that is, those elements $z \neq 0$ such that $|z| < u$ for all positive real numbers u in $R^\#$. The reciprocal $1/\imath_\mathscr{F}$ is an infinitesimal.) It is not difficult to verify that R_0 is an ideal in the ring R_1. In fact, since $x \in R_1 - R_0$ implies that $1/x \in R_1 - R_0$, it can be easily proved that R_0 is a maximal ideal in R_1.

Exercises

2.126 Prove that the cardinality of R^* is $2^{\aleph_0}$.
2.127 Prove that the set R_0 is closed under the operations of $+$, $-$ and $\times$.
2.128 Prove that, if $x \in R_1$ and $y \in R_0$, then $xy \in R_0$.
2.129 Prove that, if $x \in R_1 - R_0$, then $1/x \in R_1 - R_0$.

Let $x \in R_1$. Let $A = \{u|u \in R^\# \wedge u < x\}$ and $B = \{u|u \in R^\# \wedge u > x\}$. Then $\langle A, B\rangle$ is a 'cut' and, therefore, determines a unique real number r such that (1) $(\forall x)(x \in A \Rightarrow x \leqslant r)$ and (2) $(\forall x)(x \in B \Rightarrow x \geqslant r)$.† The difference $x - r$ is 0

†See Mendelson (1973, chap. 5).

or an infinitesimal. (Proof: Assume $x - r$ is not 0 or an infinitesimal. Then $|x - r| > r_1$ for some positive real number r_1. If $x > r$, then $x - r > r_1$. So $x > r + r_1 > r$. But then $r + r_1 \in A$, contradicting condition (1). If $x < r$, then $r - x > r_1$, and so $r > r - r_1 > x$. Thus, $r - r_1 \in B$, contradicting condition (2).) The real number r such that $x - r$ is 0 or an infinitesimal is called the *standard part* of x and is denoted $\mathrm{st}(x)$. Note that, if x is itself a real number, then $\mathrm{st}(x) = x$. We shall use the notation $x \approx y$ to mean $\mathrm{st}(x) = \mathrm{st}(y)$. Clearly, $x \approx y$ if and only if $x - y$ is 0 or an infinitesimal. If $x \approx y$, we say that x and y are *infinitely close*.

Exercises

2.130 If $x \in \mathrm{R}_1$, show that there is a unique real number r such that $x - r$ is 0 or an infinitesimal. (It is necessary to check this to ensure that $\mathrm{st}(x)$ is well-defined.)

2.131 If x and y are in R_1, prove the following.

(a) $\mathrm{st}(x + y) = \mathrm{st}(x) + \mathrm{st}(y)$
(b) $\mathrm{st}(xy) = \mathrm{st}(x)\mathrm{st}(y)$
(c) $\mathrm{st}(-x) = -\mathrm{st}(x) \wedge \mathrm{st}(y - x) = \mathrm{st}(y) - \mathrm{st}(x)$
(d) $x \geqslant 0 \Rightarrow \mathrm{st}(x) \geqslant 0$
(e) $x \leqslant y \Rightarrow \mathrm{st}(x) \leqslant \mathrm{st}(y)$

The set of natural numbers is a subset of the real numbers. Therefore, in the theory K there is a predicate letter N corresponding to the property $x \in \omega$. Hence, in R^*, there is a set ω^* of elements satisfying the wf $N(x)$. An element $f_{\mathscr{F}}$ of R^* satisfies $N(x)$ if and only if $\{j|f(j) \in \omega\} \in \mathscr{F}$. In particular, the elements $n^{\#}_{\mathscr{F}}$, for $n \in \omega$, are the 'standard' members of ω^*, whereas $\imath_{\mathscr{F}}$, for example, is a 'non-standard' natural number in R^*.

Many of the properties of the real number system can be studied from the viewpoint of non-standard analysis. For example, if s is an ordinary denumerable sequence of real numbers and c is a real number, one ordinarily says that $\lim s_n = c$ if

$$(\&) \quad (\forall \varepsilon)(\varepsilon > 0 \Rightarrow (\exists n)(n \in \omega \wedge (\forall k)(k \in \omega \wedge k \geqslant n \Rightarrow |s_k - c| < \varepsilon)))$$

Since $s \in R^{\omega}$, s is a relation and, therefore, the theory K contains a predicate letter $S(n,x)$ corresponding to the relation $s_n = x$. Hence, R^* will have a relation of all pairs $\langle n,x \rangle$ satisfying $S(n,x)$. Since $\mathscr{R}^* \equiv \mathscr{R}$, this relation will be a function that is an extension of the given sequence to the larger domain ω^*. Then we have the following result.

PROPOSITION 2.45

Let s be a denumerable sequence of real numbers and c a real number. Let s^* denote the function from ω^* into R^* corresponding to s in $\mathscr{R}^*$. Then $\lim s_n = c$ if and only if $s^*(n) \approx c$ for all n in $\omega^* - \omega$. (The latter condition can be paraphrased by saying that $s^*(n)$ is infinitely close to c when n is infinitely large.)

Proof

Assume $\lim s_n = c$. Consider any positive real ε. By (&), there is a natural number n_0 such that $(\forall k)(k \in \omega \wedge k \geqslant n_0 \Rightarrow |s_k - c| < \varepsilon)$ holds in $\mathscr{R}$. Hence, the corresponding sentence $(\forall k)(k \in \omega^* \wedge k \geqslant n_0 \Rightarrow |s^*(k) - c| < \varepsilon)$ holds in $\mathscr{R}^*$. For any n in $\omega^* - \omega$, $n > n_0$ and, therefore, $|s^*(n) - c| < \varepsilon$. Since this holds for all positive real ε, $s^*(n) - c$ is 0 or an infinitesimal.

Conversely, assume $s^*(n) \approx c$ for all $n \in \omega^* - \omega$. Take any positive real ε. Fix some n_1 in $\omega^* - \omega$. Then $(\forall k)(k \geqslant n_1 \Rightarrow |s^*(k) - c| < \varepsilon)$. So the sentence $(\exists n)(n \in \omega \wedge (\forall k)(k \in \omega \wedge k \geqslant n \Rightarrow |s_k - c| < \varepsilon))$ is true for $\mathscr{R}^*$ and, therefore, also for $\mathscr{R}$. So there must be a natural number n_0 such that $(\forall k)(k \in \omega \wedge k \geqslant n_0 \Rightarrow |s_k - c| < \varepsilon)$. Since ε was an arbitrary positive real number, we have proved $\lim s_n = c$.

Exercise

2.132 Using Proposition 2.45, prove the following limit theorems for the real number system. If s and u are denumerable sequences of real numbers and c_1 and c_2 are real numbers such that $\lim s_n = c_1$ and $\lim u_n = c_2$, then:

(a) $\lim(s_n + u_n) = c_1 + c_2$;
(b) $\lim(s_n u_n) = c_1 c_2$;
(c) If $c_2 \neq 0$ and all $u_n \neq 0$, $\lim(s_n/u_n) = c_1/c_2$.

Let us now consider another important notion of analysis, continuity. Let B be a set of real numbers, let $c \in B$, and let f be a function defined on B and taking real values. One says that f is *continuous* at c if

(¶) $(\forall \varepsilon)(\varepsilon > 0 \Rightarrow (\exists \delta)(\delta > 0 \wedge (\forall x)(x \in B \wedge |x - c| < \delta \Rightarrow |\mathrm{f}(x) - \mathrm{f}(c)| < \varepsilon)))$

PROPOSITION 2.46

Let f be a real-valued function on a set B of real numbers. Let $c \in B$. Let B^* be the subset of R^* corresponding to B, and let f^* be the function corresponding to f.† Then f is continuous at c if and only if $(\forall x)(x \in B^* \wedge x \approx c \Rightarrow f^*(x) \approx f(c))$.

Exercises

2.133 Prove Proposition 2.46.

2.134 Assume f and g are real-valued functions defined on a set B of real numbers and assume that f and g are continuous at a point c in B. Using Proposition 2.46, prove the following.

(a) $f + g$ is continuous at c.
(b) $f \cdot g$ is continuous at c.

2.135 Let f be a real-valued function defined on a set B of real numbers and continuous at a point c in B, and let g be a real-valued function defined on a set A of real numbers containing the image of B under f. Assume that g is continuous at the point $f(c)$. Prove, by Proposition 2.46, that the composition $g \circ f$ is continuous at c.

2.136 Let $C \subseteq R$.

(a) C is said to be *closed* if $(\forall x)((\forall \varepsilon)[\varepsilon > 0 \Rightarrow (\exists y)(y \in C \wedge |x - y| < \varepsilon)] \Rightarrow x \in C)$. Show that C is closed if and only if every real number that is infinitely close to a member of C^* is in C. (b) C is said to be *open* if $(\forall x)(x \in C \Rightarrow (\exists \delta)(\delta > 0 \wedge (\forall y)(|y - x| < \delta \Rightarrow y \in C)))$. Show that C is open if and only if every non-standard real number that is infinitely close to a member of C is a member of C^*.

Many standard theorems of analysis turn out to have much simpler proofs within non-standard analysis. Even stronger results can be obtained by starting with a theory K that has symbols, not only for the elements, operations and relations on R, but also for sets of subsets of R, sets of sets of subsets of R, and so on. In this way, the methods of non-standard analysis can be applied to all areas of modern analysis, sometimes with original and striking results. For further development and applications, see A. Robinson (1966), Luxemburg (1969), Bernstein (1973), Stroyan and Luxemburg (1976), and Davis (1977a). A calculus textbook based on non-standard analysis has been written by Keisler (1976) and has been used in some experimental undergraduate courses.

†To be more precise, f is represented in the theory K by a predicate letter A_f, where $A_f(x, y)$ corresponds to the relation $f(x) = y$. Then the corresponding relation A_f^* in R^* determines a function f^* with domain B^*.

Exercises

2.137 A real-valued function f defined on a closed interval $[a, b] = \{x | a \leqslant x \leqslant b\}$ is said to be *uniformly continuous* if

$$(\forall \varepsilon)(\varepsilon > 0 \Rightarrow (\exists \delta)(\delta > 0 \wedge (\forall x)(\forall y)(a \leqslant x \leqslant b \wedge a \leqslant y \leqslant b \wedge |x - y| < \delta \\ \Rightarrow |f(x) - f(y)| < \varepsilon)))$$

Prove that f is uniformly continuous if and only if, for all x and y in $[a, b]^*$, $x \approx y \Rightarrow f^*(x) \approx f^*(y)$.

2.138 Prove by non-standard methods that any function continuous on $[a, b]$ is uniformly continuous on $[a, b]$.

2.15 SEMANTIC TREES

Remember that a wf is logically valid if and only if it is true for all interpretations. Since there are uncountably many interpretations, there is no simple direct way to determine logical validity. Gödel's completeness theorem (Corollary 2.19) showed that logical validity is equivalent to derivability in a predicate calculus. But, to find out whether a wf is provable in a predicate calculus, we have only a very clumsy method: start generating the theorems and watch to see whether the given wf ever appears. Our aim here is to outline a more intuitive and usable approach in the case of wfs without function letters. Throughout this section, we assume that no function letters occur in our wfs.

A wf is logically valid if and only if its negation is not satisfiable. We shall now explain a simple procedure for trying to determine satisfiability of a closed wf $\mathscr{B}$.† Our purpose is either to show that $\mathscr{B}$ is not satisfiable or to find a model for $\mathscr{B}$.

We shall construct a figure in the shape of an inverted tree. Start with the wf $\mathscr{B}$ at the top (the 'root' of the tree). We apply certain rules for writing wfs below those already obtained. These rules replace complicated wfs by simpler ones in a way that corresponds to the meaning of the connectives and quantifiers.

†Remember that a wf is logically valid if and only if its closure is logically valid. So it suffices to consider only closed wfs.

Negation:

$$
\begin{array}{ccccc}
\neg\neg\mathscr{C} & \neg(\mathscr{C} \vee \mathscr{D}) & \neg(\mathscr{C} \Rightarrow \mathscr{D}) & \neg(\forall x)\mathscr{C} & \neg(\exists x)\mathscr{C} \\
\downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
\mathscr{C} & \neg\mathscr{C} & \mathscr{C} & (\exists x)\neg\mathscr{C} & (\forall x)\neg\mathscr{C} \\
 & \neg\mathscr{D} & \neg\mathscr{D} & &
\end{array}
$$

$$
\begin{array}{ccc}
 & \neg(\mathscr{C} \wedge \mathscr{D}) & \\
\swarrow & & \searrow \\
\neg\mathscr{C} & & \neg\mathscr{D}
\end{array}
\qquad
\begin{array}{ccc}
 & \neg(\mathscr{C} \Leftrightarrow \mathscr{D}) & \\
\swarrow & & \searrow \\
\mathscr{C} & & \neg\mathscr{C} \\
\neg\mathscr{D} & & \mathscr{D}
\end{array}
$$

Conjunction:
$$
\begin{array}{c}
\mathscr{C} \wedge \mathscr{D} \\
\downarrow \\
\mathscr{C} \\
\mathscr{D}
\end{array}
$$
Disjunction:
$$
\begin{array}{ccc}
 & \mathscr{C} \vee \mathscr{D} & \\
\swarrow & & \searrow \\
\mathscr{C} & & \mathscr{D}
\end{array}
$$

Conditional:
$$
\begin{array}{ccc}
 & \mathscr{C} \Rightarrow \mathscr{D} & \\
\swarrow & & \searrow \\
\neg\mathscr{C} & & \mathscr{D}
\end{array}
$$
Biconditional:
$$
\begin{array}{ccc}
 & \mathscr{C} \Leftrightarrow \mathscr{D} & \\
\swarrow & & \searrow \\
\mathscr{C} & & \neg\mathscr{C} \\
\mathscr{D} & & \neg\mathscr{D}
\end{array}
$$

Universal quantifier:
$$
\begin{array}{c}
(\forall x)\mathscr{C}(x) \\
\downarrow \\
\mathscr{C}(b)
\end{array}
$$
(Rule U) [Here, b is any individual constant already present.]

Existential quantifier:
$$
\begin{array}{c}
(\exists x)\mathscr{C}(x) \\
\downarrow \\
\mathscr{C}(c)
\end{array}
$$
[c is a new individual constant not already in the figure.]

Note that some of the rules require a fork or branching. This occurs when the given wf implies that one of two possible situations holds.

A *branch* is a sequence of wfs starting at the top and proceeding down the figure by applications of the rules. When a wf and its negation appear in a branch, that branch becomes *closed* and no further rules need be applied to the wf at the end of the branch. Closure of a branch will be indicated by a large cross $\times$.

Inspection of the rules shows that, when a rule is applied to a wf, the usefulness of that wf has been exhausted (the formula will be said to be *discharged*) and that formula need never be subject to a rule again, except in the case of a universally quantified wf. In the latter case, whenever a new individual constant appears in a branch below the wf, rule U can be applied with that new constant. *In addition, if no further rule applications are possible along a branch and no individual constant occurs in that branch, then we must introduce a new individual constant for use in possible applications of rule U along that branch.* (The idea behind this requirement is that, if we are trying to build a model, we must introduce a symbol for at least one object that can belong to the domain of the model.)

BASIC PRINCIPLE OF SEMANTIC TREES

If all branches become closed, the original wf is unsatisfiable. If, however, a branch remains unclosed, that branch can be used to construct a model in which the original wf is true; the domain of the model consists of the individual constants that appear in that branch.

We shall discuss the justification of this principle later on. First, we shall give examples of its use.

Examples

1. To prove that $(\forall x)\mathscr{C}(x) \Rightarrow \mathscr{C}(b)$ is logically valid, we build a semantic tree starting from its negation.

$$
\begin{array}{lll}
\text{(i)} & \neg((\forall x)\mathscr{C}(x) \Rightarrow \mathscr{C}(b)) & \\
\text{(ii)} & (\forall x)\mathscr{C}(x) & \text{(i)} \\
\text{(iii)} & \neg\mathscr{C}(b) & \text{(i)} \\
\text{(iv)} & \mathscr{C}(b) & \text{(ii)} \\
 & \times &
\end{array}
$$

The number to the right of a given wf indicates the number of the line of the wf from which the given wf is derived. Since the only branch in this tree is closed, $\neg((\forall x)\mathscr{C}(x) \Rightarrow \mathscr{C}(b))$ is unsatisfiable and, therefore, $(\forall x)\mathscr{C}(x) \Rightarrow \mathscr{C}(b))$ is logically valid.

2.
$$
\begin{array}{lll}
\text{(i)} & \neg[(\forall x)(\mathscr{C}(x) \Rightarrow \mathscr{D}(x)) \Rightarrow ((\forall x)\mathscr{C}(x) \Rightarrow (\forall x)\mathscr{D}(x))] & \\
\text{(ii)} & (\forall x)(\mathscr{C}(x) \Rightarrow \mathscr{D}(x)) & \text{(i)} \\
\text{(iii)} & \neg((\forall x)\mathscr{C}(x) \Rightarrow (\forall x)\mathscr{D}(x)) & \text{(i)} \\
\text{(iv)} & (\forall x)\mathscr{C}(x) & \text{(iii)} \\
\text{(v)} & \neg(\forall x)\mathscr{D}(x) & \text{(iii)} \\
\text{(vi)} & (\exists x)\neg\mathscr{D}(x) & \text{(v)} \\
\text{(vii)} & \neg\mathscr{D}(b) & \text{(vi)} \\
\text{(viii)} & \mathscr{C}(b) & \text{(iv)} \\
\text{(ix)} & \mathscr{C}(b) \Rightarrow \mathscr{D}(b) & \text{(ii)} \\
 & \swarrow \qquad \searrow & \\
\text{(x)} & \neg\mathscr{C}(b) \qquad \mathscr{D}(b) & \text{(ix)} \\
 & \times \qquad\quad \times &
\end{array}
$$

Since both branches are closed, the original wf (i) is unsatisfiable and, therefore, $(\forall x)(\mathscr{C}(x) \Rightarrow \mathscr{D}(x)) \Rightarrow ((\forall x)\mathscr{C}(x) \Rightarrow (\forall x)\mathscr{D}(x))$ is logically valid.

3.
$$
\begin{array}{lll}
\text{(i)} & \neg[(\exists x)A_1^1(x) \Rightarrow (\forall x)A_1^1(x)] & \\
\text{(ii)} & (\exists x)A_1^1(x) & \text{(i)} \\
\text{(iii)} & \neg(\forall x)A_1^1(x) & \text{(i)} \\
\text{(iv)} & A_1^1(b) & \text{(ii)} \\
\text{(v)} & (\exists x)\neg A_1^1(x) & \text{(iii)} \\
\text{(vi)} & \neg A_1^1(c) & \text{(v)}
\end{array}
$$

No further applications of rules are possible and there is still an open branch. Define a model M with domain $\{b, c\}$ such that the interpretation of A_1^1 holds for b but not for c. Thus, $(\exists x)\neg A_1^1(x)$ is true in M but $(\forall x)A_1^1(x)$ is false in M. Hence, $(\exists x)A_1^1(x) \Rightarrow (\forall x)A_1^1(x)$ is false in M and is, therefore, not logically valid.

4. (i)	$\neg[(\exists y)(\forall x)\mathscr{B}(x,y) \Rightarrow (\forall x)(\exists y)\mathscr{B}(x,y)]$	
(ii)	$(\exists y)(\forall x)\mathscr{B}(x,y)$	(i)
(iii)	$\neg(\forall x)(\exists y)\mathscr{B}(x,y)$	(i)
(iv)	$(\forall x)\mathscr{B}(x,b)$	(ii)
(v)	$(\exists x)\neg(\exists y)\mathscr{B}(x,y)$	(iii)
(vi)	$\mathscr{B}(b,b)$	(iv)
(vii)	$\neg(\exists y)\mathscr{B}(c,y)$	(v)
(viii)	$\mathscr{B}(c,b)$	(iv)
(ix)	$(\forall y)\neg\mathscr{B}(c,y)$	(vii)
(x)	$\neg\mathscr{B}(c,b)$	(ix)
	$\times$	

Hence, $(\exists y)(\forall x)\mathscr{B}(x,y) \Rightarrow (\forall x)(\exists y)\mathscr{B}(x,y)$ is logically valid.

Notice that, in the last tree, step (vi) served no purpose but was required by our method of constructing trees. We should be a little more precise in describing that method. At each step, we apply the appropriate rule to each undischarged wf (except universally quantified wfs), starting from the top of the tree. Then, to every universally quantified wf on a given branch we apply rule U with every individual constant that has appeared on that branch since the last step. In every application of a rule to a given wf, we write the resulting wf(s) below the branch that contains that wf.

5. (i)	$\neg[(\forall x)\mathscr{B}(x) \Rightarrow (\exists x)\mathscr{B}(x)]$	
(ii)	$(\forall x)\mathscr{B}(x)$	(i)
(iii)	$\neg(\exists x)\mathscr{B}(x)$	(i)
(iv)	$(\forall x)\neg\mathscr{B}(x)$	(iii)
(v)	$\mathscr{B}(b)$	(ii)†
(vi)	$\neg\mathscr{B}(b)$	(iv)
	$\times$	

Hence, $(\forall x)\mathscr{B}(x) \Rightarrow (\exists x)\mathscr{B}(x)$ is logically valid.

6. (i)	$\neg[(\forall x)\neg A_1^2(x,x) \Rightarrow (\exists x)(\forall y)\neg A_1^2(x,y)]$	
(ii)	$(\forall x)\neg A_1^2(x,x)$	(i)
(iii)	$\neg(\exists x)(\forall y)\neg A_1^2(x,y)$	(ii)
(iv)	$(\forall x)\neg(\forall y)\neg A_1^2(x,y)$	(iii)
(v)	$\neg A_1^2(a_1,a_1)$	(ii)†
(vi)	$\neg(\forall y)\neg A_1^2(a_1,y)$	(iv)

†Here, we must introduce a new individual constant for use with rule U since, otherwise, the branch would end and would not contain any individual constants.

(vii) $(\exists y)\neg\neg A_1^2(a_1, y)$ (vi)
(viii) $\neg\neg A_1^2(a_1, a_2)$ (vii)
(ix) $A_1^2(a_1, a_2)$ (viii)
(x) $\neg A_1^2(a_2, a_2)$ (ii)
(xi) $\neg(\forall y)\neg A_1^2(a_2, y)$ (iv)
(xii) $(\exists y)\neg\neg A_1^2(a_2, y)$ (xi)
(xiii) $\neg\neg A_1^2(a_2, a_3)$ (xii)
(xiv) $A_1^2(a_2, a_3)$ (xiii)

We can see that the branch will never end and that we will obtain a sequence of constants $a_1, a_2, \ldots$ with wfs $A_1^2(a_n, a_{n+1})$ and $\neg A_1^2(a_n, a_n)$. Thus, we construct a model M with domain $\{a_1, a_2, \ldots\}$ and we define $(A_1^2)^{\mathrm{M}}$ to contain only the pairs $\langle a_n, a_{n+1}\rangle$. Then, $(\forall x)\neg A_1^2(x, x)$ is true in M, whereas $(\exists x)(\forall y)\neg A_1^2(x, y)$ is false in M. Hence, $(\forall x)\neg A_1^2(x, x) \Rightarrow (\exists x)(\forall y)\neg A_1^2(x, y)$ is not logically valid.

Exercises

2.139 Use semantic trees to determine whether the following wfs are logically valid.

(a) $(\forall x)(A_1^1(x) \vee A_2^1(x)) \Rightarrow ((\forall x)A_1^1(x)) \vee (\forall x)A_2^1(x)$
(b) $((\forall x)\mathscr{B}(x)) \wedge (\forall x)\mathscr{C}(x) \Rightarrow (\forall x)(\mathscr{B}(x) \wedge \mathscr{C}(x))$
(c) $(\forall x)(\mathscr{B}(x) \wedge \mathscr{C}(x)) \Rightarrow ((\forall x)\mathscr{B}(x)) \wedge (\forall x)\mathscr{C}(x)$
(d) $(\exists x)(A_1^1(x) \Rightarrow A_2^1(x)) \Rightarrow ((\exists x)A_1^1(x) \Rightarrow (\exists x)A_2^1(x))$
(e) $(\exists x)(\exists y)A_1^2(x, y) \Rightarrow (\exists z)A_1^2(z, z)$
(f) $((\forall x)A_1^1(x)) \vee (\forall x)A_2^1(x) \Rightarrow (\forall x)(A_1^1(x) \vee A_2^1(x))$
(g) $(\exists x)(\exists y)(A_1^2(x, y) \Rightarrow (\forall z)A_1^2(z, y))$
(h) The wfs of Exercises 2.24, 2.31(a, e, j), 2.39 and 2.40.
(i) The wfs of Exercise 2.21(a, b, g).

PROPOSITION 2.47

Assume that Γ is a set of closed wfs that satisfy the following closure conditions: (a) if $\neg\neg\mathscr{B}$ is in Γ, then $\mathscr{B}$ is in Γ; (b) if $\neg(\mathscr{B} \vee \mathscr{C})$ is in Γ, then $\neg\mathscr{B}$ and $\neg\mathscr{C}$ are in Γ; (c) if $\neg(\mathscr{B} \Rightarrow \mathscr{C})$ is in Γ, then $\mathscr{B}$ and $\neg\mathscr{C}$ are in Γ; (d) if $\neg(\forall x)\mathscr{B}$ is in Γ, then $(\exists x)\neg\mathscr{B}$ is in Γ; (e) if $\neg(\exists x)\mathscr{B}$ is in Γ, then $(\forall x)\neg\mathscr{B}$ is in Γ; (f) if $\neg(\mathscr{B} \wedge \mathscr{C})$ is in Γ, then at least one of $\neg\mathscr{B}$ and $\neg\mathscr{C}$ is in Γ; (g) if $\neg(\mathscr{B} \Leftrightarrow \mathscr{C})$ is in Γ, then either $\mathscr{B}$ and $\neg\mathscr{C}$ are in Γ, or $\neg\mathscr{B}$ and $\mathscr{C}$ are in Γ; (h) if $\mathscr{B} \wedge \mathscr{C}$ is in Γ, then so are $\mathscr{B}$ and $\mathscr{C}$; (i) if $\mathscr{B} \vee \mathscr{C}$ is in Γ, then at least one of $\mathscr{B}$ and $\mathscr{C}$ is in Γ, (j) if $\mathscr{B} \Rightarrow \mathscr{C}$ is in Γ, then at least one of $\neg\mathscr{B}$ and $\mathscr{C}$ is in Γ; (k) if $\mathscr{B} \Leftrightarrow \mathscr{C}$ is in Γ, then either $\mathscr{B}$ and $\mathscr{C}$ are in Γ or $\neg\mathscr{B}$ and $\neg\mathscr{C}$ are in Γ; (l) if $\forall x)\mathscr{B}(x)$ is in Γ, then $\mathscr{B}(b)$ is in Γ (where b is any individual constant that occurs in some wf of Γ); (m) if $(\exists x)\mathscr{B}(x)$ is in Γ, then $\mathscr{B}(b)$ is in Γ for some individual constant b. If no wf and its negation both belong to Γ and some wfs in Γ contain individual constants, then there is a model for Γ whose domain is the set D of individual constants that occur in wfs of Γ.

Proof

Define a model M with domain D by specifying that the interpretation of any predicate letter A_k^n in Γ contains an n-tuple $\langle b_1, \ldots, b_n \rangle$ if and only if $A_k^n(b_1, \ldots, b_n)$ is in Γ. By induction on the number of connectives and quantifiers in any closed wf $\mathscr{E}$, it is easy to prove: (i) if $\mathscr{E}$ is in Γ, then $\mathscr{E}$ is true in M; and (ii) if $\neg\mathscr{E}$, is in Γ, then $\mathscr{E}$ is false in M. Hence, M is a model for Γ.

If a branch of a semantic tree remains open, the set Γ of wfs of that branch satisfies the hypotheses of Proposition 2.47. If follows that, if a branch of a semantic tree remains open, then the set Γ of wfs of that branch has a model M whose domain is the set of individual constants that appear in that branch. This yields half of the basic principle of semantic trees.

PROPOSITION 2.48

If all the branches of a semantic tree are closed, then the wf $\mathscr{B}$ at the root of the tree is unsatisfiable.

Proof

From the derivation rules it is clear that, if a sequence of wfs starts at $\mathscr{B}$ and continues down the tree through the applications of the rules, and if the wfs in that sequence are simultaneously satisfiable in some model M, then that sequence can be extended by another application of a rule so that the added wf(s) would also be true in M. Otherwise, the sequence would form an unclosed branch, contrary to our hypothesis. Assume now that $\mathscr{B}$ is satisfiable in a model M. Then, starting with $\mathscr{B}$, we could construct an infinite branch in which all the wfs are true in M. (In the case of a branching rule, if there are two ways to extend the sequence, we choose the left-hand wf.) Therefore, the branch would not be closed, contrary to our hypothesis. Hence, $\mathscr{B}$ is unsatisfiable.

This completes the proof of the basic principle of semantic trees. Notice that this principle does not yield a decision procedure for logical validity. If a closed wf $\mathscr{B}$ is not logically valid, the semantic tree of $\neg\mathscr{B}$ may (and often does) contain an infinite unclosed branch. At any stage of the construction of this tree, we have no general procedure for deciding whether or not, at some later stage, all branches of the tree will have become closed. Thus, we have no general way of knowing whether $\mathscr{B}$ is unsatisfiable.

For the sake of brevity, our exposition has been loose and imprecise. A clear and masterful study of semantic trees and related matters can be found in Smullyan (1968).

2.16 QUANTIFICATION THEORY ALLOWING EMPTY DOMAINS

Our definition in Section 2.2 of *interpretations* of a language assumed that the domain of an interpretation is non-empty. This was done for the sake of simplicity. If we allow the empty domain, questions arise as to the right way of defining the truth of a formula in such a domain.† Once that is decided, the corresponding class of valid formulas (that is, formulas true in all interpretations, including the one with an empty domain) becomes smaller, and it is difficult to find an axiom system that will have all such formulas as its theorems. Finally, an interpretation with an empty domain has little or no importance in applications of logic.

Nevertheless, the problem of finding a suitable treatment of such a more inclusive logic has aroused some curiosity and we shall present one possible approach. In order to do so, we shall have to restrict the scope of the investigation in the following ways.

First, our languages will contain no individual constants or function letters. The reason for this restriction is that it is not clear how to interpret individual constants or function letters when the domain of the interpretation is empty. Moreover, in first-order theories with equality, individual constants and function letters always can be replaced by new predicate letters, together with suitable axioms.‡

Second, we shall take every formula of the form $(\forall x)\mathscr{B}(x)$ to be true in the empty domain. This is based on parallelism with the case of a non-empty domain. To say that $(\forall x)\mathscr{B}(x)$ holds in a non-empty domain D amounts to asserting

$$(*) \qquad \text{for any object } c, \text{if } c \in D, \text{then } \mathscr{B}(c)$$

When D is empty, '$c \in D$' is false and, therefore, 'if $c \in D$, then $\mathscr{B}(c)$' is true. Since this holds for arbitrary c, $(*)$ is true in the empty domain D, that is, $(\forall x)\mathscr{B}(x)$ is true in an empty domain. Not unexpectedly, $(\exists x)\mathscr{B}(x)$ will be false in an empty domain, since $(\exists x)\mathscr{B}(x)$ is equivalent to $\neg(\forall x)\neg\mathscr{B}(x)$.

These two conventions enable us to calculate the truth value of any closed formula in an empty domain. Every such formula is a truth-functional combination of formulas of the form $(\forall x)\mathscr{B}(x)$. Replace every subformula

†For example, should a formula of the form $(\forall x)(A_1^1(x) \wedge \neg A_1^1(x))$ be considered true in the empty domain?

‡For example, an individual constant b can be replaced by a new monadic predicate letter P, together with the axiom $(\exists y)(\forall x)(P(x) \Leftrightarrow x = y)$. Any axiom $\mathscr{B}(b)$ should be replaced by $(\forall x)(P(x) \Rightarrow \mathscr{B}(x))$.

$(\forall x)\mathscr{B}(x)$ by the truth value T and then compute the truth value of the whole formula.

It is not clear how we should define the truth value in the empty domain of a formula containing free variables. We might imitate what we do in the case of non-empty domains and take such a formula to have the same truth values as its universal closure. Since the universal closure is automatically true in the empty domain, this would have the uncomfortable consequence of declaring the formula $A_1^1(x) \wedge \neg A_1^1(x)$ to be true in the empty domain. For this reason, we shall confine our attention to sentences, that is, formulas without free variables.

A sentence will be said to be *inclusively valid* if it is true in all interpretations, including the interpretation with an empty domain. Every inclusively valid sentence is logically valid, but the converse does not hold. To see this, let $\mathbf{f}$ stand for a sentence $\mathscr{C} \wedge \neg\mathscr{C}$, where $\mathscr{C}$ is some fixed sentence. Now, $\mathbf{f}$ is false in the empty domain but $(\forall x)\mathbf{f}$ is true in the empty domain (since it begins with a universal quantifier). Thus the sentence $(\forall x)\mathbf{f} \Rightarrow \mathbf{f}$ is false in the empty domain and, therefore, not inclusively valid. However, it is logically valid, since every formula of the form $(\forall x)\mathscr{B} \Rightarrow \mathscr{B}$ is logically valid.

The problem of determining the inclusive validity of a sentence is reducible to that of determining its logical validity, since we know how to determine whether a sentence is true in the empty domain. Since the problem of determining logical validity will turn out to be unsolvable (by Proposition 3.54), the same applies to inclusive validity.

Now let us turn to the problem of finding an axiom system whose theorems are the inclusively valid sentences. We shall adapt for this purpose an axiom system PP$^{\#}$ based on Exercise 2.28. As axioms we take all the following formulas (see the Logical Axioms on p. 69):

(A1) $\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{B})$
(A2) $(\mathscr{B} \Rightarrow (\mathscr{C} \Rightarrow \mathscr{D})) \Rightarrow ((\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}))$
(A3) $(\neg\mathscr{C} \Rightarrow \neg\mathscr{B}) \Rightarrow ((\neg\mathscr{C} \Rightarrow \mathscr{B}) \Rightarrow \mathscr{C})$
(A4) $(\forall x)\mathscr{B}(x) \Rightarrow \mathscr{B}(y)$ if $\mathscr{B}(x)$ is a wf of $\mathscr{L}$ and y is a variable that is free for x in $\mathscr{B}(x)$. (Recall that, if y is x itself, then the axiom has the form $(\forall x)\mathscr{B} \Rightarrow \mathscr{B}$. In addition, x need not be free in $\mathscr{B}(x)$.)
(A5) $(\forall x)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow (\mathscr{B} \Rightarrow (\forall x)\mathscr{C})$ if $\mathscr{B}$ contains no free occurrences of x.
(A6) $(\forall y_1)\ldots(\forall y_n)(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow [(\forall y_1)\ldots(\forall y_n)\mathscr{B} \Rightarrow (\forall y_1)\ldots(\forall y_n)\mathscr{C}]$

together with all formulas obtained by prefixing any sequence of universal quantifiers to instances of (A1)–(A6).

Modus ponens (MP) will be the only rule of inference.

PP denotes the pure first-order predicate calculus, whose axioms are (A1)–(A5), whose rules of inference are MP and Gen, and whose language contains no individual constants or function letters. By Gödel's completeness theorem (Corollary 2.19), the theorems of PP are the same as the logically valid formulas in PP. Exercise 2.28 shows first that Gen is a derived rule of inference of PP$^{\#}$, that is, if $\vdash_{PP^{\#}} \mathscr{D}$, then $\vdash_{PP^{\#}} (\forall x)\mathscr{D}$, and second that PP and PP$^{\#}$ have the same theorems. Hence, the theorems of PP$^{\#}$ are the logically valid formulas.

Let PPS$^{\#}$ be the same system as PP$^{\#}$ except that, as axioms, we take only the axioms of PP$^{\#}$ that are sentences. Since MP takes sentences into sentences, all theorems of PPS$^{\#}$ are sentences. Since all axioms of PPS$^{\#}$ are axioms of PP$^{\#}$, all theorems of PPS$^{\#}$ are logically valid sentences. Let us show that the converse holds.

PROPOSITION 2.49

Every logically valid sentence is a theorem of PPS$^{\#}$.

Proof

Let $\mathscr{B}$ be any logically valid sentence. We know that $\mathscr{B}$ is a theorem of PP$^{\#}$. Let us show that $\mathscr{B}$ is a theorem of PPS$^{\#}$. In a proof of $\mathscr{B}$ in PP$^{\#}$, let $u_1, \ldots, u_n$ be the free variables (if any) in the proof, and prefix $(\forall u_1)\ldots(\forall u_n)$ to all steps of the proof. Then each step goes into a theorem of PPS$^{\#}$. To see this, first note that axioms of PP$^{\#}$ go into axioms of PPS$^{\#}$. Second, assume that $\mathscr{D}$ comes from $\mathscr{C}$ and $\mathscr{C} \Rightarrow \mathscr{D}$ by MP in the original proof and that $(\forall u_1)\ldots(\forall u_n)\mathscr{C}$ and $(\forall u_1)\ldots(\forall u_n)(\mathscr{C} \Rightarrow \mathscr{D})$ are provable in PPS$^{\#}$. Since $(\forall u_1)\ldots(\forall u_n)(\mathscr{C} \Rightarrow \mathscr{D}) \Rightarrow [(\forall u_1)\ldots(\forall u_n)\mathscr{C} \Rightarrow (\forall u_1)\ldots(\forall u_n)\mathscr{D}]$ is an instance of axiom (A6) of PPS$^{\#}$, it follows that $(\forall u_1)\ldots(\forall u_n)\mathscr{D}$ is provable in PPS$^{\#}$. Thus, $(\forall u_1)\ldots(\forall u_n)\mathscr{B}$ is a theorem of PPS$^{\#}$. Then n applications of axiom (A4) and MP show that $\mathscr{B}$ is a theorem of PPS$^{\#}$.

Not all axioms of PPS$^{\#}$ are inclusively valid. For example, the sentence $(\forall x)\mathbf{f} \Rightarrow \mathbf{f}$ discussed earlier is an instance of axiom (A4) that is not inclusively valid. So, in order to find an axiom system for inclusive validity, we must modify PPS$^{\#}$.

If P is a sequence of variables $u_1, \ldots, u_n$, then by $\forall P$ we shall mean the expression $(\forall u_1)\ldots(\forall u_n)$.

Let the axiom system ETH be obtained from PPS$^{\#}$ by changing axiom (A4) into:

(A4′) All sentences of the form $\forall P[(\forall x)\mathscr{B}(x) \Rightarrow \mathscr{B}(y)]$, where y is free for x in $\mathscr{B}(x)$ and x is free in $\mathscr{B}(x)$, and P is a sequence of variables that includes all variables free in $\mathscr{B}$ (and possibly others).

MP is the only rule of inference.

It is obvious that all axioms of ETH are inclusively valid.

LEMMA 2.50

If $\mathscr{T}$ is an instance of a tautology and P is a sequence of variables that contains all free variables in $\mathscr{T}$, then $\vdash_{\text{ETH}} \forall P\mathscr{T}$.

Proof

By the completeness of axioms (A1)–(A3) for the propositional calculus, there is a proof of $\mathscr{T}$ using MP and instances of (A1)–(A3). If we prefix $\forall P$ to all steps of that proof, the resulting sentences are all theorems of ETH. In the case when an original step $\mathscr{B}$ was an instance of (A1)-(A3), $\forall P\mathscr{B}$ is an axiom of ETH. For steps that result from MP, we use axiom (A6).

LEMMA 2.51

If P is a sequence of variables that includes all free variables of $\mathscr{B} \Rightarrow \mathscr{C}$, and $\vdash_{\text{ETH}} \forall P\mathscr{B}$ and $\vdash_{\text{ETH}} \forall P[\mathscr{B} \Rightarrow \mathscr{C}]$, then $\vdash_{\text{ETH}} \forall P\mathscr{C}$.

Proof

Use axiom (A6) and MP.

LEMMA 2.52

If P is a sequence of variables that includes all free variables of $\mathscr{B}, \mathscr{C}, \mathscr{D}$, and $\vdash_{\text{ETH}} \forall P[\mathscr{B} \Rightarrow \mathscr{C}]$ and $\vdash_{\text{ETH}} \forall P[\mathscr{C} \Rightarrow \mathscr{D}]$, then $\vdash_{\text{ETH}} \forall P[\mathscr{B} \Rightarrow \mathscr{D}]$.

Proof

Use the tautology $(\mathscr{B} \Rightarrow \mathscr{C}) \Rightarrow ((\mathscr{C} \Rightarrow \mathscr{D}) \Rightarrow (\mathscr{B} \Rightarrow \mathscr{D}))$, Lemma 2.50, and Lemma 2.51 twice.

LEMMA 2.53

If x is not free in $\mathscr{B}$ and P is a sequence of variables that contains all free variables of $\mathscr{B}$, $\vdash_{\text{ETH}} \forall P[\mathscr{B} \Rightarrow (\forall x)\mathscr{B}]$.

Proof

By axiom (A5), $\vdash_{\text{ETH}} \forall P[(\forall x)(\text{B} \Rightarrow \text{B}) \Rightarrow (\text{B} \Rightarrow (\forall x)\text{B})]$. By Lemma 2.50, $\vdash_{\text{ETH}} \forall P[(\forall x)(\mathscr{B} \Rightarrow \mathscr{B})]$. Now use Lemma 2.51.

COROLLARY 2.54

If $\mathscr{B}$ has no free variables, then $\vdash_{\text{ETH}} \mathscr{B} \Rightarrow (\forall x)\mathscr{B}$.

LEMMA 2.55

If x is not free in $\mathscr{B}$ and P is a sequence of variables that includes all variables free in $\mathscr{B}$, then $\vdash_{\text{ETH}} \forall P[\neg(\forall x)\mathbf{f} \Rightarrow ((\forall x)\mathscr{B} \Rightarrow \mathscr{B})]$.

Proof

$\vdash_{\text{ETH}} \forall P[\neg\mathscr{B} \Rightarrow (\mathscr{B} \Rightarrow \mathbf{f})]$ by Lemma 2.50. By Lemma 2.53, $\vdash_{\text{ETH}} \forall P [(\mathscr{B} \Rightarrow \mathbf{f}) \Rightarrow (\forall x)(\mathscr{B} \Rightarrow \mathbf{f})]$. Hence, by Lemma 2.52, $\vdash_{\text{ETH}} \forall P[\neg\mathscr{B} \Rightarrow (\forall x)(\mathscr{B} \Rightarrow \mathbf{f})]$. By axiom (A6), $\vdash_{\text{ETH}} \forall P[(\forall x)(\mathscr{B} \Rightarrow \mathbf{f}) \Rightarrow ((\forall x)\mathscr{B} \Rightarrow (\forall x)\mathbf{f})]$. Hence, by Lemma 2.52, $\vdash_{\text{ETH}} \forall P[\neg\mathscr{B} \Rightarrow ((\forall x)\mathscr{B} \Rightarrow (\forall x)\mathbf{f})]$. Since $[\neg\mathscr{B} \Rightarrow ((\forall x)\mathscr{B} \Rightarrow (\forall x)\mathbf{f})] \Rightarrow [\neg(\forall x)\mathbf{f} \Rightarrow ((\forall x)\mathscr{B} \Rightarrow \mathscr{B})]$ is an instance of a tautology, Lemmas 2.50 and 2.51 yield $\vdash_{\text{ETH}} \forall P[\neg(\forall x)\mathbf{f} \Rightarrow ((\forall x)\mathscr{B} \Rightarrow \mathscr{B})]$.

PROPOSITION 2.56

ETH + $\{\neg(\forall x)\mathbf{f}\}$ is a complete axiom system for logical validity, that is, a sentence is logically valid if and only if it is a theorem of the system.

Proof

All axioms of the system are logically valid. (Note that $(\forall x)\mathbf{f}$ is false in all interpretations with a non-empty domain and, therefore, $\neg(\forall x)\mathbf{f}$ is true in all such domains.) By Proposition 2.49, all logically valid sentences are provable in PPS$^{\#}$. The only axioms of PPS$^{\#}$ missing from ETH are those of the form $\forall P[(\forall x)\mathscr{B} \Rightarrow \mathscr{B}]$, where x is not free in $\mathscr{B}$ and P is any sequence of variables that include all free variables of $\mathscr{B}$. By Lemma 2.55, $\vdash_{\text{ETH}} \forall P[\neg(\forall x)\mathbf{f} \Rightarrow ((\forall x)\mathscr{B} \Rightarrow \mathscr{B})]$. By Corollary 2.54, $\forall P[\neg(\forall x)\mathbf{f}]$ will be derivable in ETH + $\{\neg(\forall x)\mathbf{f}\}$. Hence, $\forall P[(\forall x)\mathscr{B} \Rightarrow \mathscr{B}]$ is obtained by using axiom (A6).

LEMMA 2.57

If P is a sequence of variables that include all free variables of $\mathscr{B}$, $\vdash_{\text{ETH}} \forall P[(\forall x)\mathbf{f} \Rightarrow ((\forall x)\mathscr{B} \Leftrightarrow \mathbf{t})]$, where $\mathbf{t}$ is $\neg\mathbf{f}$.

Proof

Since $\mathbf{f} \Rightarrow \mathscr{B}$ is an instance of a tautology, Lemma 2.50 yields $\vdash_{\text{ETH}} \forall P(\forall x)[\mathbf{f} \Rightarrow \mathscr{B}]$. By axiom (A6), $\vdash_{\text{ETH}} \forall P[(\forall x)[\mathbf{f} \Rightarrow \mathscr{B}] \Rightarrow [(\forall x)\mathbf{f} \Rightarrow (\forall x)\mathscr{B}]]$. Hence, $\vdash_{\text{ETH}} \forall P[(\forall x)\mathbf{f} \Rightarrow (\forall x)\mathscr{B}]$ by Lemma 2.51. Since $(\forall x)\mathscr{B} \Rightarrow [(\forall x)\mathscr{B} \Leftrightarrow \mathbf{t}]$ is an instance of a tautology, Lemma 2.50 yields $\vdash_{\text{ETH}} \forall P[(\forall x)\mathscr{B} \Rightarrow [(\forall x)\mathscr{B} \Leftrightarrow \mathbf{t}]]$. Now, by Lemma 2.52, $\vdash_{\text{ETH}} \forall P[(\forall x)\mathbf{f} \Rightarrow [(\forall x)\mathscr{B} \Leftrightarrow \mathbf{t}]]$.

Given a formula $\mathscr{B}$, construct a formula $\mathscr{B}^*$ in the following way. Moving from left to right, replace each universal quantifier and its scope by $\mathbf{t}$.

LEMMA 2.58

If P is a sequence of variables that include all free variables of $\mathscr{B}$, then $\vdash_{\text{ETH}} \forall P[(\forall x)\mathbf{f} \Rightarrow [\mathscr{B} \Leftrightarrow \mathscr{B}^*]]$.

Proof

Apply Lemma 2.57 successively to the formulas obtained in the stepwise construction of $\mathscr{B}^*$. We leave the details to the reader.

PROPOSITION 2.59

ETH is a complete axiom system for inclusive validity, that is, a sentence $\mathscr{B}$ is inclusively valid if and only if it is a theorem of ETH.

Proof

Assume $\mathscr{B}$ is a sentence valid for all interpretations. We must show that $\vdash_{\text{ETH}} \mathscr{B}$. Since $\mathscr{B}$ is valid in all non-empty domains, Proposition 2.56 implies that $\mathscr{B}$ is provable in ETH + $\{\neg(\forall x)\mathbf{f}\}$. Hence, by the deduction theorem,

$$(+) \quad \vdash_{\text{ETH}} \neg(\forall x)\mathbf{f} \Rightarrow \mathscr{B}.$$

Now, by Lemma 2.58,

$$(\%) \quad \vdash_{\text{ETH}} (\forall x)\mathbf{f} \Rightarrow [\mathscr{B} \Leftrightarrow \mathscr{B}^*]$$

(Since $\mathscr{B}$ has no free variables, we can take P in Lemma 2.58 to be empty.) Hence, $[(\forall x)\mathbf{f} \Rightarrow [\mathscr{B} \Leftrightarrow \mathscr{B}^*]]$ is valid for all interpretations. Since $(\forall x)\mathbf{f}$ is valid in the empty domain and $\mathscr{B}$ is valid for all interpretations, $\mathscr{B}^*$ is valid in the empty domain. But $\mathscr{B}^*$ is a truth-functional combination of $\mathbf{t}$s.

So, $\mathscr{B}^*$ must be truth-functionally equivalent to either $\mathbf{t}$ or $\mathbf{f}$. Since it is valid in the empty domain, it is truth-functionally equivalent to $\mathbf{t}$. Hence, $\vdash_{\mathrm{ETH}} \mathscr{B}^*$. Therefore by (%), $\vdash_{\mathrm{ETH}} (\forall x)\mathbf{f} \Rightarrow \mathscr{B}$. This, together with (+), yields $\vdash_{\mathrm{ETH}} \mathscr{B}$.

The ideas and methods used in this section stem largely, but not entirely, from a paper by Hailperin (1953).[†] That paper also made use of an idea in Mostowski (1951b), the idea that underlies the proof of Proposition 2.59. Mostowski's approach to the logic of the empty domain is quite different from Hailperin's and results in a substantially different axiom system for inclusive validity. For example, when $\mathscr{B}$ does not contain x free, Mostowski interprets $(\forall x)\mathscr{B}$ and $(\exists x)\mathscr{B}$ to be $\mathscr{B}$ itself. This makes $(\forall x)\mathbf{f}$ equivalent to $\mathbf{f}$, rather than to $\mathbf{t}$, as in our development.

[†]The name ETH comes from 'empty domain' and 'Theodore Hailperin'. My simplification of Hailperin's axiom system was suggested by a similar simplification in Quine (1954).

Bibliography

Listed here are not only books and papers mentioned in the text but also some other material that will be helpful in a further study of mathematical logic. We shall use the following abbreviations.

AML	for *Annals of Mathematical Logic*
AMS	for American Mathematical Society
Arch.	for *Archiv für mathematische Logik und Grundlagenforschung*
FM	for *Fundamenta Mathematicae*
HML	for *Handbook of Mathematical Logic*, Springer-Verlag
HPL	for *Handbook of Philosophical Logic*, Reidel
JSL	for *Journal of Symbolic Logic*
MTL	for *Model-Theoretic Logics*, Springer-Verlag
NDJFL	for *Notre Dame Journal of Formal Logic*
NH	for North-Holland Publishing Company
ZML	for *Zeitschrift für mathematische Logik und Grundlagen der Mathematik* (since 1993, *Mathematical Logic Quarterly*)

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Notation

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